ECONOMICS

PART 1

CPA SECTION 2
CCP SECTION 2
CIFA SECTION 2
CS SECTION 2

STUDY TEXT

GENERAL OBJECTIVES

This paper is intended to equip the candidate with knowledge, skills and attitudes that will enable him/her to apply the fundamental principles of economics in decision making

4.0 LEARNING OUTCOMES

A candidate who passes this paper should be able to:

- Apply basic mathematical and graphical techniques to analyse economic relationships and interpret the results
- Apply the knowledge of economics in decision making
- Analyse economic problems and suggest possible policy related recommendations
- Apply knowledge of economics in international trade and finance
- Apply economic principles in the development and implementation of policies in agriculture and industry
- Demonstrate an understanding of emerging economic issues.

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MICROECONOMICS

CHAPTER 1

INTRODUCTION TO ECONOMICS

DEFINITION OF ECONOMICS

The modern word "Economics" has its origin in the Greek word "Oikonomos" meaning a steward. The two parts of this word "Oikos", a house and "nomos", a manager sum up what economics is all about. How do we manage our house, what account of stewardship can we render to our families, to the nation, to all our descendants?

There is an economic aspect to almost any topic we care to mention – education, employment, housing, transport, defence etc. Economics is a comprehensive theory of how the society works. But as such, it is difficult to define. The great classical economist Alfred Marshal defined economics as the *"Study of man in the ordinary business of life*".

This, however, is rather too vague a definition. This is because any definition should take account of the guiding idea in economics which is *scarcity*. The great American economist Paul Samuelson thus defined it as: *"The study of how people and society choose to employ scarce resources that could have alternative uses in order to produce various commodities and to distribute them for consumption, now or in future amongst various persons and groups in society*. Virtually everything is scarce; not just diamonds and oil but also bread and water. The word scarcity as used in economics means that; All *resources are scarce in the sense that there are not enough to fill everyone's wants to the point of satiety*.

We therefore have limited resources, both in rich countries and in poor countries. The economist's job is to evaluate the choices that exist for the use of these resources. Thus we have another characteristic of economics; it is concerned with *choice*.

Another aspect of the problem is people themselves; they do not just want more food or more clothing they want particular types of food, specific items of clothing and so on. By want we mean; *"A materialistic desire for an activity or an item. Human wants are infinite.*

We have now assembled the three vital ingredients in our definition, People (human wants), Scarcity and choice. Thus for our purpose we could define economics as:

"The social science which is concerned with the allocation of scarce resources to provide goods and services which meet the needs and wants of the consumers"

(ii) The Scope of Economics

The study of economics begins with understanding of human "wants". Scarcity forces us to economise. We weigh up the various alternatives and select that particular assortment of goods which yields the highest return from our limited resources. Modern economists use this idea to define the scope of their studies.

Although economics is closely connected with such social sciences as ethics, politics, sociology, psychology and anthropology, it is distinguished from them by its concentration on one particular aspect of human behaviour – choosing between alternatives in order to obtain the maximum satisfaction from limited resources.

In effect, the economist limits the study by selecting four fundamental characteristics of human existence and investigating what happens when they are all found together, as they usually are.

First, the ends of human beings are without limit. Second, those ends are of varying importance.

Third, the means available for achieving those ends – human time and energy and material resources – are limited. Fourth, the means can be used in many different ways: that is, they can produce many different goods.

But no single characteristic by itself is necessarily of interest to the economist. Only when all four characteristics are found together does an economic problem arise.

Resources: The ingredients that are combined together by economists and termed economic goods i.e. goods that are scarce in relation to the demand for them.

i. Economic Goods: All things which people want are lumped together by economists and termed economic goods i.e. goods that are scarce in relation to the demand for them.
ii. Free Goods: These are goods which people can have as much as they want, e.g. air.

MICROECONOMICS AND MACROECONOMICS

Overall the study of economics is divided into two halves, microeconomics and macroeconomics.

(a) **"Micro"** comes from the Greek word meaning small, and microeconomics is the study of individual economic units or particular parts of the economy e.g. how does an individual household decide to spend its income? How does an individual firm decide what volume of output to produce or what products to make? How is price of an individual product determined? How are wage levels determined in a particular industry? It thus gives a worm's eye view of the economy.

(b) **"Macro"** comes from the Greek word meaning large, and macroeconomics is the study of "global" or collective decisions by individual households or producers. It looks at a national or international economy as a whole, e.g. Total Output, Income and Expenditure,

Unemployment, Inflation Interest Rates and Balance of International Trade, etc and what economic policies a government can pursue to influence the conditions of the national economy. It thus gives a bird's eye-view of the economy.

Economic theory

A body of economic principles built up as a result of logical reasoning, it provides the tools of economic analysis. It is pursued irrespective of whether it appears to be of any practical advantage or not.

THE METHODOLOGY OF ECONOMICS AND ITS BASIC CONCEPTS

Economics proceeds as an evolutionary discipline, looking at data, developing hypotheses, testing them and reaching sometimes uneasy consensus on how the economy works. This is called the scientific method which begins with the formulation of a theory about behaviour. For example, we may put forward the idea that the demand for a good is determined by its price. On the basis of this we may reason that as the price is increased, demand goes down, while if the prices are decreased the demand will go up. This then gives us a hypothesis which can be tested on observed behaviour. This testing of ideas on the evidence is known as *empiricism*.

Ceteris paribus

The economic world is extremely complicated. There are millions of people and firms; thousands of prices and industries. One possible way of figuring out economic laws in such a setting is by **controlled experiments**. A controlled experiment takes place when everything else but the item under investigation is held constant. This is an essential component of scientific method.

However economists have no such luxury when testing economic laws. Therefore, when formulating economic principles economists are usually careful to state that such and such will happen, **ceteris paribus** which is the Latin expression meaning **all other things remaining constant**.

ECONOMIC DESCRIPTION AND ANALYSIS

Economics is used in two important ways today. The first is to describe, explain and predict the behaviour of production, inflation, incomes etc. But for many, the fruit of such labours is found in a second task – to improve economic performance.

Thus, we first attempt to describe the hardships of poverty. We then might present programs that could reduce the extent of poverty. Or we might start with an analysis of how higher energy taxes would lead to lower energy consumption. We might then conclude that the country should raise its gasoline taxes.

In each case, we first engage in positive economics, and then in normative economics.

Positive and Normative Economics You may already have strong personal views about what sort of economic society we should have e.g. whether a free market “capitalist” economy is desirable, or whether a “communist” command economy is preferable. In our study of economics, one of the central distinctions is between a value judgement and a factual statement. Positive Economics is concerned with the objective statements about what does happen or what will happen. It limits itself to statements that can be verified by reference to facts e.g. How does a higher level of unemployment affect inflation or how will a gasoline tax affect gasoline usage? A positive approach

is more objective, and more scientific and it is the approach we shall try to take in our study of economics here.

Normative Economics, on the other hand, appreciates that in practice many economic decisions involve subjective judgements; that its, they cannot be made solely by an objective appraisal of the facts but depend to some extent on personal views in interpreting facts – ethics and value judgements. They can be argued about but they can never be settled by science or by appeal to facts, e.g. should taxation soak the rich to help the poor? Or should the defence spending grow at 3 or 5 or 10 per cent per year? They involve what ought to be and are settled by political choice.

Economic goals and problems

Whatever political party is in power, four main economic goals are:

- control of inflation
- reduction of unemployment
- promotion of economic growth
- attainment of a favourable balance of payments.

In addition to these generally agreed objectives, more "political" economic policies might be pursued, such as the redistribution of income.

SCARCITY, CHOICE, OPPORTUNITY COST AND PRODUCTION POSSIBILITY FRONTIERS AND CURVES

(i) Scarcity

To the economists all things are said to be scarce, since by "scarce" they mean simply "that there are not enough to fill everyone's wants to the point of satiety". Most people would probably like to have more of many things or goods of better quality than they possess at present: larger houses perhaps in which to live, better furnished with the latest labour-saving devices, such as electric washers, cookers, refrigeration; more visits to theatre or the concert hall; more travel; the latest models in motor cars; radios and television sets; and most women exhibit an apparently insatiable desire for clothes. People's wants are many, but the resources for making the things they want – labour, land, raw materials, factory buildings, machinery – are themselves limited in supply. There are insufficient productive resources in the world, therefore, to produce the amount of goods and services that would be required to satisfy everyone's wants fully. Consequently, to the economist all things are at all times said to be "scarce".

(ii) CHOICE AND OPPORTUNITY COST

Because there are not enough resources to produce everything we want, a choice must be made about which of the wants to satisfy. In economics, it is assumed that people always choose the alternative that will yield them the greatest satisfaction. We therefore talk of Economic Man. Choice involves *sacrifice*. If there is a choice between having guns and having butter, and a country chooses to have guns, it will be giving up butter to the guns. The cost of having guns can therefore be

regarded as the sacrifice of not being able to have butter. The cost of an item measured in terms of the alternative forgone is called its opportunity cost.

(iii) PRODUCTION POSSIBILITIES AND OPPORTUNITY COSTS

Limitations of the total resources capable of producing different commodities forces society to choose between relatively scarce commodities. This can be illustrated quantitatively by simple arithmetic examples and geometrical diagrams.

Suppose, to take an example, that a society can spend money on two products, guns and butter. The society's resources are limited; therefore there are restrictions on the amount of guns and butter that can be made, which can be shown by a "production possibility" or "transformation curve".

ALTERNATIVE PRODUCTION POSSIBILITIES

Possibilities	Butter (million pounds)	Guns(thousands)
A	0	15
B	1	14
C	2	12
D	3	9
E	4	5
F	5	0

Table 1.1: Full employment of scarce resources implies guns- butter trade off

The above possibilities can be illustrated graphically using a production possibility frontier. By production possibility frontier we mean; "A geometric representation of production possibilities of two commodities feasible within an economy, given a fixed quantity of available resources and constant technological conditions.

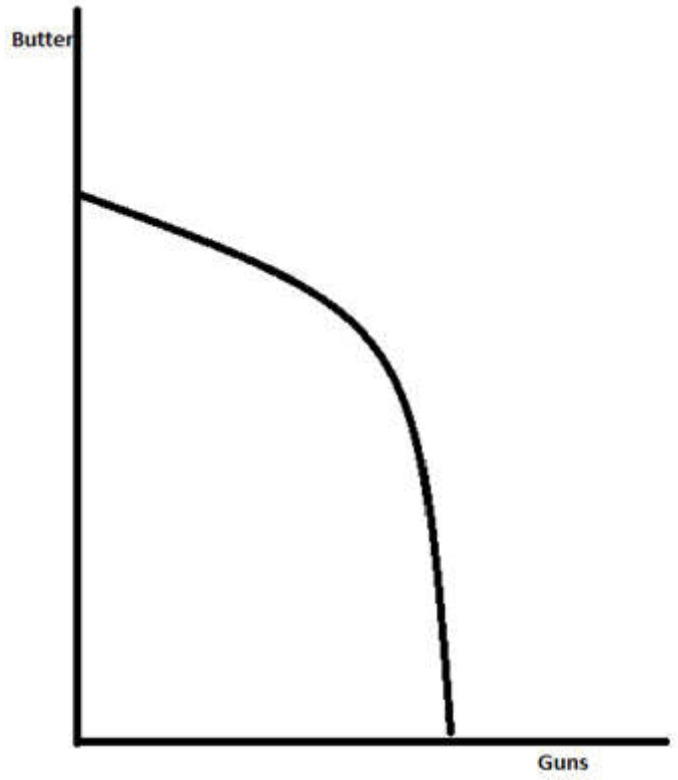


Figure Possibilities of transforming butter into guns.

The concave (to the origin) shape of the curve stems from an assumption that resources are not perfectly occupationally mobile. Points outside the P.P frontier (to the North East) are unattainable under the present technical know-how. Points inside it say, H, would be inefficient since resources are not being fully employed, resources are not being properly used, or outdated production techniques are utilized.

If production is on the frontier the resources are being fully utilized. Points on the production possibility curve such as B,C and E show the maximum possible output of the two commodities.

Output G will only become a production possibility if the country's ability to produce increases and the production possibility curve moves outwards. This can happen when there are changes such as increase in the labour force, increase in the stock of capital goods (factories power stations, transport networks, machinery) and/or an increase in technical knowledge.

(iv) Some of the uses of the possibility frontier

The production-possibility Frontier represented as a single curve can help introduce many of the most basic concepts of Economics.

The central economic problem

There are many economic problems which we encounter everyday – poverty, inflation, unemployment etc. However if we use the term *The Economic Problem* we are referring to the overall problem of the scarcity of resources. Each society has to make the best use of scarce

resources. The great American economist Paul A. Samuelson said that every economic society has to answer three fundamental questions;

What commodities shall be produced, clothes, food, cars, submarines etc. and in what quantities? *How* shall goods be produced? That is given that we have scarcity of resources of land, labour etc, how should we combine them to produce goods and services which we want?

For whom shall goods be produced? Who is to enjoy and get the benefit of the nation's goods and services? Or to put it in another way, how is national product to be divided among different individuals and families?

ECONOMIC SYSTEMS, FREE ECONOMY, PLANNED ECONOMY AND MIXED ECONOMY

While there are a million variations on answers to these questions; when we look around the world we find that there are only a limited number of ways in which societies have set about answering them. These ways or methods are called Economic systems. They are free enterprise, centrally planned and mixed economies. We will now examine these briefly.

a) THE FREE ECONOMY /ENTERPRISE: THE PRICE SYSTEM

The free market system is where the decision about what is produced is the outcome of millions of separate individual decisions made by consumers, producers and owners of productive services. The decisions reflect private preferences and interests.

For the free enterprise to operate there must be a *price system/mechanism*.

The price system is the situation where the vital economic decisions in the economy are reached through the workings of the market price.

Thus, everything – houses, labour, food, land etc come to have its market price, and it is through the workings of the market prices that the "What?", "How?", and "For whom?" decisions are taken. The free market thus gives rise to what is called **Consumer Sovereignty** – a situation in which consumers are **the ultimate dictators**, subject to the level of technology, of the kind and quantity of commodities to be produced. Consumers are said to exercise this power by bidding up the prices of the goods they want most; and suppliers, following the lure of higher prices and profits, produce more of the goods.

The features of a free market system are:

(i) Ownership of Means of Production

Individuals are free to own the means of production i.e. land, capital and *enjoy incomes from* them in the form of rent, interest and profits.

(ii) Freedom of Choice and Enterprise

Entrepreneurs are free to *invest in businesses of their choice, produce any product of their choice, workers are free to sell their labour in occupations and industries of their choice; Consumers are free to consume products of their choice.*

(iii) Self Interest as the Dominating Motive

Firms aim at maximising their profits, workers aim at maximising their wages, landowners aim at maximising their return from their land, and consumers at maximising their satisfaction

(iv) Competition

Economic rivalry or competition envisages a situation where, in the market for each commodity, there are a large number of buyers and sellers. It is the forces of total demand and total supply which determine the market price, and each participant, whether buyer or seller, must take this price as given since it's beyond his or her influence or control.

(v) Reliance on the Price Mechanism

Price mechanism is where the prices are determined on the market by supply and demand, and consumers base their expenditure plans and producers their production plans on market prices.

Price mechanism rations the scarce goods and services in that, those who can afford the price will buy and those who cannot afford the price will not pay.

(vi) Limited Role of Government

In these systems, apart from playing its traditional role of providing defence, police service and such infrastructural facilities as roads for public transport, the Government plays a very limited role in directly economic profit making activities.

Resource allocation in a free enterprise

Although there are no central committees organising the allocation of resources, there is supposed to be no chaos but order. The major price and allocation decisions are made in the markets. The market being *the process by which the buyers and sellers of a good interact to determine its* price and quantity.

If more is wanted of any commodity say wheat – a flood of new orders will be placed for it. As the buyers scramble around to buy more wheat, the sellers will raise the price of wheat to ration out a limited supply. And the higher price will cause more wheat to be produced. The reverse will also be true.

What is true of the market for commodities is also true for the markets for factors of production such as labour, land and capital inputs.

People, by being willing to spend money, signal to producers what it is they wish to be produced. **Thus what** things will be produced will is determined by the shilling votes of consumers, not every five years at the polls, but every day in their decisions to purchase this item and not that. The "How?" questions is answered because one producer has to compete with others in the market; if that producer cannot produce as cheaply as possible then customers will be lost to competitors. Prices are the signals for the appropriate technology.

The "for whom?" question is answered by the fact that anyone who has the money and is willing to spend it can receive the goods produced. Who has the money is determined by supply and demand in the markets for factors of production (i.e. land, labour, and capital). These markets determine the wage rates, land rents, interests rates and profits that go to make up people's incomes. The

distribution of income among the population is thus determined by **amounts** of factors (person-hours, Acres etc) owned and the prices of the factors (wages-rates, land-rents etc).

Advantages of a Free Market System

Incentive: People are encouraged to work hard because opportunities exist for individuals to accumulate high levels of wealth.

Choice: People can spend their money how they want; they can choose to set up their own firm or they can choose for whom they want to work.

Competition: Through competition, less efficient producers are priced out of the market; more efficient producers supply their own products at lower prices for the consumers and use factors of production more efficiently. The factors of production which are no longer needed can be used in production elsewhere. Competition also stimulates new ideas and processes, which again leads to efficient use of resources.

A free market also responds well to changes in consumer wishes, that is, it is flexible.

Because the decision happen in response to change in the market there is no need to use additional resources to make decisions, record them and check on whether or not they are being carried out. The size of the civil service is reduced.

Disadvantages of a Free Economy

The free market gives rise to certain inefficiencies called market failures i.e. where the market system fails to provide an optimal allocation of resources. These include:

Unequal distribution of wealth: The wealthier members of the society tend to hold most of the economic and political power, while the poorer members have much less influence. There is an unequal distribution of resources and sometimes production concentrates on luxuries i.e. the wants of the rich. This can lead to excessive numbers of luxury goods being produced in the economy.

It may also result to social problems like crimes, corruption, etc.

Public goods: These are goods which provide benefits which are not confined to one individual household i.e. possess the characteristic of non-rival consumption and non-exclusion. The price mechanism may therefore not work efficiently to provide these services e.g. defence, education and health services.

Externalities: Since the profit motive is all important to producers, they may ignore social costs production, such as pollution. Alternatively, the market system may not reward producers whose activities have positive or beneficial effects on society.

Hardship: Although in theory factors of production such as labour are “mobile” and can be switched from one market to another, in practice this is a major problem and can lead to hardship through unemployment. It also leads to these scarce factors of production being wasted by not using them to fullest advantage.

Wasted or reduced competition: some firms may use expensive advertising campaigns to sell "new" products which are basically the same as may other products currently on sale. Other firms, who control most of the supply of some goods may choose to restrict supply and therefore keep prices artificially high; or, with other suppliers, they may agree on the prices to charge and so price will not be determined by the interaction of supply and demand.

The operation of a free market depends upon producers having the confidence that they will be able to sell what they produce. If they see the risk as being unacceptable, they will not employ resources, including labour and the general standard of living of the country will fall..

b) PLANNED ECONOMIES

Is a system where all major economic decisions are made by a government ministry or planning organisation. Here all questions about the allocation of resources are determined by the government.

Features of this system

The command economies relies exclusively on the state. The government will decide what is made, how it is made, how much is made and how distribution takes place. The resources – factors of production – on behalf of the producers and consumers. Price levels are not determined by the forces of supply and demand but are fixed by the government.

Although division of labour and specialisation are found, the planned economies tend to be more self-sufficient and tend to take part in less international trade than market economies.

Advantages of Planned System

i. **Uses of resources**: Central planning can lead to the full use of all the factors of production, so reducing or ending unemployment.
ii. **Large scale production**: Economies of scale become possible due to mass production taking place.
iii. **Public services**: "Natural monopolies" such as the supply of domestic power or defence can be provided efficiently through central planning.
iv. **Basic services**: There is less concentration on making luxuries for those who can afford them and greater emphasis on providing a range of goods and services for all the population.
v. There are less dramatic differences in wealth and income distribution than in market economy

Disadvantages of the Planned System

The centrally planned economies suffer from the following limitations:

Lack of choice: Consumers have little influence over what is produced and people may have little to say in what they do as a career.

Little incentive: Since competition between different producers is not as important as in the market economy, there is no great incentive to improve existing systems of production or work.

Workers are given no real incentives to work harder and so production levels are not as high as they could be.

Centralised control: Because the state makes all the decisions, there must be large influential government departments. The existence of such a powerful and large bureaucracy can lead to inefficient planning and to problems of communication. Furthermore, government officials can become over privileged and use their position for personal gain, rather than for the good of the rest of the society.

The task of assessing the available resources and deciding on what to produce, how much to produce and how to produce and distribute can be too much for the central planning committee.

Also the maintenance of such a committee can be quite costly.

C) THE MIXED ECONOMY

There are no economies in the world which are entirely 'market' or planned, all will contain elements of both systems.

The degree of mix in any one economy is the result of a complex interaction of cultural, historic and political factors. For example the USA which is a typical example of a largely work-based society, but the government still plans certain areas of the economy such as defence and provides very basic care for those who cannot afford medical insurance.

FEATURES

The mixed economy includes elements of both market and planned economies. The government operates and controls the public sector, which typically consists of a range of public services such as health and education, as well as some local government services. The private sector is largely governed by the force of mechanism and "market forces", although in practice it is also controlled by various regulations and laws.

Some services may be subsidised, provided at a loss but kept for the benefit of society in general(many national railways, for example, are loss making), other services such as education or the police may be provided free of charge (though they are paid for through the taxation system). The private sector is regulated, i.e. influenced by the price mechanism but also subject to some further government control, such as through pollution, safety and employment regulation.

Advantages of the Mixed Economy

Necessary services are provided in a true market economy, services which were not able to make profit would not be provided.

Incentive: Since there is a private sector where individuals can make a lot of money, incentives still exist in the mixed economy.

Competition: Prices of goods and services in the private sector are kept down through competition taking place.

Disadvantages of Mixed Economy

Large monopolies can still exist in the private sector, and so competition does not really take place. There is likely to be a lot of bureaucracy and "red tape" due to existence of a public sector.

SPECIALIZATION AND EXCHANGE

a) **Specialization**

The economies of mass production upon which modern standards of living are based would not be possible if production took place in self-sufficient farm households or regions.

As such, many societies and individuals specialize or concentrate on only one activity or type of production.

Division of labour and specialisation

Division of labour refers to the situation in which the production process is split into very large number of individual operations and each operation is the special task of one worker. The workers then specialise on one activity. Four distinct stages can be distinguished in the development of division of labour and specialization.

- Specialisation by craft
- Specialisation by process
- Regional specialisation
- International division of labour

Advantages of Division of Labour

(i) Greater skill of worker

The constant repetition of a task makes its performance almost automatic. The workers thus acquire greater skills at their job.

(ii) A saving of time

By keeping to a single operation, a worker can accomplish a great deal more, since he wastes less time between operations. Less time, too, is required learning how to perform a single operation than to learn a complete trade.

(iii) Employment of specialists

Specialisation makes it possible for each workman to specialise in the work for which he has the greatest aptitude

(iv) Use of machinery

Specialisation permits the use of some tools specific to a particular task, which can make the life of a worker that much easier.

(v) Less fatigue

It is sometimes claimed that the worker, habituated to the repetition of simple tasks, becomes less fatigued by his work.

Disadvantages of Division of Labour and Specialisation

(i) Monotony

Doing the same work repeatedly can result in boredom, and this can offset the efficiency that would otherwise result from experience.

(ii) Decline of craftsmanship

If a person does the same kind of work repeatedly according to laid down routine, he loses initiative for innovation and this can lead to loss of job satisfaction.

(iii) Greater risk of unemployment

If a worker is highly specialised, he can be easily unemployed if something goes wrong with the product of his industry (e.g. if the product is found to have negative effects to health, and demands for it falls) or if a machine is introduced to perform his work.

(iv) Increased interdependency

Since each worker contributes only a small part towards the completion of the final product, the efficiency and success of the whole process will depend on the efficiency and co-operation of all the workers. If some of the workers are inefficient, they can frustrate the whole system even if the rest of the workers are doing their work properly.

b) Exchange

When societies or individuals specialize, they are likely to produce a flood of "surplus" goods.

They are thus bound to exchange this surplus for what they don't produce. In primitive cutlers, this exchange will take place in the form of barter. For example, it is not uncommon for food to be exchanged for weapons; or for aid in the building of a house to be exchanged for aid in cleaning a field. But exchange today in all economies – capitalist or communist takes place through he medium of money.

Rationality

One of the most important assumptions in economics and on which much **economic theory** is based, is the rationality of human behaviour. In order to make predictions about human behaviour, economists assume that human behaviour is **"rational"** and that consumers and producers act rationally e.g. in what they will decide to buy or produce at any given price.

ECONOMICS FOR ACCOUNTANTS

A few teachers and some students have questioned the rationale for including economics in a course of study for professional accountants. In order to appreciate the need for the knowledge of economics by accountants it is necessary to know something of the accountant's role. It might be necessary to provide a brief survey of accountancy before going to the value of economics to the accountant.

(i) Accountancy

In general terms accounting consists of procedures for recording, classifying and interpreting selected experiences of an enterprise to promote effective administration. More specially, the accounting function can and often is broken down into specializations, a common distinction being made between **management accounting** and **financial accounting.** Briefly put, the role of the management accountant is to provide management with the best possible information upon which decisions can be based and enable both effective use of an organisations resources. The older specialization of cost accounting is perhaps best considered as part of management accounting which establishes budgets, standard costs and actual costs of operation and processes. Financial accounting by contrast is concerned with the analysis, classification and recording of financial transactions in order to illustrate the effects on the performance and financial position of an undertaking. Both aspects of the accounting function must be executed if the organization is to have adequate information for its management to formulate policy and to plan and control operations.

(ii) The role of economic knowledge

In no type of organisation can the accountant operate in isolation, however. He/she must have a working knowledge of many other areas, which impinge on the business or undertaking. The most relevant fields of knowledge are considered to be law, management, statistics, behavioural studies, information technology and economics.

The accountant is not expected to be an expert in these subject areas but to have sufficient knowledge to relate intelligently with specialists in such areas and to know enough to appreciate when and where to go for this specialist knowledge.

As part of the management team or advisor to that team, the accountant needs to appreciate the opportunities and constrains which the economic environment offers or impose on the organisation. This is true whether the organisation is in the private or public sector. All organisations must use the **scarce resources** available to them in an effective and efficient manner if the members of the organisations and the society generally are to gain maximum benefit. Given that allocation of resources is a central concern of economics, the relevance of economics for the accountant follows.

The accountant as a key provider of financial information for planning, control and decision making purposes will be better equipped to provide relevant information if he/she is aware of the organisational objectives, and the environmental constrains within which those objective are pursued.

As a final word one can also say that accountants need economics to understand analyse and solve economic problems of the organisation and society in general.

QUESTIONS:

1. Write short notes on the following:

   a) Scarcity and choice
   b) Opportunity cost
   c) Production possibility frontier
   d) Positive and normative economics

2. Specialisation brings its benefits but it also has limitations. Discuss the statement.

CHAPTER 2

DEMAND, SUPPLY AND DETERMINATION OF EQUILIBRIUM

DEMAND ANALYSIS

Introduction

In any economy there are millions of individuals and institutions and to reduce things to a manageable proportion they are consolidated into three important groups; namely

- Households
- Firms
- Central Authorities

These are the **dramatis personae** of the economic theory and the stage on which much of their play is acted is called the MARKET (see lesson three for definition of market).

Household

This refers to all the people who live under one roof and who make or are subject to others making for them, joint financial decisions. The household decisions are assumed to be consistent, aimed at **maximizing utility** and they are the principal owners of the factors of production. In return for the factors or services of production supplied, they get or receive their income e.g.

- Labour – wages and salaries
- Capital – interest
- Land – rent
- Enterprise – profit

The firm

The unit that uses factors of production to produce commodities then it sells either to other firms, to household, or to central authorities. The firm is thus the unit that makes the decisions regarding the employment of the factors of production and the output of commodities. They are assumed to be aiming at **maximizing profits**.

Central authorities

This comprehensive term includes all public agencies, government bodies and other organisations belonging to or under the direct control of the government. They exist at the centre of legal and political power and exert some control over individual decisions taken and over markets.

DEMAND ANALYSIS

DEFINITION

Demand is the quantity per unit of time, which consumers (households) are willing and able to buy in the market at alternative prices, other things held constant.

INDIVIDUAL VERSUS MARKET DEMAND

(i) Individual and market demand schedule

The plan of the possible quantities that will be demanded at different prices by an individual is called **Individual demand schedule.** Such a demand schedule is purely hypothetical, but it serves to illustrate the **First Law of Demand and Supply that more of a commodity will be bought at a lower than a higher price**.

PRICE (SH)	QUANTITY DEMANDED PER WEEK
20	3
18	3.5
16	4
14	5
13	6
12	7
11	8
10	9

Table : The individual demand schedule

Theoretically, the demand schedule of all consumers of a given commodity can be combined to form **a composite demand schedule,** representing the total demand for that commodity at various prices. This is called the **Market demand schedule**.

PRICE(SH)	QUANTITY DEMANDED PER WEEK
20	100000
18	120000
16	135000
14	150000
13	165000
12	180000
11	200000
10	240000
9	300000
8	350000

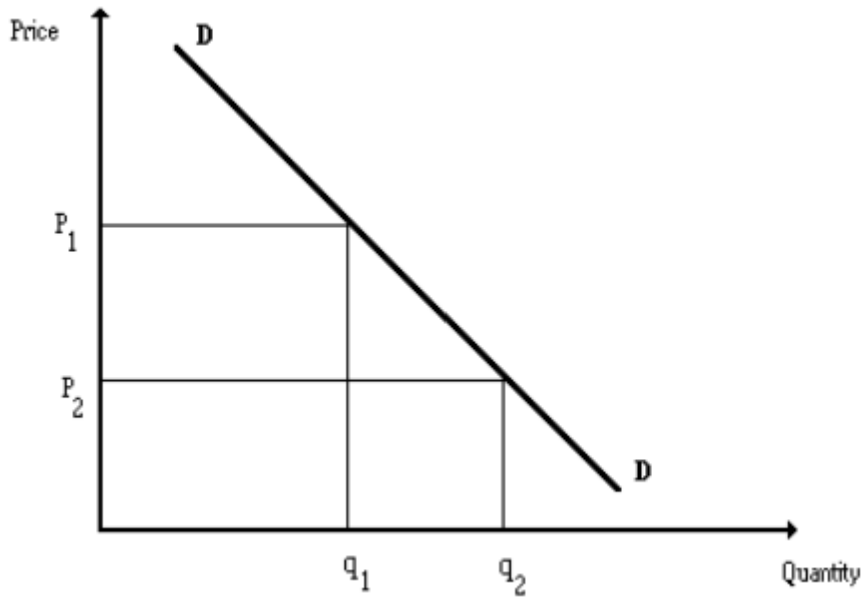
Table: The market demand schedule.

These prices are called **Demand Prices**. Thus, the demand price for 200,000 units per week is KShs 11 per unit.

(ii) The individual and market demand curves

The quantities and prices in the demand schedule can be plotted on a graph. Such a graph after the individual demand schedule is called **The Individual Demand Curve** and is downward sloping.

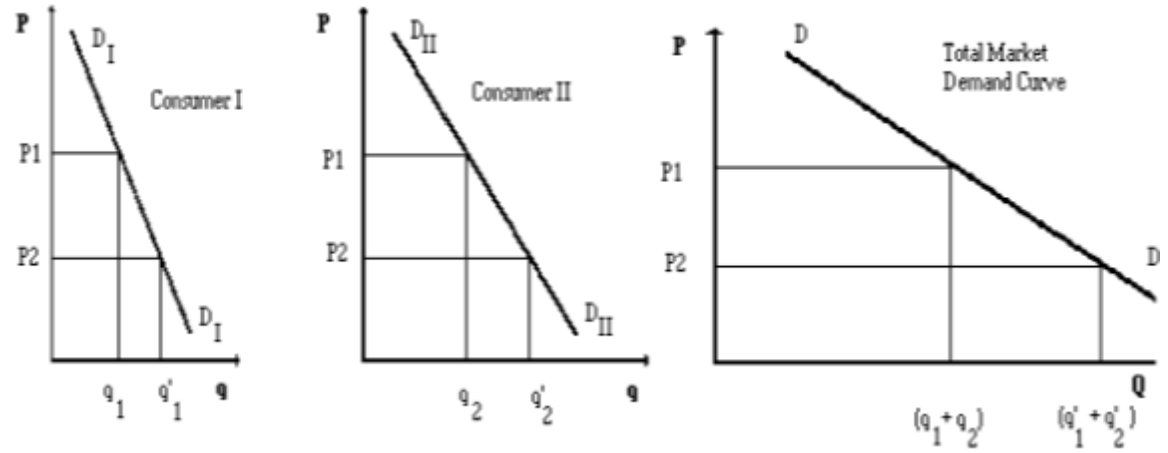
An individual demand curve is the graph relating prices to quantities demanded at those prices by an individual consumer of a given commodity



The curve can also be drawn for the entire market demand and is called a **Market Demand**

Curve:

A market demand curve is the horizontal summation of the individual demand curves i.e. by taking the sum of the quantities consumed by individual consumers at each price. Consider a market consisting of two consumers:



.At price P1in the figure above, consumer 1 demands q1, consumer II demands quantity q2, and total market demand at that price is (q1+q2). At price p2, consumer 1 demands q'1, and consumer II demands quantity q'2 and total market demand at that price is (q'1+q'2). DD is the total market demand curve.

FACTORS INFLUENCING DEMAND

These are broadly divided into factors **determining household** demand and factors affecting **market demand**.

Factors affecting household demand

- The taste of the household
- The income of the household
- The necessity of the commodity, and its alternatives if any
- The price of other goods

Factors affecting the total market demand

These are broadly divided into the determinants of demand and conditions of demand.

(a) Own price of the product

This is the most important determinant of demand. The determinants of demand other than price are referred to as the conditions of demand.

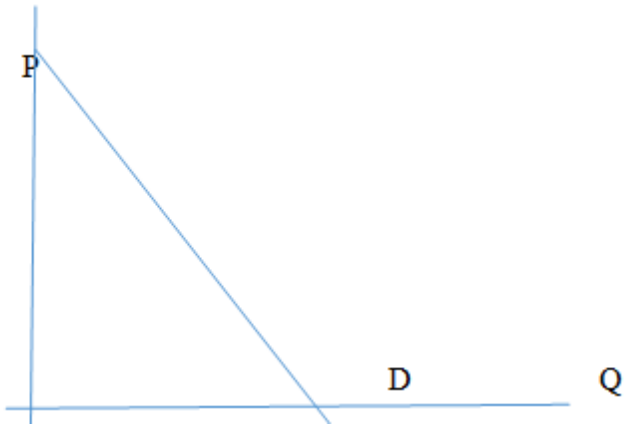
Changes in the price of a product bring about changes in quantity demanded, such that when the price falls more is demanded. This can be illustrated mathematically as follows:

Qd = a - bp

Where Qd is quantity demanded

a is the factor by which price changes

p is the price Thus, ceteris paribus, there is an inverse relationship between price and quantity demanded. Thus the normal demand curve slopes downwards from left to right as follows:



Prices of other related commodities

Related commodities can be compliments or substitutes.

(i) Compliments: The compliments of a commodity are those used or consumed with it. Suppose commodities A and B are compliments, and the price of A increased.

This will lead to a fall in the quantity demanded of A, and will in turn lead to a fall in the demand for B. Example are bread and butter or cars and petrol.

(ii) Substitutes: The substitutes of a commodity are those that can be used or consumed in the place of the commodity. Suppose commodities X and Y are substitutes. If the price of X increases, the quantity *demanded* of X falls, and the demand for Y increases.

(c) The Aggregate National Income and its distribution among the population. In normal circumstances as income goes up the quantity demanded goes up. In such a case the good is called a **normal good.** However, there are certain goods whose demand shall increase with income up to a certain point, then remain constant. In such a case the good is called a **necessity** e.g. salt. Also there are some goods whose demand shall increase with income up to a certain point then fall as the income continues to increase. In such a case the good is called **an inferior good.**

(d) Taste and preference

There is a direct relationship between quantity demanded and taste. For instance, if consumers' taste and preferences change in favour of a commodity, demand will increase.

On the other hand, if taste and preferences change against the commodity e.g. due to changes in fashion, demand will fall. Taste and preferences are influenced by religion, community background, academic background, environment, etc.

Theories of Demand and Supply and Consumer Behaviour

(a) Expectation of future price changes

If it is believed that the price of a commodity is likely to be higher in the future than at *present,* then even though the price has already risen, more of the commodity may be bought at the higher price.

(b) Climatic/seasonal factors

Seasonal variations affect the demand of certain *commodities* such as cold drinks like sodas and heavy clothing.

(c) The size and structure of population

Changes in population overtime affect the demand for a commodity. Also as population increases, the population structure changes in such away that an increasing proportion of the population

consists of young age group. This will lead to a relatively higher demand for those goods and services consumed mostly by young age group e.g. fashions, films, nightclubs, schools, toys, etc.

(d) Government influences e.g. a legislation requiring the wearing of seatbelts.

(e) Advertising especially the persuasive ones

EXCEPTIONAL DEMAND CURVES

There are exceptions when more is demanded when the price increases. These happens in the case of:

(i) Inferior goods: Cheap necessary foodstuffs provide one of the best examples of exceptional demand. When the price of such a commodity increases, the consumers may give up the less essential compliments in an effort to continue consuming the same amount of the foodstuff, which will mean that he will spend more on it. He may find that there is some money left, and this he spends on more of the foodstuff and thus ends up consuming more of it than before the price rise. A highly inferior good is called

Giffen good after Sir Robert Giffen

(ii) Articles of ostentation (snob appeal or conspicuous consumption): There are some commodities that appear desirable only if they are expensive. In such cases the consumer buys the good or service to show off or impress others. When the price rises, it becomes more impressive to consume the product and he may increase his consumption. Some articles of jewellery, perfumes- and fashion goods fall in this category.

(iii) Speculative demand: If prices are rising rapidly, a rise in price may cause more of a commodity to be demanded for fear that prices may rise further. Alternatively, people may buy hoping to resell it at higher prices. In all these three cases, the demand curve will be positively sloped i.e. the **higher** the price, the greater the quantity bought. These demand curves are called reverse **demand curves** (also called perverse or abnormal demand curve).

TYPES OF DEMAND

There are four types of demand namely Competitive Demand, Joint or Complementary Demand, Composite Demand and Derived Demand. Demand is the amount of a product buyers are willing and able to purchase at a given price over a particular period of time.

a. Competitive Demand

Commodities are substitutes if one can be used in place of the other. Substitute goods serve the same purpose and therefore compete for the consumers' income. They are said to have competitive demand because of the fact that they compete for the consumers' income. Examples of substitute goods are Milo and bournvita, butter and margarine and others. A change in the price of one affects the demand for the other. If for instance there is an increase in the price of butter, demand for

margarine will increase which will ultimately increase the price of margarine, provided the supply of margarine does not change. On the other hand a decrease in the price of butter will lead to a decrease in the demand for margarine, and hence a fall in its price, given the supply.

b. Joint or Complementary Demand

Two or more goods are said to be jointly demanded when they must be consumed together to provided a given level of satisfaction. Some examples are cars and fuel, compact disc players and CD. There are perfect complementary goods and imperfect or poor complementary goods. For perfect complementary goods, the consumer practically cannot do without the other. An example is cars and fuel. On the other hand, for imperfect complementary goods, a consumer can do without the other, so long as a substitute is obtained. For complementary demand, a change in the price of one good affects the demand for the other. If there should be an increase in the price of compact disc players, there will be a decrease in the demand for discs, other things being equal.

c. Derived Demand

When the demand for a commodity is derived from the demand for the final commodity, that commodity is said to have derived demand. Wood may be demanded for the purpose of manufacturing furniture and not for its own sake. Here, the demand for wood is derived from the demand for furniture. Demand for wood is therefore a derived demand.

Factors of production such as land, labor, and capital have derived demand. This is because an increase in the demand for a commodity will result in an increase in the factors of production used in producing the goods. The price of the factors of production will increase, other things being equal.

d. Composite Demand

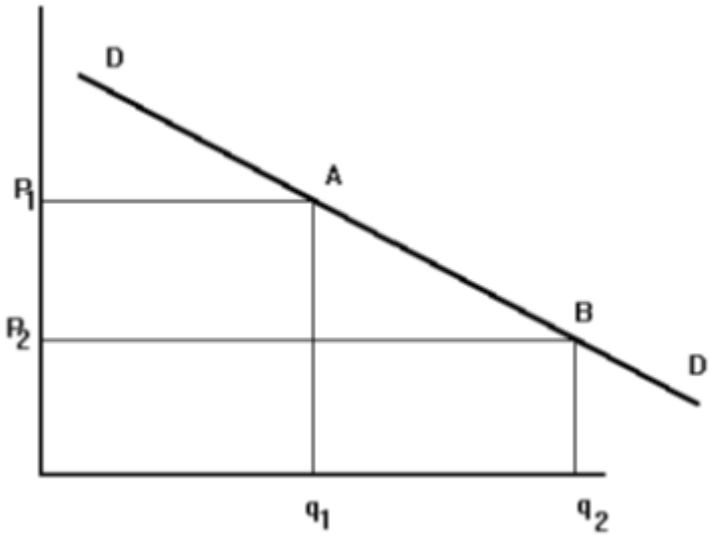
Composite demand applies to commodities which have several uses or are demanded for several and different purposes. Wood as mentioned in the example above is used for furniture – tables, chairs, beds, windows, doors and others. A change in demand for one of them will affect all others. If there is an increase in demand for table this will result in higher prices being paid for wood. The high price for wood will increase the cost of production of chairs, bed, windows and doors and any other thing for which wood is used in manufacturing

MOVEMENT ALONG AND SHIFTS OF DEMAND CURVES

There are basically two; movements in demand curves, namely:

1. Movement along the demand curve.

Movement along the demand curve are brought by changes in own price of the commodity.

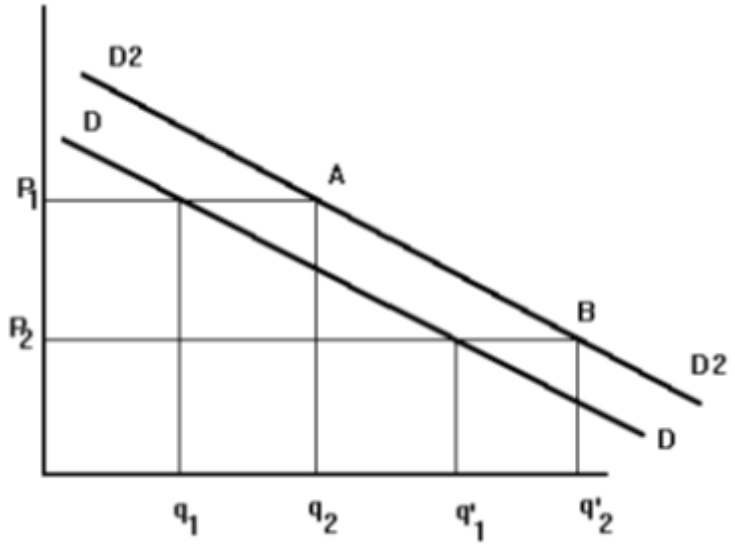


When price falls from p1 to p2, quantity demanded increases from q1 to q2 and movement along the demand curve is from A to B. Conversely when price rises from p2 to p1 quantity demanded falls from q2 to q1 and movement along the demand curve is from B to A.

2. Shifts in demand curve

Shifts in the demand curve are brought about by the changes in factors like taste, prices of other related commodities, income etc other than the price of the commodity. The change in the demand for the commodity is indicated by a shift to the right or left of the original demand curve.

In the figure below, DD represents the initial demand before the changes. When the demand increases, the demand curve shifts to the right from position DD to positions D2D2. The quantity demanded at price P1 increases from q1 to q'1. Conversely, a fall in demand is indicated by a shift to the left of the demand curve from D2D2 to DD. The quantity demanded at price P1 decreases from q1 to q1



ELASTICITY OF DEMAND

Definition of Elasticity

Is defined as the ratio of the relative change of one (dependent) variable to changes in another (independent) variable, or it's a percentage change of one variable given a one percent change in another

b) Elasticity of Demand

Measures the extent to which the quantity demanded of a good responds to changes in one of the factors affecting demand

TYPES OF ELASTICITY

The various types of the elasticity of demand are: Price Elasticity, Income elasticity and Cross Elasticity.

Price Elasticity of Demand

Is the responsiveness of the quantity demanded to changes in price; its co-efficient is Ped = Proportionate change in quantity demanded/ Proportionate change in price

Types of Price Elasticity of demand

a) Perfectly inelastic demand

Demand is said to be perfectly inelastic if changes in price have no the quantity demanded so that the demand is infinitely price elastic. This is the case of an absolute necessity i.e. one which a consumer cannot do without and must have in fixed amount e.g. analysis, insulin etc.

b) Inelastic demand

This is where changes in price bring about changes in quantity demanded in less proportion so that elasticity is less than one. This is the case of a necessity or a habit forming commodity e.g. drinks or cigarettes.

c) Unit Elasticity of demand

Is where changes in price bring about changes in quantity demanded in the same proportion and the elasticity of demand is equal to one or unity. This is for commodities, which are between a necessity and a luxury, e.g. film going.

d) Elastic demand

Demand is said to be price elastic if changes in price being about changes in quantity demanded in greater proportion so that elasticity is greater than one. This is the case of a luxury, i.e. one that can be done without or a commodity with close substitutes.

e) Perfectly Elastic demand

Demand is perfectly elastic when consumers are prepared to buy all they can obtain at some price and none at an even slightly higher price.

This is the case of perfectly competitive market i.e. where there are many producers producing the same product. Each of them is too insignificant to increase or reduce the market price.

MEASUREMENT OF ELASTICITY, POINT AND ARC ELASTICITY

a) Point Elasticity:

This measures elasticity at a particular point and is only valid or based on small movements i.e.

Ped

$$\frac{\Delta Q}{Q} \Big/ \frac{\Delta P}{P} = \frac{\Delta Q}{\Delta P} X \frac{P}{Q}$$

	Price	Quantity
Point A:	Kshs 3.01	4.00
Point B:	Kshs 3.00	4.10

$$\Delta P = 3 - 3.01 = -0.01$$

$$\Delta Q = 4.10 - 4.00 = 0.1$$

Point elasticity =

$$\frac{\Delta Q}{\Delta P} X \frac{P}{Q}$$

$$\frac{0.1}{-0.01} X \frac{3.01}{4.00}$$

$$= -10 \; x \; 0.7525 \; (-1)$$

$$= 7.5$$

It is only valid in very small changes.

b) Arc Elasticity

Is the average elasticity between two given points on the curve, i.e.

Because of the negative relationship between price and quantity demanded, price elasticity of demand is negative. We there take the absolute magnitude of the number.

Elasticity determines the shape of the demand curve.

FACTORS INFLUENCING ELASTICITY OF DEMAND

- Ease of substitution.
- Nature of the commodity i.e. whether it is a necessity of life, luxury or addictive.
- Consumers' income.
- The number of uses to which the good can be put.
- Time factor.
- The prices of other products.
- Advertisements especially the persuasive ones.
- Whether the use for the good can be postponed.
- Human and economic constraints.

2) Income elasticity of demand

The income elasticity of demand measures the degree of responsiveness of the quantity demanded of a product to changes in income. Its co-efficient is as follows:

$$\text{EY} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

This we may write as:

$$EY = \frac{\Delta Q/Q}{\Delta Y/Y}$$

Which can be simplified as:

$$EY = \frac{\Delta Q}{Q} * \frac{Y}{\Delta Y}$$

$$EY = \frac{\Delta Q}{\Delta Y} * \frac{Y}{Q}$$

$$Where\ Y = Income$$

Types of Income Elasticity of demand

Depending upon the product, demand might increase or decrease in response to a rise in income. There are thus five types of income Elasticity of demand viz:

i) Negative Income Elasticity

This is where the demand decreases as income rises and rises when income falls. This is the case of inferior goods.

ii) Zero Income Elastic

In this case, the demand does not change as income rises or falls. In this case it is said to be zero income, elasticity. This is the case of a necessity.

iii) Income Inelastic

This is where demand rises by a smaller proportion than income or falls by a smaller proportion than income.

iv) Unit Income Elasticity

This is where demand rises or falls by exactly the same proportion as income.

v) Income Elastic

Demand rises or falls by a greater proportion than income. Since income elasticity of demand can be either positive or negative, it is therefore very important to include the sign (+ or -) when stating the value of the co-efficient.

c) Cross Elasticity

Cross elasticity of demand measures the degree of responsiveness of the quantity demanded of one good (B) to changes in the price of another good (A). It is measured as follows:

$$\text{EY} = \frac{\text{Percentage change in quantity demanded of B}}{\text{Percentage change in Price of A}}$$

This may be written mathematically as follows:

$$Ex = \frac{\Delta\text{QB}/\text{QB}}{\Delta\text{PA}/\text{PA}}$$

$$EY = \frac{\Delta\text{QB}}{\text{QB}} * \frac{\text{PA}}{\Delta\text{PA}}$$

$$EY = \frac{\Delta\text{QB}}{\Delta\text{PA}} * \frac{\text{PA}}{QB}$$

In the case of complementary goods, such as cars and petrol, a face in the price of one will bring about an increase in the demand for the other. Thus we are considering a cut in price (-) bringing about a rise in demand (+). This therefore means that for complements, the Ex is negative.

Conversely, substitute goods such as butter and margarine might be expected to have a positive Ex because a rise in price of one (+) will bring about a rise in the demand for the other (+).

The value of Ex may vary from minus infinity to plus infinity. Goods which are close to complements or substitutes will tend to exhibit a high cross-elasticity of demand.

Conversely, when there is little or no relationship between goods then the Ex will be near zero.

Importance of Cross Elasticity

Knowledge of cross elasticity is necessary when the government wants to impose a tariff on an imported commodity to protect a domestic industry.

Firms need to know the cross elasticity of their products and substitute products when contemplating price rises.

Firms also need to know the cross elasticity of the products and complements to plan their production.

APPLICATION OF ELASTICITY OF DEMAND IN MANAGEMENT AND ECONOMIC POLICY DECISION MAKING

If a country is experiencing economic growth, the income of the people will increase. However, for those engaged in the production of goods with negative income elasticities, this will mean a declining demand for their product. Even products with positive income elasticities, there is a great variability of response.

Income elasticity therefore has a most important effect upon resource allocation. As such, prosperous areas of any economy are often those associated with products which have a high income elasticity. In recession the opposite will be true.

The practical importance of the measures of elasticity of demand is to be appreciated in various ways:

- From the point of view of individual consumers who tend to spend limited income on commodities with less elastic demand.
- From the point of view of business person who need to know the effects that changes in price will have on the sales revenue. For instance, if they know that demand for their product is relatively inelastic then increasing prices might help them to increase revenue. If on the other hand they are aware that source of their products have a high price elasticity of demand they will be more cautious when considering price increases for fear of losing revenue.
- From the point of view of firms in who may attempt to change the price elasticity of demand for their product through advertising, packaging, better service and other services to improve or help maintain sales.
- From the point of view of business people who may also want to know the price elasticity of demand as purchasers of inputs for use in their business.
- From the point of view of governments in trying to estimate the yield of a prospective market tax.
- From the point of view of devaluation policy aimed at improving the balance of payments.
- From the point of view of the effectiveness of price control and deregulation of some industries.
- From the point of view of wage bargaining among workers and employers and the government when fixing minimum wage legislation.
- For purposes of regulating farm incomes and to predict consequences of bumper harvests of crops.

SUPPLY ANALYSIS

DEFINITION OF SUPPLY

Supply is the quantity of goods/services per unit of time which suppliers/producers are willing and able to put on the market for sale at alternative prices other things held constant.

INDIVIDUAL VERSUS MARKET SUPPLY CURVES

(i) Firm and industry supply schedules

The plan or table of possible quantities that will be offered for sale at different prices by individual firms for a commodity is called **supply schedule**.

Price Per Unit (KShs)	Quantity offered for Sales per month (in '000)
20	10
25	20
30	30
35	40
40	50
45	60
50	70

The Firm Supply Schedule
Theoretically the supply schedules of all firms within the industry can be combined to form the market or industry supply schedule, representing the total supply for that commodity at various prices.

Price per unit (KShs)	Quantity offered for Sales per month (in '000)
20	80
25	120
30	160
35	200
40	240
45	285
50	320

Table: The Industry supply schedule
These prices are called the **supply prices**.

(ii) Individual firm and market supply curves

The quantities and prices in the supply schedule can be plotted on a graph. Such a graph is called the **firm supply curve.**

A firm supply curve is a graph relating the price and the quantities of a commodity a firm is prepared to supply at those prices.

The typical supply curve slopes upwards from left to right. This illustrates the second law of supply and demand "which states that the higher the price the greater the quantity that will be supplied".

More is supplied by the firms which could not make a profit at the lower price.

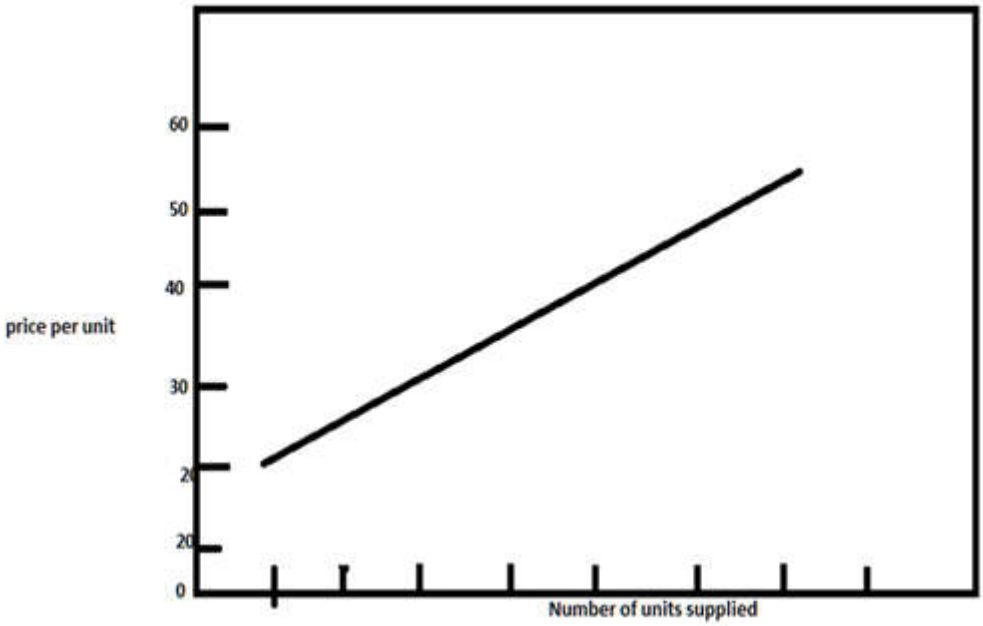


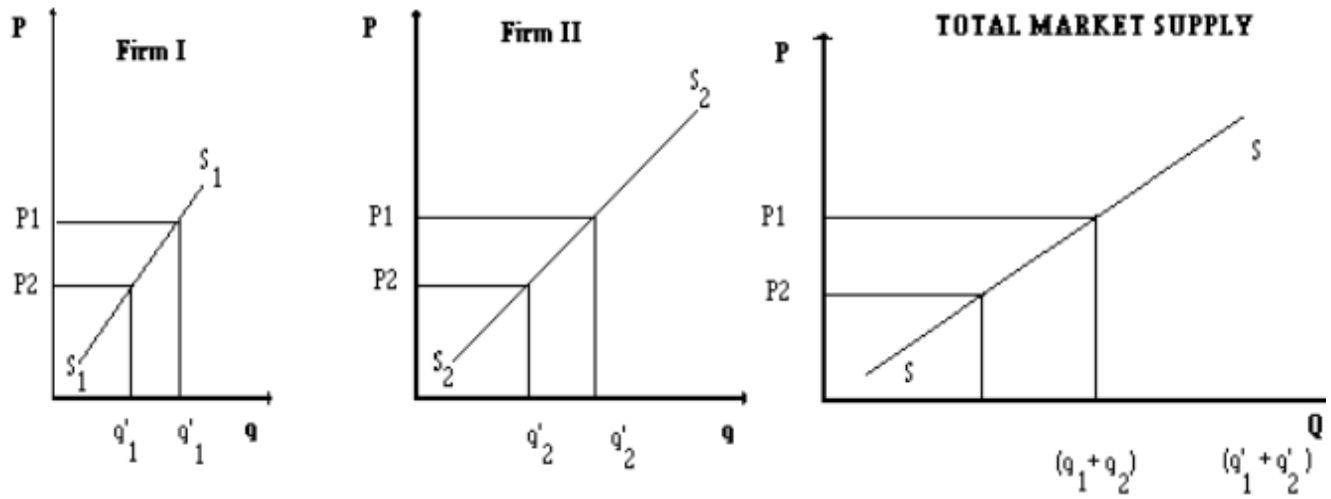
Fig 2. The firm supply curve

The market supply curve is obtained by horizontal summation of the individual firm supply curves i.e. taking the sum of the quantities supplied by the different firms at each price.

Consider, for the sake of exposition, an industry consisting of two firms. At price P1, firm I (diagram below) supplies quantity q1, firm II supplies quantity q2, and the total market supply is

q1+q2

At price P2, firm I supplies q'1, firm II supplies quantity q'2, and the total market supply is q'1+q'2,. SS is the total market supply curve.



FACTORS INFLUENCING SUPPLY

a) Own Price of the commodity

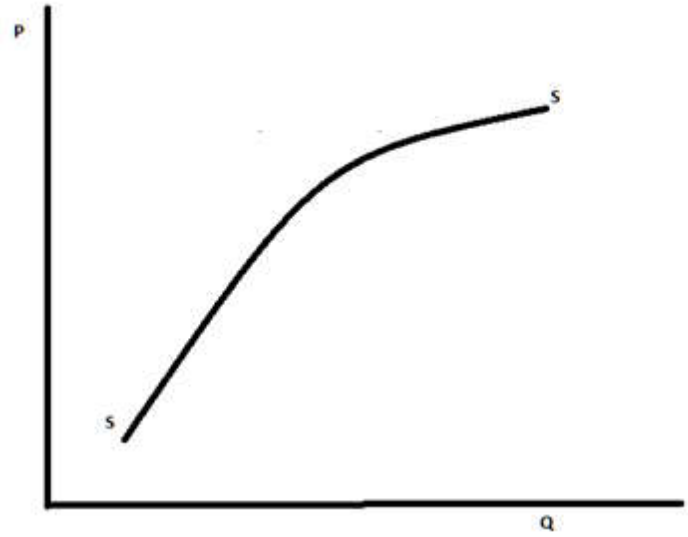
There is a direct relationship between quantity supplied and the price so that the higher the price, the more people shall bring forth to the market. Mathematically this can be illustrated as follows:

Qs = -c + dp

Where: **Qs** is the quantity supplied

-c is a constant
d is the factor by which price changes
P is the price

Thus the normal supply curve slopes upwards from left to right as follows:



organisations which could not produce profitably at the lower price would find it possible to do so at a higher price. One way of looking at his is that as price goes up, less and less efficient firms are brought into the industry.

Exceptional supply curves

In have some situations the slope of the supply curve may be reversed.

i) Regressive Supply. In this case, the higher the price within a certain range, the smaller the amount offered to the market. This may occur for example in some labour markets where above certain level, higher wages have a disincentive effect as the leisure preference becomes high. This may also occur in undeveloped peasant economies where producers have a static view of the income they receive. Lastly regressive supply curves may occur with target workers.

ii) Fixed Supply. Where the commodity is rare e.g. the "Mona Lisa", the supply remains the same regardless of price. This will be true in the short term of the supply of all things, particularly raw materials and agricultural products, since time must elapse before it is physically possible to increase output.

b) Prices of other related goods

i) Substitutes: If X and Y are substitutes, then if the price X increases, the quantity demanded of X falls. This will lead to increased demand for Y, and this way eventually lead to increased supply of Y.

ii) Complements: If two commodities, say A and B are used jointly, then an increase in the price of A shall lead to a fall in the demand for A, which will cause the demand for B to fall too.

c) Prices of the factors of production

As the prices of those factors of production used intensively by X producers rise, so do the firms' costs. This cause supply to fall as some firms reduce output and other, less efficient firms make losses and eventually leave the industry. Similarly, if the price of one factor of production would rise (say, land), some firms may be tempted to move out of the production of land intensive products, like wheat, into the production of a good which is intensive in some other factor of production.

d) Goals of the firm

How much is produced by a firm depends on its objectives. A firm which aims to maximise its sales revenue, for example, will generally supply a greater quantity than a firm aiming to maximise profits (see markets). Changes in these objectives will usually lead to changes in the quantity supplied.

e) State of technology

There is a direct relationship between supply and technology. Improved technology results in more supply as with technology there is mechanisation.

f) Natural events

Natural events like weather, pests, floods, etc also affect supply. These affect particularly the supply of agricultural products. If weather conditions are favourable, the supply of agricultural products will increase. Conversely, if weather conditions are unfavourable the supply of such products will fall.

g) Time

In the long run (with time), the supply of most products will increase with capital accumulation, technical progress and population growth so long as the last one takes place in step with the first two. This reflects economic growth.

h) Supply of Inputs

Changes in supply of inputs will affect the quantity supplied; if this falls, less shall be supplied and vice versa.

i) Changes in the supply of the product with which the product in question is in joint supply e.g. hides and skins.

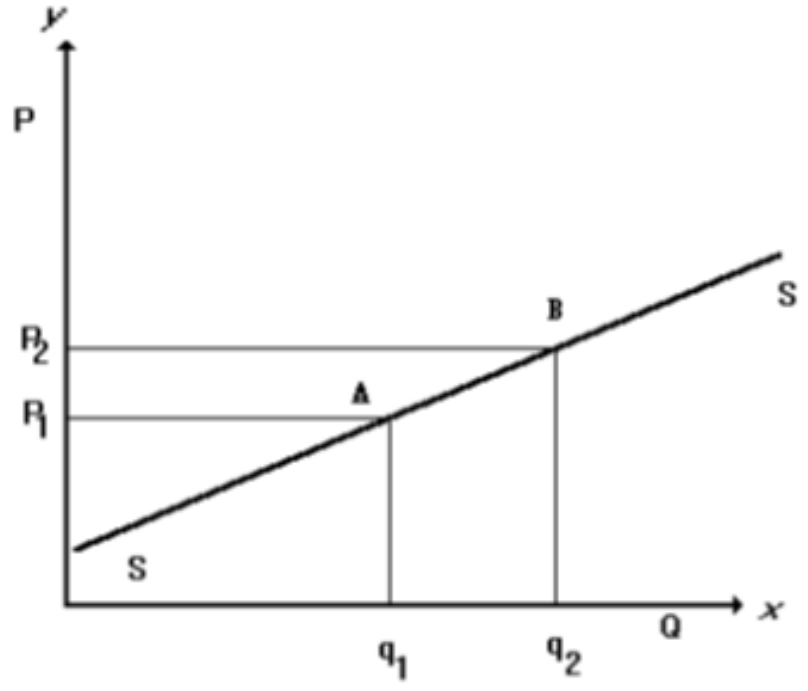
j) Taxes and subsidies

The imposition of a tax on a commodity by the government is equivalent to increasing the costs of production to the producer because the tax "eats" into the firm's profits. Hence taxes tend to discourage production and hence reduce supply. Conversely, the granting of a subsidy is equivalent to covering the costs of production. Hence subsidies tend to encourage production and increase supply.

MOVEMENT ALONG AND SHIFTS OF SUPPLY CURVES

i) Movements along the supply curve:

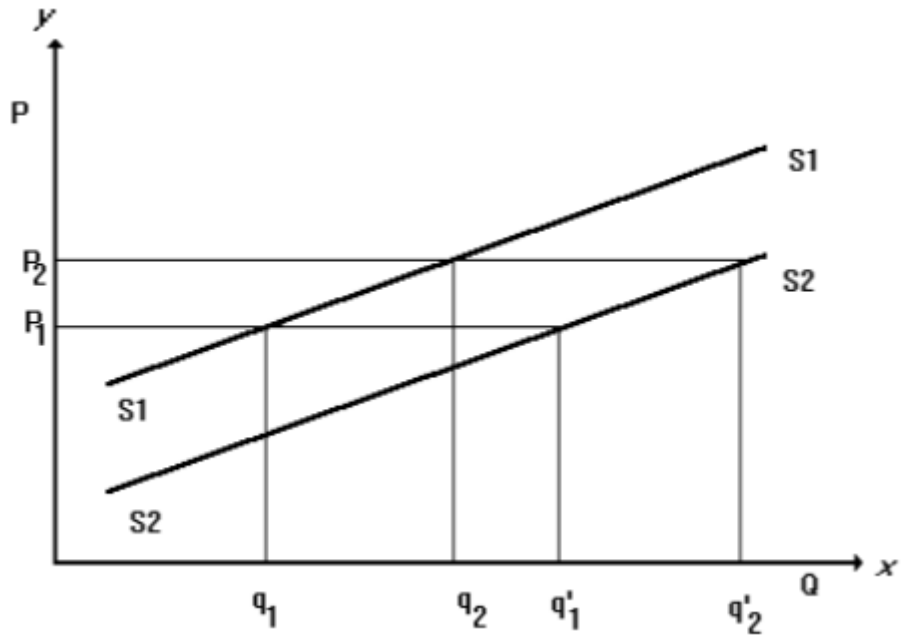
Movements along the supply curve are brought about by changes in the price of the commodity.



When price increases from P1 to P2, quantity supplied increases from Q1 to Q2 and movement along the supply curve is from A to B. Conversely when price falls from P2 to P1, quantity supplied falls from q2 to q1 and movement along the supply curve is from B to A.

ii) Shifts in the supply curve

Shifts in the supply curve are brought about by changes in factors other than the price of the commodity. A shift in supply is indicated by an entire movement (shift) of the supply curve to the right (downwards) or to the left (upwards) of the original curve.



When supply increases, the supply curve shifts to the right from S1S1 to S2S2. At price P1, supply increases from q1 to q'1 and at price P2 supply increases from q2 to q'2. Conversely, a fall in supply is indicated by a shift to the left or upwards of the supply curve and less is supplied at all prices. Thus, when supply falls, the supply curve shifts to the left from position S2S2 to position S1S1. At price p1, supply falls from q'1 to q1 and at price p2, supply falls from q'2 to q2.

ELASTICITY OF SUPPLY

It refers to the responsiveness of quantity supplied to changes in factors affecting price

PRICE ELASTICITY OF SUPPLY

Definition

Price Elasticity of supply measures the degree of responsiveness of quantity supplied to changes in price. The co-efficient of the elasticity of supply may be stated as:

$$Es = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in Price}}$$

Mathematically, this can be written as:

$$Es = \frac{\Delta Qs/Qs}{\Delta P/P}$$

$$Es = \frac{\Delta Qs}{Qs} * \frac{P}{\Delta P}$$

Symbolically it is given by the formula

$$Es = \frac{\Delta Qs}{\Delta P} * \frac{P}{Q}$$

Because of the positive relationship between price and quantity supplied, the price elasticity of supply ranges from zero to infinity.

TYPES OF ELASTICITY OF SUPPLY

i) Perfectly Inelastic (Zero Elastic) Supply:

Supply is said to be perfectly inelastic if the quantity supplied is constant at all prices. The supply curve is a vertical straight line and the elasticity of supply is equal to zero.

When price rises from P1 to P2, quantity supplied stays fixed at q, and when price falls from P2 to P1, quantity supplied stays fixed.

In the case of a price rise, this is the situation of the very **short-run** or **the momentary** period which is so short that the quantity supplied cannot be increased, e.g. food brought to the market in the morning. It is also the case where the commodity is fixed in supply e.g. land. In the case of a price fall, this is the case of a **highly perishable** commodity which cannot be stored, e.g. fresh fish.

ii) Inelastic Supply:

Supply is said to be price inelastic if changes in price bring about changes in quantity supplied in less proportion. Thus, when price increases quantity supplied increases in less proportion, and when price falls quantity supplied falls in less proportion. The supply curve is steeply sloped and the elasticity of supply is less than one.

When price increases from P1 to P2, quantity supplied increases in less proportion from q1 to

q2. This is the case when there are **limited stocks** of the product or the product takes a **long time** to produce such that when price rises, quantity supplied cannot be increased substantially.

Conversely, if price falls from P2 to P1, quantity supplied falls in less proportion from q2 to q1.

This is the case of a commodity which is perishable and cannot be easily stored, e.g. fresh foods like bananas and tomatoes. These are perishable but not so highly perishable as fresh fish. When price falls, quantity supplied cannot be drastically reduced.

iii) Unit Elasticity of Supply:

Supply is said to be of unit elasticity if changes in price bring about changes in quantity supplied in the same proportion. Thus, when price rises, quantity supplied increases in the same proportion, and when price falls, quantity supplied falls in the same proportion. The supply curve is a straight line through the origin, and the elasticity of supply is equal to **one** or **unity**.

When price rises from P1 to P2, quantity supplied increases in the same proportion from q1 to

q2. This is the case of a commodity of which there is a fair amount of stocks or which can be produced within a fairly short period of time.

Conversely, when price falls from P2 to P1, quantity supplied falls in the same proportion from q2 to q1. This is the case of a commodity which is fairly easily stockable, e.g. dry foods, like dry beans and dry maize.

iv) Elastic Supply

Supply is said to be price elastic if changes in price bring about changes in quantity supplied in greater proportion. Thus, when price increases, quantity supplied increases in greater proportion. The supply curve is not steeply sloped and the elasticity of supply is greater than one.

When price rises from P1 to P2, quantity supplied rises in greater proportion from q1 to q2.

This is the case when there are a **lot of stocks** of the commodity or the commodity can be produced within **fairly short period** of time so that when price rises, quantity supplied can be increased substantially.

Conversely, if price falls from P2 to P1, quantity supplied falls in greater proportion from q2 to

q1. This is the case of a commodity which is **easily stockable** e.g. manufactured articles.

When price falls, quantity supplied can be substantially reduced. The commodity is then stored instead of being sold at a loss or for very reduced profit.

v) Perfectly Elastic Supply

Supply is said to be perfectly or infinitely elastic if the price is fixed at all levels of demand. The demand curve has been shown in the above diagram for the sake of clarity.

If the supply is perfectly elastic, the supply curve is a horizontal straight line and the elasticity of supply is equal to infinity.

When demand increases from quantity supplied increases but price stays fixed . Conversely, if demand falls, quantity supplied falls but price stays fixed.

This is the case of Government price control.

FACTORS INFLUENCING ELASTICITY OF SUPPLY

Time

In the short run firms will only be able to increase input of labor to increase supply of commodities may not be able to increase the supply in response to the price change but the supply change will be little because other factors of production may not be increased in the same proportion and may limit the supply. However, in the long run a firm will increase the input of all factors of production and thus the supply becomes more price elastic.

Availability of resources

If the economy already using most of its scarce resources then firms will find it difficult to employ more and so output will not be able to rise. The supply of most of goods and services will therefore be price inelastic.

Number of producers

More producers mean that the output can be increased more easily. Thus supply is more elastic.

Ease of storing stocks

If goods can be stocked with ease and have a long shelf life, the supply will be elastic, otherwise inelastic. For example perishable goods such as fresh flowers, vegetables have comparatively inelastic supply because it is difficult to store them for longer periods.

Increase in cost of production as compared to output

In cases where there is a significant increase in cost of production when output is increased, supply is inelastic. This is because suppliers will have to have to do a significant investment in order to increase the output. It will take time and some suppliers may be hesitant in doing so.

Improvement in Technology

In industries where there is a rapid improvement in technology, the PES of such goods will be more elastic as compared to industries where there is not much improvement in technology.

Stock of finished goods

In industries where there are high inventories/stocks of finished goods, the suppliers can easily supply more as the price rises. Thus, the PES for these goods will be elastic.

APPLICATION OF ELASTICITY OF SUPPLY

i) If the supply of a commodity is elastic with respect to a price rise, producers will benefit by prices not rising excessively. But if the supply is inelastic with respect to a price rise, there will tend to be overpricing of the commodity to the disadvantage of the consumers. Even the producers will not benefit as much as they would if the supply was elastic because although they are charging high prices, the supply is limited.

ii) If the supply is inelastic with respect to a price fall: This increases the risk of the business because it means that producers may be forced to sell the commodity at very low prices as the commodity cannot be easily stored. But if the supply is elastic with respect to a price fall, the business is less risky as the commodity can easily be stored, and producers will not be forced to sell at low prices.

iii) The price elasticity of supply is responsible for the fact that the prices of agricultural products tend to fluctuate more than those of manufactured products.

DETERMINATION OF EQUILIBRIUM

INTERACTION OF SUPPLY AND DEMAND, EQUILIBRIUM PRICE AND QUANTITY

In perfectly competitive markets the market price is determined by the interaction of the forces of demand and supply. In such markets the price adjusts upwards or downwards to achieve a balance, or **equilibrium**, between the goods coming in for sale and those being requested by purchases. Demand and supply react on one another until a position of stable equilibrium is reached where the quantities of goods demanded equal the quantities of goods supplied. The price at which goods are changing hands varies with supply and demand. If the supply exceeds demand at the start of the week, prices will fall. This may discourage some of the suppliers, who will withdraw from the market, and at the same time it will encourage consumers, who will increase their demands. This is known as **buyers market.**

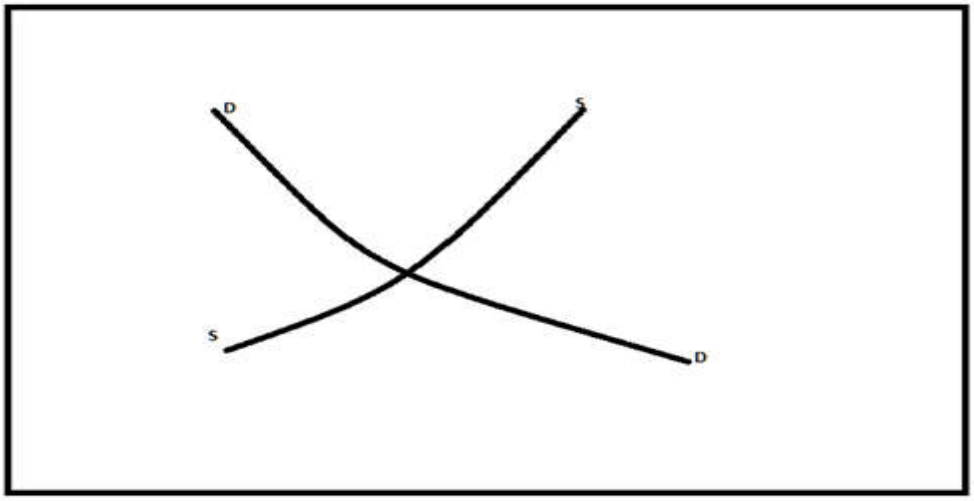
Conversely if we examine line D, where the price is £2 per kg, suppliers are only willing to supply 90kg per week but consumers are trying to buy 200 kg per week.

There are therefore many disappointed customers, and producers realise that they can raise prices. This is known as sellers market. There is thus an upward pressure on price and it will rise. This may encourage some suppliers, who will enter the market, and at the same time it will discourage consumers, who will decrease their demand.

This can be shown by comparing the demand and supply schedule below.

Price of Commodity X $/kg	Quantity demanded of commodity X(kg/week)	Quantity supplied of commodity X (kg/week)	Pressure on price
A 10	100	20	Upward
B 20	85	36	Upward
C 30	70	53	Upward
D 40	55	70	Downward
E 50	40	87	Downward
F 60	25	103	Downward
G 70	10	120	Downward

A Twin force is therefore always at work to achieve only one price where there is neither upward or downward pressure on price. This is termed the equilibrium or market price:
The equilibrium price is the market condition which once achieved tends to persist or at which the wishes of buyers and sellers coincide.



Any other price anywhere is called DISEQUILIBRIUM PRICE. As the price falls the quantity demanded increases, but the quantity offered by suppliers is reduced, since the least efficient suppliers cannot offer the goods at the lower prices. This illustrates the third "law" of demand and supply that **"Price adjusts to that level which equates demand and supply"**.

MATHEMATICAL APPROACH TO EQUILIBRIUM ANALYSIS

The demand and supply relationships explained earlier on can be expressed in mathematical form. The standard problem is one of finding a set of values which will satisfy the equilibrium condition of the market model.

Equilibrium in a single market model.

A single market model has three variables: the quantity demanded of the commodity (Qd), the quantity supplied of the commodity (Qs) and the price of the commodity (P). equilibrium is assumed to hold in the market when the quantity demanded (Qd) = Quantity Supplied (Qs) . It is assumed that

both Qd and Qs are functions. A function such as y = f (x) expresses a relationship between two variables x and y such that for each value of x there exists one and only one value of y. Qd is assumed to be a decreasing linear function of P which implies that as

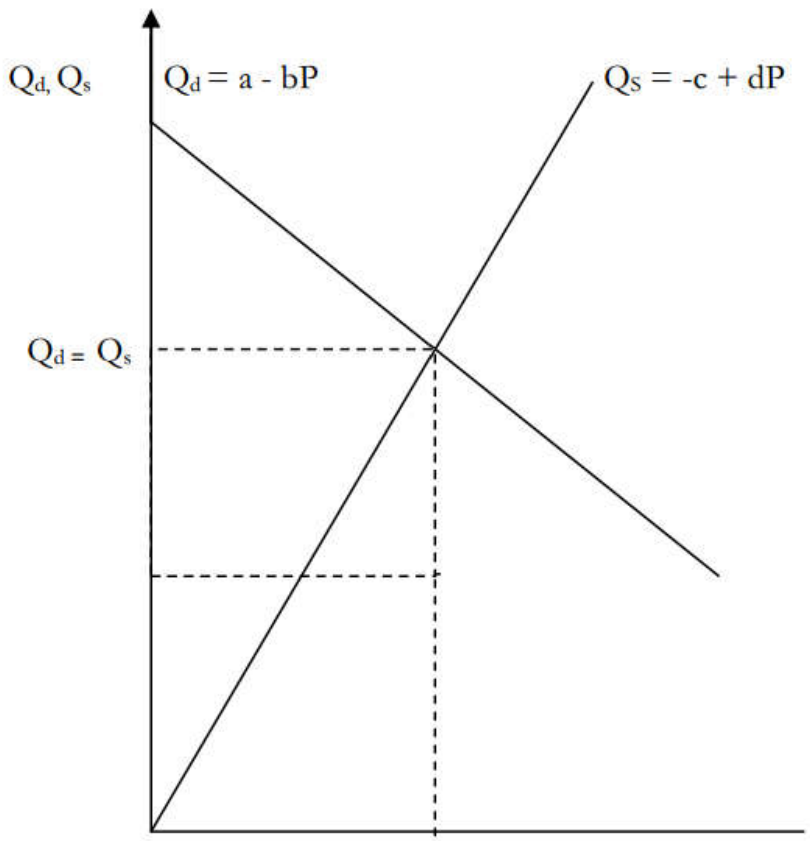
P increases, Qd decreases and Vice Versa. Qs on the other hand is assumed to be an increasing linear function of P which implies that as P increases, so does Qs.

Mathematically, this can be expressed as follows:

Qd = Qs

Qd = a – bP where a,b > 0. …….(i)
Qs = -c + dp where c,d >0. ……(ii)

Both the Qd and Qs functions in this case are linear and can be expressed graphically as follows:
Qd, Qs Qd = a - bP QS = -c + dP
Qd = Qs



Once the model has been constructed it can be solved.

At equilibrium,

Qd = Qs
∴a – bP = -c + dP

P = a + c
b + d

To find the equilibrium quantity Q, we can substitute into either function (i) or (ii).
Substituting P into equation (i) we obtain:

Q = a – b (a+c) = a (b+d) – b (a+c) = ad -bc
b + d b + d b + d

Taking a numerical example, assume the following demand and supply functions:

P = 100 – 2P
Qs = 40 + 4P

At equilibrium, Qd = Qs
∴ 100 – 2 P = 40 + 4 P

6P = 60

∴ P = 10

Substituting P = 10, in either equation.

Qd = 100 – 2 (10) = 100 – 20 = 80 = Qs

A single market model may contain a quadratic function instead of a linear function. A quadratic function is one which involves the square of a variable as the highest power. The key difference between a quadratic function and a linear one is that the quadratic function will yield two solution values.

In general, a quadratic equation takes the following form:

ax2 + bx + c = 0 where a ≠ 0.

Its two roots can be obtained from the following quadratic formula:

X1, X2 = -b + √(b2 – 4ac)

2a

Given the following market model:

Qd = 3 – P2

2 = 6P – 4

At equilibrium:

3 – P2 = 6P – 4

P2 + 6P – 7 = 0

Substituting in the quadratic formula:

a =1, b = 6, c = -7

= - 6 +√ 62 – 4 (1 x – 7)

2 x 1

= 2 1

6 62 4(1 7) $x - x - +$

P = 1 or –7 (ignoring –7 since price cannot be negative)

∴ P = 1

Substituting P = 1 into either equation:

Qd = 3 – (1)2 = 2 = Qs

∴Q = 2

Equilibrium in a two commodity market

Let us consider a two-commodity market model in which the two commodities are related to each other. Let us assume the functions for both commodities are linear. The two commodities are complementary commodities say (cars (c) and petrol (P). The functions representing the commodities are as follows:

Qdc = 820 – 10 Pc – 4Pp Qdp = 590 – 2Pc – 6Pp

Qsc = -120 + 6Pc Qsp = - 240 + 4Pp

At equilibrium,

1) Qdc = Qsc

820 – 10Pc – 4Pp = - 120 + 6Pc

940 – 16Pc – 4Pp = 0

2) Qdp = Qsp

590 – 2Pc – 6Pp = -240 + 4Pp

830 – 2Pc – 10Pp = 0

There are now therefore two equations:

940 – 16Pc – 4Pp = 0. ……(i)

830 – 2Pc – 10Pp = 0 …….(ii)

Multiply (ii) by 8 which gives (iii). Subtract (i) from (iii) to eliminate Pc and solve for Pp.

6,640 – 16 Pc – 80Pp = 0..(iii)

- (940 – 16Pc – 4Pp = 0 …….(i)

5,700 - 76Pp = 0

Pp = 75

Substituting Pp = 75 in (i) we obtain:

940 – 16Pc – 4(75) = 0

16pc = 640

∴Pc = 40

Substituting Pc = 40 and Pp = 75 into Qd or Qs for each market.

1) Qdc = 820 – 10 (40) – 4 (75)

= 820 – 400 – 300

Qdc = 120 = Qsc

2) Qdp = 590 – 2 (40) – 6 (75)

= 590 – 80 – 450

Qdp = 60 = Qsp

STABLE VERSUS UNSTABLE EQUILIBRIUM

An equilibrium is said to be stable equilibrium when economic forces tend to push the market towards it. In other words, any divergence from the equilibrium position sets up forces, which tend to restore the equilibrium.

Quality per time period

At prices above Ope, there is an excess supply which pushes the price down. At prices below

Ope there is an excess demand which pushes the price up.

Unstable equilibrium on the other hand is one such that any divergence from the equilibrium sets up forces which push the price further away from the equilibrium price. Consider the figure below which illustrates the market for good Y, which has a demand curve sloping upwards from left to right. Good Y might be an inferior good or a veblen good.

Quantity per time period

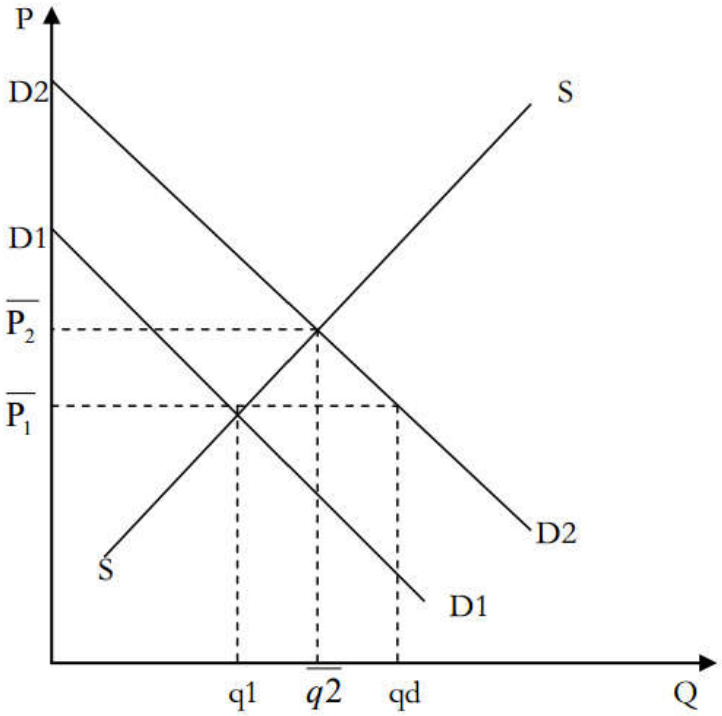
Price Ope is the equilibrium price and quantity Oqe is the equilibrium quantity. The "abnormal" demand curve means that at prices above Ope there is excess demand which pushes the price upwards and away from the equilibrium. Similarly, at prices below Ope, there

is excess supply which pushes the prices even further down.

Thus, although equilibrium are states of rest at which no economic forces exist to change the situation, it is important to remember that not all equilibria are stable. The equilibrium in the figure above is sometimes called a *knife edge* equilibrium because a small change in price sends the system well away from equilibrium.

EFFECTS OF SHIFTS IN DEMAND AND SUPPLY ON THE EQUILIBRIUM PRICE

i) Increase in demand

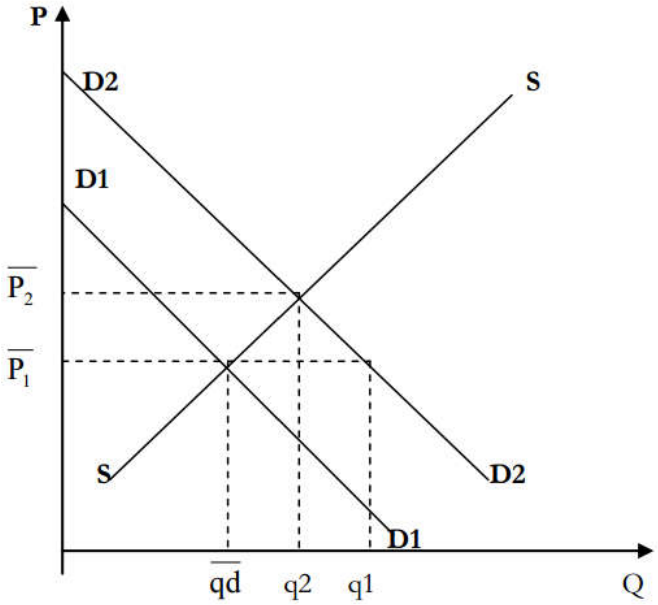


SS is the supply curve and D1D1 the initial demand curve shifts to the right, to position D2D2.

P1 is the initial equilibrium price and q1 the initial equilibrium quantity. When demand increases to D2D2, then at price P1, the quantity demanded increases from q1 to qd. But the quantity supplied at that price is still q1. This leads to excess demand over supply (qd – q1).

This causes prices to rise to a new equilibrium level P2 and the quantity supplied to rise to a new equilibrium level, q2.

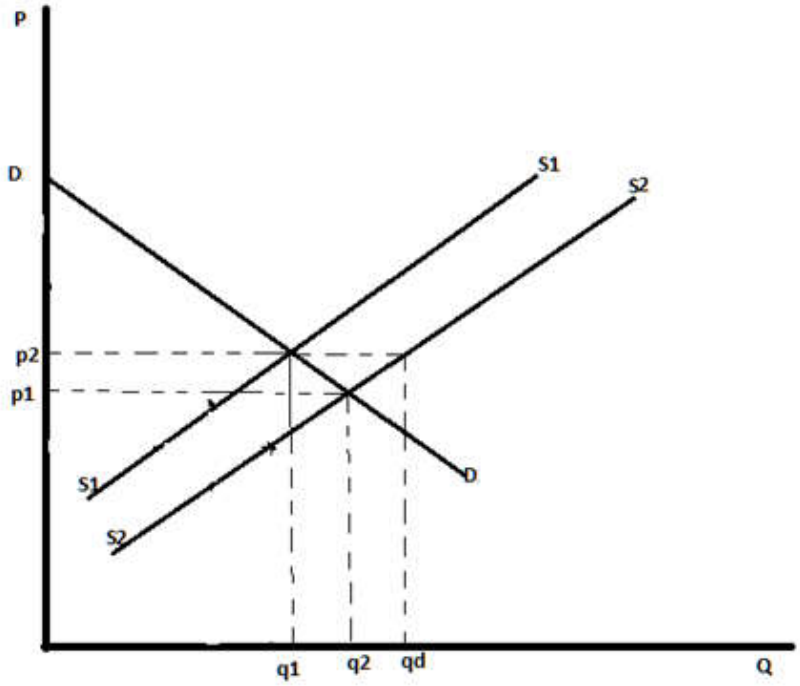
ii) Decrease in Demand



At the initial equilibrium price P1, quantity demanded falls from q1 to qd. But the quantity supplied is still q1 at this price. Hence, this creates excess of supply over demand, and this causes price to fall to a new equilibrium level P2 and quantity to fall to a new equilibrium level

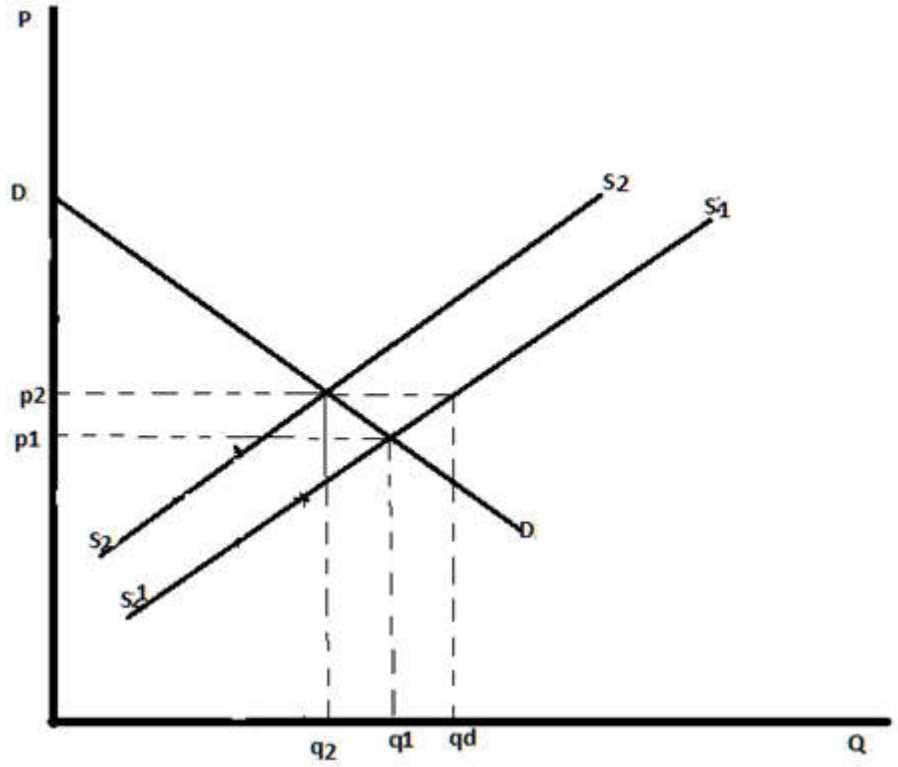
q2.

iii) Increase in Supply



DD is the demand curve and S1S1 the initial supply curve. If supply increases, the supply curve shifts to the right to position S2S2. At the initial equilibrium price P1, quantity supplied increase from q1 to q2. This creates a glut in the market and this causes the price to the new P21 and the quantity increases to a new equilibrium level q2.

iv) Fall in Supply



When the supply falls, the supply curve shifts to the left to position S1S1. At the initial equilibrium price P1, quantity supplied falls from q1 to q21 but the quantity demanded is still q1.

This creates excess of demand over supply which causes price to rise to a new equilibrium level

P21 and quantity to fall to a new equilibrium level q21 and quantity to fall to a new equilibrium level q2.

PRICE CONTROLS

Price Floors and Price Ceilings are *Price Controls*, examples of government intervention in the free market which changes the market equilibrium. They each have reasons for using them, but there are large efficiency losses with both of them.

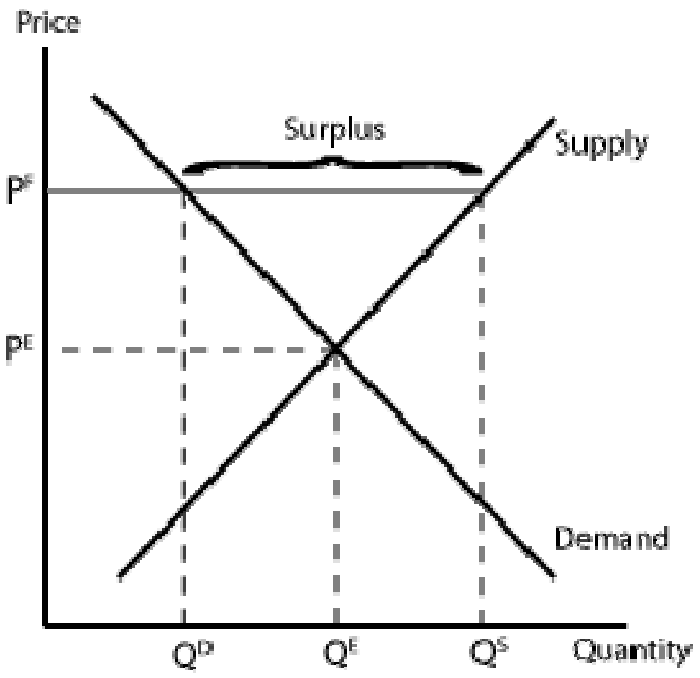
Price Floors

Price Floors are minimum prices set by the government for certain commodities and services that it believes are being sold in an unfair market with too low of a price and thus their producers deserve some assistance. Price floors are only an issue when they are set above the equilibrium price, since they have no effect if they are set below market clearing price.

When they are set above the market price, then there is a possibility that there will be an excess supply or a surplus. If this happens, producers who can't foresee trouble ahead will produce the larger quantity where the new price intersects their supply curve. Unbeknownst to them, consumers will not buy that many goods at the higher price and so those goods will go unsold.

There will be economic harm done even if suppliers can look ahead and see that there isn't sufficient demand and cut back on production in response. There is still deadweight loss associated with this reduction in quantity, reflected in the loss of consumer and producer surplus at lower levels of production. Producers can gain as a result of this policy, but only if their supply curve is relatively elastic and therefore they have no net loss. Consumers will definitely lose with this kind of regulation, as some people are priced out of the market and others have to pay a higher price than before.

There are numerous strategies of the government for setting a price floor and dealing with its repercussions. They can set a simple price floor, use a price support, or set production quotas. Price supports sets a minimum price just like as before, but here the government buys up any excess supply. This is even more inefficient and costly for the government and society as a whole than the government directly subsidizing the affected firms. Production quotas artificially raise the price by restricting production using either mandated quotas or giving businesses incentives to reduce their production. In America, this latter technique is used widely with agriculture. The government pays farmers to keep some portion of their fields fallow, thus elevating prices. Like price supports, the policy would be more efficient and less costly to society if the government directly subsidized farmers instead of setting a production restriction.



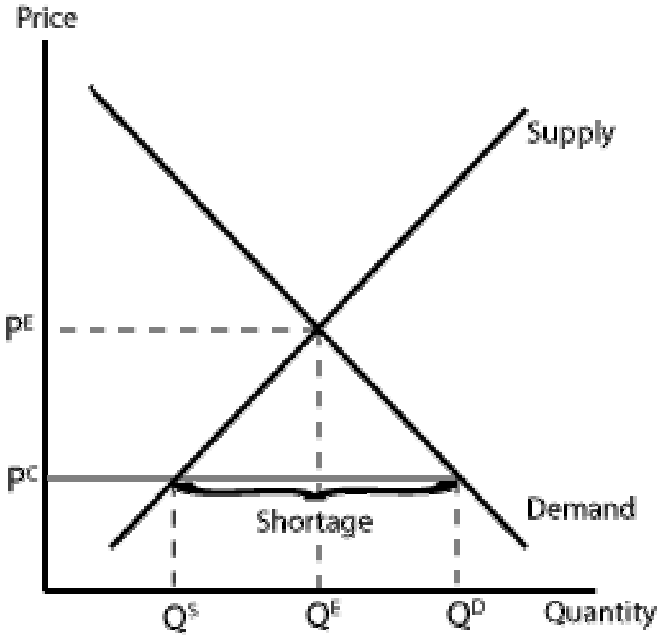
Price Ceilings

Price Ceilings are maximum prices set by the government for particular goods and services that they believe are being sold at too high of a price and thus consumers need some help purchasing them. Price ceilings only become a problem when they are set below the market equilibrium price.

When the ceiling is set below the market price, there will be excess demand or a supply shortage. Producers won't produce as much at the lower price, while consumers will demand more because the goods are cheaper. Demand will outstrip supply, so there will be a lot of people who want to buy at this lower price but can't. Still, if the demand curve is relatively elastic, then the net effect to

consumer surplus will be positive. Producers are truly harmed, as their surplus is doubly hit with a reduction in the number of firms willing to take that lower price, and those who remain in the market have to take a lower price.

The resulting shortage of goods can lead to consumers having to queue up in line to get the good, government rationing, and even the development of a black market dealing with the scarce goods. This is what occurred with the energy crisis in America during the 1970s, when cars had to line up on the street in order to just get some government rationed amount of gasoline.



Government action to stabilize farm prices and incomes

i) Buffer stocks and stabilization funds

In this case the government buys up part of the supply when output is excessive, stores this surplus, and resells it to consumers in times of shortage or reduced supply. The amounts that the government must buy or sell to stabilize incomes will therefore depend on the elasticity of demand.

In practice this normally operates through a marketing board controlling the industry, with monopoly powers to fix prices to producers. The Board will usually guarantee a minimum price for the commodity and may make an initial payment to the grower followed by an additional payment if sales by the Board subsequently realize a price in excess of the minimum. Producers of the crop are thus encouraged by the knowledge that any decrease in price during the season will be moderated by Government action.

In the stabilization Funds, the Government fix the price. When the demand is high, the government shall retain the difference, and subsidize the price to producers when demand is low.

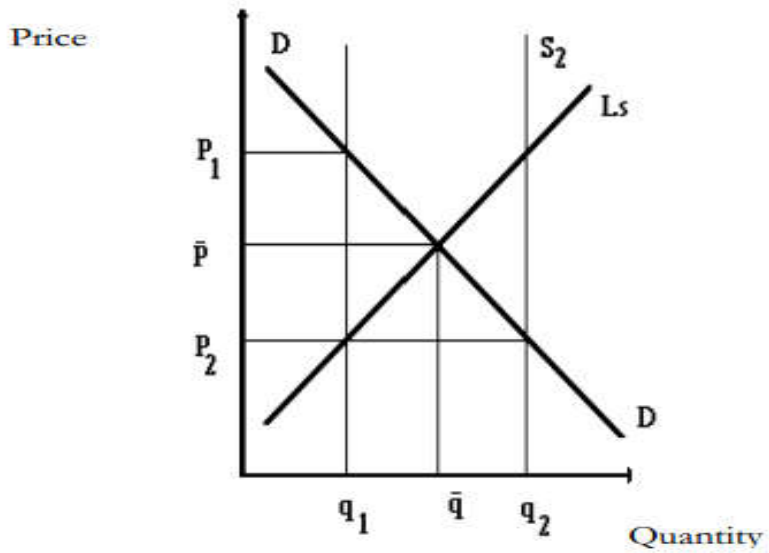
ii) The case of Diversification

This is done to reduce the uncertainty in the livelihood of the farmer.

REASONS FOR PRICE FLUCTUATIONS IN AGRICULTURE

Production depends on factors beyond the control of the producers e.g. weather, disease and pests. Actual and planned output is often not the same and the price charged is usually different from the expected price.

i) The production of these products depends on factors beyond the control of the producers e.g weather, diseases and pests, consequently, actual output is often not the same as planned output and hence the actual price is usually different from the expected price. At any one time the supply of the commodity will be perfectly inelastic.



DD is the demand curve and Ls the long run supply curve indicating what producers will be willing to produce and sell at different prices if production was entirely under their control.

Thus, P is the expected equilibrium prices and q the planned equilibrium quantity. Depending on the external factors mentioned above, actual output may be than planned output e.g. at q2 making the price p2 or less than planned output e.g. at q1 making the price P1.

The situation is made worse by the fact that these commodities are not easily stored, so that if the actual output falls short of planned output it cannot be supplemented from the stocks and if the actual output is greater than planned output it cannot be reduced which would prevent prices from being too low.

Besides the short run elasticity of supply is low, since once a given amount of the crop has been planted it is comparatively difficult to increase or decrease the resulting output. Hence high (or low) prices are likely to persist in the short term before additional supply can be made available

Furthermore the demand for these products is also price inelastic, for they are either foods or raw materials, and in the latter case they usually form a small proportion of the total inputs.

Thus, if actual output is in excess of planned output, it is difficult to sell off the excess without depressing prices excessively.

Equilibrium prices may also be difficult to attain because of lagged responses by producers to respond to price changes. In this case it is assumed that though producers are continually disappointed they never become wiser as a result and thus precipitate the price movement and that stocks of the commodity are not stored by producers or middle men in periods of low prices to be resold in periods of, otherwise high prices thus ironing out the unevenness of supply and price. This leads to the **cobweb theorem** (A dynamic model of supply and demand in which adaptive (or non-rational) expectations lead to perpetual oscillations in prices)

CHAPTER 3

THEORY OF THE CONSUMER BEHAVIOUR

APPROACHES TO THE THEORY OF THE CONSUMER

CARDINAL VERSUS ORDINAL APPROACH

Through the study of theory of consumer behaviour we can be able to explain why consumers buy more at a lower price than at a higher price or put differently why individuals or households spend their money as they do. We shall assume that the consumer is rational and aims at maximising his satisfaction, so given his income he consumes that basket of goods and services which produces maximum satisfaction. Two major theories explain the behaviour of the consumer, neither presents a totally complete picture. The first approach is the marginal utility, or cardinalist approach. The second approach centres on the indifference curve analysis or the ordinalist approach.

CARDINAL UTILITY APPROACH

The downward sloping nature of the demand curve can be explained by using the **law of diminishing marginal utility**. For instance, consider a consumer who ahs to choose between two goods, X and Y, which have prices Px and Py respectively. Assume that the individual is rational and so wishes to maximise total utility subject to the size of the income.

The consumer will be maximising total utility when his or her income has been allocated in such way that utility to be derived from the consumption of one extra shillings worth of X is equal to the utility to be derived from the consumption of one extra shillings worth of Y. In other words, when the marginal utility per shilling of X is equal to the marginal utility per shilling of Y. Only when this is true will it not be possible to increase total utility by switching expenditure from one good to another. This condition for **consumer equilibrium** can be written as follows:

Mux/PX=Muy/PY

Where *MUx* and *MUy* are the marginal utilities of X and Y respectively and *Px* and *Py* are the prices (in shillings) of X and Y respectively.

Any number of commodities may then be added to the equation. The table below gives

hypothetical marginal utility figures for a consumer who wishes to distribute expenditure of K£44 between three commodities X, Y and Z.

	Marginal utilities derived from each Kg of:		
Kg consumed	**x (£8/kg)**	**Y (£4/kg)**	**Z (£2/kg)**
1	72	60	64
2	48	44	56
3	40	32	40
4	36	24	28
5	32	20	16
6	20	8	12
7	12	4	8

In order to maximize utility, the consumer must distribute available income so that:
Mux/PX =MuY/Py =Muz/Pz

From the table you can see that this yields, a selection where the consumer buys 2 kg of X, 4 kg of Y and 6 kg of Z. Hence:

48/8= 24/4 =12/2

If the consumer wishes to spend all the K£44, it is impossible to distribute it any other way which would yield greater total quality. This theorem is called the concept of **equi-marginal utilities**.

The demand curve

Suppose that starting from a condition of equilibrium, the price of X falls relative to Y. We now have a condition where the utility from the last shilling spent on X will be greater than the utility from the last shillings spent on Y. Mathematically this can be written as:

Mux/Px >Muy/Py

In order to restore the equilibrium the consumer will buy more of X (and less of Y), thus reducing the marginal utility of X. The consumer will continue substituting X for Y until equilibrium is achieved. Thus we have attained the normal demand relationship that, ceteris paribus, as the price of X falls, more of it is bought. We have therefore a normal downward sloping demand curve. The demand curve we have derived is the individuals' demand curve for a product. The market demand curve can be then obtained by aggregating all the individual demand curves.

The explanation we have obtained here is of the price (or substitution) effect.

Market demand and consumers surplus.

Suppose that the market price of a cup of coffee is K£4 but the consumer was willing to pay £9

for the first unit, £8 for the second, £7 for the third, £6 for the fourth, £5 for the fifth and £4 for the sixth.

However, he pays the market price for all the six cups. The consumer thus earns a surplus on the first five units consumed i.e.

A measure of the difference between the value that consumers place on their total consumption of some commodity and the amount they actually pay for it.

For continuous demand curves, consumer's surplus can be measured by the area under the demand curve and above the price.

NB: The shaded area represents utility which the consumers received but did not pay for i.e. consumer surplus.

Mathematically it can be calculated as follows:

£5 + £4 + £3 + £2 + £1 = £15

ORDINAL APPROACH

In the 1930s a group of economists, including Sir John Hicks and sir Roy Allen, came to believe that cardinal measurement of utility was not necessary. They argued that demand behavior could be explained with ordinal numbers (that is, first, second, third, and so on). This is because, it is argued, individuals are able to rank their preferences, saying that they would prefer this bundle of goods to that bundle of goods and so on. Finite measurement of utility therefore becomes unnecessary and it's sufficient simply to place in order consumers preference to investigate this we must investigate indifference curves.

UTILITY ANALYSIS, MARGINAL UTILITY(MU), LAW OF DIMINISHING MARGINAL UTILITY

Utility

Utility is the amount of satisfaction derived from the consumption of a commodity or service at a particular time. Utility is not inherent but a psychological satisfaction, i.e. depends on the individual's own subjective estimate of the amount of satisfaction to be obtained from the consumption of the commodity.

Marginal Utility

The extra utility derived from the consumption of one more unit of a good, the consumption of all other goods remaining unchanged.

THE LAW OF DIMINISHING MARGINAL UTILTY

This states that as the quantity of a good consumed by an individual increases, the marginal utility of the good will eventually decrease.

Units of X consumed	Total Utility/ TU (utils)	Marginal Utility/ MU (utils)
0	0	0
1	15	15
2	25	10
3	33	8
4	38	5
5	40	2
6	40	0
7	39	-1

Consuming 1 unit of X gives 15 utils of satisfaction, consuming 2 units gives 25 utils, and so on. The figure of marginal utility decline as each successive unit is consumed. If the consumer goes on consuming more and more units, eventually he reaches a point (the sixth unit) where additional units yields no extra satisfaction at all.

LIMITATIONS OF CARDINAL APPROACH

- Utility cannot be measured objectively and the attempt to measure it by an economist Walras using utils doesn't give a satisfactory solution
- The assumption of constant marginal utility of money is unrealistic because as income increase, the marginal utility of money also increases
- The assumption of diminishing marginal utility is a psychological law that is a feeling that cannot be quantified

INDIFFERENCE CURVE ANALYSIS

In order to explain indifference curves, we will again make the simplifying assumption that the consumer buys two goods, x and y.

The table below gives a number of combinations of x and y which the consumer considers to give the same satisfaction as for example, combination c of bx and 4y is thought to give the same satisfaction as D where 7x and 2y are consumed. The consumer is thus said to be indifferent as to which combination they have hence the name given to this type of analysis. **An indifference schedule**

Combination	Units of x	Units of y
A	1	12
B	4	7
C	6	4
D	7	2

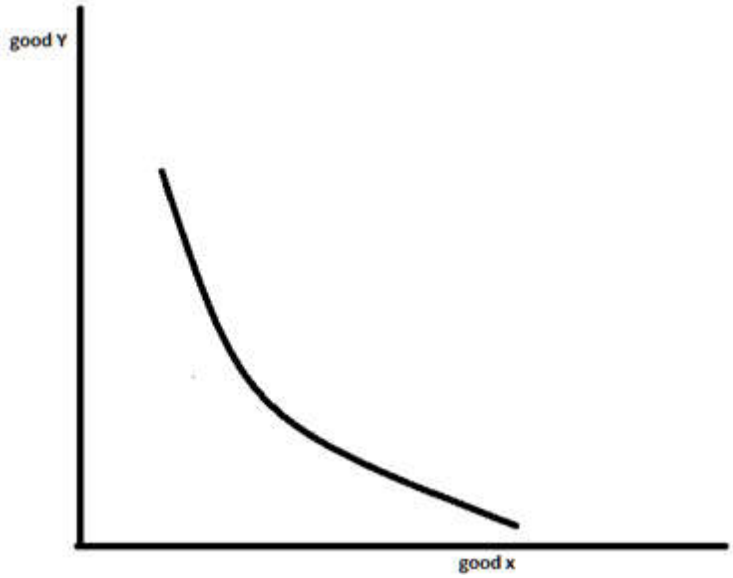


Figure gives a graphical representation of the figures

Such a graph is called an indifference curve:

An indifference curve shows the lines of combinations of the amounts of two goods say x and y such that the individual is indifferent between all combinations on that curve.

Theories of Demand and Supply and Consumer Behaviour

At each point on the indifference curve the consumer believes that the same amount of utility is received.

Properties of Indifference Curves:

- An indifference curve is usually convex to the origin.
- Indifference curves slope downwards from left to right.
- A set of indifference curves with each successive curve lying outside the previous one in a North East direction is called an indifference map.
- The curves do not cross as this would isolate the axiom of transitivity of preferences.
- Each curve is a graphical representation of a utility function expressed as:

U = f (x,y)

Where: **u** is a predetermined level of utility.

x and **y** are two commodities to be consumed in combination to guide u.

The slope of the indifference curve gives the rate at which a consumer is willing to exchange one unit of a product for units of another. This is called the **marginal rate of substitution**.

The Budget line and its economic interpretation

The indifference curve shows us consumer preferences but it does not show us the situation in the market place. Here the consumer is constrained by **income** and by the **prices** of X and Y.

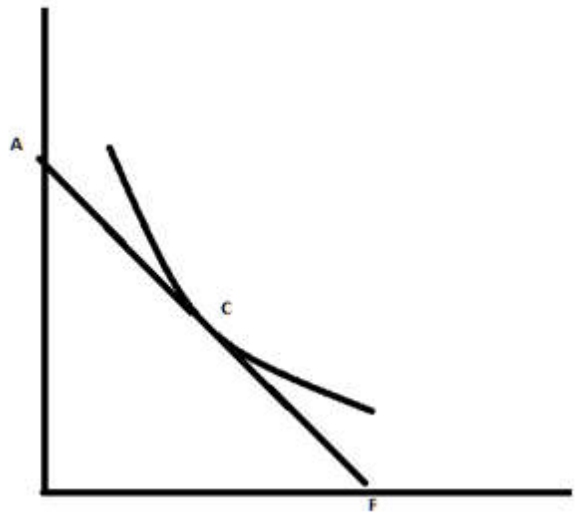
They can both be shown by a budget line. Suppose that product X costs K£2 per unit and product Y K£1 per unit and that the consumer's income is K£10.

Budget line

A budget line shows all the combinations of two products which can be purchased with a given level of income. The slope of the line shows the relative prices of the two commodities.

If the consumer is inside the budget line, e.g. at point E he is consuming less than the income.

Thus he can consume more of X or more of Y or more of both. If he is on the budget line e.g. at point C he is spending the full budget. He is said to be consuming to budget constraint. To consume more of X e.g. moving from C to D, he must consume less of Y and vice versa. For a given budget and given price, he cannot be at a point off the budget line to the right, e.g. at point F.

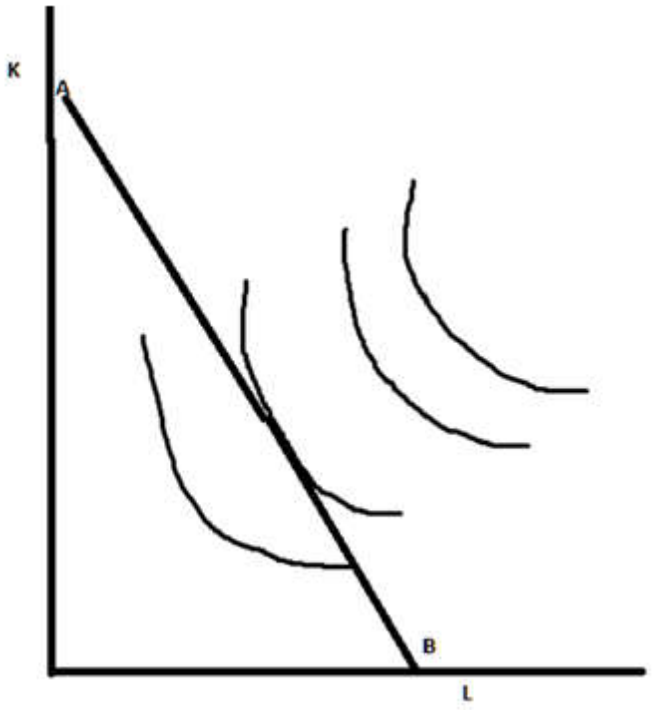


CONSUMER EQUILIBRIUM, EFFECTS OF CHANGES IN PRICES AND INCOMES ON CONSUMER EQUILIBRIUM

The Consumer Equilibrium

To demonstrate the consumer's equilibrium i.e. the point at which the consumer maximizes utility with a given budget, we need to combine the indifference map and the budget line.

The consumer obtains maximum utility from a budget of AF by choosing the combination of X and Y represented by C, where the marginal rate of substitution is equal to the relative prices of X and Y.



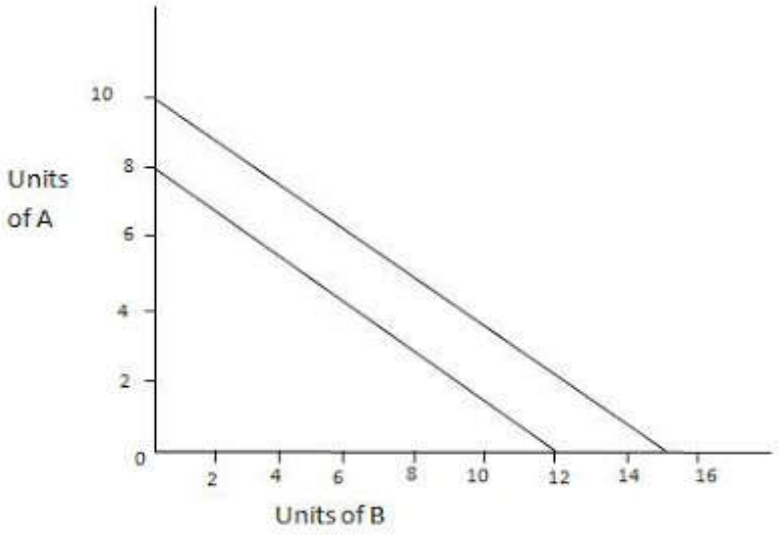
EFFECT OF CHANGES IN INCOME AND PRICES ON CONSUMER EQUILIBRIUM

Changes in income

If consumer income increases then the consumer will be able to purchase higher combinations of goods. Hence an increase in consumer income will result in a shift in the budget line. the prices of the two goods have remained the same, therefore, the increase in income will result in a parallel shift in the budget line.

Assume consumer income increased to Rs 15.So, now he can buy more quantities of both goods.

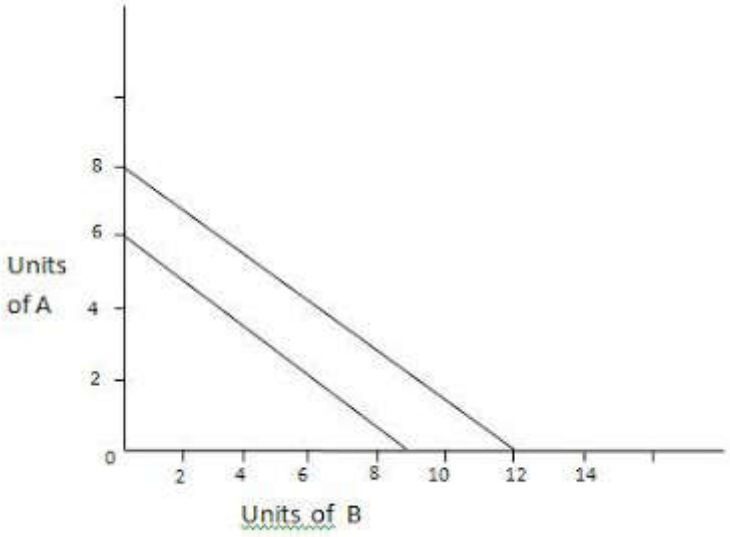
Increase in income



An Increase in income makes the purchase of more of either one or both items possible.

If consumer income falls then there would be a corresponding parallel shift to the left to represent a fall in the potential combinations of the two goods that can be purchased.Now, suppose the consumer income falls to RS. 9.

Fall in Income



An decrease in income makes the purchase of less of either one or both items possible.

Price

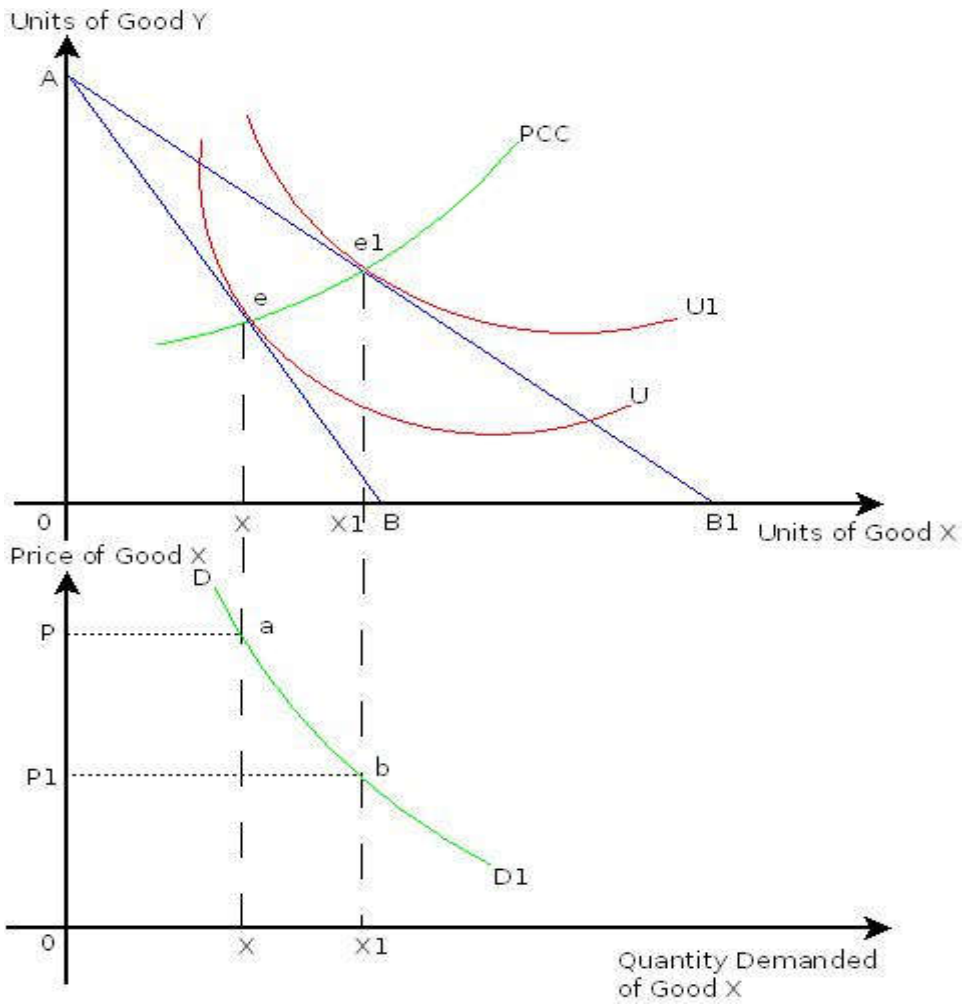
An increase in the price of one good will make the consumer buy less of that good and buy more of the other. This consequently changes the equilibrium of the consumer and raises it higher.

DERIVATION OF DEMAND CURVE

Derivation of the Consumer's Demand Curve: Normal Goods

The price consumption curve traces the effect of a change in price of a good on its quantity demanded. However, it does not directly show the relationship between the price of a good and its corresponding quantity demanded. It is the demand curve that shows relationship between price of a good and its quantity demanded. In this section we are going to derive the consumer's demand curve from the price consumption curve . Figure.1 shows derivation of the consumer's demand curve from the price consumption curve where good X is a normal good.

FIGURE.1 Derivation of the Demand Curve: Normal Goods



The upper panel of Figure.1 shows price effect where good X is a normal good. AB is the initial price line. Suppose the initial price of good X (P_x) is OP. e is the initial optimal consumption combination on indifference curve U. The consumer buys OX units of good X. When price of X (P_x) falls, to say OP_1, the budget constraint shift to AB_1. The optimal consumption combination is e_1 on indifference curve U_1. The consumer now increases consumption of good X from OX to OX_1 units. The Price Consumption Curve (PCC) is rising upwards.

Chart.1 shows the demand relationship derived form the price consumption curve.

Chart.1

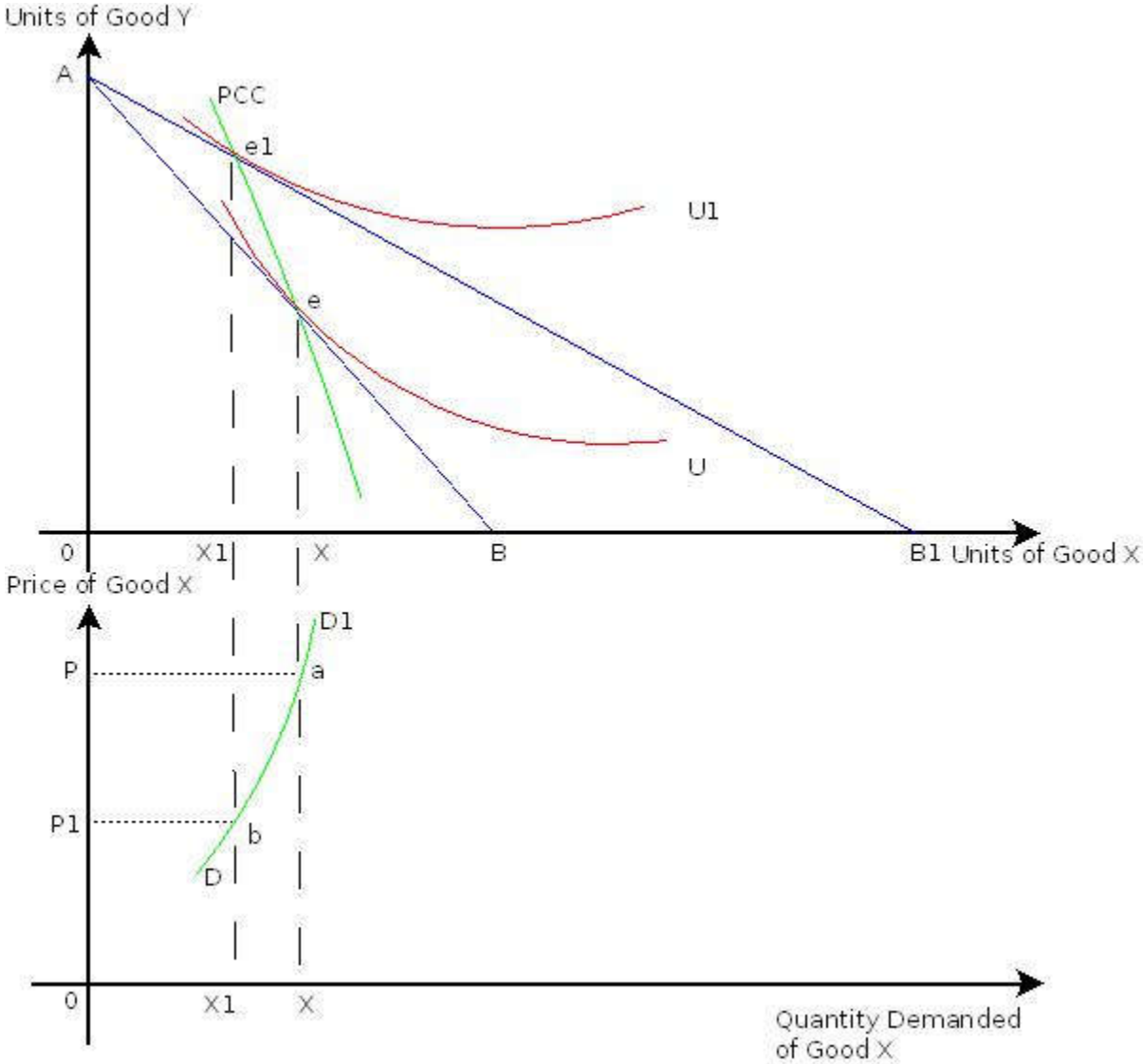
	Price of Good X	Quantity Demanded of Good X
Initial Price & Quantity of Good X	OP	OX
Fall in price & corresponding Quantity Demanded	OP_1	OX_1

The lower panel of Figure.1 shows this price and corresponding quantity demanded of good X as shown in Chart.1. At initial price OP, quantity demanded of good X is OX. This is shown by point a. At a lower price OP_1, quantity demanded increases to OX_1. This is shown by point b. DD_1 is the demand curve obtained by joining points a and b. The demand curve is downward sloping showing inverse relationship between price and quantity demanded as good X is a normal good.

Derivation of the Consumer's Demand Curve: Giffen Goods

In this section we are going to derive the consumer's demand curve from the price consumption curve in the case of inferior goods. Figure.2 shows derivation of the consumer's demand curve from the price consumption curve where good X is an inferior good.

FIGURE.2 Derivation of the Demand Curve: Inferior Goods



The upper panel of Figure.2 shows price effect where good X is an inferior good. AB is the initial price line. Suppose the initial price of good X (P_x)is OP. e is the initial optimal consumption combination on indifference curve U. The consumer buys OX units of good X. When price of X P_x) falls, to say OP_1, the budget constraint shift to AB_1. The optimal consumption combination is e_1 on indifference curve U_1. The consumer now reduces consumption of good X from OX to OX_1 units as good x is inferior. The Price Consumption Curve (PCC) is rising upwards and bending backwards towards the Y-axis.

Chart.1 shows the demand relationship derived form the price consumption curve.

Chart.2

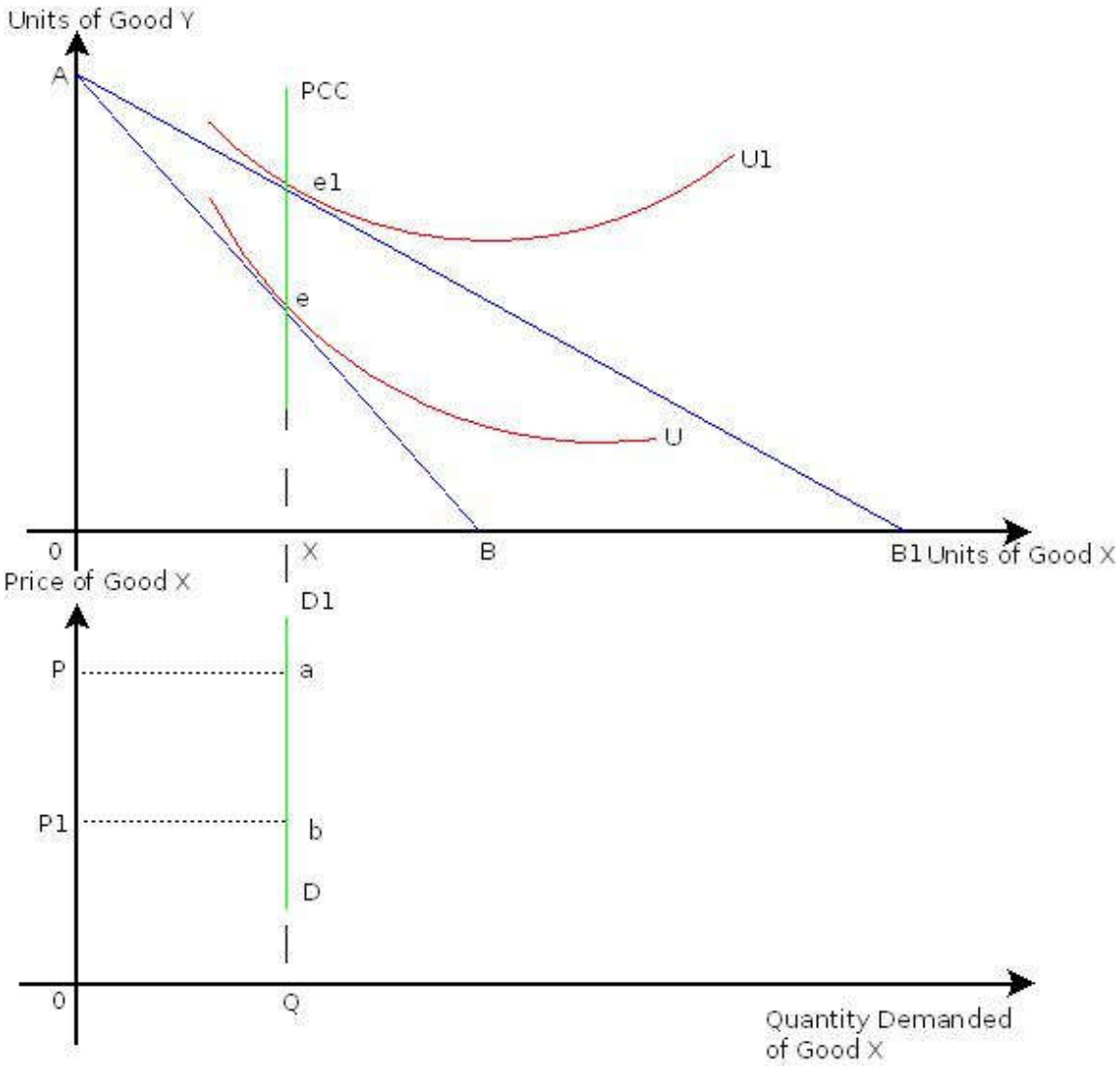
	Price of Good X	Quantity Demanded of Good X
Initial Price & Quantity of Good X	OP	OX
Fall in price & corresponding Quantity Demanded	OP_1	OX_1

The lower panel of Figure.2 shows this price and corresponding quantity demanded of good X as shown in Chart.2. At initial price OP, quantity demanded of good X is OX. This is shown by point a. At a lower price OP_1, quantity demanded decreases to OX_1. This is shown by point b. DD_1 is the demand curve obtained by joining points a and b. The demand curve is upward sloping showing direct relationship between price and quantity demanded as good X is an inferior good.

Derivation of the Consumer's Demand Curve: Neutral Goods

In this section we are going to derive the consumer's demand curve from the price consumption curve in the case of neutral goods. Figure.3 shows derivation of the consumer's demand curve from the price consumption curve where good X is a neutral good.

FIGURE.3 Derivation of the Demand Curve: Neutral Goods



The upper panel of Figure.3 shows price effect where good X is a neutral good. AB is the initial price line. Suppose the initial price of good X (P_x) is OP. e is the initial optimal consumption combination on indifference curve U. The consumer buys OX units of good X. When price of X (P_x)falls, to say OP_1, the budget constraint shift to AB_1. The optimal consumption combination is e_1 on indifference curve U_1 at which the consumer buys same OX units of good X as it is a neutral good. The Price Consumption Curve (PCC) is a vertical straight line.

Chart.3 shows the demand relationship derived form the price consumption curve.

Chart.3

	Price of Good X	Quantity Demanded of Good X
Initial Price & Quantity of Good X	OP	OX
Fall in price & corresponding Quantity Demanded	OP_1	OX_1

The lower panel of Figure.3 shows this price and corresponding quantity demanded of good X as shown in Chart.3. At initial price OP, quantity demanded of good X is OX. This is shown by point a. At a lower price OP_1, quantity demanded remains fixed at OX. This is shown by point b. DD_1 is the demand curve obtained by joining points a and b. The demand curve is a vertical straight line showing that the consumption of good X is fixed as good X is a neutral good.

APPLICATION OF INDIFFERENCE CURVE ANALYSIS

SUBSTITUTION EFFECT AND INCOME EFFECT

The **substitution effect** is one component of the effect of a change in the price of a good upon the amount of that good demanded by a consumer, the other being the income effect.

When a good's price decreases, if hypothetically the same consumption bundle were to be retained, income would be freed up which could be spent on a combination of more of each of the goods. Thus the new total consumption bundle chosen, compared to the old one, reflects both the effect of the changed relative prices of the two goods (one unit of one good can now be traded for a different quantity of the other good than before as the ratio of their prices has changed) *and* the effect of the freed-up income. The effect of the relative price change is called the *substitution effect*, while the effect due to income having been freed up is called the *income effect*.

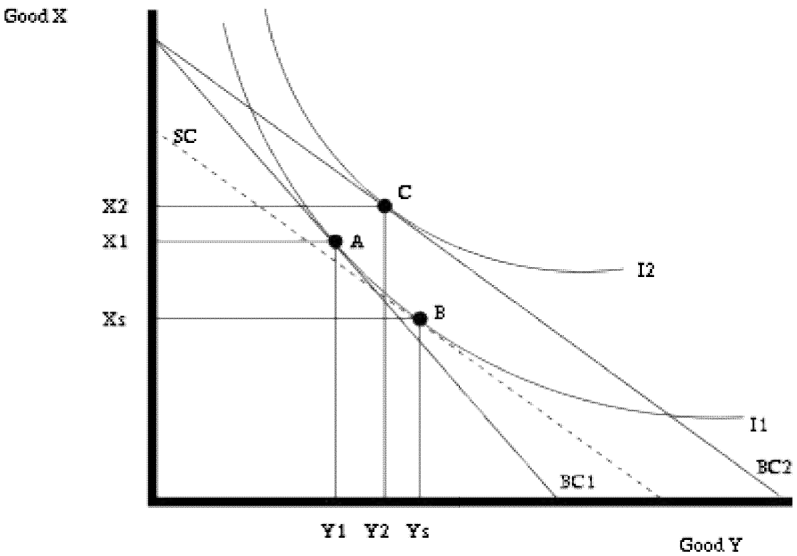
If income is altered in response to the price change such that a new budget line is drawn passing through the old consumption bundle but with the slope determined by the new prices and the consumer's optimal choice is on this budget line, the resulting change in consumption is called the **Slutsky substitution effect**. The idea is that the consumer is given enough money to purchase her old bundle at the new prices, and her choice changes are seen.

If instead, a new budget line is found with the slope determined by the new prices but tangent to the indifference curve going through the old bundle, the difference between the new point of tangency and the old bundle is the **Hicks substitution effect**. The idea now is that the consumer is given just enough income to achieve her old utility at the new prices, and how her choice changes is seen. The Hicks substitution effect is illustrated in the next section.

Some authors refer to one of these two concepts as simply the substitution effect. The popular textbook by Varian describes the Slutsky variant as the primary one, but also gives a good explanation of the distinction.

The same concepts also apply if the price of one good goes up instead of down, with the substitution effect reflecting the change in relative prices and the income effect reflecting the fact the income has been soaked up into additional spending on the retained units of the now-pricier good.

Graphical analysis



Suppose the initial situation is given by the graph (with good Y plotted horizontally) with the indicated (and never-changing) indifference curves shown and with budget constraint BC1 and with the consumer choosing point A because it puts him on the highest possible indifference curve consistent with BC1. The position and slope of the budget constraint are based on the consumer's income and on the prices of the two goods X and Y. If the price of Y falls, the budget constraint pivots to BC2, with a greater intercept of good Y because if all income were spent on Y more of it could be purchased at the now-lower price. The overall effect of the price change is that the consumer now chooses the consumption bundle at point C.

But the move from A to C can be decomposed into two parts. The **substitution effect** is the change that would occur if the consumer were required to remain on the original indifference curve; this is the move from A to B. The income effect is the simultaneous move from B to C that occurs because the lower price of one good in fact allows movement to a higher indifference curve.(In this graph Y is an inferior good)

Inferior and Giffen Goods

The substitution effect always acts in such a way that when the relative price of a good falls (real income remaining constant), more of its is purchased. The income effect, however, can work either way – when the consumer's real income rises more or less of good x may be bought.

If the less is bought, the good is said to be an **inferior good**. If the consumer buys less and the income effect is actually bigger than the substation effect so that the overall effect of the price fall is a decrease in consumption, then the good is said to be a **Giffen Good**.

Inferior Good: The negative income effect is smaller than the substitution effect

Giffen Good: The negative income effect is bigger than the substitution effect so that the net effect of a fall in the price of x is a fall in quantity demanded

a) **Uses of Indifference Curve Analysis**

Indifference curve analysis is useful when studying welfare economics as follows:

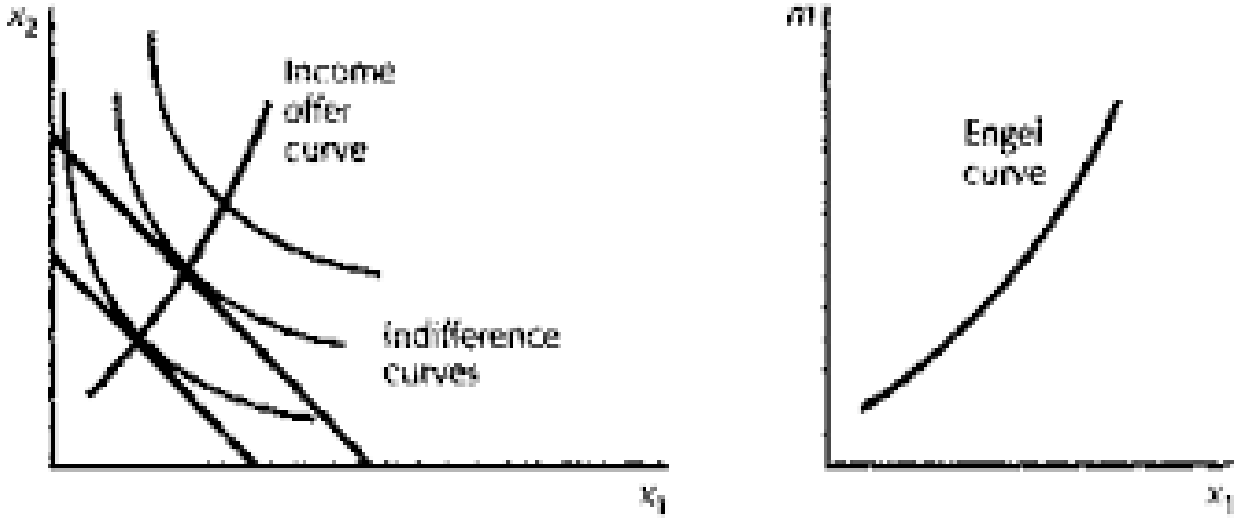
- They are used to indicate the amount of income and leisure combination that can yield a given level of satisfaction allowing for the measure of trade off between leisure and income.
- Since each indifference curve represents a given level of welfare, in an indifference map, the curve to the right represents a higher level of welfare. This is useful in analysing the effect taxation on the standard of living in an economy. A tax level may reduce the economic standard of the people and vice versa.

Theories of Demand and Supply and Consumer Behaviour

Employees use indifference curve analysis to decide whether to give employees housing facility in kind or in money allowance in a manner not to affect their welfare.

ENGELS CURVE

An Engel curve describes how a consumer's purchases of a good like food varies as the consumer's total resources such as income or total expenditures vary. Engel curves may also depend on demographic variables and other consumer characteristics. A good's Engel curve determines its income elasticity, and hence whether the good is an inferior, normal, or luxury good. Empirical Engel curves are close to linear for some goods, and highly nonlinear for others. Engel curves are used for equivalence scale calculations and related welfare comparisons, and determine properties of demand systems such as aggregability and rank.



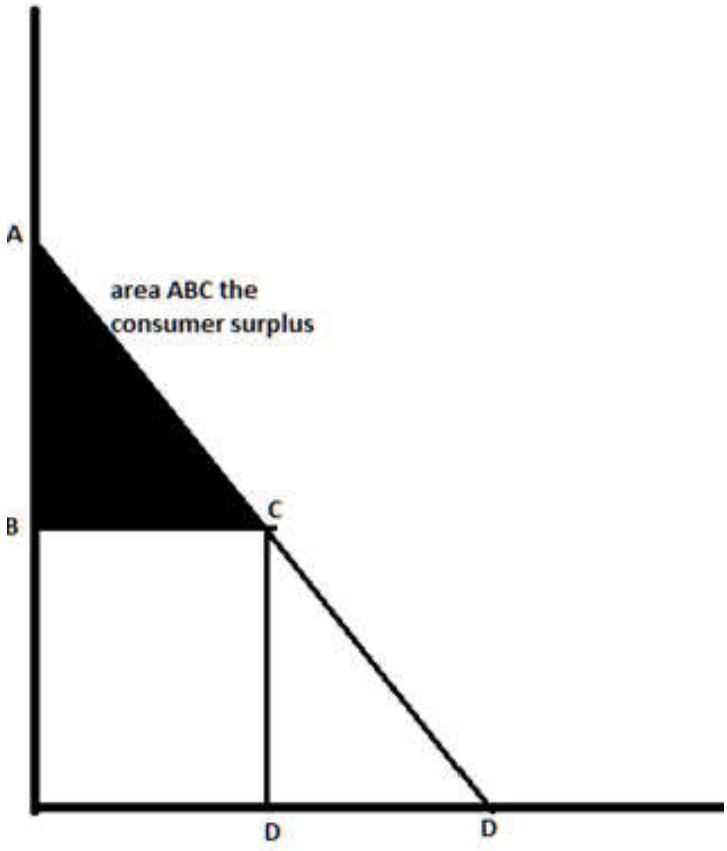
A Income offer curve B Engel curve

CONSUMER SURPLUS

Consumer surplus is the difference between the maximum price a consumer is willing to pay and the actual price they do pay. If a consumer would be willing to pay more than the current asking price, then they are getting more benefit from the purchased product than they initially paid. An example of a good with generally high consumer surplus is drinking water. People would pay very high prices for drinking water, as they need it to survive. The difference in the price that they would pay, if they had to, and the amount that they pay now is their consumer surplus. Note that the utility of the first few liters of drinking water is very high (as it prevents death), so the first few litres would likely have more consumer surplus than subsequent liters.

The maximum amount a consumer would be willing to pay for a given quantity of a good is the sum of the maximum price they would pay for the first unit, the (lower) maximum price they would be willing to pay for the second unit, etc. Typically these prices are decreasing; they are given by the individual demand curve. For a given price the consumer buys the amount for which the consumer surplus is highest, where consumer surplus is the sum, over all units, of the excess of the maximum willingness to pay over the equilibrium (market) price. The consumer's surplus is highest at the largest number of units for which, even for the last unit, the maximum willingness to pay is not below the market price

The aggregate consumers' surplus is the sum of the consumer's surplus for all individual consumers.



QUESTIONS

Question One

Write short notes on the following:

a) substitute and complementary goods
b) price controls
c) indifference curves
d) inferior and giffen goods

Question Two

a) Define the term cross price elasticity of demand and explain its value for substitutes and complementary commodities.
b) Use the data in the table below to complete income elasticity through the arc elasticity method

Quanitity	Income (Shs.)	Price (Shs.)
100	5,000	16
120	6,000	16

Question Three

Given the market model

P = Qs
Qd = 48 – 4P
Qs= -6 + 14P

Find P and Q

Question One

a) State the economic circumstances under which a perfectly competitive market may tribe.
b) In what ways does a perfect market differ from a monopoly, oligopoly and monopolistic competition?

Question Two

a) Distinguish between supply, demand and equilibrium price.
b) Explain what is meant by elasticity of supply and state the factors that determine the supply of a good in the market.
c) The table below shows the demand and supply schedules for a product:

PRICE (KShs. per kg)	DEMAND (kg)	SUPPLY (kg)
10	100	20
20	85	36
30	70	53
40	55	70
50	40	87
60	25	103
70	10	120

Draw the demand and supply curves and draw the equilibrium price and quantity.

Question Three

a) Comment on the pattern of industrial motion either in Kenya or your country and give possible explanations for such a pattern.
b) What policy changes would you recommend in order to improve industrial development in your country?

Question Four

a) What are the main factors of production?
b) What determines the supply of demand for the factors of production that you have identified in a) above?

Question Five

a) Explain the term price control.
b) In recent years many countries have abolished price controls. Discuss the short and longterm implications of decontrolled prices.

Question Six

Write short notes on the following::

i. Scarcity and choices
ii. Isoquants and isocost curves
iii. Substitute and complementary goals
iv. Opportunity cost

Question Seven

a) Explain what the term indifference curve entails.

b) By use of indifference curves diagrams, show and explain:

i) Consumers equilibrium

ii) Income and substitution effect of a fall in price and inferior good

Question Eight

a) What is a monopolist market?

b) Write short explanatory notes on the following:

i. Positive and normative economics
ii. Price discrimination
iii. Marginal rate of technical substitution

CHAPTER 4

THE THEORY OF PRODUCTION

FACTORS OF PRODUCTION

The sum total of the economic resources which we have in order to provide for our economic wants are termed as *factors of production*. Traditionally economists have classified these under four headings. They are:

i. Labour
ii. Land
iii. Capital
iv. Enterprise

The first two are termed primary factors since they are not the result of the economic process; they are, so to speak, what we have to start with. The secondary factors, however are a consequences of an economic system.

i) Land

The term land is used in the widest sense to include all the free gifts of nature; farmlands, minerals wealth such as coal mines, fishing grounds, forests, rivers and lakes.

In practise it may be very difficult to separate land from other factors of production such ascapital but, theoretically, it has two unique features which distinguish it.

Firstly, it is **fixed in supply**. As land includes the sea in definition, then we are thus talking about the whole planet, and it is obvious that we cannot acquire more land in this sense.

Secondly, land has no cost of production. The **individual** who is trying to **rent** a piece of land may have to pay a great deal of money but it never cost **society** as a whole anything to produce land.

ii) Capital

Capital as a factor of production can be defined either as the stock of wealth existing at any one time and as such, capital consists of all the real **physical** assets of society. An alternative formulation of capital is that it refers to all those goods, which are used in the production of further wealth.

Capital can be divided into **fixed capital**, which is such things as building, roads, machinery etc and **working capital** or **circulating capital** which consists of stocks of raw materials and semi-manufactured goods. The distinction is that fixed capital continues through many rounds of production while working capital is used up in one round; For example, a classroom would be fixed capital, while stocks of chalk to be used for writing would be circulating/working capital.

As stated previously, capital is a secondary factor of production, which means that results from the economics system. Capital has been created by individuals **forgoing current consumption**, i.e.

people have refrained from consuming all their wealth immediately and have saved resources which can then be used in the production of further wealth.

iii) Labour

Labour is the exercise of human, physical and mental effort directed to the production of goods and services. Included in this definition is all the labour which people undertake for reward, either in form of wages and salaries or incomes from self employment. We would not, therefore include housework or the efforts of do-it-yourself enthusiasts, even though these may be hard work.

Some aspects of labour

Labour is no doubt the most important of all factor or production, for the efficiency of any production will to a large extent depend on the **efficiency** and **supply** of the labour working in the process. Besides labour is also the end for which all production is undertaken.

Supply of labour

Supply of labour refers to the number of workers (or, more generally, the number of labour hours) available to an economy. The supply of labour will be determined by:

I. Population Size

In any given economy, the population size determines the upper limit of labour supply.

Clearly there cannot be more labour than there is population.

II. Age Structure

- The population is divided into three age groups. These are:
- The young age group usually below the age of 18, which is considered to be the minimum age of adulthood. People below this age are not in the labour supply, i.e. they are not supposed to be working or looking for work.
- The working age group, usually between 18 and 60, although the upper age limit for this group varies from country to country. In Kenya for example, for public servants, it is 55 years. It is the size of this group which determines the labour supply.
- The old age group, i.e. above 60 years are not in the labour force.

III. The Working Population

Not everybody in the working age group will be in the labour force. What is called the working population refers to the people who are in the working group, and are either working or are actively looking for work, I.e. would take up work if work was offered to them. These are sometimes called the actively active people. Hence this group excludes the sick, the aged, the disabled and (full time) housewives, as well as students. These are people who are working and are not willing or are not in a position to take up work was given to them.

IV. Education System

If the children are kept in school longer, then this will affect the size of the labour force of the country.

V. Length of the Working Week

This determines labour supply in terms of Man-hours. Hence the fewer the holidays there are, the higher will be the labour supply. This does not, however mean that if the number of working hours in the week are reduced, productivity if there is a high degree of automation.

VI Remuneration

The preceding five factors affect the supply of labour in totality. Remuneration affects the supply of labour to a particular industry. Thus, an industry which offers higher wages than other industries will attract labour from those other industries.

VII The Extent to Barriers to Entry into a Particular Occupation

If there are strong barriers to the occupation mobility of labour into a particular occupation, e.g. special talents required or long periods of training, the supply of labour to that occupation will be limited.

Efficiency of Labour

Efficiency of labour refers to the ability to achieve a greater output in a shorter time without any falling off in the quality of the work – that is to say, increase productivity per man employed. The efficiency of a country's labour force depends on a number of influences.

i. Climate

This can be an important influence on willingness to work, for extremes of temperatures or high, humidity are not conducive to concentration even on congenial tasks.

ii. Education and training

Education and training produce skills and therefore efficient labour. Education has three aspects: general education, technical education and training within industry. A high standard of general education is essential for developing intelligence and providing a foundation upon which more specialized vocational training can be based. Technical training provided in the universities, colleges and by industry itself. Training within industry is given by each firm to its employees.

iii. Working Conditions

Research has shown that if working conditions are safe and hygienic, the efficiency of labour will be higher than if the conditions were unsafe or unhygienic.

iv. Health of the worker

The efficiency of the worker is closely related to his state of health which depends on his being adequately fed, clothed, and housed.

v. Peace of Mind

Anxiety is detrimental to efficiency. People (workers) may be tempted to overwork themselves to save at the expense of health to provide for contingencies like times of sickness, unemployment and old age. Others may be worried about their work or their private problems.

vi. Efficiency of the Factors

The productivity of labour will be increased if the quality of the factors is high. The more fertile the land, the greater will be the output per mass, other things being equal. Similarly, the greater the amount and the better the quality of the capital employed, the greater will be the productivity of the labour.

Efficiency of the organisation is even more important since this determines whether the best use is being made of factors of production.

vii. Motivating factors

These are factors which boost the morale of the workers and hence increase the efficiency.

They include such things as free or subsidised housing, free medical benefits, paid sick leave, allowing workers to buy shares in the company and incorporating workers' representatives in the decision-making of the firm, In this way the workers feel that they are part and parcel of the organisation and are not being used.

viii. The Extent of Specialisation and Division of Labour

The greater the amount of specialisation, the greater will be the output per man, Division of labour increases the efficiency of labour.

ix The Entrepreneur

Land, capital and labour are of no economic importance unless they are organised for production. The entrepreneur is responsible not only for deciding what method of production shall be adopted but for organising the work of others. He has to make many other important decisions such as what to produce and how much to produce.

Functions of the Entrepreneur

1. Uncertainty Bearing

Most production is undertaken in anticipation of demand. Firms will produce those things which they believe will yield profit. They do not know that they will do so because the future is unknown.

2. Management Control

This involves responsibility for broad decisions of policy and the ability to ensure that these decisions are carried out.

MOBILITY OF FACTORS OF PRODUCTION

Factor mobility means the ease with which a factor can be moved from one form or area of employment to another. There are two aspects to mobility. Movement from one employment to another is called occupational mobility and movement from one place of employment to another is called geographical mobility. The geographical sense of the movement from one place to another is geographical mobility.

Mobility of Land

Land is geographically immobile in that a given piece of land cannot be moved from one place to another. However, land can be occupationally mobile in that it can be put to different uses, e.g. farming, grazing and building. Some forms of land have limited occupational mobility in that they can be put to a limited number of uses e.g. arid or desert areas and mountainous land.

The former may be used as grazing land by nomadic people, unless it is found to have mineral deposits, while the latter may be used as a tourist attraction or for pleasure in mountain climbing. Immobility geographically implies that it cannot be used to increase production of a particular product unless this is done at the expense of other products.

Mobility of Capital

Some forms of capital are immobile in both geographical and occupational sense e.g. heavy machinery and railway networks. Usually once such equipment has been installed on land in a particular place, it becomes uneconomical to uproot it and move it to another place. Hence, because of the heavy costs that such an operation would involve, it is for all practical purposes geographically immobile. Also such equipment can usually be put to only the use for which it was intended and it is occupationally immobile.

Other forms of capital are geographically immobile but are occupationally mobile e.g. buildings.

Once a building has been set up in a place, it cannot be moved intact to another place, but it can be converted to a hotel or bar. Other forms of capital are mobile both geographically and occupationally e.g. vehicles and hand tools which can be moved from place to place and can also be put to different uses.

Mobility geographically facilitates production. Immobility occupationally makes it difficult to increase output in the short run.

Mobility of Labour

Labour is relatively mobile geographically, but less so occupationally in that people can be moved from one place to another but find it hard to change occupations if it is highly specialized. If a person moves from one occupation to another occupation on higher level, either in terms of remuneration or in terms of status or both, this is called Vertical occupational mobility, e.g. if an accountant becomes a manager. If movement is from one occupation to another on the same level, this is called horizontal occupational mobility. Also in this case the social status of two occupations is more-or-less the same.

Although labour is relatively mobile geographically, there are factors, which act as barriers to its mobility geographically.

Barriers to Geographical Mobility of Labour

i. Cost of Movement

The cost of movement from one place to another place can be a prohibitive factor in the geographical movement of labour.

ii. Shortage of Housing

This can be a prohibitive factor especially in urban areas. Thus if person is working in a ruralarea, he may be reluctant to move to an urban area, for example, if he is living in his own house or a cheaper house in the rural area, he may not be able to find a house for himself and his family in the urban area.

iii. Education of Children

A person may be reluctant to move from one area to another because he does not want to interrupt the education of his children either because it is late in the year and the children cannot find school places in the new area or because the new area has a different education system, e.g. it is a different state of the country or a different country altogether.

iv. Social and Family Ties

If a person has been living in a place for a long time, he may be reluctant to leave and go to another place because he does not want to leave his friends and relatives (especially the aged parents) behind.

v. Geographical and Economic Factors

Some areas are known to be hostile areas, i.e. they are too hot or too cold consequently people are reluctant to go to work in such areas. Also in addition to factors of shortages of housing in urban areas referred to earlier, people may be reluctant to leave their jobs in rural areas because of higher cost of living in urban areas.

i. Personal Talents

There are some occupations which require special talents which not everybody has e.g. the occupation of engineers, statisticians and surgeons.

ii. Length of Training

It requires a long period of training to acquire the necessary skills for a particular occupation; this can act as a discouragement to go into such occupations.

iii. Capital Limitations

In order for a person to go into such a business enterprise, he has to have the necessary capital. Hence lack of capital can act as a barrier to entry into commercial activities.

iv. Class Limitations

In societies, which operate a caste system, people belonging to a particular caste may not do certain jobs. This will also happen to countries, which practice racial discrimination. People belonging to a particular race may not be allowed to do certain jobs, which are reserved for other races. The Mobility geographically facilitates production while immobility occupationally discourages it. Immobility geographically and occupationally leads to structural unemployment and unemployment due to technological progress.

The most mobile of the factors of production is probably the entrepreneur. This is because the basic functions of the entrepreneur are common to all industries. Whatever the type of economic activity there will be a need to raise capital, to organise the factors of production and to take the fundamental decisions on where, what and how to produce.

FACTOR INCOMES

The various incomes which the factors receive can be termed **factor rewards** or **factor returns**. Labour receives wages and salaries, land earns rent, capital earns interest and enterprise earns profit.

Theory of the firm

The Theory of the firm is that branch of economics which studies how firms combine various inputs to produce a stipulated output in an economically efficient manner given technology and the various costs that they must meet to produce the various levels of output.

The firm is an entity, which produces any economic good under one management with the aim of maximizing its profits. It differs from the plant in that the plant is the unit of production in an industry while the firm is the unit of ownership and control. An industry is all the firms concerned with a particular line of production.

PRODUCTION FUNCTION ANALYSIS

This deals with how firms combine various inputs to produce a stipulated output in an economically efficient manner, given technology.

a. Varying the proportions

In making a product, a firm does not have to combine the inputs in fixed proportions. Many farm crops can be grown by using relatively little labour and relatively large amounts of capital (machinery, fertilizers etc) or by combining relatively large amounts of labour with very little capital. In most cases a firm has the opportunity to vary the "input mix" However, before looking at how the firms combine the various factors we need to know some concepts which firms must take into account namely;

i. The short run: The period of time in which at least one factor is fixed in supply i.e. cannot be varied.

ii. The long run: The period, in which all factors may be varied, in which firms may enter or leave the industry.

iii. Variable (factor) Input: This is a factor of production which varies with output in the short run and is one whose quantity may be changed when market conditions require immediate change in output.

iv. Fixed Input: Is factor whose quantity in the short run cannot readily be changed when market conditions require an immediate change in output.

v. Total Physical Product (TPP): This is the total output realized by combining factors of production.

vi. Average Physical Product (APP): This is the average of the Total Physical product per unit of the variable factor of production in the short run. Thus, if the variable factor is labour, average physical product is output per unit of labour e.g.

Total Physical Product = Total Physical Product/ Number of Workers

vii. Marginal Physical Product (MPP): Is the addition to the total physical product attributed to the addition of one extra unit of the variable input to the production process, the fixed input remaining unchanged.

SHORT RUN ANALYSIS

In the short run at least one of the factors of production will be fixed and changes in output will be caused by varying only one input.

The variations in output that can result from applying more or less of the variable factor to a given quantity of a fixed factor can be illustrated by the use of a simple numerical example.

Suppose the fixed factor of production is a farm (land), say 20 hectares. Labour will be the variable factor. Table 1 sets out some hypothetical results obtained by varying the amount of labour employed.

Non-proportional returns

Table 1

1	2	3	4
No. of workers	**Total product**	**Average product**	**Marginal product**
0	0	0	
1	8	8	8
2	24	12	16
3	54	18	30
4	82	20.5	28
5	95	19	13
6	100	16.7	5
7	100	14.3	0
8	96	12	-4

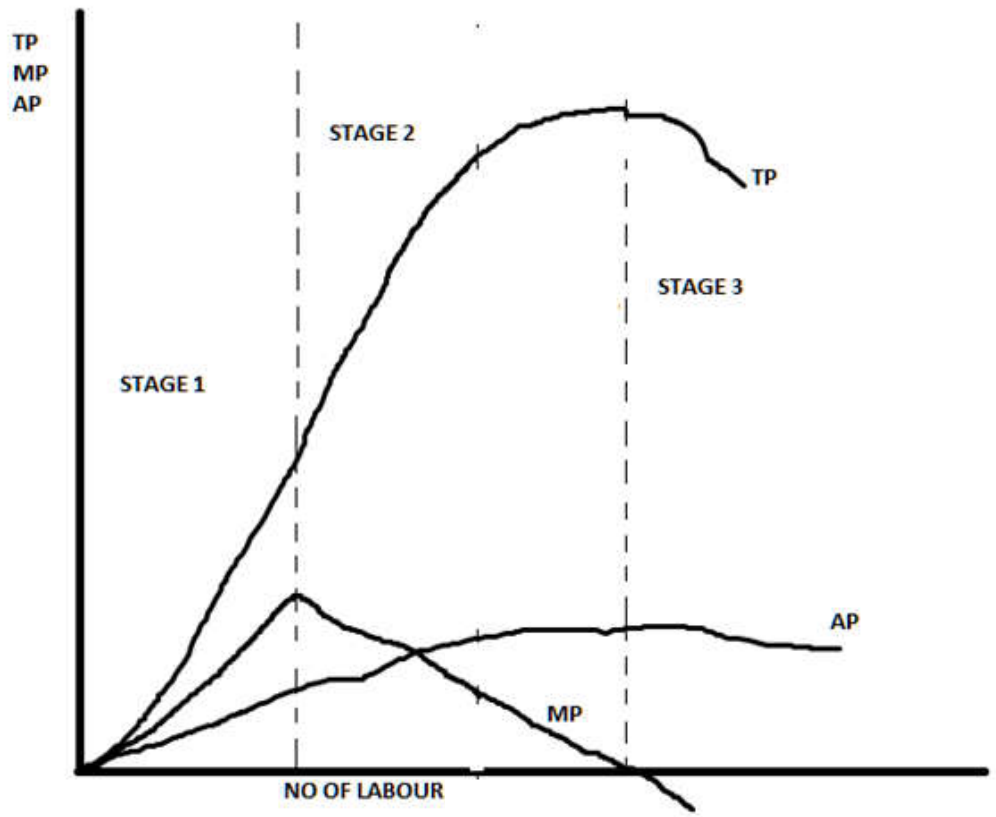
The above table illustrates some important relationships, but before we examine them we must state the assumptions on which the table is based;

1. The time period must be the short run i.e. there must be a fixed factor of production.
2. There must be a variable factor of production.
3. Successive units of the variable factors must be equally efficient.
4. There should be no changes in the production techniques

In the third column, Average Physical Product is obtained by dividing Total Physical Product by the number of workers. Thus for the first worker, the total output of 8 is divided by 1. For the second worker, the total output of 24 is divided by 2 to give 12 and so on.

In the fourth column, the marginal product for each worker is obtained by subtracting the previous Total physical Products from the Total Physical Product, when that extra worker is employed. Thus, for the first worker, Marginal Physical Product is 8. For the second worker,

Marginal Physical Product is 16 and so on.



Observations:

- It can be observed from the Total Physical Product that it begins by rising, reaches maximum and then falls.
- Total Physical Product begins by increasing at increasing rate as shown by the table of up to the third worker, beyond this it increases at a decreasing rate then reaches a maximum and falls.
- MPP begins by rising, reaches a maximum and then falls.
- When TPP is increasing at an increasing rate, MPP is raising
- When TPP is at increasing at a decreasing rate, MPP is falling
- When TPP is at the maximum i.e. increasing at a zero rate when MPP is equal to zero, measure of the rate of change in TPP.
- APP begins by rising, reaches a maximum and then falls.
- When APP is rising, MPP is above it, although MPP begins to fall earlier than APP, when APP is falling, MPP is below it. MPP is equal to APP when is at the maximum

TOTAL PRODUCT, AVERAGE AND MARGINAL PRODUCTS

Total product

This is the maximum output obtainable from different amounts of variable inputs given a specified amount of the fixed input.

Average product

It is the total product divided by the number of inputs . In other words, product per unit input.

Marginal product

It is the change in total product resulting from a unit change in a variable input.

THE LAW OF DIMINISHING RETURNS (LAW OF VARIABLE PROPORTIONS) AND STAGES OF PRODUCTION

It is one of the most important and fundamental principles involved in economics called **the law of diminishing returns or variable proportions.** We may state it thus:

The law of diminishing returns comes about because of several reasons:

1. The ability of labour to substitute for the fixed quantity of land.
2. The marginal physical output of labour increases for a time, as the benefits of specialization and division of labour make for greater efficiency.
3. Later all the advantages of specialization are exhausted.
4. The law of diminishing returns comes about because each successive unit of the variable factor has less of the fixed factor to work with. In fact, they therefore start getting in the way of others with the fixed factor with consequent decline in output.

From the figure we can see the law leads to three stages of production, namely, stage of:

1. Increasing returns
2. Diminishing returns
3. Negative returns

CHARACTERISTICS OF THE THREE STAGES

Stage I

Here the Total Physical Product, Average Physical Product and Marginal Physical Product are all increasing. However MPP later starts decreasing. The stage is called stage of increasing returns because either the APP or MPP is increasing.

Stage II

Is a stage of diminishing returns and we have:

- Diminishing Average Physical Product

- Diminishing MPP and
- Increasing Total Physical Product
- APP and MPP are declining but since the MPP is still positive, the TPP keeps on rising. The stage where MPP reaches zero, TPP reaches maximum.

Stage III

Marks a change in the direction of TPP curve. The APP continues to diminish the MPP continues to diminish too, but it is negative and is what distinguishes stage IV from II and I.

This is the stage of negative returns.

Where does the firm Operate

The firm will avoid stages I, II and III and will instead choose stage II. It will avoid stage I because this shall involve using the fixed factor inefficiently because its MPP is increasing since the variable input is spread to scarcely (thinly) over the fixed input. Expansion of the variable input will permit specialization, hence increased output because of effective use of the variable input.

The firm shall avoid stage III because MPP for the variable input is negative.

Stage II is chosen because the marginal returns for both resources is diminishing. Here the

MPP and APP are declining but the MPP of both resources is positive. With one factor fixed, and additional unit of the variable input increases total product. Therefore the firm which attempts to be economically efficient operates in stage II.

Relevance of the Law of Diminishing Returns

The law of diminishing returns is important in that it is seen to operate in practical situations where its conditions are fulfilled. Thus, in a number of developing countries with peasant agricultural economies populations are increasing rapidly on relatively fixed land, and with unchanging traditional methods of production. Consequently, productivity in terms of output per head is declining, and in some cases total productivity is falling.

Also the law of diminishing returns is important in the short run. The aim of the firm is to maximize profits. This happens when the firm is in a state of least-cost-factor-combination.

This is achieved when the firm maximises the productivity of its most expensive factor of production. Productivity is measured in terms of output per unit of the factor. Thus, if the variable factor is the most expensive factor, the firm should employ the variable factor until

APP is at the maximum. If the fixed factor is most expensive the firm should employ the variable factor up to the level when TPP is at maximum.

LONG RUN ANALYSIS

In the long run it is possible to vary all factors of production. The firm is therefore restricted in its activities by **the law of diminishing return to scale.**

The law states that successive proportionate increments in all inputs simultaneously will lead eventually to a less than proportionate increase in output.

Returns to scale refers to the rate at which output increases as all inputs are increased simultaneously. In the illustration below, labour and land are assumed to be the factors of production.

Land (units)	Labour (units)	Output (units)
30	5	41} Increasing returns
60	10	` 100
90	15	168 Constant returns
120	20	224
150	25 `	275 Decreasing returns
180	30	300

When land increases from 30 units to 60 units and labour from 5 units to 10 units, each has doubled or increased by 100%. Output increases from 41 units to 100 units i.e. by more than 100. When land increases from 60 to 90 units, each has increased by 50%. Output increases from 100 to 168 i.e. by more than 50%. In each of these cases when the inputs are increased in a certain proportion, output increases in greater proportion. We say that the firm is in a stage of increasing returns to scale.

This should not be confused with the stage of increasing returns in the short run. In the short run. In the short run the increasing returns to the variable factor, and the scale of production fixed.

When land is increased from 90 to 120 units and labour from 12 to 16 units, each has increased by 1/3 or 33 1/3% output increases from 168 to 224, i.e. by 1/3 or 33 1/3. Thus, when the input factors are increased in a certain proportion, output increases in the same proportion.

This is a stage of constant returns to scale.

When land is increased from 120 to 150 units and labour fro 20 to 25 units, each has increased by 25%. Output increases from 224 to 275 units i.e. by less than 25%. When land increases from 150 to 180 and labour from 25 to 30 units, each has increased by 20%, output increases from 275 to 300 i.e. by less than 20%. In both of these cases, when the inputs are increased in a certain proportion output increases in 20%. The firm is said to be in a stage of decreasing returns to scale not to be confused with diminishing returns to the variable factor in the short run.

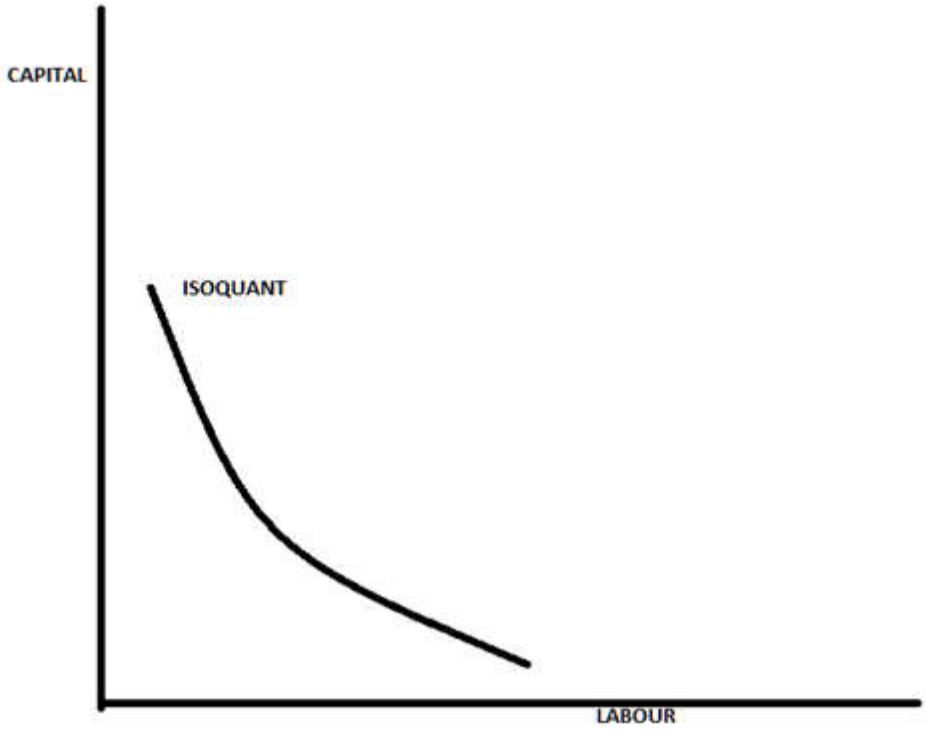
ISOQUANT AND ISOCOST LINES

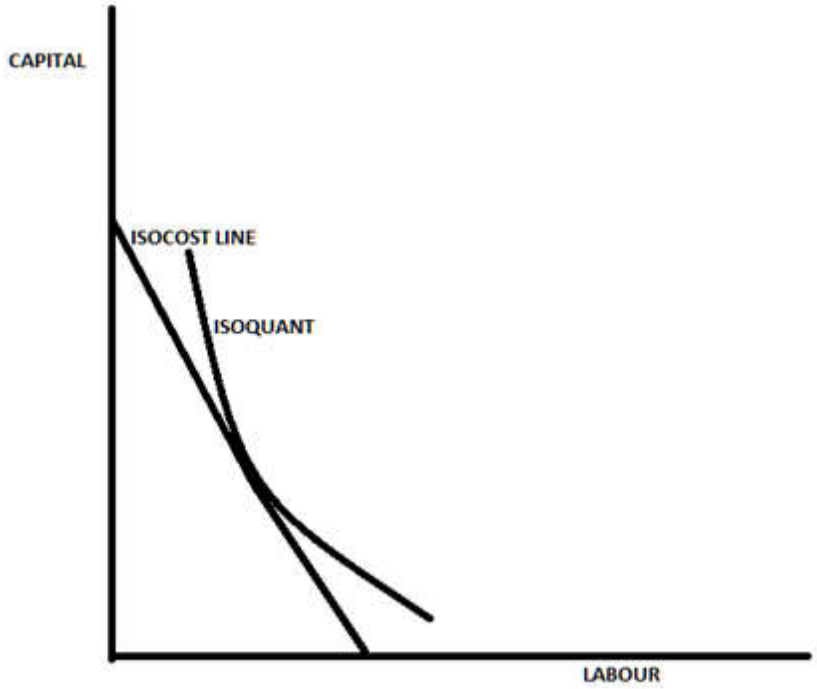
In the long run it is possible for a firm to produce the same output using different combinations of two factors of production. For instance it the two factors of production, are capital and labour, then labour may be substituted for capital or vice versa. Thus for instance an output of 69 units of X can

be produced by using units of capital and one unit of labour or six units of labour and one unit of capital.

If the various combinations of factors of production which produce the same amount of output are plotted on a graph this produces an isoquant or equal product curve.

Theoretically, we can construct any number of isoquants on the graph to produce an isoquant map. They are downward sloping because although capital can be substituted for labour or vice versa they are not perfect substitutes. Therefore as we substitute capital for labour, for example, it takes more and more units of capital to replace labour, as capital becomes a less and less perfect substitute. Like indifference curves isoquants can never intersect. The slope of the isoquant shows the substitution ratios of the factors of production.





THE CONCEPT OF PRODUCER EQUILIBRIUM AND FIRMS EXPANSION CURVE

AB, CD, EF and GH are the four isocost lines representing different levels of total cost or outlay. All isocost lines are parallel to one-another indicating that prices of the two factors remain the same". E_1, E_2, E_3 and E_4 are the points of producer's equilibrium corresponding to the point of tangencies of the above four isocost lines with the highest possible isoquant in each case.
The line joining the least cost combinations like E_1, E_2, E_3 and E_4 is called the expansion path. It is so called, because, it shows how for the given relative prices of the two inputs (the slope of factor price line), the optimal factor combinations with which the producer plans it output will alter as he expands the volume of output.
Expansion path may be defined as the locus of efficient combinations of the factors (the points of tangency between the isoquants and the iso-cost lines). It is the curve along which output or expenditure changes, when factor prices remain constant.

Hence, the optimal proportion of the inputs will remain unchanged. It is also known as scale-line, as it shows how the producer will change the quantities of the two factors, when it raises the scale of production.

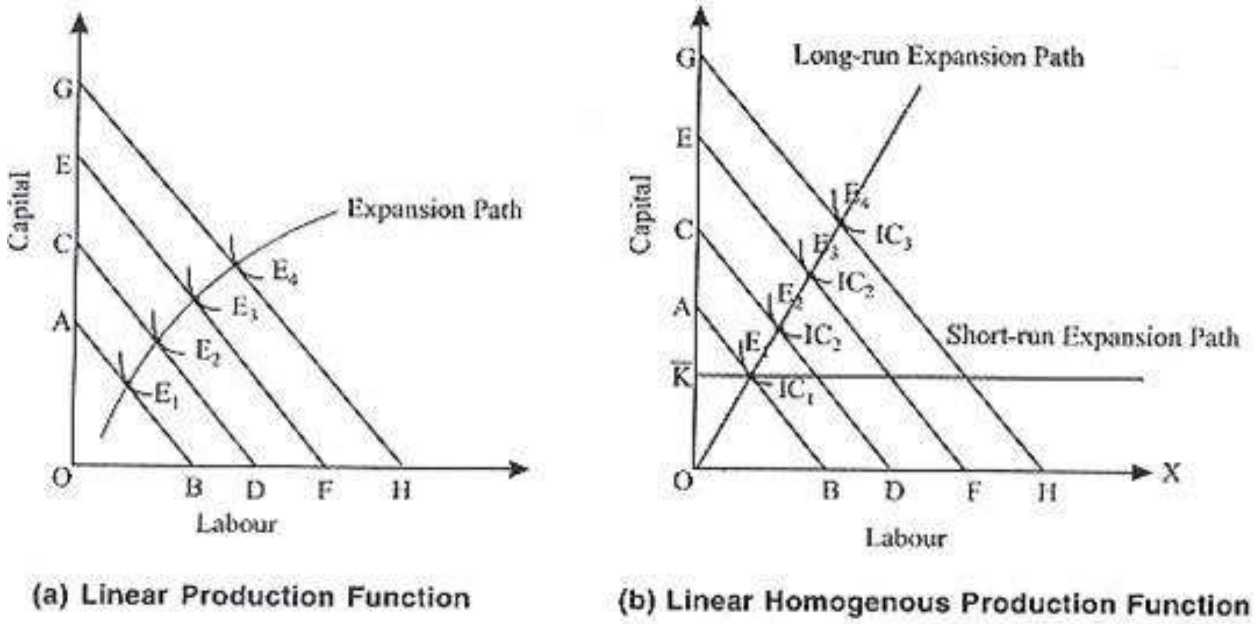


Fig. 7.12: Expansion Path

The expansion path may have different shapes and slopes depending upon the relative prices of the factors used and shape of the isoquant. In case of constant returns to scale (homogenous production function), the expansion path will be a straight line through the origin, indicating constancy of the optimal proportion of the inputs of the firm, even with changes in the size of the firm's input budget. (Fig. 7.12 (b)). In short-run, however, the expansion path will be parallel to X-axis (when capital is hold constant at K shown in Fig. 7.12 (b)).

As expansion path depicts least cost combinations for different levels of output, it shows the cheapest way of producing each output, given the relative prices of the factors. It is difficult to tell precisely the particular point of expansion path at which the producer in fact be producing, unless one knows the output which he wants to produce or the size of the cost or outlay it wants to incur.

But, this much is certain that though for a given isoquant map, there can be different expansion paths for different relative prices of the factors. Yet, when prices of the variable factors are given, a rational producer will always try to produce at one or the other point of the expansion path.

LAW OF RETURNS TO SCALE

In the long run all factors of production are variable. No factor is fixed. Accordingly, the scale of production can be changed by changing the quantity of all factors of production.

Definition:

"The term returns to scale refers to the changes in output as all factors change by the same proportion." Koutsoyiannis

"Returns to scale relates to the behaviour of total output as all inputs are varied and is a long run concept". Leibhafsky

Returns to scale are of the following three types:

1. Increasing Returns to scale.
2. Constant Returns to Scale
3. Diminishing Returns to Scale

Explanation:

In the long run, output can be increased by increasing all factors in the same proportion. Generally, laws of returns to scale refer to an increase in output due to increase in all factors in the same proportion. Such an increase is called returns to scale.

Suppose, initially production function is as follows:

P = f (L, K)

Now, if both the factors of production i.e., labour and capital are increased in same proportion i.e., x, product function will be rewritten as.

$$P_1 = f(xL, xK)$$

1. If P_1 increases in the same proportion as the increase in factors of production i.e., $\frac{P_1}{P} = x$, it will be constant returns to scale.

2. If P_1 increases less than proportionate increase in the factors of production *i.e.*, $\frac{P_1}{P} < x$, it will be diminishing returns to scale.

3. If P_1 increases more than proportionate increase in the factors of production, *i.e.*, $\frac{P_1}{P} > x$, it will be increasing returns to scale. Returns to scale can be shown with the help of table 8.

Table 8. Showing different stages of return to scale

Units of Labour	Units of capital	%age increase in Labour & Capital	Total Product	%age increase in TP	Returns to scale
1	3	–	10	–	
2	9	100%	30	200%	Increasing
3	9	50%	60	100%	
4	12	33%	80	33%	Constant
5	15	25%	100	25%	
6	18	20%	120	10%	Decreasing
7	21	16.6%	130	8.3%	

The above stated table explains the following three stages of returns to scale:

1. Increasing Returns to Scale:

Increasing returns to scale or diminishing cost refers to a situation when all factors of production are increased, output increases at a higher rate. It means if all inputs are doubled, output will also increase at the faster rate than double. Hence, it is said to be increasing returns to scale. This increase is due to many reasons like division external economies of scale. Increasing returns to scale can be illustrated with the help of a diagram 8.

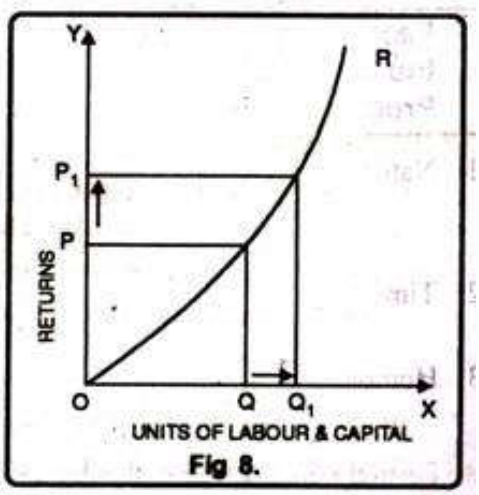


Fig 8.

In figure 8, OX axis represents increase in labour and capital while OY axis shows increase in output. When labour and capital increases from Q to Q_1, output also increases from P to P_1 which is higher than the factors of production i.e. labour and capital.

2. Diminishing Returns to Scale:

Diminishing returns or increasing costs refer to that production situation, where if all the factors of production are increased in a given proportion, output increases in a smaller proportion. It means, if inputs are doubled, output will be less than doubled. If 20 percent increase in labour and capital is followed by 10 percent increase in output, then it is an instance of diminishing returns to scale.

The main cause of the operation of diminishing returns to scale is that internal and external economies are less than internal and external diseconomies. It is clear from diagram 9.

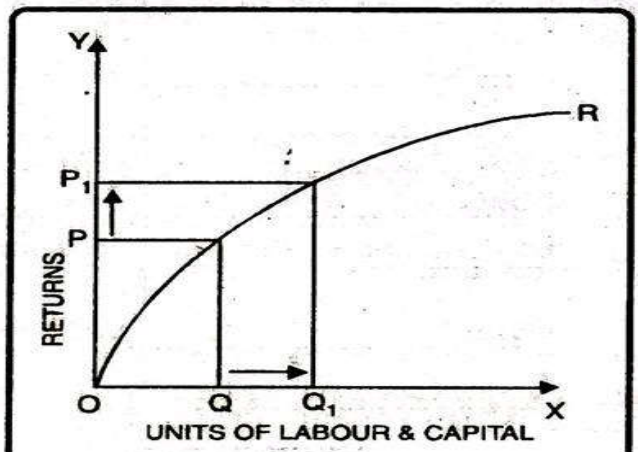


Fig. 9.

DEMAND AND SUPPLY OF FACTORS OF PRODUCTION

Demand for a Factor of Production

The demand for a factor is not a direct demand but it is an indirect or derived demand. The demand for labour, for example, is not demand for labour himself. It is in fact, demand for goods or services which the labour produces. Thus when demand for goods increases, the demand for the factors which produce those goods would also rise. If demand for goods is elastic, the demand for factors would also be elastic. Similarly when demand for goods is inelastic, the factor which produces it will also be inelastic. The demand for any given factor of production also depends upon the availability of other factors which co-operate with this factor in the process of production. Normally the demand for and price of a given factor will be higher if the co-operating factors are available in large. A third rule regarding the demand for a factor is what when more of a factor is employed, its marginal productivity is likely to fall and hence its demand and price are also likely to become lower. The demand and price of a factor also depends upon the market price of the goods for the production of which this factor has been used. If the goods are being sold at high prices the demand for the factors would also be higher.

In the Fig-A various amounts of labour employed by an individual firm at different wage rates are shown. When wages are OW_1 the firm is in equilibrium at the point E and therefore employs ON amount of labour. As the wages go down OW_2, the equilibrium position shifts to E and total employment of the factor goes up to ON'. Similarly at the wages rate OW_3 the employment of factor goes up to ON". The demand curve for labour is thus downward slopping as shown in Fig-B.

Now to obtain the demand curve for the whole industry, all the demand curves of the individual firms have to be summed up. Let us for the sake of illustration take that the industry consists of only three firms with the demand curves D_1, D_2 and D_3. The total demand curve of the industry would be summation of three demand curves

Supply of a Factor of Production

The supply of a factor of production depends upon a number of factors. Let us take the case of labour. The supply of labour depends upon the size and composition of population, its geographical and occupational distribution, efficiency of labour, expected income etc. But one thing that is generally true is that more of labour would be offered in the market when wages are higher compared to what is being offered at lower wage rate. It is only a general tendency which may not be true always. If at higher wage rate labour starts preferring leisure to work the supply of labour is likely to fall thus the supply curve of labour may be backward sloping. However such cases are very rare.' Therefore for the purpose of our analysis the supply curve for labour may be treated to be upward sloping showing that more of labour is supplied when the wages go up. The supply curve is given at Fig-D.

Determination of Market Price of a Factor

After a detailed discussion on the demand and supply aspects of labour during production, let us see how wage rate is determined by their interaction. In the Fig-E, DD represents the demand curve for the factor of production say labour and SS is the supply curve. Both curves intersect each other at point E, which is the equilibrium position in factor market at which EQ is the equilibrium wage rate. If the wage rate increases to OR, demand for labour will fall and supply will rise, which may cause competition among labourers, thus the wage rate would fall resultantly. Contrary to it if the wage rate falls to OK then supply of labour will fall and demand will rise which may cause competition among producers to employ more and more labour at lower wage rate resultantly this competition would rise the wage rate and this up and down will bring it to the equilibrium level OP or QE at OQ quantity of labour.

Criticism on Demand and Supply Theory

The theory is criticized on the basis of some of its weak assumptions which are given as

1. The aspect of increasing return in the theory of distribution or factor pricing is completely ignored.
2. As the factors of production are not close or complete substitutes of each other, therefore they cannot be substituted for one another.
3. Homogeneity in all units of a factor of production is not possible.
4. Prevalence of perfect competition in' both factor and production market is not correct because in real world it does not prevail

WAGE DETERMINATION THEORIES

Early theories about wages

The earliest theories about wage determination were those put forward by **Thomas Malthus, David Ricardo and Karl Marx.**

i**. Thomas Robert Malthus** (1766 – 1834) and the Subsistence Theory of Wages:

The germ of Malthus' Theory does come from the French "physioirats" who held that it was in the nature of things that wages could never rises above a bare subsistence level.

When wages did for a time rise much above the bare necessities of life, the illusion of prosperity produced larger families, and the severe competition among workers was soon at work to reduce wages again. In a world where child labour was the rule it was only a few years before the children forced unemployment upon the parents, and all were again reduced to poverty. Such was the subsistence theory of wages.

ii. **Ricardo and the Wages Fund Theory**:

Ricardo held that, like any other commodity, the price of labour depended on supply and demand. On the demand side, the capital available to entrepreneurs was the sole source of payment for the workers, and represented a wages fund from which they could be paid.

On the supply side, labour supply depended upon Malthus' arguments about population. The intense competition of labourers one with another, at a time when combinations of workers to withdraw their labour from the market were illegal, kept the price of labour low. The fraction:

Total wages fund (capital available)/Total population

Total population

Fixed the wages of working men.

iii**. Karl Marx** (1818 – 83) and the 'Full Fruits of Production' Theory of Wages:

Karl Marx was a scholar, philosopher, journalist and revolutionary extraordinary who spent much of his life in dedicated poverty reading in the British Museum Library. His **labour theory of value** held that a commodity's worth was directly proportional to the hours of work that had gone into making it, under the normal conditions of production and the worth the average degree of skill and intensity prevalent at that time. Because only labour created value, the worker was entitled to the full fruits of production. Those sums distributed as rent, interest and profits, which Marx called surplus values, were stolen from the worker by the capitalist class.

Modern theories of wage determination

i. Real and nominal wages

Wages are wanted only for what they will buy, **real wages** being wages in terms of the goods and services that can be bought with them. **Nominal wages** are wages in terms of money, and the term **money wages** is perhaps to be preferred. In determining nominal wages of people in different occupations; account must be taken of payments in kind, such as free uniform for policemen, railway workers and may others, free travel to and from work for those engaged in the passenger transport undertakings, the use of the car by some business executives, free board and lodging for some hotel workers and nurses. " The labourer", say Adam Smith, "is rich or poor, is well or ill rewarded, in proportion to the real, not to the nominal price of his labour."

ii. Marginal productivity theory of wages

According to the Marginal theory of distribution, the producer will pay no more for any factor of production that the value of its marginal product, since to do so will raise his costs by a greater amount than his revenue. As applied to labour this provides us with the **Marginal Productivity** theory of wages.

At this wage rate the firm will employ L units of labour. At this level of employment, R is the average revenue product. Thus, the total revenue of the firm is represented by area ORBL, and

Labour cost is represented by area OWAL. Thus, the firm makes loss (on labour above) represented by area RWAB. The firm will, therefore, not employ labour at wage rates above average revenue product. It follows, therefore, that the demand curve for labor is that part of the Marginal revenue product curve below the average revenue product curve

This theory has been criticized for following reasons:

a) It is too theoretical a concept, since it does not appear to agree with what actually takes place.
b) In practice it is impossible to calculate the amount or the value of the marginal product of any factor of production.
c) The employment of one man more or one man less may completely upset the method of production in use at the time. To employ an extra man may simply mean that there will be more labour than necessary; to take away a man may remove a vital link in the chain of production. For this reason a small rise or fall in wages is not likely to bring about an immediate change in the amount of labour employed.
d) The productivity of labour does not depend entirely on its own effort and efficiency, but very largely on the quality of the other factors of production employed, especially capital.
e) According to this theory, the higher the wage, the smaller the amount of labour the entrepreneur will employ. Surveys that have been taken appear to indicate that not all employers take account of the wage rate when considering how may men to employ, being influenced more by business prospects.
f) Lord Keynes said the theory was valid only in static conditions, and therefore, to lower the wage rate in trade depression would not necessarily increase the demand for labour.

iii. Market theory of wages

Here the approach is to regard wages as a price – the price of labour – and, therefore, like all other prices determined by the interaction of the market forces of supply and demand.

In terms of geometry, this corresponds to the point of intersection between the demand curve and the supply curve.

W is the equilibrium wage rate and L the equilibrium level of employment of labour. Atwage rates above w, there is excess of supply over demand, and hence wages will be forced downwards.

Below W, there is excess of demand oversupply and hence wages will be forced upwards. The Market Theory of Wages, however, does not run counter to the Marginal Productivity Theory. In the same way that marginal utility forms the basis of individual demand, so marginal productivity forms the basis of demand for labour and other factors of production.

iv. The institutional intervention theories

Collective bargaining provides an example of what is sometimes called bi- lateral monopoly; the trade union being the monopolist supplier and the employers' associationthe monopolist buyer of a particular kind of labour.

Those who support the Bargaining strength of the trade union concerned, so that, differences in wages in different occupations are the result of the differences of the strength of the respective trade unions.

v. The comparability principle

Associations representing workers providing services – clerical, postal, teaching, etc. – have always attempted to apply the "principle of comparability" with wages of those in similar occupations, though it is often very difficult to compare workers in different occupations, since no two jobs are alike.

vi. The effect of inventions

In the long run, new inventions will have the effect of increasing output and lowering prices, with the result that real wages of workers rise and in consequence their demand for all kinds of goods and services.

TRADE UNIONS FUNCTIONS AND CHALLENGES

Trade unions are workers' organizations whose objective is to protect the interests of their members.

Functions

i. To bargain on behalf of their members for better pay and working conditions.
ii. To persuade the government to pass legislation in favour of the working class.

Collective bargaining refers to the whole process by which trade unions and employers (or their representatives) arrive at an enforce agreements. Trade unions therefore negotiate on behalf of all their members and if agreement is not reached then they may take action collectively to enforce their demands.

Kinds of Bargaining arrangements

Basically there are three kinds of bargaining arrangements, namely:

Open Shop:

In an open shop a union represents its members, but does not have an exclusive bargaining jurisdiction for all workers of one kind. Membership in the unions is not condition of getting or keeping a job.

Closed Shop:

In this arrangement only union members may be employed and the union controls membership however it sees fit.

Union Shop:

The employer may hire anyone he chooses but every employee must join the union within a specified period.

The Basis of Wage Claims

The union's demand for higher wages is normally based on one or more of the following four arguments:

1. The cost of living argument

This is the commonest argument in that rising prices mean a fall in real incomes. Thus trade unions demand higher wages so that the workers can maintain their present living standards. It is important, however, that when trade unions use this argument they should also ensure that the productivity of the workers also increases. This is because if wages increase while productivity does not, this will mean higher costs of production for the firm and this will lead to higher prices and more wage demands leading to a wage price spiral. Thus, in addition to demanding higher wages, trade unions will also demand conditions, which go towards increasing the efficiency of labour. These are:

a) Motivating factors which boost the morale of workers like free or subsidized housing, free medical benefits, paid maternity leave, and paid sick leave.
b) Safe and hygienic working conditions
c) In-Service training or study leave to up-date the skills of workers

d) Efficient co-operating factors like efficient machinery and tools. Here, however, they must be careful because if the machinery is too efficient it may actually displace labour.

2. The productivity argument

Productivity is measured in terms of output per worker. If this increases, workers can claim higher wages with the argument that their efficiency has increased. This argument is justified if it can be proved that productivity has increases as a result of the increased efficiency of workers, e.g. through less absenteeism and less idle talking. However, if productivity increases as a result of the installation of new machinery, the benefits should go to the investor.

The argument of productivity can only be used in cases where the output of the firm can be measured. Where output cannot be used, e.g. it is difficult to measure the productivity of civil servants. In some cases productivity can fall due to factors beyond the control of the workers, and salaries of the workers cannot be reduced e.g. the efficiency of traffic police can fall due to increased traffic, but this does not mean that police should have reduced salaries.

3. The profit argument

If firm profits increases, workers can claim to have a share in them on the basis that they contributed to the increase. This argument is justified if it can be proved that the increase in profits is due to increased efficiency of the workers. If, for example, it is due to increased investment in advertising by the firm, the benefits should go to the investor.

4. The differential argument

This argument is justified if the two firms have the same profit level and if the efficiencies of the workers in the two firms are the same. Otherwise it is not justified.

Weapons of Conflict

The trade unions and the employers (or their associations) have many ways of enforcing their demands on each other. They include:

Strikes: The strike is the union's ultimate weapon. It consists of the concerted refusal to work of the members of the union. It is the strike or the threat of a strike that backs up the union's demand in the bargaining process.

Picket lines: Are made up of striking workers who parade before the entrance to their plant or firm. Other union members will not cross a 'picket line'.

The lockout: Is the employer's equivalent of a strike. By closing his plant he locks out the workers until such a time the dispute is settled.

Black list: Is an employers' list of workers who have been discharged for unions' activities and who are not supposed to be given jobs by other employers.

Strike-breakers: Are workers who are used to operate the business when union members are on strike.

FACTORS AFFECTING THE ABILITY OF TRADE UNIONS TO GAIN LARGER WAGE INCREASES FOR ITS MEMBERS

The basic factor is elasticity of demand for the type of labour concerned. The elasticity of demand for any particular type of labour will vary according to four factors:

1. The physical possibility of substituting alternative factors of production for labour:

If wages rise, labour will be relatively more expensive than the factors which will tend to be substituted for it. The extent to which this is possible will depend on technical considerations. The more substitution is possible, the greater will be the elasticity of demand of labour.

2. The elasticity of supply of alternative factors:

If substitution is technically possible, the demand for alternative factors will increase and this will result in a rise in their prices. The extent of the rise will depend on the elasticity of supply. The more elastic this is, the greater will be the increase in price, the smaller the substitution of factor for labour, hence the lower the elasticity of demand for labour itself.

3. The proportion of labour to total cost:

If the proportion is large, the demand for labour will tend to be elastic for two reasons.

First, as the percentage of total costs formed by labour is large, there will be considerable pressure to find substitutes for labour. Second, the effect of a rise in labour costs will result in a larger increase in total costs.

4. The elasticity of demand for the final product:

An increase in wages will raise the price of the final product. The extent of the price increase will be determined by the three factors above. If the demand for the good is elastic the quantity purchased will fall considerably and so will the demand for labour, which produces the good. The opposite will apply where the demand for the good is inelastic. There are some circumstances in which a wage increases need not result in a higher price for a good. Firms may be earning above-normal profits and the increase in wages may be paid out of these without raising the price of the final product.

Alternatively, the increased wage may be paid out of increased productivity.

If the demand is elastic, employment will be sensitive to wage changes and this will be a basic constraint on trade union behaviour. If the demand is inelastic, a wage increase will have relatively little effect on employment and trade unions will be able to press for, and obtain, large increases in the pay of its members. Trade unions in different industries differ in terms of their strength (as, for example, measured by the extent of their membership), degree of militancy, general approach, whether in the private or public sector, and so on.

Effectiveness of Trade Unions in Developing Countries

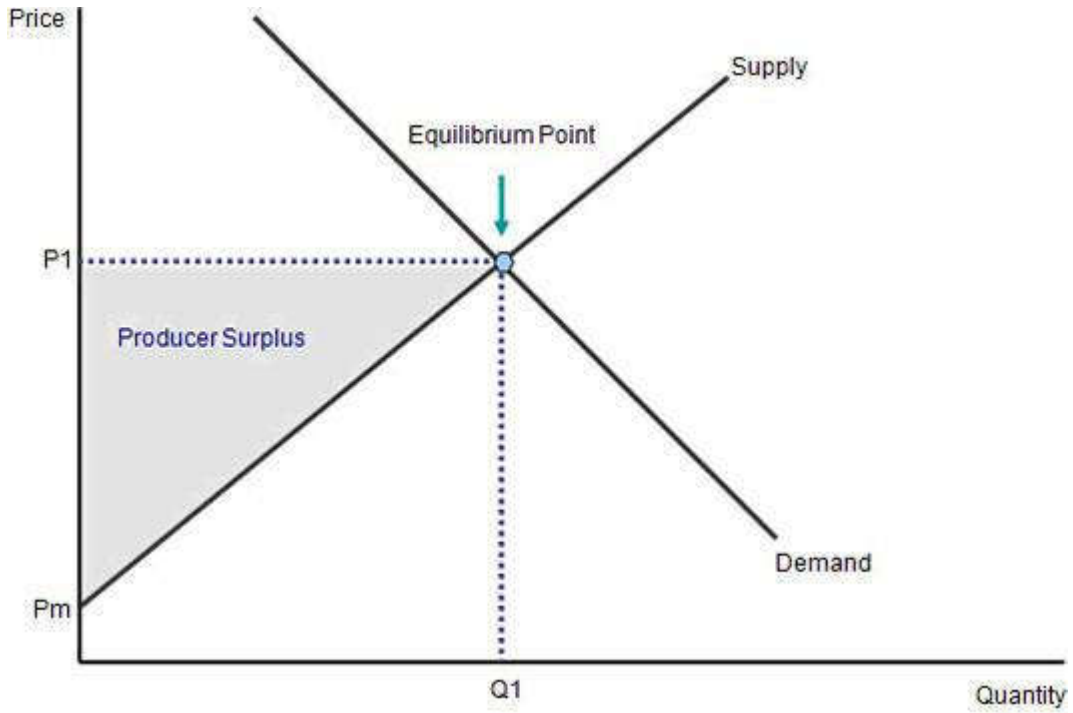
Trade Unions in developing countries tend to be less effective in their wage negotiations with employers than their counterparts in developed countries. This can be attributed to the following factors:

1. Incomes in developing countries are lower than in developed countries. Consequently, the contributions of workers to trade unions are less and the trade unions are therefore in financially weaker position to support the members while on strike.
2. As incomes in developing countries are lower, so are savings, and hence workers cannot support themselves for long periods while on strike.
3. In developing countries, there are no unemployment state benefits on which workers can depend if they are sacked for trade union activities.
4. There is more Government interference in industrial disputes in developing countries than in developed countries. This is partly because the developing countries the political structures are not strong and governments fear that too much trade union agitation may have negative political effects.
5. Labour in developing countries is mostly unskilled and semi-skilled labour and is in abundant supply. Hence striking workers can easily be replaced. For this reason trade unions in developing countries are less able to persuade their members to go on strike for long periods than their counterparts in developed countries.

PRODUCER SURPLUS

Definition: Producer surplus is defined as the difference between the amount the producer is willing to supply goods for and the actual amount received by him when he makes the trade. Producer surplus is a measure of producer welfare. It is shown graphically as the area above the supply curve and below the equilibrium price.

Here the producer surplus is shown in gray. As the price increases, the incentive for producing more goods increases, thereby increasing the producer surplus.



A producer always tries to increase his producer surplus by trying to sell more and more at higher prices. However, it is simply not possible to increase the producer surplus indefinitely since at higher prices there might be very little or no demand for goods.

THE THEORY OF COSTS

The student should explain:

- Short run cost analysis and size of the firms total cost, fied cost, average cost , variable costs and marginal costs
- Long run costs analysis
- Optimal size of the firm
- Economies and diseconomies of scale

Like in the theory of production or output, the theory of costs is concerned with the Short Run and the Long Run

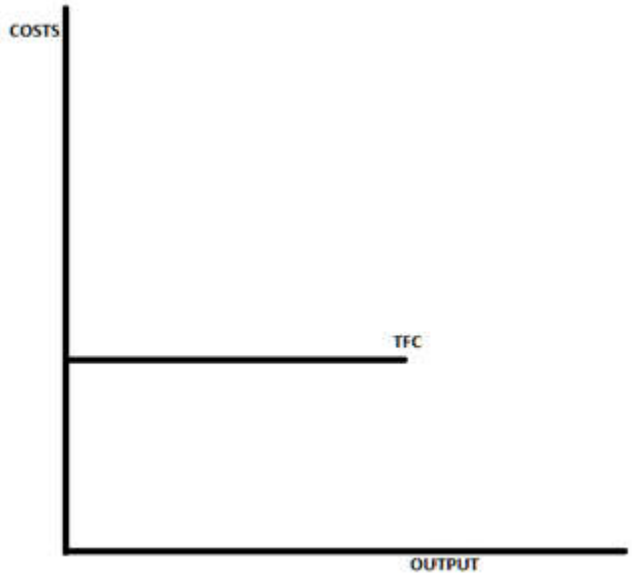
.

SHORT RUN COSTS ANALYSIS AND SIZE OF THE FIRMS TOTAL COST, FIXED COST, AVERAGE

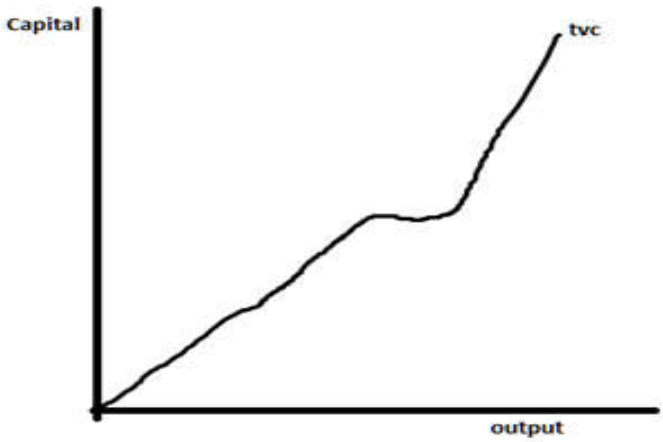
COST, VARIABLE COSTS AND MARGINAL COST.

Definitions:

a. Fixed Costs (FC): These are costs which do not vary with the level of production i.e. they are fixed at all levels of production. They are associated with fixed factors of production in the Short Run. Examples are rent or premises, interest on loans and insurance.



b. **Variable Costs (VC)**: These are costs, which vary with the level of production. The higher the level of production, the higher will be the variable costs. They are associated with variable factors of production in the Short Run. Examples are costs of materials, cost of fuels, labour costs and selling costs.

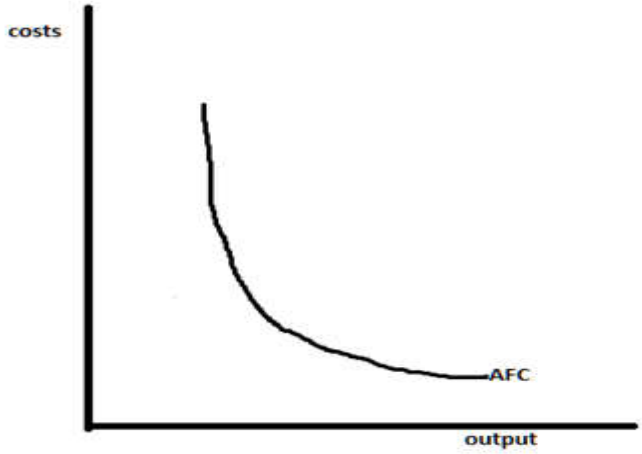


c. Total Cost (TC): This is the sum of fixed costs and variable costs i.e. TC = FC + VC.

Costs TC = TVC + TFC

d. Average Fixed Cost (AFC): This is fixed cost per unit of output, obtained by dividing fixed costs by total output i.e.

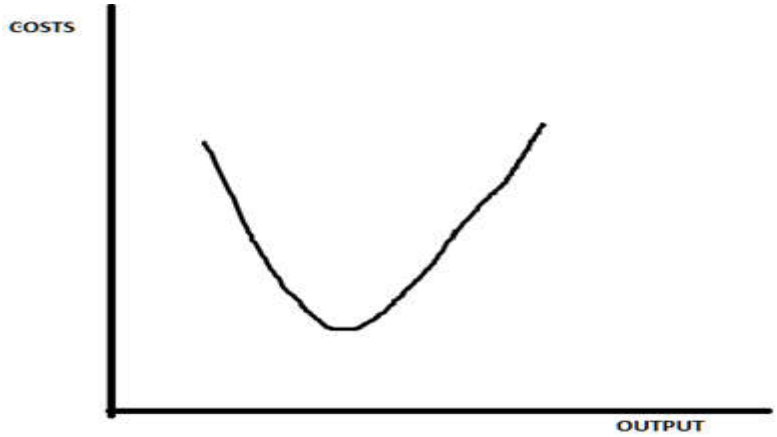
AFC = Fixed Costs/ total output



As the output increases, the Average Fixed Cost falls.

e. Average Variable Cost (AVC): This is the average cost per unit of output, obtained by dividing variable costs by total output i.e.

AVC = Variable Cost/Total output



f. Average Total Costs (ATC): This is total cost per unit of output, obtained by dividing total cost by total output i.e.

g. Marginal Cost: This is the increase in total cost resulting from the production of an extra unit of output. Thus, if TCn is the total cost of producing n units of output and

TCn-1 is the total cost of producing n-1 units of output, then the marginal cost of producing the 'nth' of unit of output is calculated as:

Marginal Cost = TCn - TCn-1

It will be observed that since fixed costs are fixed, it follows that:

Marginal Cost = VCn - VCn-1

Marginal Cost intersects the Average Total Cost at its lowest. The MC is related to the AVC in the sense that when MC is below AVC, the AVC must declining with output.

When MC is equal to AVC, the AC is at its minimum. When MC is above AVC, then

Average Cost must be rising. The AFC curve falls continuously and is asymptotic to both axes. The AVC curve falls reaches a minimum, thereafter rises. At its minimum, it's equal to MC.

As AFC curve approaches the horizontal axis asymptotically, then AVC approaches the ATC asymptotically. ATC first declines, reaches a minimum then rises thereafter.

At its minimum it is equal to the MC.

Thus, the Short Run Equilibrium Output of the firm is defined as that output at which AC is at its minimum i.e. when the cost of both inputs per unit of a product is smallest. That level of output will be defined as the most efficient output of that particular plant because the plant is used efficiently.

OPTIMAL SIZE OF A FIRM

This is the most efficient size of the firm, at which its costs of production per unit of output will be at a minimum, so that it has no motive either to expand or reduce its scale of production. Thus, as a firm expands towards the optimum size it will enjoy economies of scale, but if it goes beyond the optimum diseconomies will set in.

Optimum combination of resources

The firm can maximise output given costs. That is when the entrepreneur attains the highest isoquant given a particular Isocost.

At the point of tangency the absolute slope of Isoquant will be equal to the slope of the Isocost.

LONG RUN COST ANALYSIS

In the Long –Run, all factors of production are variable. The firm is thus constrained by economies or diseconomies to scale.

ECONOMIES OF SCALE

Economies of scale exist when the expansion of a firm or industry allows the product to be produced at a lower unit cost.

1. Internal Economies of Scale

Internal economies of scale are those obtained within the organisation as a result of the growth irrespective of what is happening outside. They take the following forms:

a) Technical Economies

i. **Indivisibilities:** These may occur when a large firm is able to take advantage of an industrial process which cannot be reproduced on a small scale, for example, a blastfurnace which cannot be reproduced on a small scale while retaining its efficiency.

ii. **Increased Dimensions:** These occur when it is possible to increase the size of the firm's equipment and hence realize a higher volume of output without necessarily increasing the costs at the same rate. For example, a matatu and a bus each require one driver and conductor. The output from the bus is much higher than that from the matatu in any given period of time, and although the bus driver and conductor will earn more than their matatu counterparts, they will not earn by as many times as the bus output exceeds the matatu output, i.e. if the bus output is 3 times that of the matatu counterparts.

iii. **Economies of Linked Processes:** Technical economies are also sometimes gained by linking processes together, e.g. in the iron and steel industry, where iron and steel production is carried out in the same plant, thus saving both transport and fuel costs.

iv. **Specialisation:** Specialisation of labour and machinery can lead to the production of better quality output and higher volume of output.

v. **Research:** A large firm will be in a better financial position to devote funds to research and improvement of its product than a small firm.

b) Marketing Economies

i. **The buying advantage:** A large-scale organisation may buy its materials in bulk and therefore get preferential treatment and buy at a discount more easily than a small firm.

ii. **The packaging advantage**: It is easier to pack in bulk than in small quantities and although for a large firm the packaging costs will be higher than for small firms, they will be spread over a large volume of output and the cost per unit will be lower.

iii. **The selling advantage:** A large-scale organisation may be able to make fuller use of sales and distribution facilities than a small-scale one. For example, a company with a large transport fleet will

probably be able to ensure that they transport mainly full loads, whereas small business may have to hire transport or dispatch partloads.

c) Organisational: As a firm becomes larger, the day-to-day organisation can be delegated to office staff, leaving managers free to concentrate on the important tasks. When a firm is large enough to have a management staff they will be able to specialise in different functions such as accounting, law and market research.

d) Financial Economies: A large firm will have more assets than a small firm. Hence, it will find it cheaper and easier to borrow money from financial institutions like commercial banks than a small firm**.**

e) Risk-bearing Economies: All firms run risks, but risks taken in large numbers become more predictable. In addition to this, if an organisation is so large as to be a monopoly, this considerably reduces its commercial risks**.**

f) Overhead Processes: For some products, very large overhead costs or processes must be undertaken to develop a product, for example an airliner. Cleary these costs can only be justified if large numbers of units are subsequently produced**.**

g) **Diversification:** As the firm becomes very large it may be able to safeguard its position by diversifying its products, process, markets and the location of the production.

2. External Economies

These are advantages enjoyed by a large size firm when a number of organisations group together in an area irrespective of what is happening within the firm. They include:

a) **Economies of concentration:** when a number of firms in the same industry band together in area they can derive a great deal of mutual advantages from one another.

Advantages might include a pool of skilled workers, a better infrastructure (such as transport, specialised warehousing, banking, etc.) and the stimulation of improvements. The lack of such external economies is serious handicap to less developed countries.

b) **Economies of information:** Under this heading we could consider the setting up of specialist research facilities and the publication of specialist journals.

c) **Economies of disintegration:** This refers to the splitting off or subcontracting of specialist processes. A simple example is to be seen in the high street of most towns where there are specialist research photocopying firms.

It should be stressed that what are external economies at one time may be internal in another.

To use the last example, small firms may not be able to justify the cost of a sophisticated photocopier, but as they expand there may be enough work to allow them to purchase their own machine.

Diseconomies of scale

Diseconomies of scale occur when the size of a business becomes so large that, rather than decreasing, the unit cost of production actually becomes greater. Diseconomies of scale flow from administrative rather than technical problems.

a) Bureaucracy: As an organisation becomes larger there is a tendency for it to become more bureaucratic. Decisions can no longer be made quickly at the local levels of management.

This may lead to loss of flexibility.

b) Loss of control: Large organisations often find it more difficult to monitor effectively the performance of their workers. Industrial relations can also deteriorate with a large workforce and a management, which seem remote and anonymous.

CHAPTER 5

MARKET STRUCTURES

DEFINITION OF A MARKET

A Market may be defined as an area over which buyers and sellers meet to negotiate the exchange of a well-defined commodity. Markets may also mean the extent of the sale for a commodity as in the phrase, "there is a wide market for this or that commodity". In a monetary economy, market means the business of buying and selling of goods and services of some kind.

Concepts to know:

i. **Average Revenue (AR):** This is the revenue per unit of the commodity sold. It is obtained by dividing Total Revenue by total quantity sold. For a firm in a perfectly competitive market, the AR is the same as price. Therefore, if price is denoted by P, then we can say:

P = AR

Because of this, the demand curve which relates prices to quantities demanded at those prices is also called Average Revenue Curve. In economic theory, the demand curve or price line is often referred to as the revenue curve.

ii. **Marginal Revenue (MR):** This is the increase in Total Revenue resulting from the sale of an extra unit of output. Thus, if TRn-1 is Total Revenue from the sale of (n-1) units and

TR n is total revenue from the sale of n units, then the marginal revenue of the nth unit is given as:

$\frac{dTR}{dQ} = P(1 - 1/Ed)$ or TRn – TRn – 1

iii. **Total Revenue:** The money value of the total amount sold and is obtained by multiplying the price by the total quantity sold.

NECESSARY AND SUFFICIENT CONDITIONS FOR PROFIT MAIMIZATION

Two conditions that however prevail, namely:

i. The necessary condition is that

w/r = MPL/MPk or MPL/PL = MPk/Pk

ii. The sufficient condition is that the Isoquant must be convex to the Isocost point of Tangency.

MATHEMATICAL APPROACH TO PROFIT MAIMIZATION

An industry which is categorized as perfectly competitive is composed of many sellers (such as porn) providing a homogeneous (having all terms/products of the same degree/kind) good to numerous buyers who are all informed about the product. Both buyers and sellers are free to enter and exit a competitive market at any time. In such a free market, individual firms have no control over price; instead these firms are forced to sell their goods at a price which is determined by the market as a whole.
If they choose to sell goods at a higher price, they will not be able to sell much (if any) because consumers could buy homogeneous products from another firm at a lower price.
If they choose to sell the good at a lower price, they would sell an infinite (if possible) number of goods, forcing all other firms to lower their price as well. This concept is what gives rise to competitive firms being referred to as "**price takers**."
Competitive firms' prices are determined by the market, and all goods are sold to all consumers at that same price. Additional revenue added by each unit sold (marginal revenue) will be equal to the price at which that last unit was sold, and because this price will be the same for that unit as any other unit in a competitive industry, price will always be equal to marginal revenue (**P=MR**).
Since competitive firms have no control over the price at which their goods are sold, their only hope to maximize profits is to choose the optimal quantity of goods to sell (**Q**). There are two methods by which one can solve for the optimal quantity for a perfectly competitive firm to produce and sell:

1. Total Revenue (**TR**) and Total Cost (**TC**)
2. Marginal Revenue (**MR**) and Marginal Cost (**MC**)

Each of these methods will be demonstrated with an example

*NOTE: To maximize any equation we must find where the slope of the equation's graph is equal to zero. Because the slope of one equation's line is equal to the derivative of that equation at a given point, we can maximize a given equation by setting the derivative of that equation equal to zero.

1. Total Revenue and Total Cost (everything expressed in dollars)

Given: $TC= 100 + 10Q + Q^2$

$P= 50$

We must first figure TR

$TR= P \times Q$ but since we know $P= 50$

$TR= 50 \times Q$

Profit (pi) function: $pi= TR - TC$

Plug in values for both TR and TC

TR= 50 x Q
TC= 100 + 10Q + Q^2
pi= TR – TC
pi= 50Q – (100 + 10Q + Q^2)
pi= 50Q – 100 – 10Q – Q^2
pi= 40Q- 100 – Q^2

Next, find the derivative of the profit function with respect to Q.
d(pi) with respect to Q = 40 – 2Q
To maximize profit, set the derivative of the profit function equal to zero and solve for Q

40 – 2Q = 0
40 = 2Q
20 = Q
To maximize profits this competitive producer will produce 20 units to sell them at a price of $50

2. Marginal Revenue and Marginal Cost (everything expressed in dollars)

We already know:

dTR/dQ = MR slope of TR curve
dTC/dQ = MC slope of TC curve
pi= TR – TC

Find the derivative of the profit function with respect to Q

dpi = dTR – dTC
To maximize profit, set the derivative of the profit function equal to zero

dTR – dTC = 0
Since we know that those above fractions are equal to MR and MC respectively we can substitute them into the equation

MR – MC = 0

MR = MC

For a competitive firm to maximize profits they must produce at an output where MR is equal to MC

Given: TC= 100 + 10Q + Q^2
P= 50
Find MR
Since this is a competitive firm MR = P

MR = P, where
P = 50. So,
MR = 50
Find MC
$\frac{dTC}{} = MC$
$TC = 100 + 10Q + Q^2$

$\frac{dTC}{} = 10 + 2Q$,. Thus, marginal cost, MC = 10 + 2Q.
For a competitive firm to maximize profits MR must be equal to MC

Set MR equal to MC and solve for Q

MR = 50
MC = 10 + 2Q
MR = MC
50 = 10 + 2Q
40 = 2Q
20 = Q

To maximize profits this competitive producer will produce 20 units to sell them at a price of $50
*As can be seen by the above examples these two methods (Total Revenue and Total Cost Approach and the

Marginal Revenue and Marginal Cost Approach) have each arrived at the same conclusion. Either one can be used on any problem and the answers derived from each method will always be in agreement.

Sample Problems
Now try these on your own! First, solve them using the Total Approach. Then, check your answers by using the Marginal Approach.

1. Given: $TC = 70 + 30Q + 2Q^2$

P= 40 find the optimal quantity produced and sold by the firm so that they are maximizing profits.

2. Given: $TC = 400 - 50Q + 5Q^2$

P= 80.

Find the optimal quantity produced and sold by the firm so that they are maximizing profits.

ANSWERS: 1. Q = 2.5, 2. Q = 13

OUTPUT, PRICES AND EFFICIENCY OF PERFECT COMPETITION, MONOPOLY , MONOPOLISTIC COMPETITION , OLIGOPOLISTIC COMPETITION

Market Structures: This refers to the nature and degree of competition within a particular market. Capitalist economies are characterised by a large range of different market structures.

These include the following:

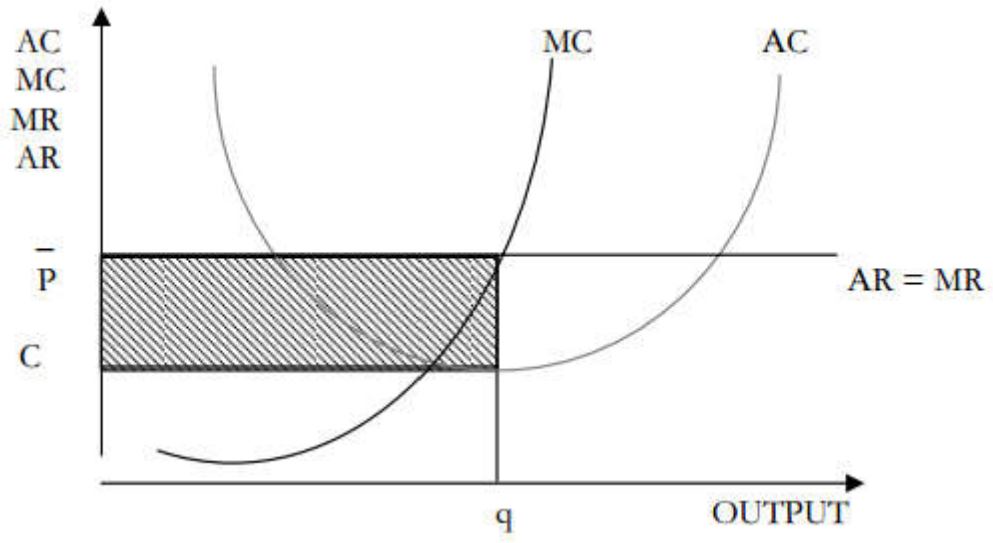
a) PERFECT COMPETITION

The model of perfect competition describes a market situation in which there are:

i. Many buyers and sellers to the extent that the supply of one firm makes a very insignificant contribution on the total supply. Both the sellers and buyers take the price as given. This implies that a firm in a perfectly competitive market can sell anyquantity at the market price of its product and so faces a perfectly price elastic demand curve.
ii. The product sold is homogenous so that a consumer is indifferent as to whom to buy from.
iii. There is free entry into the industry and exit out of the industry.
iv. Each firm aims at maximising profit.
v. There is free mobility of resources i.e. perfect market for the resources.
vi. There is perfect knowledge about the market.
vii. There is no government regulation and only the invisible hand of the price allocates the resources.
viii. There are no transport costs, or if there are, they are the same for all the producers.

SHORT RUN EQUILIBRIUM OF THE FIRM

A firm is in equilibrium when it is maximizing its profits, and can't make bigger profits by altering the price and output level for its product or service.



The firm will produce output q where Marginal Revenue is equal Marginal Cost. At this level of output, the average cost is C. Hence the firm will make *super-normal* profits shown by the shaded area.

In the Short-Run however the firm does not necessarily need to make profits or cover all its cost. It may only need to cover Total Variable Cost.

The firm's short-run supply curve will be represented by the part of the Marginal Cost curve that lie above the AVC. The firm shall not produce unless the price is equal to P1. Below the price P1 the firm minimizes its cost by shutting down.

NORMAL AND SUPERNORMAL PROFITS

Normal profit refers to the payment necessary to keep an entrepreneur in a particular line of production.

In economics, it is generally believed that any capital invested in business has an opportunity cost. The business must offer the investor a prospective return on capital at least equal to the return available on the next best alternative.

The minimum return required to keep an entrepreneur in a particular line of production is what economists call **Normal Profits**. Since it represents the opportunity cost of risk capital to the business it is treated as part of the firm's fixed cost which have to be paid if the firm is first to come into existence and then survive in the long run. Normal profits, therefore are included in the calculations which produce the AC curve. Therefore, when price exceeds average cost, the firm is said to be earning abnormal /supernormal profits – it is earning a surplus over and above what is necessary to keep it in that business (the surplus is often referred to in economics as Economic rent).

LONG RUN EQUILIBRIUM FOR THE FIRM

Since there is freedom of entry into the industry the surplus profits will attract new firms into the industry. As a result the supply of the product will increase and the price will fall. The individual firm will face a falling perfectly elastic demand curve, and the surplus profits will be reduced.

This will go on until the firm is no longer making surplus profits, i.e. when it is just covering its production costs. At this stage no more firms will be attracted to the industry. This will happen when the price is equal to the average cost and the demand curve is tangent to the average cost curve at the minimum point. The firm is said to be making normal profits.

Advantages of Perfect Market

- It achieves, subject to certain conditions, an allocation of resources which is: socially optimal" or "economically efficient" or "pareto efficient".
- Perfectly competitive firms are technically efficient in the long run, in that they produce that level of output, which minimizes their average costs, given their small capacity.

- Perfect competition achieves an automatic allocation of resources in response to changes in demand.
- The consumer is not exploited. The price of goods, in the long run will be as low as possible. Producers can only earn a normal profit, which are the minimum levels of profits necessary to retain firms in the industry, due to the existence of free entry into the markets.

Disadvantages of Perfect Competition

- There is a great deal of duplication of production and distribution facilities amongst firms and consequent waste.
- Economies of scale cannot be taken advantage of because firms are operating on such a small scale. Therefore although the firms may be highly competitive and their prices may be as low as is possible, given their scale of production, nevertheless it is a higher price that could take advantage of economies of scale.
- There may be lack of innovation in a situation of perfect competition. Two reasons account for this:
    a. The small size and low profits of the firm limit the availability of funds for research and development
    b. The assumption of free flow of information, and no barriers to entry, implies that innovations, will immediately be copied by all competitors, so that ultimately individual firms will not find it worthwhile to innovate.

REALISM OF PERFECT COMPETITION

The assumptions of perfect competition are obviously at variance with the conditions which actually exist in real world markets. Some markets approximately conform to individual assumptions, for example, the stock exchange is characterized by a fairly free-flow of information but the information requires expertise to grasp. However no markets exactly conform to the assumption of the model, with peasant agriculture probably the nearest to the mark.

We however study the model of perfect competition to enable us to see:

- How competition operates in the real world situation, within a highly simplified model.
- The advantageous features of perfect competition which governments may wish to encourage in real world markets.
- The disadvantageous features of perfect competition which the governments may wish to avoid.
- A standard against which to oil the degree of competition prevailing in a given market.

  We can discuss how closely a specific market resembles the perfectly competitive ideal
- For the student attempting a serious study of economics, a study of the perfect market is essential since no understanding of the literature of micro-economics over the century can be achieved without it.
- On a rather more mundane level, students will find themselves confronted with questions on perfect competition in examinations.

MONOPOLY

Definition: Monopoly in the market place indicates the existence of a sole seller. This may take the form of a unified business organization, or it may be association of separately controlled firms, which combine, or act together, for the purposes of marketing their products (e.g. they may charge common prices). The main point is that buyers are facing a single seller.

Sources of Monopoly:

1. Exclusive ownership and control of factors inputs.
2. Patent rights e.g. beer brands like Tusker, Soft drinks like Coca Cola etc.
3. Natural monopoly, which results from a minimum average cost of production. The firm could produce at the least cost possible and supply the market.
4. Market Franchise i.e. the exclusive right by law to supply the product or commodity e.g. Kenya Bus Service before the coming of the Matatu business in Nairobi.

PRICE AND OUTPUT LEVELS FOR A MONOPOLIST

A monopolist, being the sole (producer and) supplier of the commodity is a price maker rather than a price-taker as the price and quantity he will sell will be determined by the level of demand at that price, and if he decided on the quantity to sell, the price he will charge, will be determined by the level of demand.

The monopolist, because he is the sole seller faces a market demand curve which is downward sloping.

In this illustration, **total revenue** is obtained by multiplying price by quantity demanded or sold at that price. **Marginal revenue** is obtained by subtracting the previous total revenue from the next total revenue resulting from the sale of an extra unit of output. Thus, when the quantity demanded is one, the marginal revenue is 60 because the previous total revenue is zero. When 2 units of output are sold, the marginal revenue of the second unit of output is 50 obtained by subtracting 60 from 110. When 3 units of output are sold, the marginal revenue of the third unit of output is 40 obtained by subtracting 110 from 150 and so on.

It will be noted that average revenue values are plotted at quantities while marginal revenue values are plotted between quantities. Thus, 60 is plotted between 0 and 1. 50 is plotted between 1 and 2 because it is the increase from 1 to 2 and so on.

The graphs show that marginal revenue is below average revenue. For a straight line demand curve, the marginal revenue curve is twice as steeply sloped as the average revenue curve, and the two meet the vertical axis at the same point.

SHORT-RUN EQUILIBRIUM

All firms are assumed to aim at maximizing profits or minimizing losses. The monopolist controls his output or price, but not both.

The monopoly maximizes profits where: MR = MC (the necessary condition of profit maximisation)

He cannot produce at less than Qo because MR will be greater than MC. The monopolist will determine his output at Q X o and set the price at Po and his total Revenue is OQo X OPo and the to total cost will be OCo X bQo and abnormal profits Po CO AB

MONOPOLISTIC PRACTICES

The following practices may be said to characterize monopolies.

Exclusive dealing to supply and collective boycott

Producers agree to supply only to recognized dealers, normally only one dealer in each area, on condition that the dealer does not stock the products of any producer outside the group (or trade association). Should the dealer break the agreement, all members of the group agree to withhold supplies from the offender. This practice has proved a very effective restriction on competition for it ensures that any new firms would find it extremely difficult to secure market outlets for their products.

Barriers

The creation of barriers to ensure that there is no competition against them. E.g price undercutting, individual ensure that actual text printed collective boycott and exclusive holding of patent rights.

Resale Price Maintenance (PRM)

A monopolistic firm may dictate to wholesalers and retailers the price at which its products would be sold. This is another way of ensuring that other firms are not attracted into the industry, if such firms can sell their products at more competitive prices.

Consumer Exploitation

Perhaps the most notorious practice for which monopolists are known is that of exploiting consumers by overcharging their products. There are three ways in which the monopolist can overcharge his products.

i. Profit maximization:

The price charged by the monopolists in order to maximize his profits is higher than would be the case if competitive firm was also maximizing its profits because in thecase of the monopolist, supply cannot exceed what he has produced.

ii. Cartels:

A cartel is a selling syndicate of producers of a particular product whose aim is to restrict output so that they can overcharge for the product. Thus, they collectively act as a monopoly and each producer is given his quota of output to produce.

iii. Price discrimination:

There are two forms of price discrimination:

a) The practice employed by firms of charging different prices to different groups of buyers and
b) That of charging the same consumer different prices for different units of the same good.

In the first case, each group of buyers has a different price elasticity of demand. The firm can by equalizing the marginal revenue generated by each group earn a higher level of profits than would be the case of a uniform f price were charged.

The preconditions for the successful operation of this form of discrimination are

i. Ability of the monopolistic firm to identify different segments of the market a according to price elasticity of demand and
ii. Prevention of resale by those customers who buy at a lower price.

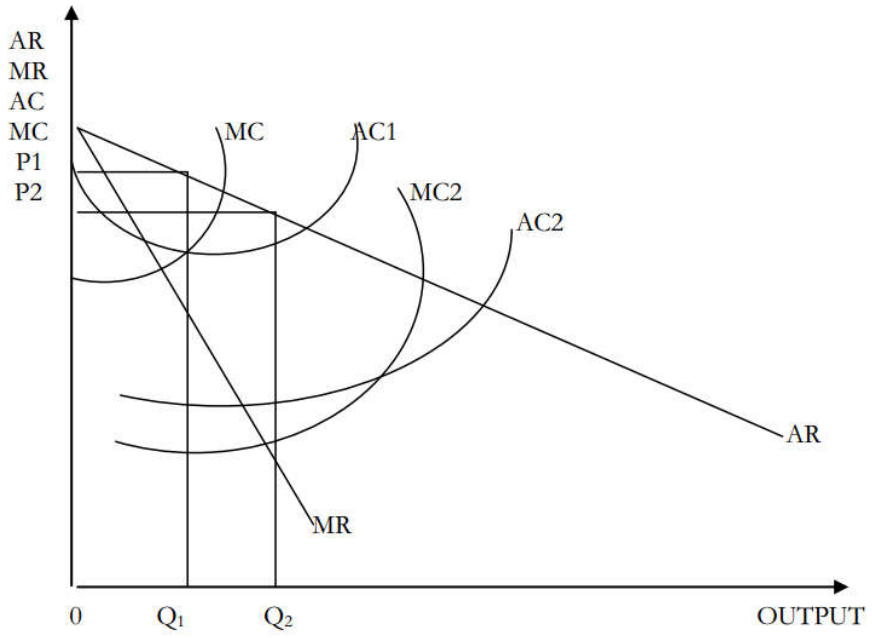
In the second case, the operation involves the firm appropriating all the consumer surplus that each consumer would have got if the price were constant. This can be achieved by setting the price of each unit equal to the maximum amount an individual would be willing to pay as given by the individual's demand curve and is therefore to be employed.

ARGUMENTS FOR AND AGAINST MONOPOLIES

Although monopolies are usually hated mainly because their practice of consumer exploitation, there are some aspects of monopolies which are favourable. The following arguments can be put forward in favour of monopoles:

1. Economies of Scale

As it has the whole market to itself, the monopolistic firm will grow to large size and exploit economies of large scale production. Hence its product is likely to be of higher quantity than product of a competitive firm that has less changes of expanding and lowering of the long run average cost (LRAC) of the firm. The price charged by the monopolistic firm may not be as high as is usually assumed to be the case.



When the MC and AC curves are MC1 and AC1, the price charged is P1 and the output is q1. When the cost curves the lower at MC2 and AC2 a lower price P2 charged and the level of output is higher.

2. No wastage of resources

As there is no competition from other firms, the monopolistic firm does not waste resources in product differentiation and advertising in an effort to capture consumers from rival firms.

3. Price stability

Since the monopolist is price maker, prices under a monopoly tend to be more stable than in competition where they are bound to change due to changes in supply and demand beyond the control of the individual firm.

4. Research

A monopolistic firm is in a better financial position to carryout research and improve its products than a competitive firm.

However monopolies have been accused of the following weaknesses.

1. Diseconomies of scale

While the monopolistic firm can grow to large size and exploit economies of scale, there is danger that it eventually suffers from diseconomies of scale. This will raise its LRAC and hence also raise its price.

2. Inefficiency

Since there is no competition, the firm can be inefficient as it has no fear of losing customers to rival firms.

3. Lack of innovation

Although the firm is in a better financial position to carry out research and improve its product than a firm in a competitive market, it may NOT actually do so because of the absence of competition.

4. Exploitation

Exploitation of consumer is the most notorious practice for which monopolists are known as in over-pricing so as to maximize profits, and price discrimination.

IMPERFECT COMPETITION

Assumptions of Monopolistic Competition:

Monopolistic competition as the name implies, combines features from both perfect competition and monopoly. It has the following features from **perfect competition.**

i. There are many producers and consumers. The producers produce differentiated substitutes. Hence there is competition between them. The difference from perfect competition is that the products area not homogeneous.
ii. There is freedom of entry into the industry so that an individual firm can make surplus profits in the short-run but will make normal profits in the long-run as new firms enter the industry.

Monopolistic competition has the following features from **monopoly**:

As the products are differentiated substitutes, each brand or type has its own sole seller e.g. each brand of toilet soap is produced by only one firm.

If one firm raises its price it is likely to lose a substantial proportion of its customers to its rivals.

If it lowers price it is likely to capture a proportion of customers from its rivals. But in the first case some of its customers will remain loyal to it and in the second case some customers will remain loyal to their traditional suppliers. Hence, as in monopoly the demand curve for the firm slopes downwards but it is more elastic than in monopoly

SHORT RUN OUTPUT AND PRICE

In monopolistic competition, it's the product differentiation that permits its price without losing sales.

Due to brand loyalty consumers will continue buying a particular product as preferred to all other brands in spite of increases in the price of that product.

If one firm lowers its price it may capture a few more customers therefore expanding its sales over and above the traditional customers. Besides the product differentiation need not be physical, only the customers need to feel the products are different.

Generally the demand for one seller's product will be price elastic due to close substitutes. If one firm raises its prices, TR will go down. If the price is reduced there are possibilities of substantial increase in revenue because of capturing some customers from rivals.

The level of elasticity will depend on the strength of product differentiation.

PRODUCT DIFFERENTIATION

Product differentiation describes a situation in which there is a single product being manufactured by several suppliers, and the product of each supplier is basically the same.

However, the suppliers try to create differences between their own product and the products of their rivals. It can be achieved through quality of service, after sales service, delivery dates, performance, reliability, branding, packaging, advertising or in some cases the differences maybe more in the minds of the customers rather than real differences, but a successful advertising can create a belief that a service or product is better than others and thus enable one firm to sell more and at higher price than its competitors.

Advantages of Product Differentiation

We can distinguish between those advantages for the firm itself and those for the consumer:

a. For the firm.

i. The ability to increase prices without losing loyal consumers
ii. The stability of sales, due to brand loyalty. The firm will not be subject to the risks and uncertainties o f intense price competition.

b. For the Customer

i. Consistent Product quality
ii. Wide consumer choice, between differentiated products.

Disadvantages of product differentiation

a) Product differentiation generally reduces the degree of competition in the market. It does this in two ways:
   i. It reduces competition amongst existing firms because consumers are reluctant to substitute one product for another, since they have developed brand loyalty.
   ii. It makes it more difficult for new firms to enter the industry in the long run if the consumers are already loyal to existing products.
b) All the effort and expense that the firms put into product differentiation are wasteful. Too much is spent on packaging, advertising and design changes. The price of goods could have been reduced instead.
c) Too many brands on the market, produced by large number of firms, could prevent the realization of full economies of scale in the production of goods.

d) Since the firms cannot expand their output to the level of minimum average cost output without making a loss, the "excess capacity theorem" predicts that industries marked by monopolistic competition will always tend to have excess capacity i.e. output is at less than capacity and price is above the average cost.

LONG RUN OUTPUT

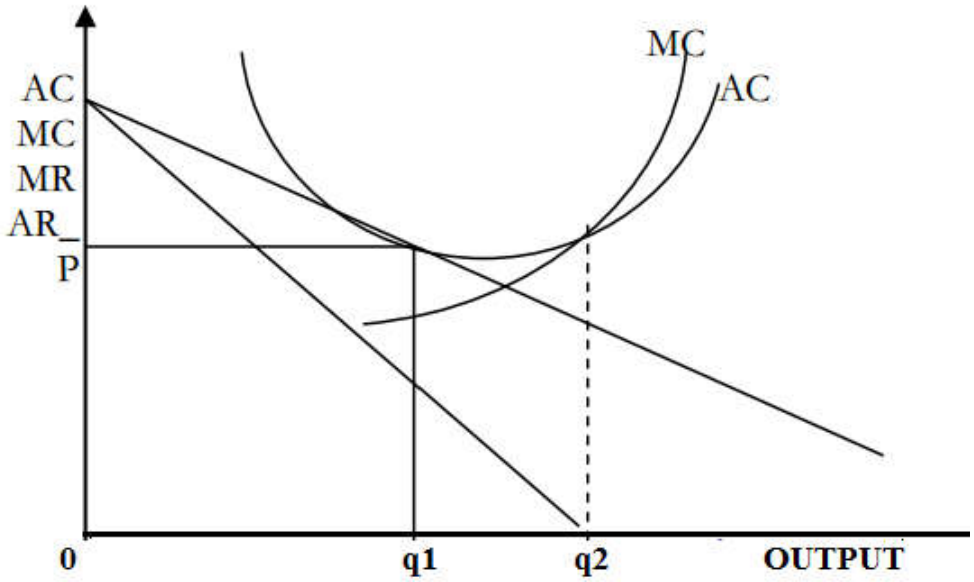
In the LR whether or not the firm makes profit will depend on the conditions of entry. For example, when surplus profits exist, there will be new entrants because they will make profit.

But as new firms enter, the market share enjoyed by each firm dwindles and their curves will shift to the Left. The costs will also be affected by the new entrants in three ways:

1. The new firms might make the cost resources to go up.
2. The cost curves might also be unaffected.
3. As new firms enter the cost curves might shift downwards because many sellers might force costs of resources downward.

But in the LR increasing costs are mostly likely. If there are increasing costs then existing profits will be squeezed. Because of the reduction in an individual firm's product, there will be a reduction in profits.

As long as there area profits in the industry, more firm's entry will stabilize when profits are ZERO. The losses might also cause exit of the firm's. The incentive to withdraw ceases when losses have been eliminated



Zero profits imply that LRAC = LRAR. Therefore LRAC is tangent to AR at Q1. But Q2 is the LR optimum output for the firm but with a negatively sloped AR curve. Zero profits imply that each firm will utilize a scale of plant smaller than optimum. Hence free entry leads to EXCESS capacity for each plant.

WASTE IN IMPERFECT COMPETITION

Monopolistic competition involves some degree of waste in two aspects.

When new firms enter the industry and the demand for the individual firm's product falls it will be forced to reduce productions. This means that part of its plant equipment will be unused. It is said to be operating under conditions of excess of the demand or the market for its product.

In practice, the firm will not allow a situation where it is reduced to a state of lower than normal profits. It will try to maintain its customers against new firms through product differentiation and advertising in an effort to convince customers that its products are the best. This wastage of resources, which could be used to expand and exploit economies of scale.

b. OLIGOPOLY

Oligopoly in the market describes a situation in which:

- Firms are price makers
- Few but large firms exist
- There are close substitutes
- Non-price competition exist like the form of product differentiation
- Supernormal profits re earned both in the short run and long run.

Because the sellers are few, then the decisions of sellers re mutually inter-dependent and they cannot ignore each other because the actions of one will affect the others.

PRICING AND OUTPUT DECISIONS OF THE FIRM

The price and output shall depend on whether the firm operates in:

a) Pure oligopoly or
b) Differentiated oligopoly

a) PURE OLIGOPOLY

Oligopolists normally differentiate their products. But this differentiation might either be weak or strong. Pure oligopoly describes the situation where differentiation of the product is weak. Pricing and output in pure oligopoly can be collusive or non-collusive.

COLLUSIVE OLIGOPOLY

Collusive oligopoly refers to where there is co-operation among the sellers i.e. co-ordination of prices. Collusion can be Formal or Informal.

i. FORMAL COLLUSIVE OLIGOPOLY

This is where the firms come together to protect their interests e.g. cartels like OPEC.

In this case the members enter into a formal agreement by which the market is shared among them. The single decision maker will set the market price and quantity offered for sale by the industry. There is a central agency which sets the price and quarters produce by the firms and all firms aside by the decisions of the central agency. The maximized joint profits are distributed among firms based on agreed formula.

ii. INFORMAL COLLUSIVE OLIGOPOLY

Informal collusive oligopoly can arise into two cases, namely:

- Where the cartel is not possible may be because it's illegal or some firms don't want to enter into an agreement or lose their freedom of action completely.
- Firms may find it mutually beneficial for them not to engage in price competition.
- When in outright cartel does not exist then firms will collude by covert gentlemanly agreement or by spontaneous co-ordination designed to avoid the effects of price war.
- One such means by which firms can agree is by price leadership. One firm sets the price and the others follow with or without understanding. When this policy is adopted firms enter into a tacit market sharing agreement.

There are two types of price leadership, namely:

By a low-cost firm

When there is a conflict of interests among oligopolists arising from cost differentials, the firms can explicitly or implicitly agree on how to share the market in which the low-cost firm sets the price. We can assume that the low cost firm takes the biggest share of the market.

Price leadership by a large firm

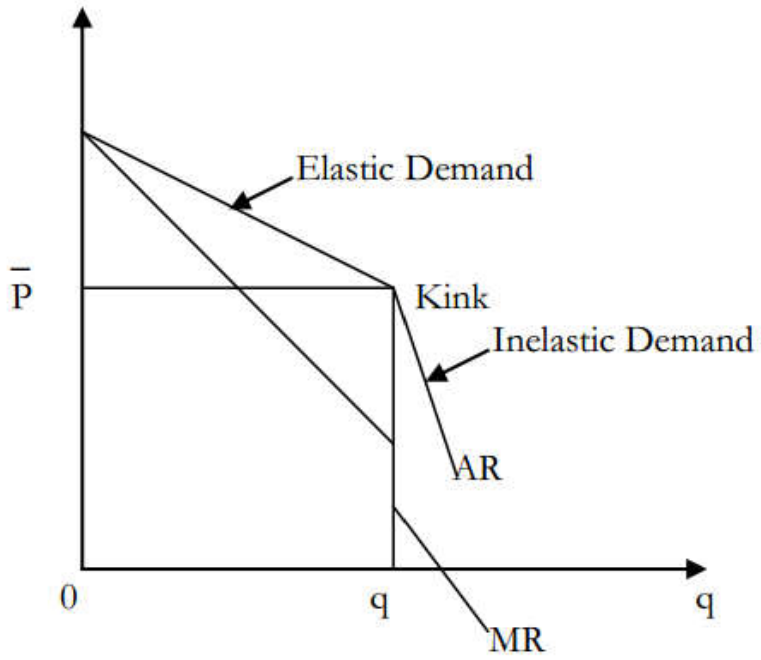
Some oligopolists consist of one large firm and a number of smaller ones. In this case the larger firm sets the price and allows the smaller firms to sell at that price and then supplies the rest of the quantity. Each smaller firm behaves as if in a purely competitive market where price is given and each firm sells without affecting the price because each will sell where MC = P = MR = AR

NON-COLLUSIVE OLIGOPOLY

Operates in the absence of collusion and in a situation of great uncertainty. In this case if one firm raises price, it is likely to lose a substantial proportion of customers to its rivals. They will not raise price because it is the interests to charge a price lower than that of their rivals.If the firm lower price it will attract a large proportion of customers from other firms. Theother firms are likely to retaliate by lowering price either to the same extent or a large extent.

The first firm will retaliate by lowering the price even further. As the firms will always expect a counter-strategy from rival firms, each price and outputdecision the firms comes up with is a tactical move within the framework of a broader strategy.

This then leads to a price war. If it goes on there will come a time when the prices are so low that if one firm lowers price, the consumers will see no point in changing from their traditional suppliers. Thus, the demand for the product of the individual firm will start by being elastic and it will end by being inelastic. The demand curve for the product of the individual firm thus consists of two parts, the elastic part and the inelastic part. It is said to be "kinked' demand curve as shown below.



If the firm is on the inelastic part and it raises price, the others will not follow suit. But on this part prices are so low that is likely to retain most of its customers. If it raises price beyond the kink, it will lose most of its customers to rivals. Hence the price p will lose most of its customers to rivals. Hence the price p will be the stable price because above it prices are unstable in that rising price means substantial loss of customers and lowering price may lead to price war. Below p prices are considered to be too low.

Elastic Demand

Barriers to entry in pure oligopoly

The barriers to entry can be artificial or natural.

Artificial Barriers

This can be acquired through:

- State protection through issuance of exclusive market (franchise) licences and patent rights.
- Control of supply of raw materials
- Threat of price war, if financial resources can sustain loses temporarily the cartel or price leader can threaten the new entrant by threatening to lower prices sufficiently to scare new firms.

A MATHEMATICAL APPROACH TO REVENUE AND COST FUNCTIONS

Recall that TR = P x Q

This implies that $P(AR) = \frac{TR}{Q}$

For example, assuming that the AR function is given by:

$$AR = 20 - \frac{1}{3}Q$$

$$\therefore TR = P \, x \, Q$$

$$= 200 - \frac{1}{3}Q^2$$

Marginal revenue is measure of the instantaneous rate of change of total revenue with respect to output Q. (Refer to the basic rules of differentiation in Appendix 1 of Modern Economics by Mudida)

$$MR = \frac{d\,TR}{dQ}$$

Thus, for example, given the following TR function:

$$TR = 2Q - \frac{1}{2}Q^2$$

$$AR = 2 - \frac{1}{2}Q$$

$$MR = \frac{d\,TR}{dQ}$$

$$= 2 - Q$$

The cost concepts studied earlier can also be expressed in functional form. Cubic functions are commonly used to represent cost functions. For example, a cost function may take the form:

$$TC = a + bQ + cQ^2 + dQ^3$$

Average cost refers to the cost per unit of output.

$$AC = \frac{TC}{Q}$$

$$= a + \frac{b}{Q} + cQ + dQ^2$$

Marginal cost refers to the instantaneous rate of change of the total cost function with respect to output.

$$MR = \frac{d\,TC}{dQ}$$

Given that

$$TC = a + b\,Q + cQ^2 + d\,Q^3$$

$$MR = \frac{d\,TC}{dQ}$$

$$= b + 2cQ + cQ^2 + 3dQ^3$$

For example, given a total cost function

$$TC = Q^3 - 8Q^2 + 68Q + 4$$

$$MC = \frac{d\,TC}{68dQ} = 3Q2 - 16Q$$

An application of revenue and cost concepts to profit maximisation

The functional concepts relating to revenue and costs can in turn be applied to provide a mathematical approach to profit maximisation.

(Refer to Appendix 2 of Modern Economics by Mudida for a treatment of maxima and minima).

The first and necessary condition for profit maximisation is that the first derivative of the function to the function to be maximised must be equal to zero.

Formally,

$$\frac{dy}{0dx} = 0$$

Thus, for example, given the following profit function:

$$\Pi = -Q^2 + 12Q - 20$$

$$\frac{d\Pi}{dQ} = -2Q + 12 = 0$$

$$2Q = 12$$

$$Q = 6$$

The second and sufficient condition for a maximum is that the second derivative be negative for relative maximum.

Formally,

$$\frac{d^2y}{dx^2} < 0$$

Thus for the profit function:

$$\pi = -Q^2 + 12Q - 20$$

$$\frac{d^2\Pi}{dQ^2} = \frac{d}{dQ}(-2Q + 12) = -2 < 0$$

$$\therefore Q = 6 \text{ provides a relative maximum.}$$

An application of profit maximisation to price discrimination

An optimisation problem can arise when a monopolistic firm sells a product in two or more separate markets. This firm has to decide upon the quantities to be supplied to the respective markets in order to maximise profits.

Assume that a monopolist is faced with the following demand functions:

$$Q_1 = 48 - 0.4\,P_1$$

$$Q_2 = 20 - 0.1\,P_2$$

Where

$$TC = 70 + 80\,Q$$

We can compute the prices that the monopolist will charge in the different markets.

With

$$Q_1 = 48 - 0.4\,P_1$$

$$P_1 = 120 - 2.5Q_1$$

$$TR_1 = (120 - 2.5Q_1)\,Q_1$$

$$= 120Q_1 - 2.5{Q_1}^2$$

$$MR_1 = \frac{dTR_1}{dTQ_1}$$

$$= 120 - 5Q_1$$

The firm will maximize profits where MC = MR_1 = MR_2

Recall that

$$TRC = 70 + 80\,Q$$

$$MC = \frac{d\,TC}{dQ}$$

$$= 80$$

∴ When MC = MR1

$$80 = 120 - 5Q_1$$

$$Q_1 = 8$$

When

$$Q_1 = 8$$

$$P_1 = 120 - 2.5(8)$$

$$= 120 - 20$$

$$= 100$$

In the second market

$$Q_2 = 20 - 0.1\,P_2$$

$$P_2 = 200 - 10Q_2$$

$$TR_2 = (200 - 10Q_2)\,Q_2$$

$$Q_2 = 200 - 10{Q_2}^2$$

$$MR_2 = \frac{dTR_2}{dTQ_2}$$

$$= 200 - 20Q_2$$

When MR_2= MC

$$80 = 200 - 20Q_2$$

$$\therefore\ Q_2 = 6$$

When Q_2 = 6

$$P_2 = 200 - 10\,(6)$$

$$= 140$$

The producer therefore charges a higher P_2 in market 2.

QUESTIONS:

1. What is monopolist market?

2. What is meant by an optimum size of a firm?

3. With the help of a well-labelled diagram, explain the relationship between the average fixed cost, average variable cost, total cost and marginal cost curves.

4. Discuss the necessary and sufficient conditions for profit maximization by a firm. Support your answer with appropriate illustrations.

5. (i) Maximize the following total profit function

$$\pi = -3Q^2 + 33Q - 72$$

(Determine that both the necessary and sufficient conditions for profit maximization are satisfied)

ii) Evaluate the function in (i) at the desired critical value.

MACROECONOMICS

CHAPTER 6

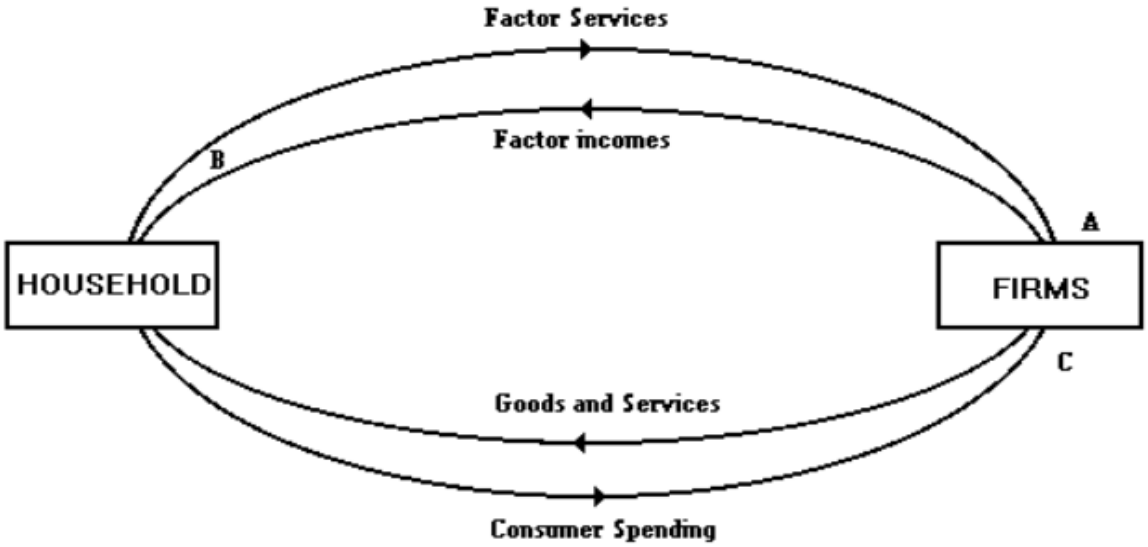
NATIONAL INCOME

DEFINITION OF NATIONAL INCOME

National Income is a measure of the money value of goods and services becoming available to a nation from economic activities. It can also be defined as the total money value of all final goods and services produced by the nationals of a country during some specific period of time – usually a year – and to the total of all incomes earned over the same period of time by the nationals.

THE CIRCULAR FLOW OF INCOME AND EXPENDITURE

This is an economic model illustrating the flow of payments and receipts between domestic firms and domestic households. The households supply **factor services** to the firms. In return, they get **factor incomes**. With factor incomes, they buy **goods and services** from the firms. These flows can be illustrated diagrammatically as follows:



The points at which flows from one sector meets the other sector and generate other flows are called **critical points**. In the above diagram, the critical points are A, B and C. At A, the flow of factor services from the households sector meets the firm sector and generates the **flow of factors incomes** from the firms to the households. At B, the flow of factor incomes meets the household sector and

generates the flow of **consumer spending**. At C, the flow of consumer spending meets the firms sector and generates the flow of **goods and services**.

APPROACHES TO MEASURING NATIONAL INCOME

The compilation of national income statistics is a very laborious task. The total wealth of a nation has to be added up and there are millions of nationals. Moreover, in order to double check and triple check the statistics, the national income statistician has to work out the figures out in three different ways, each way being based on a different aspect. The three aspects are:

a. The national output: - The creation of wealth by the nation's industries. This is valued at factor cost, so it must be the same as b) below.

b. The national income: - The incomes of all the citizens.

c. The national expenditure because whatever we receive we spend, or lend to the banks to invest it, so that the addition of all the expenditure should come to the same as the other two figures.

Put in its simplest form we can express this as an identity:

National output ≡ National Income ≡ National Expenditure.

(i) Using Total Expenditure for Calculating National Income

The expenditure approach centres on the components of **final demand** which generate production. It thus measures GDP as the total sum of expenditure on **final goods and services** produced in an economy. It includes all consumers' expenditure on goods and services, except for the purchase of new houses which is included in gross **fixed capital formulation**. Secondly we included all general **government final consumption**. This includes all current expenditure by **central and local** government on goods and services, including wages and salaries of government employees. To these we add **gross fixed capital formation or expenditure on fixed assets** (buildings, machinery, vehicles etc) either for replacing or adding to the stock of existing fixed assets. This is the major part of the investment which takes place in the economy. In addition we add the **value of physical increases in the stocks, or inventories,** during the course of the year. The total of all this gives us **Total domestic expenditure** (TDE). We then add expenditure on exports to the

TDE and arrive at a measure known as **Total Final Expenditure**. It is so called because it represents the total of all spending on final goods. However, much of the final expenditure ison **imported goods** and we therefore **subtract spending on imports**. Having done this we arrive at a measure known **as gross domestic product at market prices**. To gross domestic product at market price we **subtract the taxes on expenditure** levied by the government and add on the amount of subsidy. When this has been done we arrive at a figure known **as Gross Domestic Product at factor cost.** National Income however is affected **by rent, profit interest** and dividends paid to, or received from, overseas. This is added to GDP as net property income from abroad. This figure may be either **positive or negative.** When this has been taken into account we arrive at the gross national product at factor cost. As production takes place, the capital stock of a country wears out. Part of the gross fixed capital formation is therefore, to replace worn out capital and is referred to as **Capital Consumption**. When this has been subtracted we arrive at a figure known as the **net national product**. Thus, summarising the above, we can say:

Y = C + I + G + (X – M)

Calculating National Income from Total Expenditure

Country Y National Expenditure (in £millions)		1999
Expenditure of Consumers		
Food		27,148
Alcoholic drink		13,372
Tobacco		6,208
Housing		27,326
Fuel and light		9,395
Clothing		12,114
Household goods and services		12,274
Transport and communications		31,475
Recreation		16,541
Other goods and services		23,356
Total		179,209
Less: Adjustment of non-profit making bodies		(443)
		178,766
		178,766
Add: Expenditure of non-profit making bodies		3,661
		182,427
Central Government expenditure		40,623
Local Government expenditure		25,236
Capital formation		49,559
Growth in stocks		267
Total Domestic expenditure at market prices		298,112
Deduct: Taxes on expenditure		(49,865)
		248,247
Add: Net result exports-imports	3,186	
Subsidies	6,056	
Net property income from abroad	1,948	11,190
		259,437
Less: Estimated depreciation on capital assets		36,490
		222,947m

(ii) Using Factor Incomes for Calculating National Income

A second method is to **sum up all the incomes** to individuals in the form of **wages, rents, interests** and **profits** to get domestic incomes. This is because each time something is produced and sold someone obtains income from producing it. It follows that if we add up all incomes we should get the value of total expenditure, or output. Incomes earned for purposes other than rewards for producing goods and services are ignored. Such incomes are **gifts, unemployment or relief benefits, lottery, pensions, grants** for students etc. These payments are known as **transfer income (payments)** and including them will lead to **double counting.** The test for inclusion in the national income calculation is therefore that there should be a **"quid pro quo"** that the money should have been paid against the exchange of a good or service. Alternatively, we can say that there should be a **"real"** flow in the opposite direction to the money flow. We must also include income obtained from **subsistence output.** This is the opposite case from transfer payments since there is a flow of

real goods and services, but no corresponding money flow. It becomes necessary to "impute'' values for the income that would have been received. Similarly workers may, in addition to cash income, receive **income in kind**; if employees are provided with **rent free housing**, the rent which they would have to pay for those houses on the open market should, in principle, be "imputed" as part of their income from employment. The sum of these incomes gives **gross domestic product GDP**. This includes incomes earned by foreigners at home and excludes incomes earned by nationals abroad. Thus, to **Gross Domestic Income** we add **Net property Income** from abroad. This gives **Gross National Income.** From this we deduct **depreciation** to give **Net National Income.**

Calculating National Income from Factors Incomes

Country Y National Income rewards to factors (in £ millions)			**1999**
Incomes from employment			
Wages and salaries			143,348
Pay in cash and kind of HM Forces			3,121
Employers' contribution to National Health Insurance		10,632	
Employers' contribution to other funds		12,971	
			170,072
Income from self-employment			23,123
Other Incomes			
Profits of companies		41,530	
Surpluses of public corporations		9,661	
Surpluses of other public enterprises (-)		(109)	
Rent			17,424
Imputed charge for consumption of capital		2,456	
			264,157
Less: Stock appreciation			(4,326)
			259,831
Add: net property income from abroad			1,948
			261,779
Less: Residual error	2,342		
Estimated depreciation on capital assets	36,490		(38,832)
			222,947m

Note: The residual error is a small error (about 1%) in the collection of these figures.

(iii) Using the National Output for Calculating National Income

A final method which is more direct is the **"output method"** or the **value added approach**.

This involves adding up the total contributions made by the various sectors of the economy.

"Value Added" is the **value added by each industry to the raw materials or processed products that it has bought from other industries before passing on the product to the next stage in the production process**. This approach therefore centres on **final products.** Final products will include **capital goods** as well as **consumer goods** since while **intermediate goods**

are used up during the period in producing other goods, capital goods are not used up (apart from "wear and tear" or depreciation) during the period and may be thought of as consumer goods

"stored up" for future periods. Final output will include **"subsistence output",** which is simply **the output produced and consumed by households themselves.** Because subsistence output is not sold in the market, some assumption has to be made to value them at some price. We also take into account the **final output of government,** which provides services such as education, medical care and general administrative services. However, since state education and other governmental services are not sold on the market we shall not have market prices at which to value them. The only obvious means of doing this is to value public services at what it costs the government to supply them, that is, by the **wages bill spent on teachers, doctors, and the like.** When calculating the GDP in this matter it is necessary to avoid **double counting**.

Calculating National Income from National Output

Country Y National Product by industry (£ millions)		1999
Agriculture		5,535
Energy and water supply		29,645
Manufacturing		62,258
Construction		15,319
Distribution, hotels, catering, repairs		35,002
Transport		11,543
Communications		7,092
Insurance, banking and finance		31,067
Ownership of dwellings		15,761
Public administration, defence and social security		18,027
Public health and education services		24,021
Other services		16,415
Total domestic output		271,685
Deduct: Residual error	2,342	
Adjustment for financial services	11,854	
Estimated depreciation on capital assets	36,490	- 50,686
		220,990
Add: Net property income from abroad		1,948
Aggregate net national product		222,947 m

CONCEPTS OF NATIONAL INCOME

Gross Domestic Product

The money value of all goods and services produced within the country but excluding net income from abroad.

Gross National Product

The sum of the values of all final goods and services produced by the nationals or citizens ofa country during the year, both within and outside the country.

Net National Product

The money value of the total volume of production (that is, the gross national product) after allowance has been made for depreciation (capital consumption allowance).

Nominal Gross National Product

The value, at current market prices, of all final goods and services produced within some period by a nation without any deduction for depreciation of capital goods.

Real Gross National Product

This is the national output valued at the prices during some base year or nominal GNPcorrected for inflation.

National income accounting

This refers to the measuring of the total flow of output (goods and services) and of the total flow of inputs (factors of production) that pass through all of the markets in the economy during the same period.

To see exactly what national income includes, how it is measured, and what it can tell us, westart with economic models: By economic models we mean:

A simplification of a real world or a practical situation aimed at explaining that situation within a set of assumptions.

PROBLEMS OF MEASUREMENT; USES OF NATIONAL INCOME STATISTICS AND THEIR LIMITATIONS

National Income Accounting is beset with several difficulties. These are:

a. What goods and services to include

Although the general principle is to take into account only those products which change hands for money, the application of this principle involves some arbitrary decisions and distortions. For example, unpaid services such as those performed by a housewife are not included but the same services if provided by a paid housekeeper would be.

Many farmers regularly consume part of their produce with no money changing hands. An imputed value is usually assigned to this income. Many durable consumer goods render services over a period of time. It would be impossible to estimate this value and hence these goods are included when they are first bought and subsequent services ignored.

Furthermore, there are a number of governmental services such as medical care and education, which are provided either 'free' or for a small charge. All these provide a service and are included in the national income at cost. Finally, there are many illegal activities, which are ordinary business and produce goods and services that are sold on the market and generate factor incomes.

b. Danger of Double Counting

The problem of double counting arises because of the inter-relationships between industries and sectors. Thus we find that the output of one sector is the input of another.

If the values of the outputs of all the sectors were added, some would be added more than once, giving an erroneously large figure of national income. This may be avoided either by only including the value of the final product or alternatively by summing the values added at each stage which will give the same result.

Some incomes such as social security benefits are received without any corresponding contribution to production. These are **transfer payments** from the taxpayer to the recipient and are not included. **Taxes** and **subsidies** on goods will distort the true value of goods. To give the correct figure, the former should not be counted as an increase in national income for it does not represent any growth in real output.

c. Inadequate Information

The sources from which information is obtained are not designed specifically to enable national income to be calculated. Income tax returns are likely to err on the side of understatement. There are also some incomes that have to be estimated. Also, some income is not recorded, as for example when a joiner, electrician or plumber does a job in his spare time for a friend or neighbour. Also information on foreign payments or receipts may not all be recorded.

Factors affecting the size of a National Income

The size of a nation's income depends upon the quantity and quality of the factor endowments at its disposal. A nation will be rich if its endowments of natural resources are large, its people are skilled, and it has a useful accumulation of capital assets. The following points are of interest:

a) Natural Resources

These include the minerals of the earth; the timber, shrubs and pasturage available; the agricultural potential (fertile soil, regular rainfall, temperature or tropical climate); the fauna and flora; the fish; crustacea etc of the rivers and sea; the energy resources, including oil, gas, hydro-electric, geothermal, wind and wave power.

b) Human Resources

A country is likely to prosper if it has a large population; literate and numerate sophisticated and knowledgeable about wealth creating processes. It should be well educated and skilled, with a nice mixture of theory and practice. It should show enterprise, being inventive, energetic and determined in the pursuit of a better standard of living.

c) Capital Resources

A nation must create and then conserve capital resources. This includes not only tools, plant and machinery, factories, mines, domestic dwellings, schools, colleges, etc, but a widespread infrastructure of roads, railways, airports and ports. Transport creates the utility of space. It makes

remote resources accessible and high-cost goods into low-cost goods by opening up remote areas and bringing them into production.

d) Self-sufficiency

A nation cannot enjoy a large national income if its citizens are not mainly self-supporting.

If the majority of the enterprises are foreign –owned there will be a withdrawal of wealth in the form of profits or goods transferred to the investing nation.

e) Political Stability

USES OF NATIONAL INCOME FIGURES

- We need national income statistics to measure the size of the "National cake' of goods and services available for competing uses of private consumers, government, capital formation and exports (less imports).
- National Income statistics are also used in comparing the standard of living of a country over time
- And also the standards of living between countries.
- National Income Statistics provide information on the stability of performance of the economy over time e.g. a steadily increasing income would be indicative of increasing national income.
- If National Income Statistics are disaggregated it would enable us to assess the relative importance of the various sectors in the economy. This is done by considering the contribution of the various sectors to Gross National Product over time. Such information is crucial for planning purposes for it reveals to planners where constraints to economic development lie. It therefore becomes possible to design a development strategy that eventually would overcome these problems. This central contribution could be in the form of employment or the production of goods and services.
- By assessing exports and imports as a percentage of Gross national Product i.e. using national statistics, it is possible to determine the extent to which a country depends on external trade.
- National Income Statistics also help in estimating the saving potential and hence investment potential of a country.

Real Vs Nominal GNP: "Deflating" by a price Index

One of the problems that confront economists when measuring GNP is that they have to use money as the measuring rod. These days however, inflation sends the general price level up and up clearly this means that the yardstick stretches in their hands everyday. Economists repair most of the damage wrought by the elastic yardstick by using a price index.

The price index used to remove inflation (or "deflate'' the GNP) is called the GNP deflator. The GNP deflator is defined as the ratio of nominal GNP to real GNP. It is constructed as follows:

GNP Deflator = Nominal GNP

Real GNP

Real GNP = Value at current Price CPI

Where CPI is Consumer Price Index

The GNP deflator is useful because it includes prices on all goods and services in GNP.

PER CAPITA INCOME

By National Income is meant the value of outputs produced within a year. Income per capita is simply the National Income divided by the population of the country in a year.

INCOME PER CAPITA = National Income

Population

It shows the standard of living a country can afford for its people. The level of income per capita is determined by the size of a country's population. The higher is the rate of growth of population, the lower is the rate of growth of income per capita.

Per capita income is a theoretical rather than a factual concept. It shows what the share of each individual's National Income would be if all citizens were treated as equal. In a real world situation there exists considerable inequality in the distribution of income especially in the third world countries

NATIONAL INCOME AND WELFARE

The relationship between National Income and Welfare is best explained in terms of economic growth (By economic growth is meant capacity expansion). The effect of economic growth is an increase in the National Income. This increase in National

Income has several effects on a country's citizens.

1. Assuming a fair distribution of income, the average citizen would be in a position to enjoy a higher living standard.
2. The ordinary households or persons could be able to afford luxury commodities. NB: luxury differs in its definition from one country to another and the determining factor being the level of income. e.g. clothing can be a luxury for some people.
3. It enables the ordinary household to afford leisure which may be regarded as luxury i.e. reducing working hours.

Points 1, 2 and 3 are based on assumption that there exists a fair distribution of the Nationalcake. This may not be the case in fact it is disastrous to rely on GNP, its growth rate and GNP per capita as indicators of economic well being. GNP per capita e.g. gives no indication of how

National Income is actually distributed and who is benefiting most from the growth of production. A rising level of absolute and per capita GNP may camouflage the fact that the poor are not better than before. In fact the calculation of GNP, and especially its rate of growth is in reality largely a calculation of the rate of growth, of the incomes of upper 20% who receive a disproportionately large share of the National Product. It is, therefore, unrealistic to use GNP growth rates as an index of improved economic welfare for the general public.

Example: Assuming a 10 people economy and assuming 9 of them had no income and the 10th person receives 100 units of income the GNP for this economy would be 100 units of income and per capita income would be 10 units. Suppose everyone's income increases by 20% so that

GNP rises to 120 units per capita income would rise to 12 units. However for the 9 people without income before and currently such a rise in per capita income provides no cause for rejoicing since the one rich individual still has the income. In this case we observe that GNP instead of being a welfare index of a society as a whole is merely increasing the welfare of a single individual.

This exchange though an extreme case is indicative of what happens in the real life situation where incomes are very unequally distributed.

Costs of Economic Growth (Increase in National Income)

1. People living in industrial towns suffer from the effects of a polluted atmosphere.
2. The manufacture of intoxicants together with urbanization and urban housing problems leads to an increase in crimes. This creates for the state the additional costs of the maintenance of prisons and a large police force to maintain law and order. Thus social welfare would be increased if the production and sale of intoxicants are curtailed.
3. While the expansion of the National Income owes a great deal to scientific research the application of research to new means of destruction add nothing to social welfare.
4. It leads to employment of women in industry leading to children being left without care or simply maternal care.

Arguments for and against Uneven Distribution of Income and Wealth

The basic economic argument to justify large income inequality was the assumption that high personal and corporate incomes were necessary conditions for saving which made possible **investments** and economic growth through mechanism such as the Harrod-Domar Model. In this argument it is maintained that the rich save and invests a significant proportion of their incomes while the poor spend all their incomes on consumer items, and since GNP growth is assumed to be directly related to the proportion of National Income saved then an economy characterised by highly unequal distribution of income would save more and grow faster than one with more equitable distribution of income. It was also assumed that eventually National per capita income would be high enough to allow for a sizeable distribution of income via Taxes and subsidies but until such time is reached, any attempt to redistribute income significantly could only serve to lower growth rate and delay the time when a large income cake would be cut up into smaller sizes for all population group.

Limitations of the argument (Against)

Unlike the historical experience of the now developed countries, the rich in contemporary Third World Countries are not noted for the desire to save and invest substantial proportions of their

income in the local economy. Instead businessmen, politicians and other elites are known to squander much of their income on imported goods, luxury houses, foreign travel and investment in gold, jewellery and foreign banking countries. Such savings and investments do not add to the National Productive resources. Instead they represent substantial drains on these resources in that the income so derived is extracted from the sweat and toil of common uneducated unskilled labourers thus the rich do not necessarily save and invest a significantly large proportion of their income than the poor.

A growth strategy based on sizeable and growing income inequalities may in reality be nothing more than an opportunistic myth designed to perpetuate the vested interests and maintain "status quo" of the economic and political elite of the 3rd world, often at the expense of the great majority of the general population.

1) The low income and low levels of living for the poor which are manifested in poor health, nutrition and education can lower their economic productivity and thereby lead directly and indirectly to a slower growing economy. Therefore strategies to lift the living standard and incomes of say the bottom 40% would contribute not only to their material well being, but also to the productivity and income of the economy as a whole.

2) Raising the income level of the poor will stimulate an overall increase in the demand for locally produced necessity products like food and clothing. Rising demand for local goods provided a greater stimulus for local production i.e. stimulates local production, employment and investment. This creates a broader popular participation in that growth.

The rich, on the other hand, tend to spend more of their additional income on imported luxuries.

3) A more equitable distribution of income achieved through the reduction of mass poverty can stimulate healthy economic expansion by acting as a powerful material and psychological incentive to widespread public participation in the development process. Wide income disparities and substantial absolute poverty on the other hand can act as a powerful and psychological disincentive to economic progress. In the extreme, it may create conditions for its ultimate rejection by the masses of frustrated and politically exploitive people notably the educated.

NATIONAL INCOME AND STANDARDS OF LIVING

Standard of living refers to the quantity of goods and services enjoyed by a person. These goods may be provided publicly, such as in the case of health care or education or they may be acquired by direct purchase. It also includes the less easily quantifiable aspects of living such as terms and conditions of employment and general living environment.

National Income figures can be used to measure the standard of living at a particular point of time and over time. This is done by working out the per capita income of the country. **By per capita income we mean: the value of goods and services received by the average man.** Per capita income is obtained by dividing the National Income by the Total population. If the per capita income is high, it can be deduced that the standard of living is high.

PROBLEMS OF USING PER CAPITA INCOME TO COMPARE STANDARD OF LIVING OVER TIME

1) The composition of output may change. e.g. more defence-related goods may be produced and less spent on social services, more producer goods may be made and less consumer goods, and there may be a surplus of exports over imports representing investment overseas. Standards of living depend on the quantity of consumer goods enjoyed.
2) Over time prices will change. The index of retail prices may be used to express the GNP in real terms but there are well known problems in the use of such methods.
3) National Income may grow but this says nothing about the distribution of that income. A small group may be much better off. Other groups may have a static standard of living or be worse off.
4) Any increase in GNP per capita may be accompanied by a decline in the general quality of life. Working conditions may have deteriorated. The environment may have suffered from various forms of pollution. These non-monetary aspects are not taken into account in the estimates of the GNP.
5) Finally the national income increases when people pay for services which they previously carried out **themselves**. If a housewife takes an office job and pays someone to do her housework, national income will increase to the extent of both persons' wages. Similarly a reduction in national income would occur if a man painted his house rather than paying a professional painter to do the same. Changes of the above type mean that changes in the GNP per capita will only imperfectly reflect changes in the standard of living.

PER CAPITA INCOME AND INTERNATIONAL COMPARISONS

Per capita income figures can also be used to compare the standards of living of different countries. Thus if the per capita income of one country is higher than that of another country, the living standard in the first country can be said to be higher. Such comparisons are made by aid giving international agencies like the United Nations and they indicate the relevant aid requirements of different countries. But there are major problems in using **real income per head** (per capita income) to measure the standard of living in different countries. First there is the whole set of **statistical problems** and, secondly, there are a number of difficult **conceptual problems** or problems of interpretation.

i. **Inaccurate estimates of population:** The first statistical problem in calculating income per head particularly in less developed countries is that we do not have very accurate population figures with which to divide total income.
ii. **Specific items which are difficult to estimate:** Another data problem, as already mentioned, is that data for depreciation and for net factor income from abroad are generally unreliable. Hence although we should prefer figures for “the’ national income, we are likely to fall back on GDP, which is much less meaningful figure for measuring income per head. Inventory investment and work-in-progress are also difficult items to calculate.
iii. **Non-marketed subsistence output and output of government:** some output like subsistence farming and output of government are not sold in the market. These are measured by taking the cost of the inputs. In one country, however, salary of doctors for instance, might be higher and their quality low compared to another country. Although the

medical wage bill will be high, the "real consumption" of medical care in the former might be lower. Since "public consumption" is an important element in national income, this could affect comparisons considerably. Also in making international comparisons it is assumed that the complied national income figures of the countries being compared are equally accurate. This is not necessarily the case. If, for example, in one country there is a large subsistence sector, a lot of estimates have to be made for self-provided commodities. The national figures of such a country will, therefore, be less accurate than those of a country whose economy is largely monetary or cash economy.

iv. **Different degrees of income distribution**: If the income of one country is evenly distributed, the per capita income of such a country may be higher than that of another country with a more evenly distributed income, but this does not necessarily mean that most of its people are at a higher living standard.

v. **Different Types of Production:** If one country devotes a large proportion of its resources in producing non-consumer goods e.g. military hardware, its per capita income may be higher than that of another country producing largely consumer goods, but the standard of living of its people will not necessarily be higher.

vi. **Different forms of Published National Income figures**: The per capita income figures used in international comparisons are calculated using the published figures of national income and population by each country. For meaningful comparisons, both sets of national income figures should be in the same form i.e. both in real terms or both in money terms, the latter may give higher per capita income figures due to inflation, and thus give the wrong picture of a higher living standard. On the other hand, if both sets are in money terms the countries being compared should have the same level of inflation. In practice, this is not necessarily the case.

vii. **Exchange Rates:** Every country records its national income figures in its own currency. To make international comparisons, therefore, the national income figures of different countries must have been converted into one uniform currency. Using the official exchange rates does this. Strictly speaking, these apply to internationally traded commodities, which normally form a small proportion of the national production. The difficulty is that these values may not be equivalent in terms of the goods they buy in their respective commodities i.e. the purchasing power of the currencies may not be the same as those reflected in the exchange rate.

viii. **Difference in Price Structures:** Differences in the relative prices of different kinds of goods, due to differences in their availability, mean that people can increase their welfare if they are willing to alter their consumption in the direction of cheaper goods. The people in poor countries probably are not nearly as badly off as national income statistics would suggest, because the basic foodstuffs, which form an important part of their total consumption, are actually priced very low.

ix. **Income in relation to Effort:** The first conceptual problem in calculating income per head is to look at goods and services produced in relation to the human effort that has gone into producing them. Obviously if people work harder, they will be able to get more goods; but they may prefer the extra leisure. Indeed, the amount of leisure that people want depends in part on their level of income. Strictly, therefore, we should take income per unit of labour applied. It is largely because this would be statistically awkward that economists prefer to take real income per head.

x. **Differences in size:** A problem which is both conceptual and statistical is due to the transport factor. If two countries are of different sizes, the large country may devote a large

proportion of its resources in developing transport and communication facilities to connect the different parts of the country. This will be reflected in its national income, but the standard of living of its people will not necessarily be higher than that of smaller country, which does not need these facilities to the same extent.

xi. **Differences in Taste:** Another formidable difficulty is that tastes are not the same in all countries. Also in different countries the society and the culture may be completely different thus complicating comparisons of material welfare in two countries. Expensive tastes are to some extent artificial and their absence in poor countries need not mean a corresponding lack of welfare. Tastes also differ as regards the emphasis on leisure as against the employment of the fruit of labour: if in some societies people prefer leisure and contemplation, who is to say this reduces their welfare as compared to those involved in the hurly-burly of life and labour in modern industry?

xii. **Different climatic zones:** If one country is in a cold climate, it will devote a substantial proportion of its resources to providing warming facilities, e.g. warm clothing and central heating. These will be reflected in its national income, but this does not necessarily mean that its people are better off than those in a country with a warm climate.

xiii. **Income per head as index of economic welfare:** We cannot measure material welfare on an arithmetic scale in the same way as we measure real income per head. For instance, if per capita income increases, material welfare will increase; but we cannot say by how much it has increased, and certainly that it has increased in proportion.

ANALYSIS OF CONSUMPTION, SAVING AND INVESTMENT AND THEIR INTERACTION IN A SIMPLE ECONOMIC METHOD

Aggregate demand refers to the total planned or desired spending in the economy as a whole in a given period. It is made up of consumption demand by individuals, planned investment demand, government demand and demand by foreigners of the nationals output.

CONSUMPTION

The consumption function is the relationship [expressed in mathematical or diagrammatic form] between planned consumption and other independent variables, particularly income.

The consumption function is one of the most important relations in Macro-economics.

Consumption is the largest single component of aggregate expenditure and if we are to predict the effects of income and employment of variations in private investment and in government spending, we must know how consumption varies in response to changes in income. Thus it is important to take a closer look at the consumption.

Other Determinants

1. Rate of Interest

Is contained in the argument of the classified economists who argued that rational consumers will save more and consume less if the rate of interest is high

2. Relative Prices Influences the aggregate consumption. If relative prices are high, the level of consumption will be low

3. Capital Gains

Keynes observed that there is a possibility of windfall gains or losses influencing consumption. He says consumption of the wealth owning group may be extremely susceptible to unforeseen changes in the money value of their wealth. This is true of the stock minded speculative economy.

4. Wealth

The possession of liquid assets influences the amount that you have to save. It stems from the Diminishing Marginal Utility of Wealth. The larger the stock of wealth, the lower its

Marginal Utility and consequently the weaker the desire to add to future wealth by curtailing present consumption

In this case, the more wealth an individual has, the weaker will be the desire to accumulate still more savings at that particular time.

5. Money Stock or Liquid Assets:

Possession of liquid assets boosts consumption in that they can be changed into cash and thus consumed.

6. Availability of Consumer Credit:

Normally influences spending of the consumer of durables.

7. Attitudes and Expectations of the Consumer

A change in the consumer attitudes will affect consumer behaviour. The expectations attained by the consumer about income increases will affect the consumer behaviour. If in the face of price increases they expect further price increases; they shall increase their purchases further. N.B. These things might be true of an individual, but not the [aggregate] society.

8. The money Illusion

Some people look at money at the face value. Consumption will be affected if customers are subject to money illusion. The phenomenon of Money illusion occurs when despite proportional changes in the prices of goods and services and then their money incomes which keeps real incomes unchanged, consumers make a change in their real consumption pattern. It is know as Pigou Effect which talks of real balance. With a change in nominal income, people behave in the same way as though their real income has gone up.

Suppose price and Money Income increases by 10%, for the families which regard their real income unchanged and do not suffer from money illusion they would take their real incomes as unchanged and would only increase their consumption by 10%.

9. Distribution of Income

If the Marginal Propensity to consume among the poor is high, then redistribution of wealth from the rich to the poor leads to higher consumption.

10. Composition of the Population:

In sex and age

THE KEYNESIAN THEORY OF CONSUMPTION FUNCTION

The theory was developed during the Great Depression which plagued Europe and America.

During this time, there was excess capacity and idle resources and no effective demand i.e. people were unemployed and had no purchasing power. The determination of aggregate demand, then, was of crucial significance in Keynes analysis.

DEFINITIONS

i. Average Propensity to Consume:

The average Propensity to Consume **[APC]** is defined as the fraction of aggregate national income which is devoted to consumption. If consumption is denoted by **C** and income by **Y**, then:

APC = YC

The Average Propensity to Consume decreases in Keynes model as income increases.

ii. Average Propensity to save

The Average Propensity to Save **[APS]** is defined as the fraction of aggregate national income which is devoted to savings. Thus if S denotes savings then,

APS = YS

In a closed ungoverned economy, where income is spent or saved, APC = APS = 1

iii. Marginal Propensity to Save

The Marginal Propensity to Save is the fraction of an increase in income that is saved. Thus, if

ΔS denoted changes in savings, and **ΔY** change in income, then,

MPC = ΔC/Δ y

An increase in income is partly consumed and partly saved. Thus

$\Delta C + \Delta S = \Delta Y$

Dividing through by **ΔY,** we get

$\Delta c/\Delta Y + \Delta S/\Delta Y = 1$

Therefore **$\Delta C + \Delta S = 1$,** and **$S = 1 - C$**

INVESTMENT

Investment is the process of increasing the productive capital stock of a country, or can be defined as the production of goods not for immediate consumption. The investment function is the relationship [expressed in mathematical or diagrammatic form] between planned investment and the real interest rate.

Definitions

Induced Expenditure

Also called endogenous Expenditure, is any expenditure that is determined by, and thus varies with, economic variables within our theory.

Autonomous Expenditure:

Also called Exogenous expenditure, is any expenditure that is taken as a constant or unaffected by any economic variables within our theory. For instance, in the simple theory of the determination of national income, investment is assumed to vary directly with national income.

Actual income and Full employment income:

Full employment income (Also called Potential National) is the national income that could be produced when the country's factors of production are fully employed. This concept is given the symbol YF. Actual national income, symbolized by Y, can be below or equal to YF and, by working resources overtime and otherwise harder than normal, it can occasionally rise above YF.

Intended or planned Investment:

Expenditure on investment depends on business expectations on the chance of making profits and on the availability of funds for the purchase of producer goods. Although business expectations re complex in nature, a rough approximation is that the expectations on profits rise or fall in direct response to movements in the GNP this year increases businessmen expect an increase in planned consumption expenditures, and hence profits to increase next year.

SAVING

Saving is income not spent on goods and services for current consumption. Both households and firms can save. Households save when they elect not to spend part of their current income on goods and services for consumption. Firms save when they elect not to pay out to their owners some of the

profits that they have earned. **Distributed profits** are profits actually paid out to the owners of firms, and **undistributed profits** are profits held back by firms for their own uses.

INVESTMENT

Investment is defined as the production of goods not for immediate consumption. All such goods a are called **investment goods**. They are produced by firms and they may be bought either by firms or by households. Most investment is done by firms, and firms can invest either in **capital goods**, such as plant and equipment, or **inventories.**

The total investment that occurs in the economy is called **Gross Investment**. The amount necessary for replacement is called the **Capital consumption Allowance** and is often loosely referred to as **Depreciation**. The remainder is called **NET Investment**.

The current production of final commodities in the frugal economy can be divided into two sorts of output. First, there are consumption goods and services actually sold to households. Second, there are investment goods that consist of capital goods plus inventories of semi-finished commodities still in the hands of firms. The symbols **C** and **I** can be used to stand for **currently produced consumption goods** and **currently produced investment goods respectively.**

In an economy that uses capital goods, as does the Frugal economy, it is helpful to distinguish between two concepts of National Income (or National Product).

GROSS NATIONAL INCOME (or Gross National Product, GNP); It is the sum of the values of all final goods produced for consumption and investment, and thus it is also the sum of all

Firms factor incomes earned in the process of producing the National output.

NET NATIONAL INCOME (or Net National Product, NNP) is GNP minus the capital consumption allowance. NPP is thus a measure of the Net output of the economy after deducting from gross output an amount necessary to maintain the existing stock of capital intact.

THE DETERMINATION OF EQUILIBRIUM NATIONAL INCOME

National income is said to be in equilibrium when there is no tendency for it either to increase or for it to decrease. The actual National Income achieved at that point is referred to as the equilibrium National Income. For there to be equilibrium, firm spending must be equal to firm's receipts. If this were not the case, the firms will receive less and lose money until there is no more money in the system. Hence, for there to be equilibrium:

Factor Incomes = Consumer Spending

INCOME MODELS

1) The Spendthrift Economy:

This assumes a circular flow of income in a closed economy with no Government sector and no foreign trade. It also assumes the existence of **two sectors,** namely the sector of **households** and the sector of **firms.** Firms make the commodities that households consume. They purchase the services of factors of production from the household that own them, paying wages, rent, interest and profits in return, and then use the factors to make commodities.

It is assumed firms sell all of their output to households and receive money in return. All of the money received is in turn **paid out to households**. Part goes to households that sell factor services to firms, and the rest is profit paid out as Dividends to the owners of the firm. In short, neither households nor firms save anything in the spendthrift economy; everything that one group receives goes to buy goods and services from the other group. **Expenditure is the rule of the day!**

Now, suppose we wish to calculate the Total Value of the economy's output. We can do this based on either side of the circular flow shown in the figure above. The output-expenditure approach uses calculations based on the flows on the right hand side of the figure, while the input factor income approach uses calculations based on the flows on the left-hand side of the figure.

2) The Frugal Economy:

In the Frugal economy, households and firms look to the future, and as a result undertake both Saving and **Investment.**

Equilibrium National Income in a Frugal Economy

Saving and investment are examples of two categories of expenditure called withdrawals and injections. A WITHDRAWAL is any income that is not passed on in the circular flow. Thus if households can earn income and not spend it on domestically produced goods and services, this is a withdrawal from the circular flow. Similarly, if firms receive money from the sale of goods and do not distribute it as payments to factors, this is a withdrawal from the circular flow.

AN INJECTION is an addition to the incomes of domestic firms that does not arise from the expenditure of domestic households or arise from the spending of domestic firms.

The effects of withdrawals and injections are to interfere with Equilibrium income. Withdrawals by reducing expenditure exert a contractionary force on national income. If, for example, households decide to increase their savings and correspondingly reduce the amount they used to spend buying consumption goods from firms, this reduces the incomes of firms, and reduces the payments they will make to factors of production. Injections, by raising expenditure, exert an expansionary force on national income. If, for example, firms sell machines to other firms, their incomes and payments to household for factor services will rise without there having been an increase in household expenditure.

Thus for equilibrium National Income to exist, firm spending should be equal to firm receipts.

Thus, denoting consumption by **C**, saving by **S** and Investment by **I**, there is equilibrium if:

C + S = C + I

Or

S = I

i.e. there is equilibrium when savings are equal to investments.

- savings

+ Investments

To measure the National Income in a frugal economy, through the output and Expenditure approach, the National Income Accountant includes production of goods for inventories as part of total expenditure since the firm certainly spends money on the factor services necessary to produce goods for its own inventories. The accountant calculates the economy's total output as the actual expenditure on final goods and services sold, plus the market value of final commodities currently produced and added inventories. This definition makes total expenditure the same thing as the value of all final commodities produced and thus ensures that the measured value of expenditure is identical with the value of total output in any economy.

3) THE GOVERNED ECONOMY:

The governed economy contains central authorities often simply called "the government" – who levy taxes on firms and households and which engages in numerous activities such as defending the country, making and enforcing the laws, building roads, running schools, and predicting weather.

When the government produces goods and services that households desire such as roads and air traffic control, it is obviously engaged in a useful activity and is obviously adding to the sum total of valuable output. The National Income Statistician count as part of the GNP every government expenditure on goods and services, whether it is to build **a scud missile to promote police protection, or to pay a civil servant to file and re-file papers from a now defunct ministry.**

Definitions:

Transfer Payments:

Are any payments made to households by the government that are not made in return for the services of factors of production i.e. there is no Quid pro Quo. Such payments do not lead directly to any increase in output and for this reason they are not included in the nation GNP.

Disposable Income:

This is the income which households actually have available to spend or to save. To calculate disposal income, which is indicated by Ya, the statistician must make several adjustments to GNP.

First, all those elements of the value of output that are not paid out to households must be **deducted:** business savings represent receipts by firms from the sale of output that are withheld by firms for their own uses, and corporation taxes are receipts by firms from the sale of output that are paid over to the government. Secondly, personal income taxes must be **deducted** from the income paid to

households in order to obtain the amount households actually have available to spend or save. Finally, it is necessary to **add** government transfer payments to households.

Although these are not themselves a part of GNP, they are made available to households to spend and save, and are thus a part of disposable Income. Thus disposal income is:

GNP minus any part of it that is not actually paid over to households, minus the personal income taxes paid by households, plus transfer payments received by households.

Real and nominal measures

Output, Expenditure and Income can be valued at current market price in which case we speak, for example, of **money** or **Nominal NNP, or NNP** valued at current prices. Changes from one year to another are then a compound of changes in physical quantities and prices. Output,

Expenditure and Income can also be valued at the prices ruling in some base year. In this case, each year's quantity is priced at its base-year prices and then summed. We then speak, for example, of GDP at constant prices, or REAL GDP. Changes in constant-price GDP give a measure of real or quantity changes in total output.

Equilibrium Income

In this model, aggregate desired expenditure has three components: Consumption, Investment and Government Expenditure:

E = C + I + G

However, in the Governed Economy, taxes levied by the government are a second withdrawal. If the government taxes firms, some of what firms earn is not available to be passed on to households. If the government taxes households, some of what households earn is not available to be passed on firms. Whatever subsequently happens to money raised, taxes withdraw expenditure from the circular flow.

In the Governed Economy, however, government expenditure is a second injection. Such expenditure creates income for firms that does not arise from the spending of households, and it creates income for households that does not arise from the spending of firms. Whatever the source of funds, government spending injects expenditure into the circular flow.

Letting **G** stand for Government Expenditure, **T** for Taxes, **J** for injections and **W** for withdrawals, we can say the National Income is in equilibrium when total withdrawals, savings plus taxes, is equal to **total injections, investment plus government expenditure.** The equilibrium condition for national income can thus be written as:

W = J, or S + T = G + I

Government purchases G -

+ S

Personal

Income Taxes

- Taxes on commodities

+

Taxes on I

Business - +

Government Purchases From firms

Open Economy:

None of the three economies considered so far are engaged in trade with Foreign Countries. Such economies are often referred to as Closed Economies. In contrast, open economies engage in significant amounts of foreign trade, so that some of the goods produced at home are sold a broad while some of the goods sold at home are produced abroad. The model is more applicable in real life.

A mathematical approach to national income equilibrium.

Equilibrium analysis also has applications in the area of national income. A simple Keynesian national income model may be expressed as follows:

$Y = C + IO + GO$ …… (i)

$C = a + bY$ …….…… (ii)

$(a > o, o < b < 1)$

Y and C are both endogenous variables since they are determined within the model. Io

And Go , on the other hand, represent exogenously determined investment and government expenditure respectively. Exogenously determined variables are those whose values are not determined within the model. $C = a + b Y$ represents a consumption function where a and b stand for autonomous consumption and the marginal propensity to consume, respectively.

If we substitute equation (ii) into (i) we obtain:

$Y = a + bY + Io + Go$

$(1 – b) Y = a + Io + Go$

–

Equilibrium national income is represented by Y.

–

$Y = a + Io + Go$ …….. (iii)

$1 – b$

The equilibrium level of consumption can be obtained by substituting equation (iii) into equation (ii).

C = a + bY = a + b (a + Io + Go)

1 – b

= a (1 - b) + b (a + Io + Go)

1 – b

A Numerical Example

Assume a simple two sector model where Y = C + I C = a + bY and I = Io . Assume in addition, that a = 85, b = 0.45 and Io = 55. this implies that Y = a + bY + Io = 85 + 0.45Y + 55

Y – 0.45Y = 140

0.55Y = 140

Y = 255

This simple model can be extended to include government expenditure and foreign trade. It may take the following general form:

Y = C +I + G+ (X – M)

Where C = a +bY

And M = mo + mY

Mo represents autonomous imports and m represents induced imports (imports dependent on _the level of income). Equilibrium national income in this case is represented by

Y = a + Io + Go+ Xo - Mo

1 – b + mo

Numerical Example

Assume that Io = 360, Go= 260, Xo= 320, Mo = 120, a = 210, b = 0.8 and m = 0.2

The equilibrium level of national income can be computed as follows:

Y = 360 + 260 + 320 + 210 – 120 = 2,575

1 – 0.8 + 0.2

INFLATIONARY AND DEFLATIONARY GAPS

It is useful and important to understand the concept of inflationary gap because with it we are able to know the main cause of the rise in general level of prices. The equilibrium of an economy is established at the level of full-employment when aggregate demand or total expenditure is equal to the level of income corresponding to full-employment. This happens when the amount of investment is equal to the saving gap corresponding to full-employment level of income.

Consider Fig. 5.10 where OY_F is national income corresponding to the level of full-employment. Equilibrium at national income OY_F would be established only when aggregate demand or total expenditure (C + / + G) is equal to YF_E (YF_E is equal to OY_F).
Real national income cannot increase beyond OY_F because when all means of production including labour are fully employed, there is no possibility of further rise in production or real national income. Thus when aggregate demand is greater than the aggregate demand YF_E which is required to establish the equilibrium at OY_F level of national income, the equilibrium would not be established at OY_F

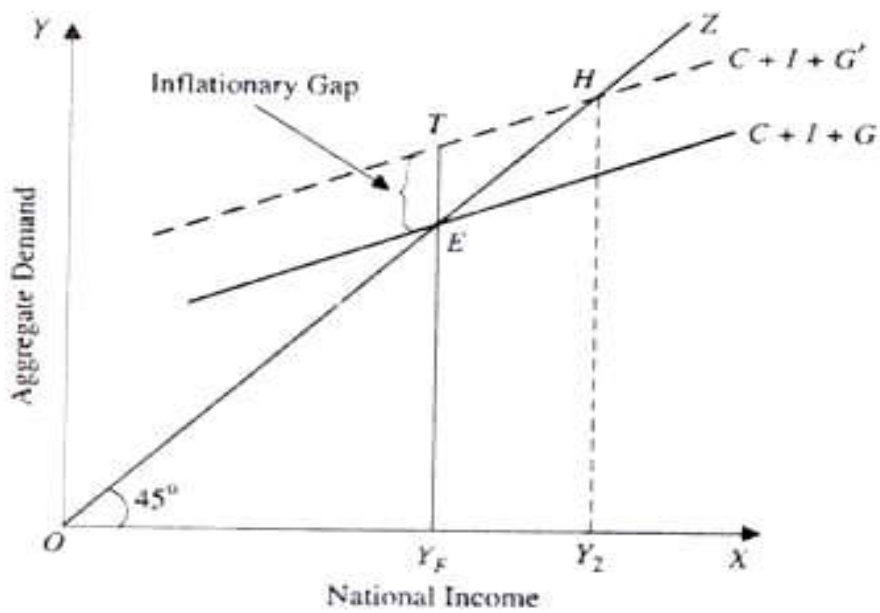


Fig. 5.10. *Inflation Gap*

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It would be seen from Fig. 5.10 that aggregate demand Y_FT is greater than aggregate demand YFE which is required to maintain the equilibrium at OY_F. Thus with the level of aggregate demand (C + I + G'), which is obtained by adding expenditure ET to the aggregate demand curve C + I + G, equilibrium would not be established at OY_F which corresponds to full-employment level of income. The actual aggregate demand being greater than YF_E by the amount ET the level of nation income would be greater than OY_F.
Since OY_F is full-employment level of national income, actual production cannot increase beyond that but there would be rise in prices which would raise the money value of OY_F production. The amount by which the actual aggregate demand exceeds the level of national income corresponding to full employment is known as inflationary gap because this excess of aggregate demand causes inflation or rise in prices in the country.
In Fig. 5.10 this excess of aggregate demand or inflationary gap is equal to ET. It would be seen from Fig. 5.10 that the aggregate demand curve C + I + G' intersects 45° line (OZ line) at H so that equilibrium level of national income would be OY_2.
It should be carefully understood that there is no difference between OY_F and OY_2 in terms of real income or actual production; only as a result of rise in price level, national income has increased

from OY_F to OY_2 in money terms. Inflationary gap represents excess demand in relation to aggregate production or supply of output which brings about demand-pull inflation.
J.M. Keynes in his revolutionary book "General Theory of Employment, Interest and Money" did not discuss the concept of inflationary gap because he was then preoccupied with the analysis of the state of depression and deflation. During the Second World War when the problem of inflation cropped up, then Keynes applied his macroeconomic analysis to explain inflation as well and in this connection he put forward the concept of inflationary gap.

Deflationary Gap:
In the theory of income and employment, the concept of deflationary gap occupies an important place, since in a capitalist economy unemployment and depression occur due to this gap. According to Keynesian theory of income and employment, equilibrium at the level of full- employment is established when aggregate demand consisting of consumption demand plus investment demand plus government demand (C + I + G) is equal to the national income at the level of full-employment.

This happens when investment and government demand is equal to the saving gap at full-employment level of national income. If aggregate demand is less than the full-employment level of national income, i.e., when investment and government demand is less than the saving gap at full-employment level of income, the deficiency of aggregate demand occurs due to which national income and employment will fall below the full-employment level causing unemployment and depression in the economy.

The concept of deflationary gap is illustrated in Fig. 5.11 in which along the X-axis national income is measured and along the K-axis level of aggregate demand is measured. Suppose national income at the level of full-employment is equal to OY_F.
Now the equilibrium level of income and employment would be established at $OY_F$when aggregate demand (consumption demand plus investment demand) is equal to YF_E (which is equal to national income OY_F). But in the real world if aggregate demand is less than the full-employment level of income OY_F or it is less than YF_E, then the problem of deficiency of aggregate demand will arise. Therefore, EK in Fig. 5.11 represents deflationary gap. Hence, deflationary gap represents the difference between the actual aggregated demand and the aggregate demand which is required to establish the equilibrium at full-employment level of income.

It should be carefully understood that due to the deflationary gap EK, the level of national income and employment will decline. The decline in national income and employment will not only be equal to the deflationary gap EK but it will be much greater than this. The decline in national income is determined by the value of the multiplier.

In Fig. 5.11 when aggregate demand curve is C + I+ G' deflationary gap is equal to EK. The aggregate demand curve C + I + G' (dotted) which cuts the 45° line at point Q asp result of which equilibrium is established at OY_F level of national income. It will be seen from Fig. 5.11 that OY_1, is less than full-employment level of income OY_F. Deflationary gap represents the situation of deficient demand in the economy

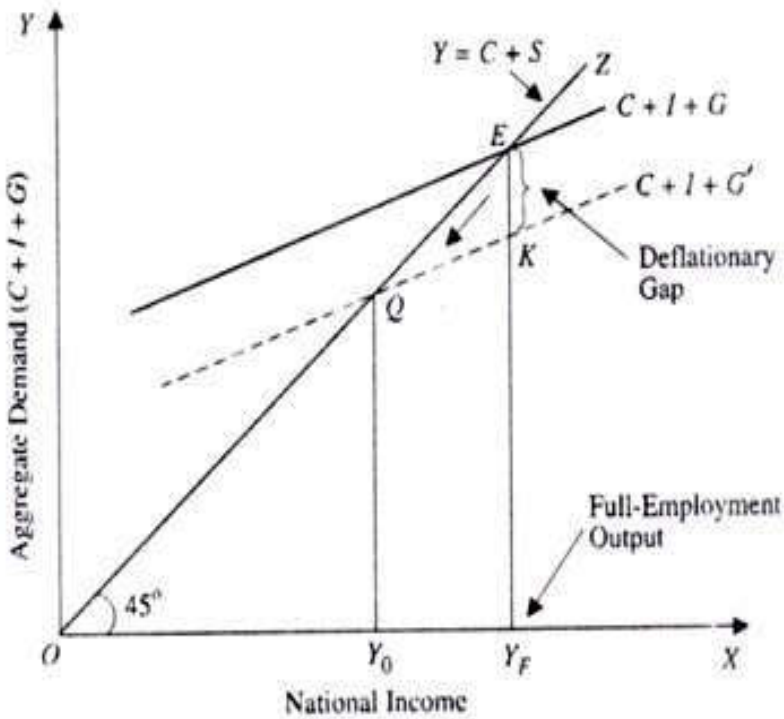


Fig. 5.11. *Deflationary Gap*

This deficiency in aggregate demand causes fall in national output and level of employment. As a result, involuntary unemployment in the economy emerges. The depression of 1929-33 in capitalist countries was caused by the emergence of deflationary gap or by demand deficiency in these economies.

THE MULTIPLIER AND ACCELERATOR CONCEPTS

In his theory Keynes asserted that consumption is a function of income, and so it follows that a change in investment, which we may call ΔI, meaning an increment in I will change Y by more than ΔI. For while the initial increase in Y, ΔY, will equal ΔI, this change in Y itself produce a change in C, which will increase Y still further. The final increase in income thus exceeds the initial increase in investment expenditure which is therefore magnified or "multiplied". This process is called the **multiplier process.**

The Operation of the "Multiplier"

The multiplier can be defined as the coefficient (or ratio) relating a change in GDP to the change in autonomous expenditure that brought it about. This is because the Multiplier can be defined as the coefficient (or ratio) relating a change in GDP to the change in autonomous expenditure that brought it about. This is because a change in expenditure, whatever its source, will cause a change in national income that is greater than the initial change in expenditure.

For example, suppose there is an autonomous increase in investment which comes about as a result of decisions by businessmen in the construction industry to increase the rate of house building by, say, 100 houses, each costing £1,000 to build, investment will increase by £100,000.

Now this will be paid out as income to workers of all kinds in the building industry, to workers in industries which supply materials to the building industry, and others who contribute labour or capital or enterprises to the building of the houses; these people will in turn wish to spend these incomes on a wide range of consumer goods, and so on. There will thus be a series of further rounds of expenditure, or **Secondary Spending**, in addition to the initial **primary spending**, which constitutes further increases in GDP.

This is because those people whose incomes are increased by the primary increase in autonomous expenditure will, through their propensity to consume, spend part of their increase in their incomes. GDP increases through the Expenditure – Income – Expenditure cycle.

How and where does the Multiplier Stop?

The multiplier concept can erroneously give the impression that an initial increase in autonomous spending would lead to an indefinite increase in GDP. This does not happen because each secondary round of increased expenditure gets progressively smaller, which is explained by the fact that the Marginal Propensity to spend the national income is less than one.

This is the ratio which scales down each successive round of expenditure and causes the GDP to converge to a new equilibrium level.

Suppose in our example, an average of three fifths of any increase in income is spent by the people receiving it:

The Marginal Propensity to consume or save will be 3/5 or 2/ 5 respectively. Since ΔI = 1000,000, the increase in Y converge at the level 250,000. This is because for any value z between 0 and 1, the series

1 + z + z2 z3 …..tends to the value 1/1-z. In our example we have the series (in thousands) **100 + 60 + 36 + 21.6 + ….**

Or

100 { 1 + (3/5) + (3/5) + …..} which thus equals:

1 1 100 = --------- = 100 ------- = 250

1 – 3/5 2/5

This result can be generalized, using our notation, as

1 1

Δ1 ---------- = Δ1 ---------- = ΔY

1 - Δc/ΔY Δs/ΔY

Dividing by Δ**I**, we obtain

ΔY 1 1

----- = ----------- = ----------

ΔI 1 – Δc/ΔY Δs/ΔY

The ratio, ΔY/ΔI, of the total increase in income to the increase in investment which produce it, is known as the MULTIPLIER, K. If we write c for ΔC/ΔT and s for ΔS/ΔY, we have

ΔY 1 1

k = ----------- = --------- = --------

ΔI 1 – c s

The multiplier is thus the reciprocal of the MPS (Marginal Propensity to Save).

Relevance of Multiplier

The Keynesian Model of the Multiplier however is a Short Run Model, which puts more emphasis on consumption than on savings. It is not a long run model of growth since savings are the source of investment funds for growth. It is appropriate for mature capitalist economies where there is excess capacity and idle resources, and it is aimed at solving the unemployment problem under those conditions – (i.e. problem of demand deficiency with the level of investment too low, because of lack of business confidence, to absorb the high level of savings at full employment incomes).

It is not a suitable model for a developing economy because:

i. In less developed economies exports rather than investment are the key injections of autonomous spending.
ii. The size of the export multiplier itself will be affected by the economies dependence on two or three export commodities.
iii. In poor but open economies the savings leakage is likely to be very much smaller, and the import leakage much greater than in developed countries.
iv. The difference, and a fundamental one, in less developed countries is in the impact of the multiplier on real output, employment and prices as a result of inelastic supply.

THE ACCELERATOR PRINCIPLE

Suppose that there is a given ratio between the level of output **Yt** at any **time t**, and the capital stock required to produce it **Kt** and that this ratio is equal to **α,** hence:

Kt =αYt

The coefficient is the capital-output ratio, **α,** = K/Y and is called the **accelerator co-efficient**.
If there is an **autonomous** increase in investment, ΔI this through the multiplier process will lead to increased employment resulting in an overall increase in income**, ΔY**. This may lead to further investment called **Induced Investment** in the production of goods and services. This process is called **acceleration.**

The ratio of induced investment to the increase in income resulting from an initial autonomous increase in investment is called the **accelerator**. Thus, if the included investment is denoted by ΔI1, and the accelerator by β, then:

ΔI1

---------- = β, ΔI1 = βΔY

ΔY

Thus another way of looking at the accelerator is as the factor by which the increase in income resulting from an initial autonomous increase in investment is multiplied by the induced investment.

From the Keynesian model ΔY = **Δ1.1/s** we c an write

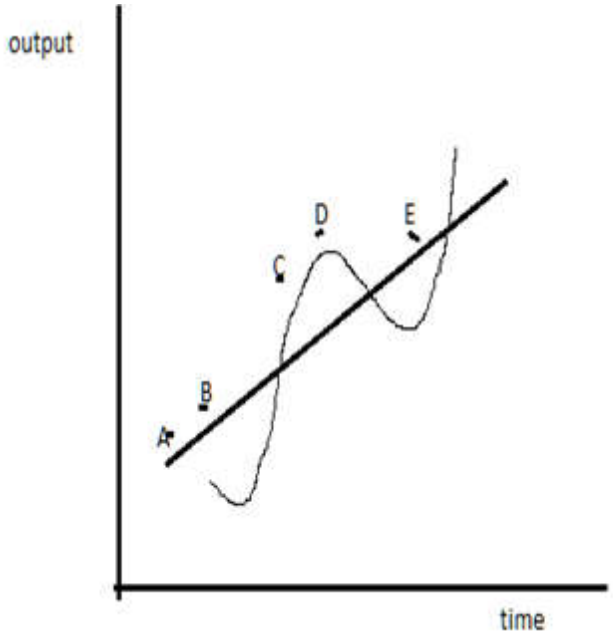
Δ11 = β, Δ11 .1/s

Thus, the higher the multiplier and the higher the accelerator, the higher will be the level of induced investment from an initial autonomous increase.

BUSINESS CYCLES/ CYCLICAL FLUCTUATIONS

Meaning:

The business cycle is the tendency for output and employment to fluctuate around their long-term trends. The figure below presents a stylised description of the business cycle. The continuous line shows the steady growth in trend output over time, while the broken line indicate the actual output over the time period.



Point A represents a slump, the bottom of a business cycle while point B suggests the economy has entered the recovery phase of the cycle. As recovery proceeds the output rises to a point C above the trend path; we call this a boom. Then as the line dips via D towards the trend line with output growing less quickly during the recovery and least quickly (perhaps even falling), during a recession.

Causes:

There are a number of explanations of the business cycle but changes in the level of investment seem to be the most likely. In the simplest Keynesian model an increase in investment leads to a larger increase in income and output in the short run. Higher investment not only adds directly to aggregate demand but by increasing income adds indirectly to consumption demand. A process known as the multiplier. The reasons for change in investment may be explained as follows.

Firms invest when their existing capital stock is smaller than the capital stock they would like to hold. When they are holding the optimal capital stock, the marginal cost of another unit of capital just equals its marginal benefit, this is the present operating profits to which it is expected to give rise over its lifetime. This present value can be increased either by a fall in interest rates at which the stream of expected profits is discounted or by an increase in the future profit expected. In practice, it is generally believed that changes in expectations about future profits are more important than interest rate changes. If real interest rates and real wages change only slowly, the most important source of short term changes in beliefs about future profits is likely to be beliefs about future levels of sales and real output. Other things being equal, higher expected future output is likely to raise expected future profits and increase the benefits from a marginal addition to the current capital stock. This kind of explanation is known as the **accelerator model of investment**. In this theory it is assumed that firms estimate future profits by **extrapolation** of past growth of output. While constant output growth leads to a constant rate of growth of capital stock, it takes accelerating output growth to increase the desired level of investment. Though the accelerator model is acknowledged to be a simplification of a complex process its usefulness has been confirmed by empirical research.

Just how firms respond to changes in output will depend on a number of things including the extent to which firms believe that current growth in output will be maintained in the future and the cost of quickly adjusting investment plans. The more costly it is to adjust quickly, the more likely are firms to spread investment over a long time period.

The underlying idea of the multiplier-accelerator model is that it takes an accelerating output growth to keep increasing investment, but it must be noted that once output growth settles to a constant level investment also becomes constant. Finally if output falls then the level of investment must fall also.

The limits of the fluctuations around the trend path of output are referred to as ceilings and floors.

If we assume that the circular flow of income is in equilibrium at less than full employment and there is an increase in investment, the effect of this will be to raise national income by more than an equivalent amount because of the effect of the multiplier. This will in turn produce a more than proportionate increase in investment because of the effect of the accelerator which will produce a more than proportionate rise in incomes and so on. This cumulative growth of income will continue until the economy's full employment ceiling is reached. The process then goes into reverse with an accelerated decline in the absolute level of net investment, followed by a multiplied reduction in income and so on. The bottom of 'floor', of the recession will come when withdrawals once more equal the reduced level of injections.

It is argued that modern economies do not fluctuate as much as they did in the past because of **built in stabilizers** which operate automatically and the use of **discretionary measures** which are available to governments. **The taxation system** is said to act as a stabilizer that operates automatically and the use of discretionary measures which are available to governments. The taxation system is said to act as a stabilizer in the following way: As income rises a progressive

taxation system takes larger and larger proportions of that increased income; when income falls revenue drops more than proportionately. Other built-in stabilizers are **unemployment benefits** and **welfare payments** because expenditures on these rise and fall with the unemployment rate.

Despite these built-in stabilizers and the actions of government in their use of discretionary measures to stabilize the economy, the cycle is still with us as recent experience has demonstrated.

In conclusion, it must be added that the causation of business cycles is a complex matter and the above is only one of a number of possible explanations.

QUESTIONS:

1. A hypothetical closed economy has a national income model of the form y = C + I + G where C = 30 + 0.8Y and I and G and private investment and government expenditure are exogenously determined t 50 and 80 units respectively. Compute the national equilibrium level of income for this economy using aggregate income equals aggregate expenditure and withdrawals and equal injection methods.
2. What are some of the limitations using Gross National Product as a measure of economic performance?
3. a) What are the main factors of production?
   b) What determines the supply and demand for the factors of production that yu have identified in a) above?
4. What is meant by mobility of factors of production? To what extent are these factors mobile and what is the significance of mobility of production?
5. Explain what is meant by diminishing returns and what is its relevance?
6. With the help of diagrams, explain the market price differential between a perfectly competitive and a monopolistic competitive market?
7. Outline the various factors which lead to monopoly power. Compare the advantages and disadvantages which consumers may derive from production by a monopolist. How far would your arguments be modified if monopolists were public enterprises?
8. a) Define the term National Income.
   c) List and explain different methods of estimating national income of a country and state some of the problems which are being experienced in computing national income.
9. From the data below calculate the gross national product at factor cost:

   £Million
   Imports 42,000
   Value of physical increase in stock 1,300
   Exports 43,000
   Capital formation 25,000
   Government expenditure 29,000
   Consumer expenditure 84,000
   Taxes on expenditure 20,000
   Subsidies 3,000
   Net property income from abroad 400
10. a)With the help of a well labeled diagram, explain the relationship between the average fixed cost, average variable cost, total cost and marginal cost curves.
    b) Discuss the necessary and sufficient conditions for profit maximization by a firm.

    Support your answer with appropriate illustrations.

11. Derive the equation for the equilibrium level of national income in an open economy with no taxes.

    b) Given a = 675, b = 0.2, Io = 375, Go = 150, Mo = 200 and m =0.4, find the equilibrium level of national income.

CHAPTER 7

ECONOMIC GROWTH, ECONOMIC DEVELOPMENT AND ECONOMIC PLANNING

THE DIFFERENCE BETWEEN ECONOMIC GROWTH AND ECONOMIC DEVELOPMENT

Economic Growth is a narrower concept than economic development .It is an increase in a country's real level of national output which can be caused by an increase in the quality of resources (by education etc.), increase in the quantity of resources & improvements in technology or in another way an increase in the value of goods and services produced by every sector of the economy. Economic Growth can be measured by an increase in a country's GDP (gross domestic product).

Economic development is a normative concept i.e. it applies in the context of people's sense of morality (right and wrong, good and bad). The definition of economic development given by Michael Todaro is an increase in living standards, improvement in self-esteem needs and freedom from oppression as well as a greater choice. The most accurate method of measuring development is the Human Development Index which takes into account the literacy rates & life expectancy which affects productivity and could lead to Economic Growth. It also leads to the creation of more opportunities in the sectors of education, healthcare, employment and the conservation of the environment.It implies an increase in the per capita income of every citizen.

Economic Growth does not take into account the size of the informal economy. The informal economy is also known as the black economy which is unrecorded economic activity. Development alleviates people from low standards of living into proper employment with suitable shelter. Economic Growth does not take into account the depletion of natural resources which might lead to pollution, congestion & disease. Development however is concerned with sustainability which means meeting the needs of the present without compromising future needs. These environmental effects are becoming more of a problem for Governments now that the pressure has increased on them due to Global warming.

Economic growth is a necessary but not sufficient condition of economic development.

Comparison

	Economic Development	**Economic Growth**
Implications	Economic development implies an upward movement of the entire social system in terms of income, savings and investment along with progressive changes in socioeconomic structure of country (institutional and technological changes).	Economic growth refers to an increase over time in a country`s real output of goods and services (GNP) or real output per capita income.

	Economic Development	Economic Growth
Factors	Development relates to growth of human capital indexes, a decrease in inequality figures, and structural changes that improve the general population's quality of life.	Growth relates to a gradual increase in one of the components of Gross Domestic Product: consumption, government spending, investment, net exports.
Measurement	Qualitative.HDI (Human Development Index), gender- related index (GDI), Human poverty index (HPI), infant mortality, literacy rate etc.	Quantitative. Increases in real GDP.
Effect	Brings qualitative and quantitative changes in the economy	Brings quantitative changes in the economy
Relevance	Economic development is more relevant to measure progress and quality of life in developing nations.	Economic growth is a more relevant metric for progress in developed countries. But it's widely used in all countries because growth is a necessary condition for development.
Scope	Concerned with structural changes in the economy	Growth is concerned with increase in the economy's output

ACTUAL AND POTENTIAL GROWTH

Actual economic growth represents the fact that there is an actual increase in output; whereas potential economic growth represents an increase in the productive capacity of the economy.

Actual growth is defined as an increase in the output that an economy produces over a period of time, the minimum being two consecutive quarters. It economic growth is an increase in what an economy can produce if it is using all its scarce resources. An increase in an economy's productive output can be shown by an outward shift in the economy's production possibility frontier (PPF).

The simplest way to show economic growth is to bundle all goods into two basic categories, *consumer* and *capital* goods. An outward shift of a PPF means that an economy has increased its capacity to produce.

Potential growth is defined in the as the level of output that an economy can produce at a constant inflation rate. Although an economy can temporarily produce more than its potential level of output, that comes at the cost of rising inflation. Potential output depends on the capital stock, the potential labour force (which depends on demographic factors and on participation rates), the non-accelerating inflation rate of unemployment (NAIRU), and the level of labour efficiency.

THE BENEFITS AND COSTS OF GROWTH

Economic growth means an increase in real GDP. This increase in real GDP means there is an increase in the value of national output / national expenditure.

The benefits of economic growth include:

1. **Higher average incomes**. This enables consumers to enjoy more goods and services and enjoy better standards of living.
2. **Lower unemployment** With higher output and positive economic growth firms tend to employ more workers creating more employment.
3. **Lower government borrowing.** Economic growth creates higher tax revenues and there is less need to spend money on benefits such as unemployment benefit. Therefore economic growth helps to reduce government borrowing. Economic growth also plays a role in reducing debt to GDP ratios.
4. **Improved public services**. With increased tax revenues the government can spend more on public services, such as the NHS and education e.t.c.
5. **Money can be spent on protecting the environment.** With higher real GDP a society can devote more resources to promoting recycling and the use of renewable resources
6. **Investment.** Economic growth encourages investment and therefore encourages a virtuous cycle of economic growth.

Despite the benefits of economic growth, there are also potential costs, such as inflation, a current account deficit, environment costs and widening inequality.

However, the costs of economic growth will depend on the type of growth that we see.

Potential costs of economic growth include:

1. Inflation. If AD increases faster than AS then economic growth will be unsustainable. Economic growth tends to cause inflation when the growth rate is above the long run trend rate of growth. It is when demand increases too quickly that we get a positive output gap and firms push up prices.

2. Boom and bust economic cycles. If economic growth is unsustainable then high inflationary growth may be followed by a recession. This occurred in the UK in the late 1980s and early 1990s.

In the 1980s there was an economic boom with growth of over 5% a year. However this caused inflation to rise to over 10%. To reduce inflation the government increased interest rates, this caused the economy to slow down and then enter into a recession.

- However if economic growth is at a sustainable rate this will not occur. For example, between 1993 and 2007, both economic growth and inflation were at a sustainable rate.

3. Current account deficit
Increased economic growth tends to cause an increase in spending on imports therefore causing a deficit on the current account.

4. Environmental costs
Increased economic growth will lead to increased output and therefore increased pollution and congestion. This will cause health problems such as asthma and therefore will reduce the quality of life. Economic growth also means greater use of raw materials and can speed up depletion of non-renewable resources.

5. Inequality
Higher rates of economic growth have often resulted increased inequality because growth can benefit a small section of society more than others. For example, those with assets and wealth will see a proportionally bigger rise in the market value of rents and their wealth. Those unskilled without wealth may benefit much less from growth.

However it depends upon things such as tax rates and the nature of economic growth. Economic growth can also be a force for reducing absolute and relative poverty.

DETERMINANTS OF ECONOMIC DEVELOPMENT

Economic development is a complex process. It is influenced by a number of factors such as natural resources, capital, human resources, technology, social attitude of the people, political condition in the country. All the factors having strong bearing on economic growth are divided into two categories, economic and non-economic. We discuss these determinants or factors of economic development under separate heads now.

(A) Economic Factors.

Economic factors play a very important role in the development of a country. The aggregate output rises or falls mainly due in changes to them. The main economic determinants are (1) Natural Resources (2) Capital (3) Labour (4) Power (5) Transport and Communication (6) Human Capital. These economic factors are discussed in brief.

(1) Natural Resources. The natural resources is the principal factor which affects the development of an economy. If a country is rich in natural resources, it is then able to make rapid progress in growth. In case a country is deficient in forest wealth, mineral resources, water supply, fertility of land etc., it is then normally not in a position to develop rapidly.

The pity with the most of LDC's is that their natural resources are under-utilized, unutilized or misutilized. This is one of the reasons for their backwardness. **It may here be noted that presence of rich resources is not a precondition for economic development.** There are countries in the world which do not have abundant resources, yet they have made rapid progress in growth by superior technology, new researches and higher knowledge. Japan, Switzerland, South Korea are resource poor countries, yet they have made rapid progress in economic growth through advanced technology and new discoveries.

(2) Capital Formation. Capital accumulation or capital formation is an important factor in the economic growth of a country. Capital formation refers to the process of adding to the stock of capital over time. The stock of capital can be built up and increased through three different resources which are as under:-

Sources of capital formation:

a) An act of saving.
b) Capital market.
c) An act of investment.

(a) An act of saving. An act saving involves the postponing of consumption whether voluntarily or involuntarily so that funds thus made available be used for investment. In developing countries the saving potential is low. A large majority of the people hardly keep their body and soul together with the meager income at their disposal. Saving is a luxury and far beyond their reach. The middle income group attracted by the superior levels of consumption of the developed world (demonstration effect). spends the resources on consumption items like fridges. VCR, colour television etc and so less saving is coming from them. The rich section of the society spends money mostly on luxury goods, foreign travels, real estates etc and therefore little saving is available for investment. The saving potential of underdeveloped countries, therefore, normally does not **exceed 13% to 10%** of the national income which keeps their stock of capital low.

(b) Capital market. The capital market consists of financial institutions, like development banks, stock exchanges and investment banks. In low income countries, the capital market is less developed. As such it is not able to mobilize saving to the desired extent.

(c) An act of investment. In less developed countries, whatever meagre saving are available with households and with the businessmen, is not all channelized for investment in capital goods. The businessmen usually hesitate to invest their resources due to political and social instability in the country, fear of nationalization of industries, limited domestic market, poor roads, etc.

How to increase capital formation

In developing countries, the capital formation is around 5%. It should be raised to the level of at-least 20% by adopting the following measures:

1) Sine voluntary saving is not forth-coming in low income countries, the government should resort to forced saving. The various methods of forced saving are (a) taxation (b) deficit financing and (c) borrowing
2) Obtaining of external resources in the form of loans and grants for development progress.
3) In creasing exports and reducing imports can also provide funds for capital formation.

3. Human resources..

Human resources of a country is an important factor in economic development. If the population of a country is educated, efficient, patriot, skilled, healthy, it makes significant contribution to economic development.

On the other hand, if a country is overpopulated, labour force is unemployed, uneducated, unskilled, unpatriotic, it can put serious hurdles on the path of economic development.

4. Power.

Power resources are the foundation of economic development. They are derived mainly from two types of sources (1) Commercial and (2) Non-commercial Commercial resources of power are (a)

oil, gas, coal, hydel, thermal electricity and nuclear. Non- commercial sources include animal power, fuel, wood, cow dung

The power resources are vital to economic growth of a country. Its importance has been changing with the passage of time. Before industrial Revolution, the energy for operating the machines was mainly supplied by animals, human power and wind. With the scientific advancement, coal, oil, gas, and water falls are used as the principal sources of energy. In developed countries of the world, the nuclear power and solar energy are being increasingly used for generating electricity.

The various sources of energy are helpful in ;

a. giving an initial push to the raising of production in all sectors of the economy.
b. Quickly bridging the development gap.
c. providing economies of scale.
d. ensuring high quality standard.
e. reducing material wastages in all sectors of the economy.

The high income oil exporting developing countries like Saudi Arabia, Bahrain etc. are producing energy according to their needs from their own cheap source i.e.,.oil.

The other developing countries are giving highest priority to energy. Their main stress is on (a) accelerated exploitation of coal, hydel and nuclear power etc. (b) intensification of exploration for oil and gas (c) energy consideration.

5. The Means of Transport and Communication

The means of transport and communication have an important bearing on the economic growth of a country. If a country is well connected with rail road, sea ports and has a developed means of communication including information technology, it then helps in improving the productive capacity of the various sectors of the economy. An efficient transport and communication network contributes to improving the quantity and quality of goods due to competition and reduction in production costs.

COMMON CHARACTERISTICS OF DEVELOPING COUNTRIES

Following are the economic characteristics of UDC's:

1- General Poverty and Low Living Standard

Poverty cannot be described, it can only be felt. The most of the less developed countries (LDC) are facing the major problem of general as well as absolute poverty and low standard of living. Most of the people in developing nations are ill-fed, ill-housed, ill-clothed and ill-literate. In LDCs almost 1/3 population is much poor. But in Pakistan, 21.0 % population is living below poverty.

2- Burden of Internal and External Debts

Under developed countries (UDC) are loans and grants receiving nations. Most of the developing countries of the world are depending on foreign economic loans. An amount of foreign

loans is increasing as the years pass. Their foreign trade and political structure is also dependent on the guidance of foreigners. The outstanding total public debts are Rs. 10020 billion (55.5 % of GDP) and the value of external debts and liabilities is $ 59.5 billion and its services charges are $ 7.8 billion in 2010-11 in Pakistan.

3- Low Per Capita Income

Due to low national income and huge population growth rate, per capita income in developing countries is very low. At constant prices (Base Year 1959-60) per capita income of Pakistan was Rs. 985 and according to the Economic Survey of Pakistan 2010-11 per capita income of Pakistan is $ 1254.

4- Over Dependence on Agriculture

61% Population of Pakistan is living in more than 50,000 villages. Backward agriculture is the major occupation of the population. Agriculture sector is backward due to old and traditional methods of cultivation, in-efficient farmers, lack of credit facilities; un-organized agriculture market etc. 66.7% population is directly or indirectly depending on agriculture sector in Pakistan. It contributes to GDP 20.9 % while in advanced nations it is less than 10 %. It employed 45.0 % of labour force while it is less than 5 % in developed countries.

5- Backward Industrial Sector

Backward industrial sector is an additional feature of under developed countries. Industrial sector of Pakistani economy is backward since independence. Pakistan got only 34 (3.7 % of total industrial units) industrial units out of 921 units in sub-continent in 1947. Small and backward industrial sector is based on low level of capital formation, technology, training and education and over dependence on agriculture sector. 13.2 % labour force is attached with industrial sector in Pakistan. Its share to GDP is 25.8 % and to exports is about 60 %.

6- Unemployment

An outstanding problem of developing countries is their high rate of un-employment, under-employment and disguised-unemployment. More than 3.05 million people are unemployed in Pakistan. There is 16 % underemployed and 20 % disguised unemployed of total labour force. Unemployment rate is 5.6 %; it is mainly due to high population growth rate, which is 2.1 %.

7- Low level of Productivity

The productivity level is very low in under developed countries as compared to developed countries. Low level of productivity is due to economic backwardness of people, lack of skill, illiteracy and ill-training. Value of annual productivity of labour is about $ 100 while it is more than $ 2500 in advanced nations in Pakistan.

8- Deficit Balance of Payment

Third world countries have to import some finished and capital goods to make economic development, on the other hand they have no products to export but raw material. During July-March, its exports were $ 24 billion and imports were $ 32.3 billion In case of Pakistan. So, its deficit balance of payment was $ 8.3 billion in 2010-11.

9- Dualistic Economy

Dualistic economy refers to the existence of advanced & modern sectors with traditional & backward sectors. Pakistani economy is also a dualistic economy as other developing countries on the following grounds: Co-existence of modern and traditional methods of production in urban and rural areas, Co-existence of wealthy, highly educated class with a large number of illiterate poor classes and Co-existence of very high living standard with very low living standard.

10- Deficiency of Capital

Shortage of capital is another serious problem of poor nations. Lack of capital leads to low per capita income, less saving and short investment. Domestic saving is 9.5% of GDP and total investment is 13.4% of GDP in Pakistan. Rate of capital accumulation is very low as 5%. On the other hand, capital output ratio (COR) is very high which is not desirable for economic development.

11- In-appropriate Use of Natural Resources

Mostly there is shortage of natural resources in developing nations and this is also a cause of their economic backwardness. Natural resources are available in various poor countries but they remain un-utilized, under-utilized or mis-utilized due to capital shortage, less efficiency of labour, lack of skill and knowledge, backward state of technology, improper government actions and limited home market. Natural resources contribute to the GDP about 1%.

12- Market Imperfection

Market is imperfect in accordance with market conditions, rules and regulations in the most of developing nations. There exist monopolies, mis-leading information, immobility of factors; hoarding and smuggling etc. that cause the market to remain imperfect.

13- Limited Foreign Trade

Due to backwardness, developing countries have to export raw material because the quality of their products is not according to international standard ISO etc. Lower developing nations have to import finished and capital goods. Imports of Pakistan are $ 32.3 billion and exports are $ 24 billion that cause into unfavourable balance of payment.

14- Vicious Circle of Poverty

According to vicious circle of poverty, less developed nations are trapped by their own poverty. Vicious circle of poverty is also applied in case of Pakistani economy. Due to poverty, national income of Pakistan is low which causes low saving and low investment. So, rate of capital formation is very low results in *"a country is poor because she is poor"*.

15- Inflation

High rate of inflation causes economic backwardness in poor nations. Due to high level of price, purchasing power, value of money and saving of the consumers tend to decrease. Rate of inflation (CPI) is 14.1% in 2010-11 in Pakistan.

B. DEMOGRAPHIC CHARACTERISTICS

Following are the demographic characteristics:

16- Backward Population Explosion

Another common feature of lower developing nations is population pressure due to high growth rate and reduction in death rate. Population of the Pakistan is 177.1 million with the rapid growth rate of 2.1 % and death rate 0.73 % in 2010-11. Pakistan is at 6th number in the list of the most populous nations. Basic needs like food, clothing, housing, education, sanitations and health facilities are not available for the huge portion of population in these countries.

17- Poor Health and Diseases

M. P. Todaro in his "Economic Development" states, "Many people in developing countries fight a constant battle against malnutrition, diseases and ill health". Average life expectancy in Pakistan is 66.04 year against 78 years in developed countries. One Doctor is for 1222 persons and one Nurse is for 2369 persons, number of hospitals is 972 and one hospital bed is available for 1701 persons. The total expenditure on health sector is just 0.23 % of the GDP.

18- Pollution

There is too much pollution in poor countries. On the one side huge existing population is not provided basic facilities of life, like sanitation, clean water, infrastructure etc. but on the other side due to rapid population growth, industrialization and transportation air, water and earth pollution is increasing. Industries are causing pollution because of non-installation of treatment plants. Number of continuous air pollution monitoring stations is only 7 in Pakistan. Pakistan is at number 29th at the chart of the most polluted nations and at number 6th in Asian countries.

19- Brain Drain

An outflow of the best, brightest and talented student from poor nations to rich nations is called brain drain. There is less reward for the talent, which causes an outflow of best brain in the backward countries. Reward is not paid in accordance with the capability, skill and efficiency in less developed countries.

20- Inadequate Infrastructure

Adequate infrastructure is needed which is not available in poor economies to enhance the process of economic development. Roads, transport, telecommunications, sanitation, health and education facilities are not at their best level in these nations;. Government has reserved an amount of Rs. 133 billion to develop the infrastructure.

C. CULTURAL AND POLITICAL CHARACTERISTICS

Following are the cultural and political characteristics of LDC's:

21- High Degree of Illiteracy

Illiteracy rate is very high in poor countries while it is almost zero in rich countries. There is lack of technical education and training centers, which is necessary for economic growth and development. Literacy rate in Pakistan is 57.7 % during 2010-11. Expenditure on education sector is just 1.8 % of GDP.

22- Low Level of Organization

There is absence of developed minded leadership in economic activities in third world nations. Decision making power of entrepreneur is very low due to illiteracy, less training and backward techniques. Most of educational institutions are producing employees rather than employers.

23- Low Self-esteem

There is less respect, honour and dignity of people in the lower developed countries. People are honoured due to their powers, relations and castes instead of capabilities. There is poverty, poor health, poor education and shortage of other social services. Government and population of poor countries are under the external influence.

24- Un-productive Expenditures

Population mostly copies the styles of population of developed nations due to demonstration action in poor economies. Their consumption activities not only move around their income but also depend upon the relatives, friends and locality. They spend more on birth, death, marriages and various other ceremonies etc. which reduces their savings and investment.

25- Political Instability

There is political instability in the most of the developing countries. There are a lot of clashes between government and the opposition that is a cause to reduction in domestic as well as foreign investment. Political instability keeps low the level of economic development.

26- Influence of Feudal Lords

The poor class is under the influence of feudal lords and tribal heads in lower developed nations,. The feudal lords want to keep the people backward and do not appreciate the development of the poor. About 50.8% poor borrow from landlords and 57.4 % poor are working for feudal lords without wages in Pakistan.

27- Unproductive Use of Funds

The unproductive expenditures are rising day by day in developing countries like Pakistan due to socio-economic and administrative reasons. During the year 2009-10, Rs. 343 billion were spent for defence. About 75 % of the budget is spent on defence, administration, repayments of loan and interest charges in Pakistan.

28- Govt. Control by Wealthy Persons

Wealthy persons, landlords and elite class not only control the government but also they have full control over all the major sectors of the economy in poor countries. This rich class is not interested to solve the problems of the poor for their welfare but they make government policies for their own improvement.

29- Frequent Changes in Fiscal Policy

Revenues and expenditures policy of government is not stable in developing countries. Government has to change the fiscal policy according to the will of its own people. Industrialists are the main controller of the government and they adjust the fiscal policy in accordance with their own benefits.

30- Violation of Law and Order

Law and order conditions are at their poor stage in Pakistan like other developing countries. A huge portion of saving of people is wasted in costly and lengthy legal process. As in case of Iftikhar Muhammad Chohdery (CJP), he himself has to wait for justices for a long period.

D. TECHNOLOGICAL AND MISCELLANEOUS CHARACTERISTICS

Following are the technological and miscellaneous characteristics of developing countries:

31- Backward State of Technology

Use of modern techniques of production is not adopted in developing countries. It may cause further unemployment. Use of advanced technology is impossible due to shortage of capital, lack of skill and training, high cost of production and lack of foreign exchange reserves. Backward state of technology is results in low production, high cost and wastage of time.

32- Social Aspects

Under developed countries have also some factors such as joint family system, caste system, cultural and religious views, beliefs and values that badly affect their economic development. 32.17 % population is working population and remaining 67.83 % population is depending on them in Pakistan.

33- Un-fair Wealth and Income Distribution

There are not only regional inequalities in developing countries but also wealth and income inequalities. There is unfair wealth and income distribution in less developed nation. 20 % extremely rich population has 50.02 % of national resources, while 20 % poorest population has just 6.37 % of national resources in Pakistan. The difference between rich and poor is increasing day by day.

34- Lack of Experts and Skilled Persons

People have to move abroad for advanced study due to illiteracy and lack of training institutes. They adjust them in foreign countries due to low remuneration and less self-esteem. So, there is scarcity of experts, skilled and trained staff that causes the poor nation to remain backward.

35- Dependence on External Resources

The international trade, political activities and other economic activities are under the influence of other advanced countries in less developing countries. Their development plans are financed by the loan giving countries; these plans are made to serve the interests of foreign countries. So, poor nations are loans and grants receiving nations.

Conclusion:

We conclude that all above characteristics are unfavourable for the developing economies. These features are obstacles in way of economic development. All these features are cause of low rate of capital formation, poverty and creation of vicious circle of poverty.

ROLE OF AGRICULTURE AND INDUSTRY IN ECONOMIC DEVELOPMENT

Importance of agricultural sector can be shown as below:

1- Increase in Per Capita Income

Per capita income is the annual average income of the individuals of a nation. We derive it by dividing national income on total population. Agricultural sector provides more jobs to unemployed people. It increases the individuals as well as national income. Agricultural sector is essential to increase the PCI, which is $ 1254 at present.

2- Major Source of Employment

Agriculture sector provides employment to major portion of our labour force. More than 45.0% of our labour force is directly involved in agriculture sector, while 66.7% of our rural population is dependent on agriculture. This is the major sector, which provides employment to a large portion of our population. Agriculture sector is helpful to reduce the unemployment and disguised unemployment.

3- Reduction in Poverty

Agriculture development has significant impact on rural development. If productivity increases in agriculture it reduces poverty and stimulates non-farm employment, too. They are able to get basic services of life such as water supply, sanitation, provision of health and educational facilitates. About 21.0% population of Pakistan is much poor.

4- Supply of Food

Food is the first in basic necessities of the life. The agriculture sector is the sole provider of all type of food like wheat, rice, corn, sugarcane vegetables and fruits etc., to the population engaged in various sectors of the economy. Agricultural sector also provides food to those animals that provide milk, cheese, butter and meat to population to maintain the efficiency. According to economic survey of Pakistan, exports of food group are $ 2007.3 million.

5- Source of Forex Earning

Agriculture sector is the main source of foreign exchange (forex) earning which is used to import capital goods. Total forex resources of Pakistan is $ 17.1 billion out of which share of agriculture is $2007.3million. Major cash crops of Pakistan are cotton, rice, tobacco etc. that is helpful to correct the balance of payment.

6- Supply of Surplus Labour

About 61% population is living in more than 50,000 villages in Pakistan. This sector is the major source for supply of labor force to industrial sector due to farm mechanization and disguised unemployment. In this way an increase in the income of the people leads to improved living standard.

7- Source of National Income

At the time of partition, contribution of agriculture sector to GNP was 60%, which decreased to 29.4% in 1980-81. Now its share to GDP is 20.9% in 2010-11. Accordingly, agricultural sector is the major contributor to Pakistan's national income.

8- Supply of Raw Material

Agricultural sector provides not only food but also provides cotton, sugarcane, tobacco, rice, oil-seed, meat and milk to various agro-based, small scale and large-scale industries as a raw material.

9- Development of Industrial Sector

There is inter-dependence and inter-relationship between agricultural and industrial sector, both are helpful to develop each other. Due to use of modern techniques of production in agricultural sector, when income of the agriculturists increases, they will demand for industrial articles, like threshers, harvesters, tractors, tub-wells and various other consumer goods, which results in industrial growth and development.

10- Balance of Payment Position

Promotion of agriculture sector will provide surplus production for exports and our foreign exchange reserves will be increased. This will be helpful in the improvement of balance of payment. At present Pakistan is facing a deficit of $ 8.3 billion in its balance of payment.

11- Improvement in Living Standard

There is an increase in the income of the farmers due to development of agricultural sector. It will improve their living standard by constructing better houses, demanding luxuries of life, television, computer, mobile, motor-cycles and a lot of other items. So, growth of agricultural sector will cause to improve the standard of living of the population.

12- Extension in Market Size

There is more output in agricultural sector due to farm mechanization. It enlarges the size of market. If there is more production, then surplus production can be export to the other countries of the world. So, agricultural sector expands the market at national and international level.

13- Capital Formation

Role of agricultural sector in capital formation cannot ignore. It increases the incomes of the people that lead to more saving and more investment. Here, more investment leads to rapid capital formation. Capital formation rate in Pakistan is 5%.

14- Increase in Investment

Total investment is only 13.4% of GDP in Pakistan. This low investment is mainly due to low income and low savings. Agricultural sector increases the income of people that caused in more saving and more investment. Accordingly, agricultural sector is helpful in increasing the investment.

15- Economic Development

Economic development, progress and prosperity cannot achieve without agricultural sector. It provides employment opportunities to a lot of people. It produces exportable items, which increases the foreign exchange resources. So, agricultural sector is helpful to make rapid economic development. Real GDP growth rate is 2.4%.

16- Self Reliance Policy

Agriculture sector is helpful to achieve the self-sufficiency. Our country will not only become self-sufficient in food but supply of raw materials for industries will also expand our industrial sector.

17- Direct Foreign Investment

Now a day, use of advanced technologies in farming is common. It boost-up the industries like tractors, harvesters, thrashers, chemicals etc. It has increases the foreign direct investment in our country. It provides employments to our population and also increases the national income. Foreign investment is $ 1.8 billion in Pakistan.

18- Controlling Inflation

Inflation refers to the increase in general price level. It may be due to increase in demand and shortage in supply. Agricultural sector is very helpful to control the inflation. It produces more goods and maintains the equilibrium in demand and supply. Rate of inflation is 14.1% in Pakistan.

19- Reduction in Regional Disparities

Agriculture sector is the life-blood of our economy. Development of agriculture sector will increase the living standard of the rural population. This leads to reduce the urban and rural differences in the country.

20- Demand for Industrial Goods

As the agriculture productivity increases, the income of farmers goes up. With the rise in income the demand for both agricultural goods (tractor, fertilizer, pesticides, tube-wells etc.) and industrial goods (television, mobile, computer etc.) will increase in rural areas.

21- Balanced Growth Economy

There exists close inter-dependence between agricultural sector and industrial sector. Agricultural sector will develop the industrial sector also. Income received by cultivator will develop the demand for the industrial goods. Its results are in balance growth of the economy.

22- Increase in Govt. Revenue

There is no direct tax on the agricultural income. But indirectly government receives a huge amount of tax form agricultural sector. It imposed some duties on imported technologies applied in farming. It imposes tax on chemicals etc. So agriculture sector is also a source of income for the government.

The role of industry includes:

- **Increase in National Income:**

Industrial sector also contributes in gross domestic product (GDP) of country According to year 2004-05 the contribution of industrial sector in GDP is 18.3 Percent. So in this ways this sector increases national GDP.
Development of industrial sector means more investment, employment and production. Increase in production will increase the national income. If we study the history of economic development we find that the growth in national income and per capita income has been accompanied by a relative decline in their dependence on agriculture. At present the share of industrial sector to GDP is 6.8%.

- **Higher standard of Living**

Industrialization helps in increasing the value of output per worker. The income of the labor, due to higher productivity increases. The rise in income raises the living standard of the people.

- **Economic Stability:**

Industrial sector also helps government to stabilize price in country when goods are available in sufficient quantity, there is n0 change of increasing prices of these items.
Industrialization is the best way of providing economic stability to the country. A nation which depends upon the production and export of raw material alone cannot achieve a rapid rate people.

- **Improvement in balance of payments**

Industrial sector earns a good amount of foreign exchange for the country which is used for repairing national debt and for imp0rtant of machinery etc.
Industrialization brings structural changes in the pattern of foreign trade of the country. It helps in increasing the export of manufactured goods and thus earns foreign exchange. On the other hand The processing of raw material at home curtails the import of goods and thereby helps in conserving foreign exchange.

- **Agricultural Development**

Industrial Sector is also important and helpful in agricultural development. Because it provides fertilizers pesticides, seeds, machinery etc, agricultural sector these things promotes agricultural development
Industrialization provides machinery like tractors, thrashers, harvesters, bulldozers, aerial spray etc.. To be used in the farm sector. The increased use of modem inputs has increased the yield of crops

per hectare. The increase in the income of the farmers has given boost to economic development in the country.

- **More Exports**

Industrial Sector plays very important role in increasing exports of country major exports of our industrial sector are

- Textile Items
- Sports Goods
- Leather. Etc

- **Increase in Employment**

Industrial Sector provides employment to majority of population. This sector provides employment to both skilled and unskilled labour. According to one estimate more that 50% of total labor force is employed in industry.
Kenya is a developing country and there population growth rate is 1.573% in 2011. but the rate of increase in G.N.P is relatively low, therefore, there is general unemployment (25 %) in 2009, and disguised unemployment is increase in agriculture sector. Developed industrial sector can absorb millions of unemployed which will increase not only the income level but also the standard of living.

- **Modern Defense**

Modem defense cannot be put on a sound footing unless its basic needs are met from within the country itself and these basic needs are mainly supplied by modem industry, in the present times it have become more important as we surrounded by enemies.

- **Stimulates progress of other industries**

Development of one industry leads to the development and expansion of other industries. For instance.. The construction of a transistor radio plant develops the small battery industry (backward linkage). The construction of milk processing plants adds to its line of production ice cream. Cone cream plants, etc…

- **Promotes Specialization**

Industrialization promotes specialization of labor. The division of work increases the marginal value product of labor. The income of worker in the industrial sector is therefore, higher than that of a worker in agricultural sector.

- **Increased Saving and Investment**

Industrialization increases the income of the workers. It enhances their capacity to save. The voluntary savings stimulate industrial growth and by cumulative effect lead to further expansion of industry.

- **Large scope for Technological progress**

Industrialization provides larger scope for on the job-training and technological progress. The use of advanced technology increases the scale of production, reduces cost of production. Improves quality of the product and helps in widening of the market.

Increase in the Government Revenue

Industrial sector increases revenue of Government because industrial sector pays different type of taxes to government like

- Excise Duty
- Custom Duty
- Corporate Tax
- Sales Tax

Industrialization increases the supply of goods both for internal and external markets. The export of goods provides foreign exchange. The customs excise duties and other taxes levied on the production of goods increase the revenue of the state.

- **Easy to control Industrial Activity**

The industrial activity compared to agricultural is easy to control. The industrial production can be expanded or cut down according to the price, cost and demand of the product.

Reduction in the Rate of population Growth

Industrialization leads to migration of surplus labor from farm sector to the industries mostly situated in urban centers. In cities, improved facilities of sanitation and health care are available. People through the adoption of family planning measures, reduce the rate of population growth.

- **Diversification of Economy**

Industrialization eliminates dependence on agriculture only. By exporting finished and industrial goods we can earn a lot of money as compared to exporting raw materials.

- **Import Substitutes**

With developed industrial sector we cannot only produce substitute of imported goods but we can export them to other countries. Therefore dependence on other countries and deficit in balance of trade and balance of payments can be decreased and we may be able to save foreign exchange.

- **Extension of Market**

Extension of market depends on the quantity of demand. There is a demand of finished products within and outside the country. With increase in industrial production we can supply goods to far-off places and to other countries and extend market which may help in economic development.

- **Increase in Rate of Economic Development**

Agriculture, minerals, transport and communication trade and other sector of the economy mainly depend on industrial sector. Developed industrial sector means development and economic welfare of the individuals.

- **Change in people's Attitude**

In an agrarian country people in general are illiterate. Customary traditional and backward in thinking industrial starts running on the path of development.

- **Political Stability**

In international market, those countries are enjoying higher living standards which we industrially developed. They not only receive higher prices for their exports and earn foreign exchange but their financial and political condition also become more stable.

- **Increase in social welfare**

Industrial sector is also plays great role in social welfare this sector meets different needs of society like food and fiber requirement of population.
Industrial development increases the revenue of the government which is spent on the welfare of people. New roads, dams, buildings and parks are constructed.

- **Use of Raw Material**

Due to industrialization in Pakistan now cotton and sugarcane is used in the domestic industries. It has increased the income of the people and there is no need to export the raw material.

- **Reduces Imports**

Imports of country can be reduced by establishing industries of those goods that are being imported from other countries in this way national foreign exchange service can be saved.

- **Scale of Production**

Increased Production will expand the scale of production with the result cost of production will be decreased and the profit will be increased for the business people.

- **Efficiency of Labour**

Industries will make the people hardworking and will go a long way to increase their efficiency.

- **National Defense Requirement**

Industrial Sector also meets national Defense requirement. Defense is very important requirement of every country industrial sector provide modern arms and other equipment's for defense.

- **Conclusion of Importance of Industrial Sector**

From the above discussion it is clear that no doubt Industrial sector has much importance for economic development without industrial sector no country can make progress in this modern world.

- **Optimum Diversification**

We should establish every type of industry in every line so that we will be able to produce every product in our country and in this we can maximum Imports.

- **Labor Intensive Industries**

Our Country should establish labor intensive industries instead of capital intensive industries because we have cheap labor and we have shortage of capital.

- **Use Foreign AID**

Government should obtain maximum foreign AID to compensate low rate of saving so that maximum investment will take place in country.

- **Tax Concessions**

Governments should charge Minimum taxes from industrialists. So that they may not be discount maximum tax concessions should be given to businessman.

- **Industrial Cities & Loons**

Government should establish maximum industrial cities and zones where every facility should be provided to industrialists easily and at low rates.

THE NEED FOR DEVELOPMENT PLANNING

Why Planning is Important?

Now let's understand why planning is important for organization:

1. **Increases efficiency**: Planning makes optimum utilization of all available resources. It helps to reduce wastage of valuable resources and avoids their duplication. It aims to give the highest returns at the lowest possible cost. It thus increases the overall efficiency.

2. **Reduces business-related risks**: There are many risks involved in any modern business. Planning helps to forecast these business-related risks. It also helps to take the necessary precautions to avoid these risks and prepare for future uncertainties in advance. Thus, it reduces business risks.

3. **Facilitates proper coordination**: Often, the plans of all departments of an organization are well coordinated with each other. Similarly, the short-term, medium-term and long-term plans of an organization are also coordinated with each other. Such proper coordination is possible only because of efficient planning.

4. **Aids in Organizing**: Organizing means to bring together all available resources, i.e. 6 Ms. Organizing is not possible without planning. It is so, since, planning tells us the amount of resources required and when are they needed. It means that planning aids in organizing in an efficient way.

5. **Gives right direction**: Direction means to give proper information, accurate instructions and useful guidance to the subordinates. It is impossible without planning. It is because planning tells us what to do, how to do it and when to do it. Therefore, planning helps to give a right direction.

6. **Keeps good control**: With control, the actual performance of an employee is compared with the plans, and deviations (if any) are found out and corrected. It is impossible to achieve such a control without right planning. Therefore, planning becomes necessary to keep a good control.

7. **Helps to achieve objectives**: Every organization has certain objectives or targets. It keeps working hard to fulfill these goals. Planning helps an organization to achieve these aims, but with some ease and promptness. Planning also helps an organization to avoid doing some random (done by chance) activities.
8. **Motivates personnel**: A good plan provides various financial and non-financial incentives to both managers and employees. These incentives motivate them to work hard and achieve the objectives of the organization. Thus, planning through various incentives helps to motivate the personnel of an organization.
9. **Encourages creativity and innovation**: Planning helps managers to express their creativity and innovation. It brings satisfaction to the managers and eventually success to the organization.
10. **Helps in decision-making**: A manager makes many different plans. Then the manager selects or chooses the best of all available strategies. Making a selection or choosing something means to take a decision. So, decision-making is facilitated by planning

SHORT TERM, MEDIUM TERM AND LONGTERM PLANNING TOOLS

Planning Characteristics

Many economies develop strategic planning within a short-term, medium-term and long-term framework. Short-term usually involves processes that show results within a year. Economies aim medium-term plans at results that take several years to achieve. Long-term plans include the overall goals of the country set four or five years in the future and usually are based on reaching the medium-term targets. Planning in this way helps you complete short-term tasks while keeping longer-term goals in mind.

Short term planning

The process of setting smaller, intermediate milestones to achieve within closer time frames when moving toward an important overall goal. Many business operators will engage in short term planning that typically covers time frames of less than one year in order to assist their company in moving gradually toward its longer term goals.

Short-term planning looks at the characteristics of the country in the present and develops strategies for improving them.

Medium-Term

Medium-term planning applies more permanent solutions to short-term problems. If training courses for employees solved problems in the short term, companies schedule training programs for the medium term. If there are quality issues, the medium-term response is to revise and strengthen the company's quality control program. Where a short-term response to equipment failure is to repair the machine, a medium-term solution is to arrange for a service contract. Medium-term planning implements policies and procedures to ensure that short-term problems don't recur.

Long-Term

In the long term, companies want to solve problems permanently and to reach their overall targets. Long-term planning reacts to the competitive situation of the company in its social, economic and political environment and develops strategies for adapting and influencing its position to achieve long-term goals. It examines major capital expenditures such as purchasing equipment and facilities, and implements policies and procedures that shape the company's profile to match top management's ideas. When short-term and medium-term planning is successful, long-term planning builds on those achievements to preserve accomplishments and ensure continued progress.

The tools in planning include:

1. **Affinity Diagram**: organizes a large number of ideas into their natural relationships.
2. **Relations Diagram:** shows cause-and-effect relationships and helps you analyze the natural links between different aspects of a complex situation.
3. **Tree Diagram:** breaks down broad categories into finer and finer levels of detail, helping you move your thinking step by step from generalities to specifics.
4. **Matrix Diagram:** shows the relationship between two, three or four groups of information and can give information about the relationship, such as its strength, the roles played by various individuals, or measurements.
5. **Matrix Data Aanalysis:** a complex mathematical technique for analyzing matrices, often replaced in this list by the similar prioritization matrix. One of the most rigorous, careful and time-consuming of decision-making tools, a prioritization matrix is an L-shaped matrix that uses pairwise comparisons of a list of options to a set of criteria in order to choose the best option(s).
6. **Arrow Diagram:** shows the required order of tasks in a project or process, the best schedule for the entire project, and potential scheduling and resource problems and their solutions.
7. **Process Decision Program Chart (PDPC):** systematically identifies what might go wrong in a plan under development.

LIMITATION OF PLANNING IN DEVELOPING COUNTRIES

There are a number of reasons that may make a country to fail or miss achieving the set development goals. **Developing countries** more often are faced with some unseen problems which automatically keep them out the ring of proper development. There are reasons therefore that may limit planning in developing countries. These may include among others;

1. Insufficient data.

The available data in the country cannot be relied on since it is insufficient and cannot support the possible requirement by statistical systems. The data that should be available in the country is not adequately sieved to meet the possibly needed data.

2. Unreliable data.

The data available is also inadequate. Collection methods of data are still poor and this end up giving unreliable data.

3. Lack of appropriately qualified and experienced personnel.

The appropriate and qualified personnel who should be considered relevant persons to collect and analyze data are also not still fully qualified for this function.

4. Lack of necessary technical facilities.

The facilities that should be used to analyze and make accurate conclusions on data collected plus those facilities to store adequately the collected data are not adequate.

5. High levels of illiteracy.

The high level of illiteracy in Kenya has been another challenge that limits planning in the developing countries.

6. Unanticipated economic disturbances.

Both external and internal economic disturbances may pose unanticipated problems to the country and this poses a problem.

7**. Institutional weaknesses.**

Such weaknesses such as lack of coordination, political and bureaucratic corruption, excessive bureaucratic procedures and private objectives by political leaders are among other limitations that may affect planning in the developing countries.
Other limitations that may limit planning in the developing countries include among others,

8. **Deficiencies in plans and their implementation**
9. **Lack of political will**, lack of commitment and political will since they have higher-level decision making.
10. Limited range and effectiveness of policy instruments.

CHAPTER 8

MONEY AND BANKING

MONEY

THE NATURE AND FUNCTIONS OF MONEY

The development of money was necessitated by specialization and exchange. Money was needed to overcome the shortcomings and frustrations of the barter system which is system where goods and services are exchanged for other goods and services.

Disadvantages of Barter Trade

- It is impossible to barter unless A has what B wants, and A wants what B has. This is called double **coincidence** of wants and is difficult to fulfill in practice.
- Even when each party wants what the other has, it does not follow they can agree on a fair exchange. A good deal of time can be wasted sorting out equations of value.
- The indivisibility of large items is another problem. For instance if a cow is worth two sacks of wheat, what is one sack of wheat worth? Once again we may need to carry over part of the transaction to a later period of time.
- It is possible to confuse the **use value** and **exchange value** of goods and services in a barter economy. Such a confusion precludes a rational allocation of resources and promotion of economic efficiency.
- When exchange takes place over time in an economy, it is necessary to store goods for future exchange. If such goods are perishable by nature, then the system will break down.
- The development of industrial economies usually depends on a division of labour, specialization and allocation of resources on the basis of choices and preferences.

Economic efficiency is achieved by economizing on the use of the most scarce resources. Without a common medium of exchange and a common unit of account which is acceptable to both consumers and producers, it is very difficult to achieve an efficient allocation of resources to satisfy consumer preferences.

For these reasons the barter system is discarded by societies which develop beyond **autarky** to more specialized methods of production. For such peoples a money system is essential.

Money may be defined as anything generally acceptable in the settlement of debts.

The Historical development of money

For the early forms of money, the intrinsic value of the commodities provided the basis for general **acceptability**: For instance, corn, salt, tobacco, or cloth were widely used because they had obvious value themselves. These could be regarded as **commodity money.**

Commodity money had uses other than as a medium of exchange (e.g. salt could be used to preserve meat, as well as in exchange). But money commodities were not particularly convenient to use as

money. Some were difficult to **transport,** some **deteriorated overtime**, some **could not be easily divided and some were valued differently by different cultures.**

As the trade developed between different cultures, many chose **precious metal's** mainly gold or silver as their commodity money. These had the advantage of being **easily recognizable, portable, indestructible and scarce** (which meant it preserved its value over time).

The value of the metal was in terms of weight. Thus each time a transaction was made, the metal was weighed and payment made. Due to the inconvenience of weighing each time a transaction was made, this led to the development of coin money. The state took over the **minting** of coins by stamping each as being a particular weight and purity (e.g. one pound of silver). They were later given a rough edge so that people could guard against being cheated by an unscrupulous trade filling the edge down.

It became readily apparent, however, that what was important was public confidence in the "currency" of money, it's ability to run from hand to hand and circulate freely, rather than its intrinsic value. As a result there was deliberately reduced below the face value of the coinage.

Any person receiving such a coin could afford not to mind, so long as he was confident that anyone to whom he passed on the coin would also **"not mind".** Debasement represents an early form of fiduciary issue, i.e. issuing of money dependent on the "faith of the public" and was resorted to because it permitted the extension of the supply of money beyond the availability of gold and silver.

Paper Money

Due to the risk of theft, members of the public who owned such metal money would deposit them for safe keeping with **goldsmiths** and other **reliable merchants** who would issue a receipt to the depositor. The metal could not be withdrawn without production of the receipt signed by the depositor. Each time a transaction was made, the required amount of the metal would be withdrawn and payment made.

It was later discovered that as long as the person being paid was convinced the person paying had gold and the reputation of the goldsmith was sufficient to ensure acceptability of his promise to pay, it became convenient for the depositor to pass on the goldsmith's receipt and the person being paid will withdraw the gold himself. Initially, the gold would be withdrawn immediately after the transaction was made. But eventually it was discovered that so long as each time a transaction was made the person being paid was convinced that there was gold, the signed receipt could change hands more than once. Eventually, the receipts were made payable to the bearer (rather than the depositor) and started to circulate as a means of payment themselves, without the coins having to leave the vaults. This led to the development of **paper money,** which had the added advantage of lightness.

Initially, paper money was backed by precious metal and convertible into precious metal on demand. However, the goldsmiths or early bankers discovered that not all the gold they held was claimed at the same time and that more gold kept on coming in (gold later became the only accepted form of money). Consequently they started to issue more bank notes than they had gold to back them, and the extra money created was lent out as loans on which interest was charged. This became lucrative business, so much so that in the 18th and 19th centuries there was a bank crisis in England when the banks failed to honour their obligations to their depositors, i.e. there were more demands than there was gold to meet them. This caused the government to intervene into the baking system so as to

restore confidence. Initially each bank was allowed to issue its own currency and to issue more currency than it had gold to back it.

This is called **fractional backing,** but the Bank of England put restrictions on how much money could be issued.

Eventually, the role of issuing currency was completely taken over by the Central Bank for effective control. Initially, the money issued by the Central Bank was backed by gold (fractionally), i.e. the holder had the right to claim gold from the Central Bank. However, since money is essentially needed for purchase of goods and services, present day money is not backed by gold, but it is based on the level of production, the higher the output, the higher is the money supply. Thus, present day money is called **TOKEN MONEY** i.e. money backed by the level of output.

Over time, therefore, it became clear that for an item to act as money it must possess the following characteristics.

• **Acceptability**

If money is to be used as medium of exchange for goods and services, then it must be generally accepted as having value in exchange. This was true of metallic money in the past because it was in high and stable demand for its ornamental value. It is true of paper money, due to the good name of the note-issuing authority.

• **Portability**

If an item is to be used as money, it must be easily portable, so that it is a convenient means of exchange.

• **Scarcity**

If money is to be used in exchange for scarce goods and services, then it is important that money is in scarce supply. For an item to be acceptable as money, it must be scarce.

• **Divisibility**

It is essential that any asset which is used as money is divisible into small units, so that it can be used in exchange for items of low value.

• **Durability**

Money has to pass through many different hands during its working life. Precious metals became popular because they do not deteriorate rapidly in use. Any asset which is to be used as money must be durable. It must not depreciate over time so that it can be used as a store of wealth.

• **Homogeneity**

It is desirable that money should be as uniform as possible.

Functions of money

a. **Medium of exchange:** Money facilitates the exchange of goods and services in the economy. Workers accept money for their wages because they know that money can be exchanged for all the different things they will need. Use of money as an intermediary in transactions therefore, removes the requirement for double coincidence of wants between transactions. Without money, the world's complicated economic systems which are based on specialization and the division of labour, would be impossible. The use of money enables a person who receives payment for services in money to obtain an exchange for it, the assortment of goods and services from the particular amount of expenditure which will give maximum satisfaction.

b. **Unit of account:** Money is a means by which the prices of goods and services are quoted and accounts kept. The use of money for accounting purposes makes possible the operation of the price system and automatically provides the basis for keeping accounts, calculating profit and loss, costing etc. It facilitates the evaluation of performance and forward planning. It also allows for the comparison of the relative values of goods and services even without an intention of actually spending (money) on them e.g. "window shopping".

c. **Store of Wealth/value:** The use of money makes it possible to separate the act of sale from the act of purchase. Money is the most convenient way of keeping any form of property which is surplus to immediate use; thus in particular, money is a store of value of which all assets/property can be converted. By refraining from spending a portion of one's current income for some time, it becomes possible to set up a large sum of money to spend later (of course subject to the time value of money). Less durable or otherwise perishable goods tend to depreciate considerably over time, and owners of such goods avoid loss by converting them into money.

d. **Standard of deferred payment:** Many transactions involve future payment, e.g. hire purchase, mortgages, long term construction works and bank credit facilities. Money thus provides the unit in which, given the stability in its value, loans are advanced/made and future contracts fixed. Borrowers never want money for its own sake, but only for the command it gives over real resources. The use of money again allows a firm to borrow for the payment of wages, purchase of raw materials or generally to offset outstanding debt obligations; with money borrowing and lending become much easier, convenient and satisfying. It's about making commerce and industry more viable.

Only money, of all possible assets, can be converted into other goods immediately and withoutcost.

The Determination of the Value Money

Since money is primarily a medium of exchange, the value of money means what money will buy. If at one time a certain amount of money buys fewer things than at a previous time, it can be said that the value of money has fallen. Since money itself is used as unit of account and a means of measuring the "value" of other things, its own value can be seen only through the prices of other things. Changes in the value of money, therefore, are shown through changes in prices.

DEMAND AND SUPPLY OF MONEY

i. Demand for money

The demand for money is a more difficult concept than the demand for goods and services. It refers to the desire to **hold one's assets as money rather than as income-earning assets (or stocks).**

Holding money therefore involves a loss of the interest it might otherwise have earned. There are two schools of thought to explain the demand for money, namely the Keynesian Theory and the Monetarist Theory. **(See pp 18 – 26)**

The demand for money and saving

The demand for money and saving are quite different things. Saving is simply that part ofincome which is not spent. It adds to a person's wealth. Liquidity preference is concerned with the form in which that wealth is held. The motives for liquidity preference explain why there is desire to hold some wealth in the form of cash rather than in goods affording utility or in securities. (See pp 18 – 26)

ii. The supply of money

Refers to the total amount of money in the economy.

Most countries of the world have two measures of the money stock – **broad money supply** and **narrow money supply**. Narrow money supply consists of all the purchasing power that is immediately available for spending. Two narrow measures are recognized by many countries.

The first, M0 (or monetary base), consists of notes and coins in circulation and the commercial banks' deposits of cash with the central banks.

The other measure is M2 which consists of notes and coins in circulation and the NIB (noninterest-bearing) bank deposits – particularly current accounts. Also in the M2 definition are the other interest-bearing retail deposits of building societies. Retail deposits are the deposits of the private sector which can be withdrawn easily. Since all this money is readily available for spending it is sometimes referred to as the "transaction balance".

Any bank deposit which can be withdrawn without incurring (a loss of) interest penalty is referred to as a "sight deposit".

The broad measure of the money supply includes most of bank deposits (both sight and time), most building society deposits and some money-market deposits such as CDs (certificates of deposit).

Legal Tender

Legal tender is anything which must be by law accepted in settlement of a debt.

Determinants of the money supply

Two extreme situations are imaginable. In the first situation, the money supply can be determined at exactly the amount decided on by the Central Bank. In such a case, economists say that the money supply is exogenous and speak of **an exogenous money supply.**

In the other extreme situation, the money supply is completely determined by things that are happening in the economy such as the level of business activity and rates of interest and is wholly out of the control of the Central Bank. In such a case economists would say that there was an Endogenous money supply, which means that the size of the money supply is not imposed from outside by the decisions of the Central Bank, but is determined by what is happening within the economy.

In practice, the money supply is partly endogenous, because commercial banks are able to change it in response to economic incentives, and partly exogenous, because the Central Bank is able to set limits beyond which the commercial banks are unable to increase the money supply.

THEORIES OF DEMAND FOR MONEY

THE QUANTITY THEORY

In the 17th Century it was noticed that there was a connection between the quantity of money and the general level of prices, and this led to the formulation of the Quantity Theory of Money. In its crudest form is stated that an increase in the quantity of money would bring about an appropriate rise in prices. If the quantity of money was doubled, prices would double and so on. Algebraically, this could be stated as:

$P = a M$

Where a is constant, P the price level, and M the supply of money. If the supply of money doubled, to 2M, the new price level P will equal $a(2m) = 2(aM) = 2P$ that is, double the old price level.

After being long discarded, the theory was revived in the 1920s by Professor Irving Fisher, who took into account the volume of transactions, that is to say, the amount of "work" that the money supply had to do as a medium of exchange. That is the velocity of circulation. Money circulated from hand to hand. If one unit of money is made to serve four transactions, this is equivalent to four units of money, each being used in only one transaction.

As modified by Irving Fisher, the quantity theory came to be expressed by the equation of exchange.

$MV = PT$

The symbol M represents the total amount of money in existence – bank notes etc, and bank deposits.

The symbol V represents the velocity of circulation, i.e. the number of times during the period each unit of money passes from hand to hand in order to affect a transaction. Thus if the amount of money in the hands of the public during the year was an average $1,000,000 and each dollar on average was used five times, the total value of transactions carried out during the year must have been $5,000,000. MV therefore represents the amount of money used in a period.

On another side of the equation, P stands for the general price level, a sort of average of the price or all kinds of commodities-producers' goods as well s consumer's goods and services.

The symbol T is the total of all transactions that have taken place for money during the year.

The equation of exchange shows us that the price level, and, therefore, the value of money, can be influenced not only by the quality of money but also by:

i. the rate at which money circulates, and
ii. the output of goods and services.

Thus prices may rise without any change taking place in the quantity of money if a rise occurred in the velocity of circulation. On the other hand, prices might remain stable in spite of an increase in the quantity of money if there was corresponding increase in the output of goods and services.

Even in its revised form, however, the Quantity Theory has been subjected to the following criticisms:

a) It is not a theory at all, but simply a convenient method of showing that there is certain relationship between four variable quantities – M, V, P and T. it shows that only the total quantity of money, as determined by the actual amount of money in existence the velocity of circulation, is equal to the value of total trade transactions multiplied by their average price. As such it is obviously a truism, since the amount of money spent on purchases is obviously a truism, since the amount of money spent on purchases is obviously equal to the amount received from sales. Not only must MV be equal to PT, but MV is PT, since they are only two different ways of looking at the same thing.
b) Even if the equation of exchange is only a truism, it would not be quite correct to say that it demonstrates nothing. For example, it shows that it is possible for there to be an increase in the quantity o f money without a general rise in prices. It informs us, too, that if there is to be a change in one or more of the variables of the equation, there must be a change in one or more of the other variables. Cleary, it would be wrong to read into it more than this.
c) The four variables, M, V, P and T, are not independent of one another as the equation of exchange implies. For example, a change in M is likely of itself to bring a change in V or T or both. It is probable that a rise in pries will follow an increase in the quantity of money, but this will most likely be brought about because the increase in the quantity of money stimulates demand and production.
d) A serious defect is to allow the symbol P to represent the general price level. Price changes do not keep in step with one another. In its original form the equation was criticized because it implied that an increase in the quantity would automatically bring about a proportionate increase in all prices. A study of price changes shows that some prices increased by many times while others by fewer times. Clearly, then, there is no general price level, but instead, as the index of Retail Price shows, a number of sectional price levels, one for food, another for clothing, another for fuel and light, and so on.
e) The Quantity Theory only attempts to explain changes in the value of money, and does not show how the value of money is in the first place determined.
f) The Quantity Theory approaches the question of the value of money entirely from the supply perspective.

THE KEYNESIAN LIQUIDITY PREFERENCE THEORY

Also called the **Monetary Theory of Interest**, was put forward by the Lord John Maynard Keynes in 1936. In the theory, he stated that the rate of interest is determined by the supply of money and the desire to hold money. He thus viewed money as a liquid asset, interest being the payment for the loss of that liquidity.

Keynes formulated derived from three motives for holding money, namely:

- Transactions;
- Precautionary; and
- Speculative.

Thus Keynes contended that an individual's aggregate demand for money in any given period will be the result of a single decision that would be a composite of those three motives.

a. Transactions demand for money

Keynes argued that holding money is a cost and the cost is equal to the interest rate foregone.

People holding money as assets could also buy Government bonds to earn interest. But money's most important function is as medium of exchange. Consumers need money to purchase goods and services and firms need money to purchase raw materials and hire factor services. People therefore hold money because income and expenditure do not perfectly synchronize in time. People receive income either on monthly, weekly, or yearly basis but spend daily, therefore money is needed to bridge the time interval between **receipt of income** and its **disbursement over time**. The amount of money that consumers need for transactions will depend on their **spending habits, time interval after which income is received** and **Income.** Therefore holding habit and Interval Constant, the higher the income level the more the money you hold for transactions. Keynes thus concluded that transactions demand for money is Interest Inelastic.

b. Precautionary Demand for Money

Individuals and businessmen require money for unseen contingencies, Keynes hypothesized that individuals' demand and institutional factors in society to be considered in the short run.

Money demanded for these two motives is called active balances, because it is demanded to be put to specific purposes. The demand for active balances is independent of the rate of interest. Hence the demand curve for active balances is perfectly inelastic.

c. Speculative Demand for Money

Finally, money is demanded for speculative motives. This looks at money as a store of value i.e. money is held as an asset in preference to an income yielding asset such s government bond. Keynes thus explained the Speculative motive in terms of the buying and selling of Government Securities or Treasury Bills on which the government pays a fixed rate of interest.

According to Keynes, securities can be bought and sold on the free market before the government redeems them, and the price at which they are sold does not have to be equal to their face value. It can be higher or lower than the face value depending on the level of demand for securities.

THE BANKING SYSTEM

DEFINITION OF COMMERCIAL BANKS

A Commercial Bank is a financial institution which undertakes all kinds of ordinary banking business like accepting deposits, advancing loans and is a member of the clearing house i.e. operates or has a current account with the Central Bank. They are sometimes known as Joint Stock Banks.

THE ROLE OF COMMERCIAL BANKS AND NON-BANKING FINANCIAL INSTITUTIONS IN THE ECONOMY

In modern economy, commercial banks have the following functions:

a. They provide a safe deposit for money and other valuables.
b. They lend money to borrowers partly because they charge interest on the loans, which is a source of income for them, and partly because they usually lend to commercial enterprises and help in bringing about development.
c. They provide safe and non-inflationary means for debt settlements through the use of cheques, in that no cash is actually handled. This is particularly important where large amounts of money are involved.
d. They act as agents of the central banks in dealings involving foreign exchange on behalf of the central bank and issue travellers' cheques on instructions from the central bank.
e. They offer management advisory services especially to enterprises which borrow from them to ensure that their loans are properly utilized.

Some commercial banks offer insurance services to their customers eg. The Standard Bank (Kenya) which offers insurance services to those who hold savings accounts with it. Some commercial banks issue local travellers' cheques, e.g. the Barclays Bank (Kenya). This is useful in that it guards against loss and theft for if the cheques are lost or stolen, the lost or stolen numbers can be cancelled, which cannot easily be done with cash. This also safe if large amounts of money is involved.

Bank Deposit

Bank notes and coins together constitute the currency in circulation. But they form only a part of the total money supply. The larger part of the money supply in circulation today consists of bank deposits. Bank deposits can either be a current account or deposit account. These are created by commercial banks and the process is called credit creation.

CREDIT CREATION

The ability of banks to create deposit money depends on the fact that bank deposits need to be only fractionally backed by notes and coins. Because the bank does not need to keep 100 per cent reserves, it can use some of the money deposited to purchase income-yielding investments.

Illustration

i. A Single Monopoly Bank

Consider first a country with only one bank (with as many physical branches as is necessary) and assume that the bank has found from experience that it needs only to hold 10% of cash s a proportion of total deposits – proportion of transactions that customers prefer to settle by means of cash, rather than cheque. Now imagine the balance sheet of the bank look like this:

Initial Position of single bank

Liabilities	£1,000	Assets	£1,000
Deposits	1,000	Cash	100
		Loans	900
Total	1,000	Total	1,000

Deposits are shown as liabilities, since the bank can be called up to repay in cash any amounts credited to customers in this way. Assets consist of cash held by the bank, plus loans, which represents the obligations of borrowers towards the bank. The cash ratio is the ratio of cash

held (£100,000) to its liabilities ((£1,000,000), and is 10 per cent in this case.

Suppose now a customer deposits (liabilities) in this initial position will be:

120 x 100 = 11.8%

1,020

This is unnecessarily high, nearly 12 per cent compared to the conventional ratio of 10 per cent. The bank can therefore safely make additional interest-bearing loans. If it lends an extra £180,000, according deposits will rise from £1,020,000 to £1,200,000, so that the 10 per cent ratio of cash to deposit is restored. The final position is as shown below, and indicates that bank deposits have been created to the extent of ten times the new cash deposit.

Addition of cash deposit raises cash reserves and cash ratio

Liabilities	£1,000	Assets	£1,000
Deposits	1,020	Cash	120
		Loans	900
Total	1,020	Total	1,020

This is a stable position. Borrowers will make out cheques to other people in payment for goods and services supplied. But these others must be customers of the same bank, since there is only one bank. There will follow no more than a book transaction within one bank, the bank deposits being transferred from one customer to another. Total deposits, total cash, and the cash ratio will not be affected.

Comparing the initial position in the first table with the final position in the table below, we can see that the increase in bank deposits, which we can call **ΔD** is 200 and the increase in cash held by the banks, which we can write as **ΔC**, is 20. Thus **ΔD** is ten times **ΔC**, obviously because

1/r, where r is the cash ratio used, is 10.

Thus **ΔD = ΔC/r**

Restoration of Conventional cash ratio by creation of additional bank deposits

Liabilities	£1,000	Assets	£1,000
Deposits	1,200	Cash	120
		Loans	1,080
Total	1,200	Total	1,200

ii. Many Banks

A single bank with a "monopoly" of credit creation is rarely found in real world. What is usually found is where the bank receiving the new deposit is one of several independent banks.

In that case the bank will not seek immediately to expand deposits to the number of times the cash ratio, by extending loans. It will know that the borrowers will use the credit granted to them to pay for goods and services, or to repay debts; and that therefore they will be making cheques out to other individuals who by now have accounts in other banks. The bank can thus expect to lose cash to other banks. Either the borrowers will withdraw cash directly, with which to pay individuals who then deposit this cash with other banks, or if they pay by cheque these cheques will be deposited with other banks, and the other banks themselves will present them for cash at the first bank.

Suppose in our example above, (illustration 1), the bank made the extreme assumption that none of the borrowers' cheques would be paid to its own customers. It will create only a relatively small amount of extra deposits, just sufficient to restore its cash ratio. It will, in fact, make £18,000 worth of additional loans and retain cash of £2,000. That will restore its cash ratio as shown below.

Initial round of credit creation by first bank

Liabilities (£1,000) Assets (£1,000) Cash ratio

	Deposits	Cash	Loans	%
Original position	1,000	100	900	10.0
Add cash deposit	+20	+20	-	- -
Second position	1,020	120	900	11.8
Increase loans	+18	-	+18	- -
Cash drain by cheques	-18	-18	-	- -
Final position	1,020	102	918	10.0
Net change	+20	+2	+18	- -

However, the £18,000 lost in cash through cheques drawn by borrowers will be received by other banks who in turn will find themselves with excess cash reserves, and in turn create additional loans. There will thus be second generation of bank deposit creation, each bank again retaining only 10 per cent of the new cash received, and creating loans in the ratio of nineto one. This new drain of cash

will generate more deposits, and so on, each new round being nine-tenths of the value of the previous one as follows (£'000s).
20 +18+ 16.20 + 14.85 + 13.12 + …which can be written as

{1 + 9/10 + (9/10) 3 + (9/10) 4 +……..}

Each successive round of deposit creation is smaller than the previous one, so that the series converges. Mathematically, the series will eventually add up to converge to 200. This is because for any value of between 0 and 1 the series tends to the value 1/1-z.

In an example z = 9/10, which is between 0 and 1, so on the formula applies. Hence

20{1 + 9/10) 2 + (9/10) 3 + ….} = { 1 } = 200

1 – 9/10

if we use **ΔD** to refer to the final increase or increment in bank deposits, **ΔC** too the initial increase in cash received, and "r" to the **cash ratio,** then **ΔD = 200, ΔC = 20,**

r = 1/10.

Since 1 – r = 1 – 1/10 = 9/10, we have

ΔD = ΔC {1 + (1 – r) + (1 – r) 2 + (1 + r) 3 + …..}

ΔD = ΔC = ΔC

1 – (1 – r) r

Given the increase in cash received, the additional deposits created will depend on the fraction of cash retained as backing. The ration ΔD/ΔC of deposits created to increase in cash is referred to as the **bank deposit multiplier.**

Limits on the process of bank deposit creation

On the **demand side**, there may be a lack of demand for loans, or at least of borrowers who are **sufficiently credit worthy**. On the supply side, the volume of bank deposits will not in

general rise and fall as a result of changes in the amount of cash held or deposited by the public, since the **public's currency requirements tends to be fairly stable**, and roughly proportional to the volume of transactions.

DEFINITION OF CENTRAL BANK

Banks Consists of all those institutions which determine the supply of money. The main element of the Banking System is the Commercial Bank (in Kenya). The second main element of Banking System is the Central Bank and finally most Banking Systems also have a variety of other specialized institutions often called Financial Intermediaries. **It's the** governments bank that controls the monetary and fiscal policies in an economy.

ROLE OF CENTRAL BANK

i. Government's banker: Government's need to hold their funds in an account into which they can make deposits and against which they can draw cheques. Such accounts are usually held by the Central Bank

ii Banker's Bank: Commercial banks need a place to deposit their funds; they need to be able to transfer their funds among themselves; and they need to be able to borrow money when they are short of cash. The Central Bank accepts deposits from the commercial banks and will on order transfer these deposits among the commercial banks. Consider any two banks A and B. On any given day, there will be cheques drawn on A for B and on B for A. If the person paying and the person being paid bank with the same bank, there will be a transfer of money from the account or deposit of the payee. If the two people do not bank with the same bank, such cheques end up in the central bank. In such cases, they cancel each other out. But if there is an outstanding balance, say in favour of A, then A's deposit with the central bank will go up, and B's deposit will go down. Thus the central bank acts as the Clearing House of commercial banks.

iii. Issue of notes and coins: In most countries the central bank has the sole power to issue and control notes and coins. This is a function it took over from the commercial banks for effective control and to ensure maintenance of confidence in the banking system.

iv. Lender of last resort: Commercial banks often have sudden needs for cash and one way of getting it is to borrow from the central bank. If all other sources failed, the central bank would lend money to commercial banks with good investments but in temporary need of cash. To discourage banks from over-lending, the central bank will normally lend to the commercial banks at a high rate of interest which the commercial bank passes on to the borrowers at an even higher rate. For this reason, commercial banks borrow from the central bank as the lender of the last resort.

v. Managing national debt: It is responsible for the sale of Government Securities or Treasury Bills, the payment of interests on them and their redeeming when they mature.

vi. Banking supervision: In liberalized economy, central banks usually have a major role to play in policing the economy.

MONEY MARKETS

The expression **"money markets"** is used to refer to the set of institutions and individuals who are engaged in the borrowing and lending of large sums of money for short periods of time (overnight to three months). The money market is not located in a place – it is rather a network of brokers, buyers and sellers.

Most money market transactions are concerned with the sale and purchase of near money assets such as bills of exchange and certificates of deposit.

Function of Money Markets

The money markets are the place where money is "wholesaled". As such the supply of money and interest rate which are of significance to the whole economy is determined there.

It is also used by the central bank to make its monetary policy effective.

CAPITAL MARKETS

Markets in which financial resources (money, bonds, stocks) are traded i.e. the provision of longer term finance – anything from bank loans to investment in permanent capital in the form of the purchase of shares. The capital market is very widespread.

It can also be defined as the institution through which, together with financial intermediaries, savings in the economy are transferred to investor.

Interest and the Keynesian Liquidity Preference Theory

Interest is a factor income in that it is considered to be payment to or return on capital in the sense that it is payment to those who provide **loanable funds**, which are used for the purchase of capital assets. The payment of interest to the providers of loanable funds may be justified on the following grounds:

• The lender postpones present consumption and enjoyment and interest is paid as **persuasion** for him/her to make this sacrifice.

• There is risk of default in that the borrower may fail to pay back and interest is paid as persuasion for the lender to undertake this risk.

• There is loss of purchasing power due to increases in prices over time, and interest is paid as compensation for this loss.

• The borrower earns income from the investment, and the tender can justifiably claim a share in that income.

MONETARY POLICY, DEFINITION, OBJECTIVE, INSTRUMENTS AND LIMITATIONS

DEFINITION

Monetary policy is the regulation of the economy through the control of the quantity of money available and through the price of money i.e. the rate of interest borrowers will have to pay. Expanding the quantity of money and lowering the rate of interest should stimulate spending in the economy and is thus expansionary, or inflationary. Conversely, restricting the quantity of money and raising the rate of interest should have a restraining, or deflationary effect upon the economy.

OBJECTIVES OF MONETARY POLICY

The following are the principal objectives of monetary policy:

1. Full Employment:

Full employment has been ranked among the foremost objectives of monetary policy. It is an important goal not only because unemployment leads to wastage of potential output, but also because of the loss of social standing and self-respect.

2. Price Stability:

One of the policy objectives of monetary policy is to stabilise the price level. Both economists and laymen favour this policy because fluctuations in prices bring uncertainty and instability to the economy.

3. Economic Growth:

One of the most important objectives of monetary policy in recent years has been the rapid economic growth of an economy. Economic growth is defined as "the process whereby the real per capita income of a country increases over a long period of time."

4. Balance of Payments:

Another objective of monetary policy since the 1950s has been to maintain equilibrium in the balance of payments.

INSTRUMENTS AND LIMITATIONS

a) Open Market Operations**:** The Central Bank holds government securities. It can sell some of these, or buy more, on the open market, buying or selling through a stock exchange or money market. When the bank sells securities to be bought by members of the public, the buyers will pay by writing cheques on their accounts with commercial banks. This means a cash drain for these banks to the central bank, represented by a fall in the item "bankers" deposits' at the central bank, which forms part of the commercial banks' reserve assets. Since the banks maintain a fixed liquidity (or cash) ratio, the loss of these reserves will bring about multiple contraction of bank loans and deposits.

By going into the market as a buyer of securities, the central bank can reverse the process, increasing the liquidity of commercial banks, causing them to expand bank credit, always assuming a ready supply of credit-worthy borrowers.

Conversely, if the central bank wanted to pursue an expansionary monetary policy bymaking more credit available to the public, it would buy bonds from the public. It would pay sellers by cheques drawn on itself, the sellers would then deposit these with commercial banks, who would deposit them again with the central bank. This increase in cash and reserve assets would permit them to carry out a multiple expansion of bank deposits, increasing advances and the money supply together.

b) Discount Rate (Bank Rate) This is the rate on central bank advances and is also called official discount rate or "minimum lending rate". When commercial banks find themselves short of cash they may, instead of contracting bank deposits, go to the central bank, which can make additional cash available in its capacity as "lender of last resort", to help the banks out of their difficulties. The Central Bank can make cash available on a short-term basis in either of two ways; by lending cash directly, charging a rate of interest which is referred to as the official "discount rate", or by buying approved short-term securities from the commercial banks. The central bank exercises regulatory powers as a lender of last resort by making this help both more expensive to get and more difficult to get. It can do the former by charging a very high "penal" rate of interest, well above other short-term rates ruling in the money market. Similarly, when it makes cash available by buying approved short-term securities, it can charge a high effective rate of interest by buying them at low prices. The effective rate of interest charged when central bank buys securities (supplying cash) is in fact a re-discount rate, since the bank is buying securities which are already on the market but at a discount.

The significance of this rate of interest charged by the central bank in one way or the other to commercial banks, as a lender of last resort, is that if this rate goes up the commercial banks, who find that their costs of borrowing have increased, are likely to raise the rates of interest on their lending to businessman and other borrowers. Other interest rates such as those charged by building societies on house mortgages, are then also likely to be pulled up.

c) Variable Reserve Requirement

(Cash and Liquidity Ratios)

The Central Bank controls the creation of credit by commercial banks by dictating cash and liquidity ratios. The cash ratio is:

Cash Reserves÷ Deposits

The Central Bank might require the commercial banks to maintain a certain ratio, say 1/10.

Hence:

Cash Reserves/ Deposits = 1/10

10

Deposits = 10 x Cash Reserves

This means that the banks can create deposits exceeding 8 times the value of its liquid assets. The liquidity ratio can be rewritten as:

Cash + Reserves Assets = Cash + Reserves Assets

Deposits Deposits Deposits

= Cash Ratio + Reserve Assets Ratio

If the liquidity ratio is 12.5, then:

Cash + Reserved Assets = 1

Deposits Deposits 8

Deposits = 10 x cash + 2.5 x Reserve Assets.

In most countries the Central Bank requires that commercial banks maintain a certain level of Liquidity Ratio i.e. Cash reserves (in their own vaults and on deposit with the Central

Bank) well in excess of what normal prudence would dictate. This level shall be varied by the Central Bank depending on whether they want to increase money supply or decrease it.

This is potentially the most effective instrument of monetary control in less developed countries because the method is direct rather than via sales of securities or holding bank loans and advances. The effects are immediate. This method moreover does not require the existence of a capital market and a variety of financial assets. However, increased liquidity requirements may still be offset in part if the banks have access to credit from their parent companies. A further problem is that a variable reserve asset ratio is likely to be much more useful in restricting the expansion of credit and of the money supply than in expanding it: if there is a chronic shortage of credit-worthy borrowers, the desirable investment projects, reducing the required liquidity.

Ratio of the banks may simply leave them with surplus liquidity and not cause them to expand credit. Finally, if the banks have substantial cash reserves the change in the legal ratio required may have to be very large:

d) Supplementary Reserve, Requirements/Special Deposit

If the Central Bank feels that there is too much money in circulation, it can in addition require commercial banks to maintain over and above cash or liquid assets some additional reserves in the form of Special Deposits. The commercial banks are asked to maintain additional deposits in their accounts at the central bank, deposits which cease to count among their reserve assets as cover for their liabilities.

e) Direct control and Moral Suasion

Without actually using the above weapons, the central bank can attempt simply to use “moral suasion” to persuade the commercial banks to restrict credit when they wish to limit monetary expansion. Its effectiveness depends on the co-operation of the commercial banks.

f) General and Selective Credit Control

These are imposed with the full apparatus of the law or informally using specific instructions to banks and other institutions. For instance, the central bank can dictate a ceiling value to the amount of deposits the bank can create. This is more effective in controlling bank lending than the cash and liquidity ratio. It can also encourage banks to lend more to a certain sector of the economy (e.g. agriculture) than in another (estate building). Selective controls are especially useful in less developed investment away from less important sectors such as the construction of buildings, the commercial sector, or speculative purchase of land, towards more important areas.

LIMITATION OF MONETARY POLICY

1. Large Non-monetized Sector:

There is a large non-monetized sector which hinders the success of monetary policy in such countries. People mostly live in rural areas where barter is practised. Consequently, monetary policy fails to influence this large segment of the economy.

2. Undeveloped Money and Capital Markets:

The money and capital markets are undeveloped. These markets lack in bills, stocks and shares which limit the success of monetary policy.

3. Large Number of NBFLs:

Non-bank financial intermediaries like the indigenous bankers operate on a large scale in such countries but they are not under the control of the monetary authority. The factor limits the effectiveness of monetary policy in such countries.

4. High Liquidity:

The majority of commercial banks possess high liquidity so that they are not influenced by the credit policy of the central bank. This also makes monetary policy less effective.

5. Foreign Banks:

In almost every underdeveloped country foreign owned commercial banks exist. They also render monetary policy less effective by selling foreign assets and drawing money from their head officers when the central bank of the country is following a tight monetary policy.

6. Small Bank Money:

Monetary policy is also not successful in such countries because bank money comprises a small proportion of the total money supply in the country. As a result, the central bank is not in a position to control credit effectively.

7. Money not deposited with Banks:

The well-to-do people do not deposit money with banks but use it in buying jewellery, gold, real estate, in speculation, in conspicuous consumption, etc. Such activities encourage inflationary pressures because they lie outside the control of the monetary authority.

On account of these limitations of monetary policy in an under-developed country, economists advocate the use of fiscal policy along-with it.

DETERMINATION OF INTEREST RATESAND THEIR EFECTS ON THE LEVEL OF INVESTMENT, OUTPUT, INFLATION AND EMPLOYMENT

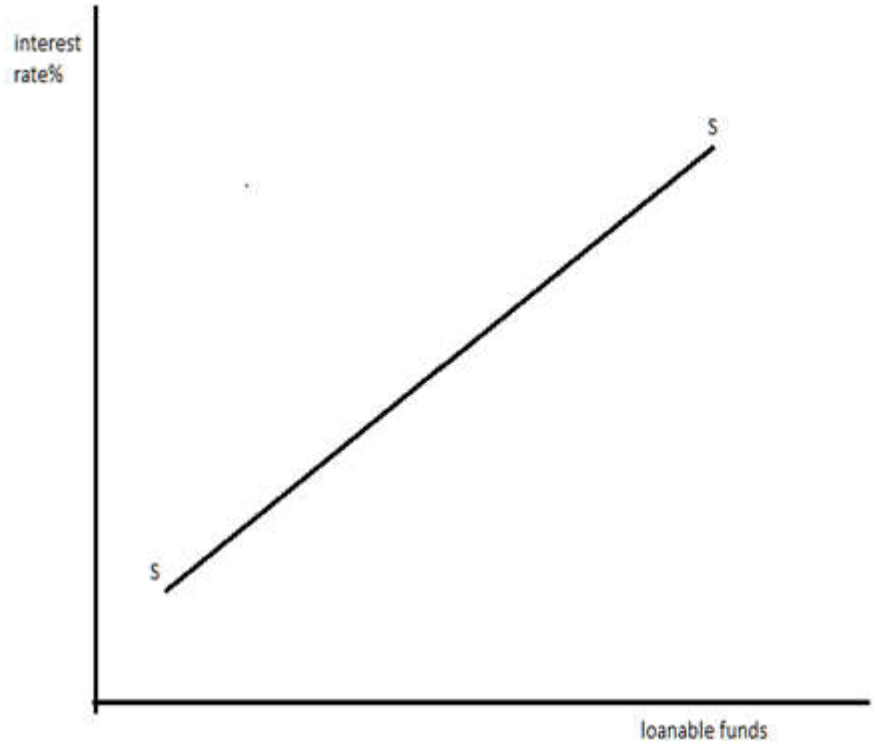
Interest rates, refers to payment, normally expressed as a percentage of the sum lent which is paid over a year, for the loan of money. There are many rates of interest depending on the **degree or risk involved, the term of the loan**, and the **costs of administration,** namely, real, nominal and pure rate of interest.

Pure rate of interest is one from which factors like **risk involved**, the **term of the loan** and the **cost of administration** has been removed. All rates of interest are related to each other and if one rate changes so will others.

There are two theories as to how the rate of interest is determined – the **loanable funds** and **the liquidity preference** theories.

a. The Loanable Funds Theory

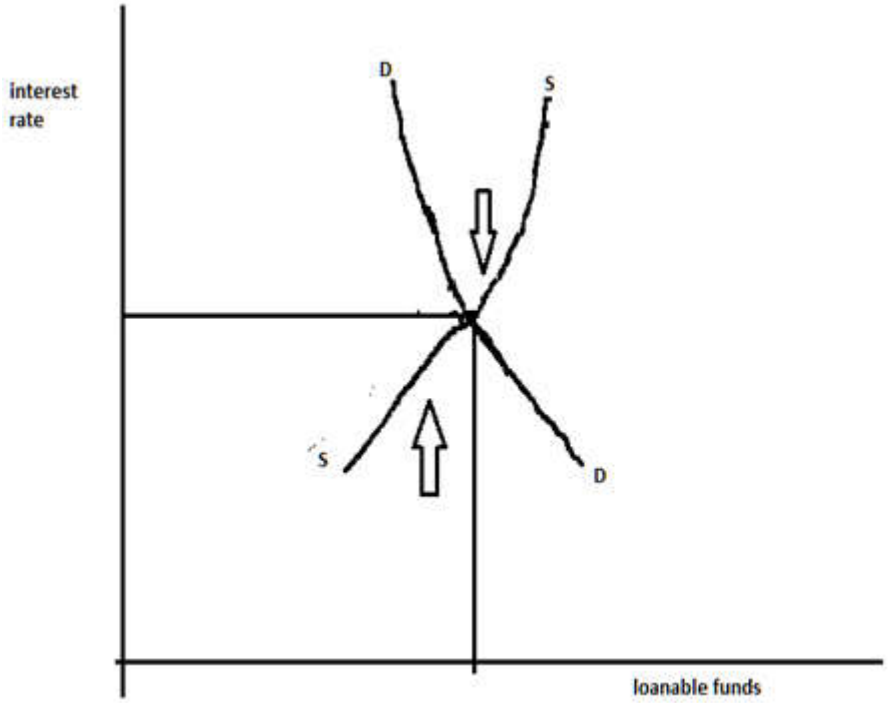
Also called the classical theory of interest, was developed at the time of classical economists like Adam Smith, David Ricardo and Thomas Malthus, who held the view that economic activities were guided by some kind of invisible hand i.e. through the self interest motive and the price mechanism, and that Government interference was unnecessary and should be kept at minimum.



They therefore explained the rate of interest in terms of the **demand** for money and **supply of loanable funds.** The demand comes from firms wishing to invest. The lower the rate of interest the larger the number of projects which will be profitable. Thus, the demand curve for funds will slope downwards from left to right.

The supply of loanable funds comes from savings. If people are to save they will require a reward-interest – to compensate them for forgoing present consumption. If the interest rate is high, people will be encouraged to save and lend. If the interest rate is low, people will bediscouraged from saving and lending. Hence, the supply curve of loanable funds slopes upwards.

The market rate of interest is therefore determined where the demand for and supply of **loanable funds** are equal. Geometrically this corresponds to the point of intersection betweenthe supply curve and the demand curve for loanable funds.



i is the equilibrium market rate of interest and L the equilibrium level of **loanable** funds.
Above i, there is excess of supply over demand, and interest rates will be forced downwards.
Below i there is excess of demand over supply and interest rates will be forced upwards.

Changes in demand or supply will cause shifts in the relevant curves and changes in the equilibrium rate of interest.
Although this theory has a certain amount of **validity,** it has been criticized on the following grounds:

1. It assumes that money is borrowed entirely for the purchase of capital assets. This is not true because money can be borrowed for the purchase of consumer goods (e.g. cars or houses)
2. It assumes that the decision to borrow and invest depends entirely on interest. This is not the case, for business expectations play more important role in the decision to invest. Thus if business expectations are high, investors will borrow and invest, even if the rate of interest is high and if business expectations are low investors will not borrow and invest even if the rate of interest is low.
3. It assumes that the decision to save depends entirely on the rate of interest. This is not true for people can save for purposes other than earning interest, e.g. as precaution against expected future events like illness or in order to meet a certain target (this is called target savings) or simply out of habit.

It follows therefore, that when the market price of securities is high the market interest rate will low. Also if the market price of securities (high holders of securities) will sell them now and hold money. Hence the demand for money is high when the interest rate is low. On the other hand when the market price of securities is low, the market rate of interest will be high. Also if the market price of securities is low, it can be expected to rise. Hence people will buy securities at a low price, hoping to sell them at higher prices. In buying securities, people part with money. Hence the demand for money is low when interest rate is high. It follows, therefore, that the demand curve for money for the speculative motive slopes downwards as shown on the next page. It flattens out at the lower end because there must be a minimum rate of interest payable to the people to persuade them to part with money. This perfectly elastic part is called **LIQUIDITY TRAP.**

HARMONISATION OF FISCAL AND MONETARY POLICIES

This occurs through the following

1) The Partner States under take to adopt policy measures in accordance with an agreed macro-economic policy framework.
2) For the purposes of paragraph 1 of this Article, the Partner States undertake to:

   (a) remove all exchange restrict ions on imports and exports within the Community;
   (b) maintain free market determined exchange rates and enhance the levels of their international reserves;
   (c) adjust their fiscal policies and net domestic credit to the government to ensure monetary stability and the achievement of sustained economic growth;
   (d) liberalise their financial sectors by freeing and deregulating interest rates with a view to achieving positive real interest rates in order to promote savings for investment within the Community and to enhance competition and efficiency in their financial systems; and
   (e) harmonise their tax policies with a view to removing tax distortions in order to bring about a more efficient allocation of resources within the Community.

THE IS – LM MODEL

IS – LM analysis aims to find the level of income and rate of interest at which both the commodity market and money market will be in equilibrium.

There IS curve is locus of points representing all the different combinations of interest rates and income levels consistent with equilibrium in the goods or commodity market.

The IS curve is shown in the diagram below:

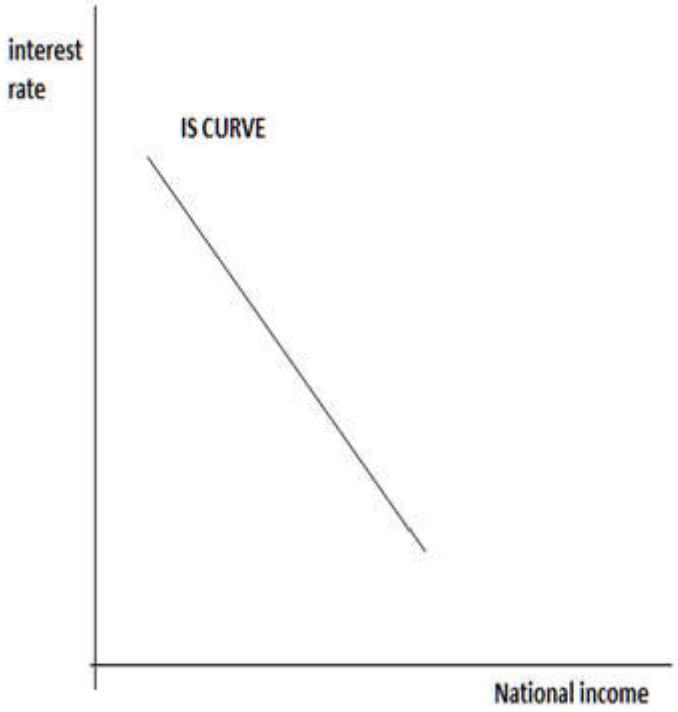


Figure: The IS curve

The IS curve is a linear function in the two variables Yand I.

The LM curve is a locus of points representing all the different combinations of interest rates and income levels consistent with equilibrium in the money market. The LM curve is shown in the following diagram.

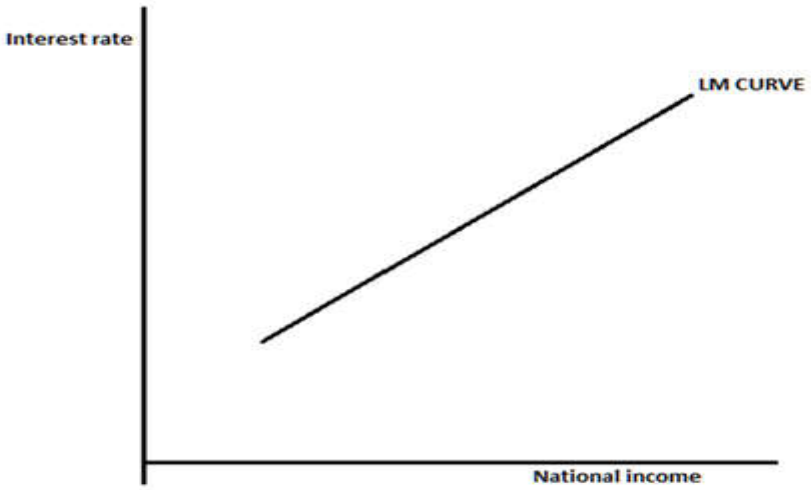


Figure: The LM Curve

The LM curve is also a function which is linear in the two variables I and Y.

IS – LM analysis aims at obtaining simultaneous equilibrium in both the commodity and the money markets. Graphically, this situation can be represented as follows:

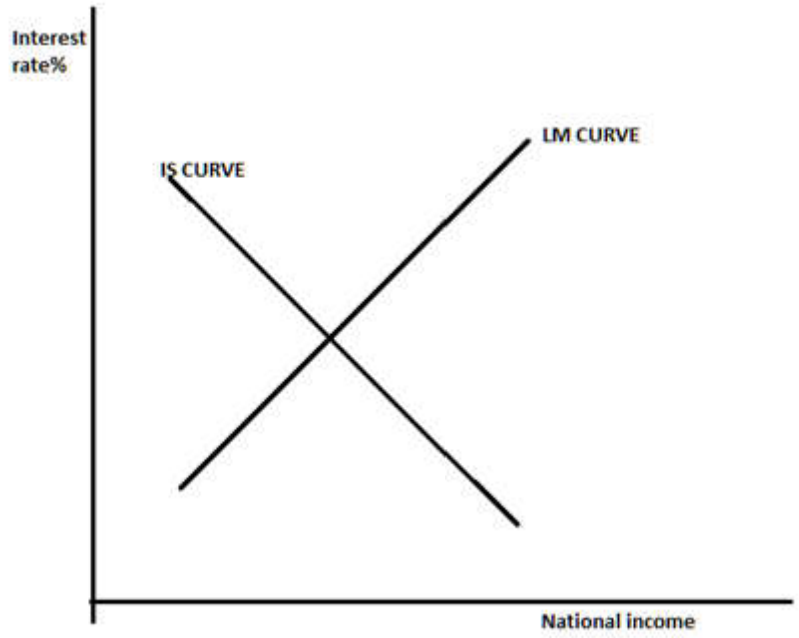


Figure: Equilibrium in both the commodity and money markets.

The equilibrium in the two markets is represented graphically by the intersection of the IS and LM curves.

The commodity market for a simple two – sector economy is in equilibrium when Y = C + I. The money market, on the other hand, is in equilibrium when the supply of money (Ms) equals the demand for money (Md). The demand for money is in turn made up of the transaction – precautionary demand (MDT) and speculative demand for money (MDS).

Numerical example.

Assume that:

C = 178 + 0.6 Y

I = 240 – 300 i

MS = 550

MDT = 0.2 Y

MDS = 480 – 500i

Commodity market equilibrium (IS) exists where;

$Y = C + I$

$= 178 + 0.6Y + 240 – 300i$

$Y – 0.6Y = 418 – 300 i$

$0.4Y + 300 i – 418 = 0$

Monetary equilibrium (LM) exists where

$MS = MDT + MDS$

$550 = 0.2y + 480 – 500 i$

$0.2Y – 500 i – 70 = 0$

Simultaneous equilibrium in both markets requires that :

$0.4Y + 300 i – 418 = 0$ …… (i)

$0.2Y + 500i – 70 = 0$ …… (ii)

Multiply (i) by 5 and (ii) by 3 in order to eliminate i.

$2 Y + 1500 i – 2,090 = 0$

$0.6Y + 1500 i - 210 = 0$

$2.6Y$ _= 2,300

$Y =2,300$

$= 885$

_substitute $Y = 885$ into (i) or (ii)

$0.4 (885) + 300i – 418 = 0$

$354 + 300i – 418 = 0$

$300 i = 64$

$i = 0.21$

PARTIAL EQUILIBRIUM AND GENERAL EQUILIBRIUM

Partial equilibrium -focuses on the effects that one change, such as a shift in consumer preferences, has on a one or two markets, disregarding the wider ripples. A partial equilibrium analysis makes sense when the wider ripples are unlikely to be earth-shakingly large. To be precise, when cross-price, cross-income, cross-everything elasticities are at or near zero. For example, with a little imagination you may find a reason why a change in American preference towards low fat food might have in some way or another a effect on the Italian shoe industry or the demand for Dutch tulips. Who knows? Producers of low fat food may like Italian shoes more than the producers of

high fat foods. And Italian shoe makers may use their extra earnings to buy more tulips. Ripples can move in funny and unexpected ways. However, the effects on Italian shoes and Dutch tulips will be so small that they are better ignored to keep the analysis simple. Partial equilibrium theory, therefore, asks you to limit the scope of your analysis, reasonably.

Partial equilibrium theory considers the effects of a change on one or a few markets only.

But some problems require a general equilibrium theory. If Saddam Hussein, the deposed ruler of Iraq, had succeeded some years ago in grabbing the oil riches of the Persian Gulf, then the price of oil would have gone up dramatically. In this case the ripple effects would be big and far-reaching. After all, the Persian Gulf counts for enough of production to significantly affect the price of oil. And the price of oil counts for enough in the economy of the world that a rise in its price has big ripple effects. Not ripples; waves. For one, you will pay more at the pump. You also may stay forsake a trip home for Thanksgivings because of a steep rise in the price of a ticket. Your rent may go up as well because a rise in heating expenses. Consequently, an analysis of the effects of Hussein's putative action would have to consider many markets and determine to which new equilibrium all these markets would move. We might even go so far and aspire to include all possible markets to determine the new equilibrium all around, that is, the general equilibrium. In 2005-2006 the price of oil did go up abundantly, rippling across distant foreign markets, so much so that politicians in remote areas of the United States began talking (irrationally, we might add) about price controls.

General equilibrium economics considers all markets in an economy to account for all possible effects of a change.

A general equilibrium model of the economy shows the economy as an interconnected system of numerous markets, in accordance with the vision of one of the great economists, Leon Walras (1834-1910). The economy hangs together as a whole, said Walras. In contemporary economics his vision lives on in the so-called Walrasian general equilibrium theory. The technical workshop below uses diagrams to show you how that theory works.

QUESTIONS:

1. State clearly the meaning of Liquidity Preference as applied to an individual and acommercial bank.
2. How does a commercial bank reconcile the need for security, liquidity and profitability indistribution of the assets?

CHAPTER 9

INFLATION AND UNEMPLOYMENT

DEFINITION AND TYPES OF INFLATION

Meaning

The word inflation has at least four meanings.

- A persistent rise in the general level of prices, or alternatively a persistent falls in the value of money.
- Any increase in the quantity of money, however small can be regarded as inflationary.
- Inflation can also be regarded to refer to a situation where the volume of purchasing power is persistently running ahead of the output of goods and services, so that there is a continuous tendency of prices – both of commodities and factors of production – to rise because the supply of goods and services and factors of production fails to keep pace with demand for them. This type of inflation can, therefore, be described as persistent/creeping inflation.
- Finally inflation can also mean runaway inflation or hyper-inflation or galloping inflation where a persistent inflation gets out of control and the value of money declines rapidly to a tiny fraction of its former value and eventually to almost nothing, so that a new currency has to be adopted.

Demand-pull inflation is when aggregate demand exceeds the value of output (measured in constant prices) at full employment. The excess demand of goods and services cannot be met in real terms and therefore is met by rises in the prices of goods. Demand-pull inflation could be caused by:

Causes of Inflation

At present three main explanations are put forward: cost-push, demand-pull, and monetary.

Cost-push inflation occurs when he increasing costs of production push up the general level of prices. It is therefore inflation from the supply side of the economy. It occurs as a result of increase in:

a. Wage costs: Powerful trade unions will demand higher wages without corresponding increases in productivity. Since wages are usually one of the most important costs of production, this has an important effect upon the price. The employers generally accede to these demands and pass the increased wage cost on to the consumer in terms of higher prices.

b. Import prices: A country carrying out foreign trade with another is likely to import the inflation of that country in the form of intermediate goods.

c. Exchange rates: It is estimated that each time a country devalues it's currency by 4 per cent, this will lead to a rise of 1 per cent in domestic inflation.

d. Mark-up pricing: Many large firms fix their prices on unit cost plus profit basis. This makes prices more sensitive to supply than to demand influences and can mean that they tend to go up automatically with rising costs, whatever the state of economy.

e. Structural rigidity: The theory assumes that resources do not move quickly from one use to another and that wages and prices can increase but not decrease. Given these conditions, when patterns of demand and cost change, real adjustments occur only very slowly.

Shortages appear in potentially expanding sectors and prices rise because slow movement of resources prevent the sector and prices rise because of slow sectors keep factors of production on part-time employment or even full time employment because mobility is low in the economy. Because their prices are rigid, there is no deflation in these potentially contracting sectors. Thus the process of expanding sectors leads to price rises, and prices in contracting sectors stay the same. On average, therefore, prices rise.

f. Expectational theory: This depends on a general set of expectations of price and wage increases. Such expectations may have been generated by a continuing demand inflation.

Wage contracts may be made on a cost plus basis.

- **Increases in general level of demand of goods and services**. A rise in aggregate demand in a situation of nearly full employment will create excess demand in may individual markets, and prices will be bid upward. The rise in demand for goods and services will cause a rise in demand for factors and their prices will be bid upward as will. Thus, inflation in the pries of both consumer goods and factors of production is caused by a rise in aggregate demand.
- **General shortage of goods and services**. If there is a general shortage of commodities e.g. in times of disasters like earthquakes, floods or wars, the general level of prices will rise because of excess demand over supply.
- **Government spending:** Hyper-inflation certainly rises as a result of government action. Government may finance spending though budget deficits; either resorting to the printing press to print money with which to pay bills or, what amounts to the same thing, borrowing from the central bank for this purpose. Many economists believe that all inflation is caused by increases in money supply.

Monetarist economists believe that "inflation is always and everywhere a monetary phenomenon in the sense that it can only be produced by a more rapid increase in the quantity of money than in output" as Friedman wrote in 1970.

The monetarist's theory is based upon the identity:

M x V = P x T

And thus this was turned into a theory by assuming that V and T are constant. Thus, we would

THE EFFECTS OF INFLATION AND HOW TO CONTROL INFLATION

Inflation has different effects on different economic activities on both micro and macro levels.

Some of these problems are considered below:

i. During inflation money loses value. This implies that in the lending-borrowing process, lenders will be losing and borrowers will be gaining, at least to the extent of the time value of money. Cost of capital/credit will increase and the demand for funds is discouraged in the economy, limiting the availability of investable funds. Moreover, the limited funds available will be invested in physical facilities which appreciate in value over time. It's also impossible the diversion of investment portfolio into speculative activities away from directly productive ventures.
ii. Other things constant, during inflation more disposable incomes will be allocated to consumption since prices will be high and real incomes very low. In this way, marginal propensity to save will decline culminating in inadequate saved funds. This hinders the process of capital formation and thus the economic prosperity to the country.
iii. The effects of inflation on economic growth have inconclusive evidence. Some scholars and researchers have contended that inflation leads to an expansion in economic growth while others associate inflation to economic stagnation. Such kind of inflation if mild, will act as an incentive to producers to expand output and if the reverse happened, there will be a fall in production resulting into stagflation i.e. a situation where there is inflation and stagnation in production activities.
iv. When inflation imply that domestic commodity prices are higher than the world market prices, a country's exports fall while the import bill expands. This basically due to the increased domestic demand for imports much more than the foreign demand for domestic produced goods (exports). The effect is a deficit in international trade account causing balance of payment problems for the country that suffers inflation.
v. During inflation, income distribution in a country worsens. The low income strata get more affected especially where the basic line sustaining commodities' prices rise persistently. In fact such persistence accelerates the loss of purchasing power and the vicious cycle of poverty.
vi. Increased production-It is argued that if inflation is of the demand-pull type, this can lead to increased production if the high demand stimulates further investment. This is a positive effect of inflation as it will lead to increased employment.
vii. Political instability-When inflation progresses to hyper-inflation, the unit of currency is destroyed and with it basis of a free contractual society.
viii. Inflation and Unemployment-For many years, it was believed that there was a trade-off between inflation and unemployment i.e. reducing inflation would cause more unemployment and vice versa.

Measures to control inflation

An inflationary situation can effectively be addressed/tackled if the cause is first and foremost identified. Governments have basically three policy measures to adopt in order to control inflation, namely:

Fiscal Policy: This policy is based on demand management in terms of either raising or lowering the level of aggregate demand. The government could attempt to influence one of the components C + I + G (X – M) of the aggregate demand by reducing government expenditure and raising taxes. This policy is effective only against demand-pull inflation.

Monetary Policy: For many years monetary policy was seen as only supplementary to fiscal policy. Neo-Keynesians contend that monetary policy works through the rate of interest while monetarists' viewpoint is to control money supply through setting targets for monetary growth.

This could be achieved through what is known s medium term financial strategy (MTFs) which aims to gradually reducing the growth of money in line with the growth of real economy – the use of monetary policy instruments such as the bank rate, open market operations (OMO) and variable reserve requirement (cash & liquidity ratios).

Direct Intervention: Prices and incomes policy: Direct intervention involves fixing wages and prices to ensure there is almost equal rise in wages and other incomes alongside the improvements in productivity in the economy. Nevertheless, these policies become successful for a short period as they end up storing trouble further, once relaxed will lead to frequent price rises and wage fluctuations. Like direct intervention, fiscal and monetary policies may fail if they are relied upon as the only method of controlling inflation, and what is needed is a combination of policies.

QUESTIONS:

1. a) Outline the functions of taxation .
   b) What are the possible disadvantages of a progressive income tax system?
2. State and explain Adam Smith's canons of taxation. Give local example as appropriate in each.
3. a) What is meant by inflation
   b) What are the major causes of inflation?
   c) Explain the economic problems that arise from a high rate of inflation.

DEFINITION OF UNEMPLOYMENT

Total number of able men and women of working age seeking paid work. Unemployment statistics vary according to how unemployment is defined and who is deemed to be part of the workforce. Traditional methods for collecting unemployment data are based, typically, on sampling or the number of unemployment benefit requests. International labor organization (ILO) computes unemployment on the basis of number of people who have looked for employment in the last four weeks and are available to start work within two weeks, plus those who are waiting to start working in a job already obtained.

Types and Causes of Unemployment

i. **Transitional unemployment**: Transitional unemployment is that situation which prevails due to some temporary reasons. The main reason for this type of unemployment are:

- **Turnover unemployment:** Some individuals leave their present jobs and make efforts to secure better ones and in this way, they remain unemployed for some time.
- **Casual unemployment**: Casual workers are employed for a specific job and when the job is competed, such workers become eventually unemployed. E.g. shipping or building construction workers.
- **Seasonal** unemployment**:** Some industries, for instance have seasonal demand and their produce is manufactured for a specific period of time (a specific period of the year). The workers of such industries remain unemployed for that time e.g. ice factories may remain closed during winter.

ii. **Structural unemployment:** Caused by structural changes such that there exist:

- **Cyclical unemployment**: During depression, prices are too low and profit margins remain distinctively low. In this case, investment decreases and unemployment increases.
- **Technological unemployment:** Due to inappropriate technology. Technology is not inappropriate per se but in relation to the environment in which it is applied. In most developing countries, most production structures tend to be labour saving (capital-intensive), which is not appropriate as these countries experience high labour supply. Capital – labour ratios tend to be high in these countries implying that less labour is absorbed compared to capital in production undertakings causing unemployment.
- **Industrial change:** The establishment of new industries decreases the demand for the products of existing industries e.g. the rapid increase in the demand for Japanese industrial products is one reason for greater unemployment in some European countries.
- **Keynesian unemployment:** According to Keynesian theory of income and employment, unemployment occurs due to lack of effective demand. If effective demand is less, production of goods and services will fall which will further result in the unemployment of labour. Another feature of Keynesian unemployment is that unemployment of labour is associated with unemployed capital such as plant and machinery which tend to be idle during depression.
- **Urban unemployment**: Due to availability of more facilities in urban areas, more and more people tend to move to these areas. The employment opportunities are not sufficient to absorb all those people who settled in the urban areas. This kind of unemployment is therefore due to rural-urban migration.
- **Disguised unemployment**: Situation where some people are employed apparently, but if they are withdrawn form this job, total production remains the same. In most developing countries this type of unemployment is estimated at 20 to 30% andmeasures should be taken to employ such people in other sectors of the economy.

iii **Insufficient Capital**: Shortage of capital is a hindrance in the establishment of more industries and other productive installations, and due to this reason, more employment opportunities are not created.

iv. **Nature of education system:** Education systems for most developing countries are white-collar oriented, yet the nature of productive capacities of these economies are not sufficiently supportive. Moreover, inadequate education and training facilities render(s) most people unable to secure those job opportunities that require high skills and specialized training.

v. **Rapidly increasing population:** The rate of growth in population exceeds the amount of job opportunities that the economy can generate.

Thus in summary, of the causes of unemployment in developing countries can be said to include:

- Rapidly increasing population
- Inappropriate technology
- Insufficient capital base
- Demand deficiency/structural changes
- Presence of expatriates
- Education Systems – white-collar orientation
- Rural-urban migration
- One person for more than one job
- Corruption and general mismanagement
- Inadequate knowledge on market opportunities

Cost of Unemployment

Unemployment is a problem because it imposes costs on society and the individual. The cost of unemployment to a nation can be categorized under three headings: the social costs, the cost to the exchequer and the economic cost.

The Social Cost of Unemployment

iii. For the individual, there is the demoralizing effect which can be devastating particularly when they are old. This is because as some job seekers become more and more pessimistic about their chances of finding a job, so their motivation is reduced and their changes of succeeding in finding jobs become even more remote.
iv. Many of the longer-term unemployed become bored, idle, lose their friends and suffer from depression
v. There is also evidence of increased family tension leading in some cases to violence, infidelity, divorce and family breakups.
vi. Unemployment may also lead to homelessness, as in some circumstances building societies may foreclose on a mortgage if the repayments are not kept up.
vii. Long-term unemployment may also lead to vandalism, football, hooliganism and increases in the crime rate and insecurity in general.

The cost to the exchequer (Ministry of Finance)

There is increasing dependency ratio on the few who are employed in the form of:

- The loss of tax revenues which would otherwise have been received: This consists mostly of lost income tax but also includes lost indirect taxes because of the reduction in Spending.
- The loss of national insurance contributions which would otherwise have been received.

The economic cost

Unemployment represents a terrible waste of resources and means that the economy is producing a lower rate of output than it could do if there were full employment. This leads to an output gap or the loss of the output of goods and services as a result of unemployment.

CONTROL MEASURES FOR UNEMPLOYMENT

The measures appropriate as remedies for unemployment will clearly depend on the type and cause of unemployment. Broadly they can be divided into:

- Demand management or demand side policies
- Supply side policies.

Demand management policies

These policies are intended to increase aggregate demand and, therefore the equilibrium level of national income. They are sometimes called fiscal and monetary policies. The principal policy instruments are:

- Supporting declining industries with public funds
- Instituting proper demand management policies that increase aggregate demand including exploiting foreign and regional export markets. This can be done by increasing government expenditure, cutting taxation or expanding the money supply.
- Promoting the location of new industries in rural areas which will require an improvement of rural infrastructure.

Supply-side policies

Supply-side policies are intended to increase the economy's potential rate of output by increasing the supply of factor inputs, such as labour inputs and capital inputs, and by increasing productivity. They include:

- Increasing information dissemination on market opportunities.
- Reversing rural-urban migration by making rural areas more attractive and capable of providing jobs. This particularly is the case in developing countries where rural-non-farm opportunities offer the longest employment opportunities.
- Changing attitude towards work i.e. eliminating the white-collar mentality and creating positive attitudes towards agriculture and other technical vocational jobs.
- Provision of retraining schemes to keep workers who want to acquire new skills to improve their mobility.

- Assistance with family relocation to reduce structural unemployment. This is done by giving recreational facilities, schools, and the quality of life in general in other parts of the country even the provision of financial help to cover moving costs and assist with home purchase.
- Special employment assistance for teenagers many of them leave school without having studied work-related subjects and with little or no work experience.
- Subsidies to firms which reduce working hours rather than the size of the workforce.
- Reducing welfare payments to the unemployed. There are many economists who believe that welfare payments have artificially increased the level of unemployment.
- Reduction of employee and trade union rights.

RELATIONSHIP BETWEEN INFLATION AND UNEMPLOYMENT

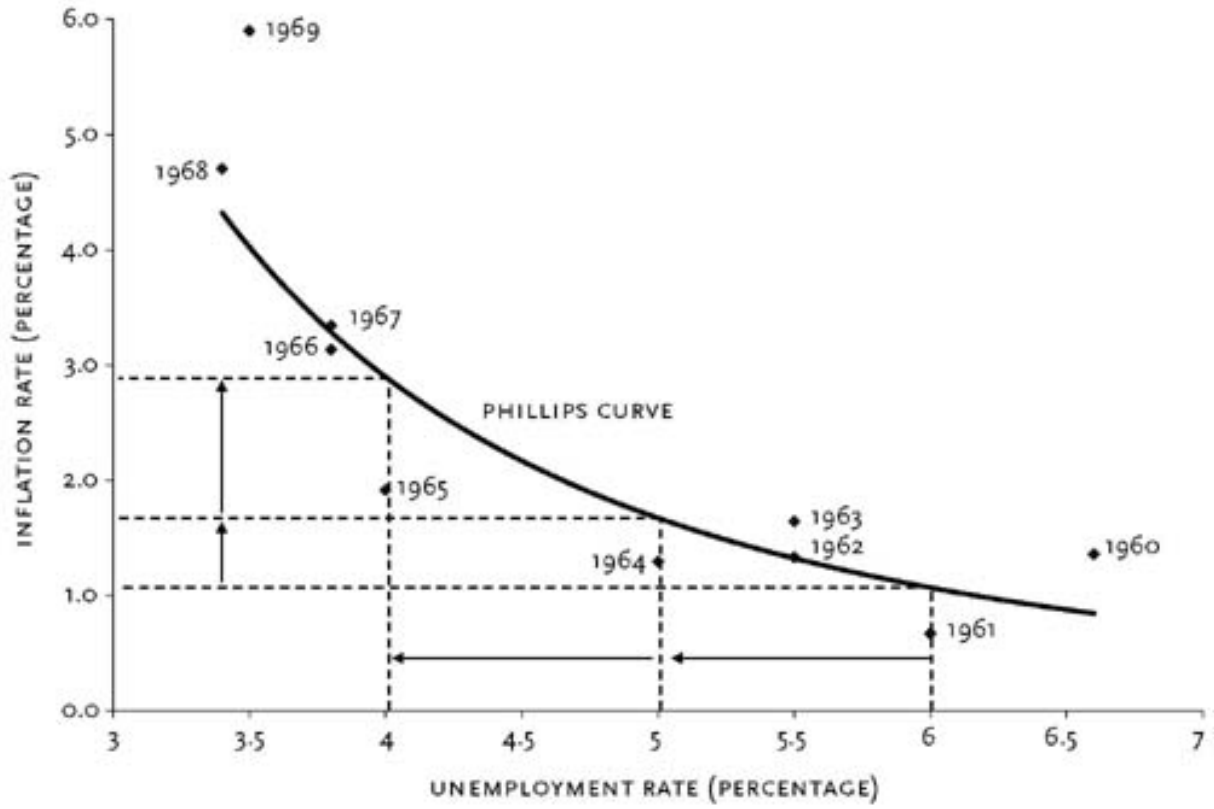
The Phillips curve represents the relationship between the rate of inflation and the unemployment rate. Although he had precursors, A. W. H. Phillips's study of wage inflation and unemployment in the United Kingdom from 1861 to 1957 is a milestone in the development of macroeconomics. Phillips found a consistent inverse relationship: when unemployment was high, wages increased slowly; when unemployment was low, wages rose rapidly.

Phillips conjectured that the lower the unemployment rate, the tighter the labor market and, therefore, the faster firms must raise wages to attract scarce labor. At higher rates of unemployment, the pressure abated. Phillips's "curve" represented the average relationship between unemployment and wage behavior over the business cycle. It showed the rate of wage inflation that would result if a particular level of unemployment persisted for some time.

Economists soon estimated Phillips curves for most developed economies. Most related general price inflation, rather than wage inflation, to unemployment. Of course, the prices a company charges are closely connected to the wages it pays. Figure 1 shows a typical Phillips curve fitted to data for the United States from 1961 to 1969. The close fit between the estimated curve and the data encouraged many economists, following the lead of Paul Samuelson and Robert Solow, to treat the Phillips curve as a sort of menu of policy options. For example, with an unemployment rate of 6 percent, the government might stimulate the economy to lower unemployment to 5 percent. Figure 1 indicates that the cost, in terms of higher inflation, would be a little more than half a percentage point. But if the government initially faced lower rates of unemployment, the costs would be considerably higher: a reduction in unemployment from 5 to 4 percent would imply more than twice as big an increase in the rate of inflation—about one and a quarter percentage points.

At the height of the Phillips curve's popularity as a guide to policy, Edmund Phelps and Milton Friedman independently challenged its theoretical underpinnings. They argued that well-informed, rational employers and workers would pay attention only to real wages—the inflation-adjusted purchasing power of money wages. In their view, real wages would adjust to make the supply of labor equal to the demand for labor, and the unemployment rate would then stand at a level uniquely associated with that real wage—the "natural rate" of unemployment.

Figure 1 The Phillips Curve, 1961–1969



Note: Inflation based on the Consumer Price Index.

Both Friedman and Philps argued that the government could not permanently trade higher inflation for lower unemployment. Imagine that unemployment is at the natural rate. The real wage is constant: workers who expect a given rate of price inflation insist that their wages increase at the same rate to prevent the erosion of their purchasing power. Now, imagine that the government uses expansionary monetary or fiscal policy in an attempt to lower unemployment below its natural rate. The resulting increase in demand encourages firms to raise their prices faster than workers had anticipated. With higher revenues, firms are willing to employ more workers at the old wage rates and even to raise those rates somewhat. For a short time, workers suffer from what economists call money illusion: they see that their money wages have risen and willingly supply more labor. Thus, the unemployment rate falls. They do not realize right away that their purchasing power has fallen because prices have risen more rapidly than they expected. But, over time, as workers come to anticipate higher rates of price inflation, they supply less labor and insist on increases in wages that keep up with inflation. The real wage is restored to its old level, and the unemployment rate returns to the natural rate. But the price inflation and wage inflation brought on by expansionary policies continue at the new, higher rates.

Friedman's and Phelps's analyses provide a distinction between the "short-run" and "long-run" Phillips curves. So long as the average rate of inflation remains fairly constant, as it did in the 1960s, inflation and unemployment will be inversely related. But if the average rate of inflation changes, as it will when policymakers persistently try to push unemployment below the natural rate, after a period of adjustment, unemployment will return to the natural rate. That is, once workers' expectations of price inflation have had time to adjust, the natural rate of unemployment is compatible with any rate of inflation. The long-run Phillips curve could be shown on Figure 1 as a

vertical line above the natural rate. The original curve would then apply only to brief, transitional periods and would shift with any persistent change in the average rate of inflation. These long-run and short-run relations can be combined in a single "expectations-augmented" Phillips curve. The more quickly workers' expectations of price inflation adapt to changes in the actual rate of inflation, the more quickly unemployment will return to the natural rate, and the less successful the government will be in reducing unemployment through monetary and fiscal policies.

The 1970s provided striking confirmation of Friedman's and Phelps's fundamental point. Contrary to the original Phillips curve, when the average inflation rate rose from about 2.5 percent in the 1960s to about 7 percent in the 1970s, the unemployment rate not only did not fall, it actually rose from about 4 percent to above 6 percent.

Most economists now accept a central tenet of both Friedman's and Phelps's analyses: there is some rate of unemployment that, if maintained, would be compatible with a stable rate of inflation. Many, however, call this the "nonaccelerating inflation rate of unemployment" (NAIRU) because, unlike the term "natural rate," NAIRU does not suggest that an unemployment rate is socially optimal, unchanging, or impervious to policy.

QUESTIONS:

1. What are likely to be the effects of increased technological change on employment?

2. Why should demographic analyses and information be of interest to Kenyan planners?

Discuss the problems and opportunities which rapid population growth poses for the economy of Kenya.

CHAPTER 10

INTERNATIONAL TRADE AND FINANCE

DEFINITION OF INTERNATIONAL TRADE

It is the exchange of goods and services between one country and another.

International Trade can be in goods, termed visibles or in services, termed invisibles e.g. trade in services such as tourism, shipping and insurance.

Reasons for the Development of International Trade

a) Some goods **cannot be produced** by the country at all. The country may simply not possess the raw materials that it requires; thus it has to buy them from other countries. The same would apply to many foodstuffs, where a different climate prevents their cultivation.
b) Some goods cannot be produced as **efficiently** as elsewhere. In many cases, a country could produce a particular good, but it would be much less efficient at it than another country.
c) It may be better for the country to give up the production of a good (and import it instead) in order to specialize in something else. This is in line with the **principle of comparative advantage.**
d) In a free market economy, a consumer is free to choose which goods to buy. A foreign good may be more to his or her liking. This is in line with the **principle of competitive forces** and the exercise **of choice.**
e) **Shortages:** At a time of high domestic demand for a particular good, production may not meet this demand. In such a situation, imports tend to be bought to overcome the shortage.

THEORY OF ABSOLUTE ADVANTAGE AND COMPARATIVE ADVANTAGE

THEORY OF ABSOLUTE ADVANTAGE

International trade allows a country to specialize in the production of commodities where it more efficient than other countries. For instance, if we take a situation in which each country in a simple two country model has an absolute advantage in producing either fruits or beef but is able to produce the other commodity only if required (for simplicity we assume constant returns to scale and full utilization of resources). Suppose that each country has equal resources and devotes half its limited resources to citrus fruit and half to beef and the production totals are:

	Units of Citrus fruits	**Units of Beef**
Country X	10	5
Country Y	**5**	**10**
World total	**15**	**15**

The relative or comparative costs of citrus production is lower in country X than in country Y, but the situation is reserved in the case of beef production. Country X has an absolute advantage in citrus fruit production and Y has an absolute advantage in beef production. If each country

specializes in the production of the commodity in which it is most efficient and possesses absolute advantage, we get:

	Units of Citrus fruits	Units of Beef
Country X	20	0
Country Y	0	20
World total 20 20		

The gains from trade are obvious with five units more of fruit and five more of beef – provided we assume that transport costs are not so enormous as to rule out gains made.

THEORY OF COMPARATIVE ADVANTAGE

In his theory put forward in a book published in 1817, David Ricardo argued that what was needed for two countries to engage in international trade was comparative advantage.

He believed that 2 countries can still gain, even if one country is more productive then the other in all lines of production. Using the Labour Theory Value, Ricardo's contribution was to show that a sufficient basis for trade was a difference, not in absolute costs. He illustrated his theory with 2 countries and two commodities, I and II and A and B respectively.

COUNTRY	**COST OF PRODUCING I UNIT (In Manhours)**	
	A	B
I	8	9
II	12	10

We can observe that country I has complete absolute advantage in the production of both commodities since it can produce them with a lower level of resources. Country I is more efficient than country II.
Ricardo believed that even then there could still be a basis for trade, so long as country II is not equally less productive, in all lines of production. It still pays both countries to trade. What is important is the Comparative Advantage. A country is said to have comparative advantage in the production of a commodity if it can produce at relatively lower opportunity costs than another country. (The Law of **Comparative Advantage** states that a nation should specialize in producing and exporting those commodities which it can produce at **relatively** lower costs, and that it should import those goods in which it is a relatively high cost producer). Ricardo demonstrated this by introducing the concept of **Opportunity Cost.**

The opportunity Cost of good A is the amount of other goods which have to be given up in order to produce one unit of the good. To produce a unit of good A in country I, you need 8 man hours and 9 man hours to produce good B in the same country. It is thus more expensive to produce good B then A. The opportunity costs of producing a unit of A is equivalent to 8/9 units of good B. One unit of B is equal to 9/8 units of A.

In country II, one unit of A is equal to 12/10 of B and one unit of B = 10/12 units of A.

Therefore he felt that: -

Opportunity cost of producing one unit of:

COUNTRY	A	B
I	9/8 (1.25) B	8/9 (0.89) A
II	10/12 (0.83) B	12/10 (1.2) A

B is cheaper to produce in country II in terms of resources as opposed to producing it in country I. The opportunity costs are thus lower in country II than in country I. Consider commodity A valued in terms of B. A cheaper in country I than country II.
A country has comparative advantage in producing commodity if the opportunity cost of producing it is lower than in other counties. Country I has a lower opportunity cost in producing A than B and II has a lower opportunity cost in the production of B than A. In country I, they should specialize in the production of A and Import B.

Limitations of Comparative advantage

This doctrine is valid in the case of a classical competitive market characterized by a large number of informed buyers and sellers and homogenous products in each market, with world market places serving as efficiency determinants for global allocation of resources to their most suitable uses. Unfortunately, world markets and their prices are largely inefficient showing influences of trade barriers, discrimination and market distortions. Individual countries systematically aim at maximizing their potential gains from trade rather than with optimizing the allocation of world resources.

By pursuing gains from trade in the short run young nations may jeopardize long term development prospects because:

i. It is important to protect infant industries to acquire new skills, technology and home markets that are necessary in the early years of industrial development;
ii. Concentrating on short term comparative advantage may lead to internalizing wrong externalities e.g. promoting use of illiterate peasants and primary sector production;
iii. Long term movements in commodity terms of trade disfavour primary commodities as their prices rise more slowly than those of industrial manufactures (income elasticity of demand for primary commodities is lower than for manufactures and as world incomes rise demand for the latter rises more rapidly affecting their relative world prices).

WORLD TRADE ORGANISTION (WTO) AND CONCERNS OF DEVELOPING COUNTRIES

There are many opportunities for small businesses in developing countries. Exports to developing countries may be eligible for aid finance and imports may benefit from preferential UK and EU customs duties under the Generalised System of Preferences (GSP). Small businesses can also trade with developing countries as subcontractors to larger businesses.

The World Trade Organization (WTO) sets a global trading framework for its 153 member countries, two-thirds of which are developing countries. The WTO's open market policies have led to changes in the EU's main trade and aid agreement with 79 developing countries - the Cotonou

Agreement. The WTO has also recognised the need to make greater provision for developing countries and also for small business.

This guide covers the main WTO agreements and special measures affecting developing countries - the changes to the Cotonou Agreement - to aid finance and developing country debt, and explains how the GSP works.

How developing countries work within the WTO

The WTO has reached several agreements to reduce and eliminate barriers to global trade, including:

- the General Agreement on Tariffs and Trade (GATT - goods)
- the General Agreement on Trade in Services (GATS)
- the Trade-Related Aspects of Intellectual Property Rights (TRIPS)
- dispute resolution between member governments
- specific product/service and exporter/importer agreements

The WTO also allows developing countries and 'least developed countries' (LDCs) to adapt more slowly to free trade.

There are 50 LDCs as defined by the UN - 32 of which are WTO members. These include self-defined developing countries, and two of the world's largest economies - China and India.

These developing countries have successfully agreed certain changes in WTO agreements, including:

- a programme of Technical Assistance and Capacity Building - for which the UK has pledged £45 million
- a change in the TRIPS rules on patented medicines to allow developing countries to use cheaper medicines under certain circumstances

There are also other changes being considered by the WTO that would benefit developing countries, such as:

- special measures for LDCs and small economies
- changes to the relationship between trade, debt and finance
- the possibility of technology transfers
- the relationship between patents and development

There are also some issues which have not been resolved due to disagreements between developed and developing countries. These issues include:

- agricultural commodities and subsistence farming
- trade, debt and finance
- the lack of safeguard mechanisms for developing countries during an economic crisis

WTO agreements and their changes have created a vast range of new business opportunities. Rules preventing special help for small business in public procurement - including development aid contracts - have been effectively removed in the EU and other developed countries, allowing many opportunities in these markets.

PROTECTION OF INTERNATINAL TRADE

Despite the arguments of the "classical" theory of free trade, the twentieth century has seen the gradual movement away from free trade, with governments increasingly imposing restrictions on trade and capital flows. All have adopted, to varying extents, various forms of restrictions to protect some of their industries or agriculture.

Reasons for Protection

Cheap Labour

It is often argued that the economy must be protected from imports which are produced with cheap, or 'sweated", labour. Some people argue that buying foreign imports from low wage countries amounts not only to unfair competition, but continues to encourage the exploitation

of cheap labour in those countries as well as undermining the standard of living of those in high wage economies.

Infant Industry Argument

Advocates of this maintain that if an industry is just developing, with a good chance of success once it is established and reaping economies of sale, then is it necessary to protect it from competition temporarily until it reaches levels of producdton and cost which allow it to compete with established industries elsewhere, until it can "stand on its own feet". The argument is most commonly used to justify the high level of protection that surrounds the manufacturing industry in developing countries, as they attempt to replace foreign goods with those made in their own country ("import substitution").

Structural Unemployment

The decline of the highly localized industry due to international trade causes great problems of regional (structural) unemployment. If it would take a long time to re-locate the labour to other jobs, then this can put the government, under considerable political and humanitarian pressure, to restrict the imports that are causing the industry to decline.

Dumping

If goods are sold on a foreign market below their cost of production this is referred to as dumping. This may be undertaken either by a foreign monopolist, using high profits at home to subsidize exports for political or strategic reasons. Countries in which such products are "dumped" feel justified in protecting themselves. This is because dumping could result in the elimination of the home industry, and the country then becomes dependent on foreign goods which are not as cheap as they had appeared.

Balance of Payments

Perhaps the most immediate reason for bringing in protection is a balance of payment deficit.

If a country had a persistent deficit in its balance of payments, it is unlikely to be able to finance

International Trade and Finance

These deficits from its limited reserves. If therefore becomes necessary for it adopt some form of restriction on imports (**e.g. tariffs, quotas, foreign exchange restrictions**) or some means of boosting its exports (**e.g. export subsidies**).

Danger of over-specialising

A country may feel that in its long-term interests it should not be too specialized.

A country may not wish to abandon production of certain key commodities even though the foreign product is more competitive, because it is then too dependent on imports of that good.

In the future, its price or supplies may diminish. It is for this reason that countries wish to remain largely self-sufficient in food.

An exporting country may not wish to become overspecialized in a particular product. Such over specialization may make sense now, but in the future, demand may fall and the country will suffer disproportionally. It is for this reason that many developing countries choose not to rely solely on their comparative advantage; they wish to diversify into other goods as an: insurance policy".

Strategic Reasons

For political or strategic reasons, a country may not wish to be dependent upon imports and so may protect a home industry even if it is inefficient. Many countries maintain industries for strategic reasons. The steel industry, energy industries, shipping, agriculture and others have used this strategic defence argument.

Bargaining

Even when a country can see no economic benefit in protection, it may find it useful to have tariffs and restrictions bargaining gambits in negotiating better terms with other nations.

Ways of Restricting International Trade

The most common means of restricting international trade is through import restrictions. The main forms are:

Tariffs

This is a tax on each unit imported. The effect of the tax is to raise the price of imported varieties of a product in relation to the domestically produced, so that the consumers are discouraged from buying foreign goods by means of the price mechanism. Such a tax may be **ad valorem**, representing a certain percentage of the import price, or **specific** that is, an absolute charge on the physical amount imported as, for example, five shillings a ton.

Quotas

The most direct way of offering protection is by limiting the physical quantity of a good which may be imported. This can be done by giving only a limited number of import licenses and fixing a quota on the total amount which may be brought in during the period. The quota may be imposed in terms of physical quantities or in terms of the value of foreign currency, so that a maximum of so many "shillings-worth" may be imported.

Foreign exchange restriction

Exchange controls work much the same way as physical controls. Foreign exchange is not made available for all desired imports. It can be severely restricted to whatever the government decides it wants to see imported. Alternatively, the exchange rate may be fixed in such a way as to "overprice" foreign currency (compared to what would have been the free market price), so that importers have to pay more for foreign currency (in terms of domestic currency). This makes all imports dearer and thus gives protection "across the board" to all domestic production for the home market.

Procurement policies by government

The government itself, together with state corporations, is an important purchaser of goods; in its "procurement" policies, therefore, it can either buy goods from the cheapest source, whether domestic or foreign, or it can give preference to domestic producers. This could amount to a substantial advantage, or protection.

Other restrictions

Governments can devise health or safety requirements that effectively discriminate against the foreign good. Also where the country has state import agencies they can choose not to import as much as their citizens would require. The government can also introduce cumbersome administrative procedures that make it almost impossible to import.

Arguments against protectionism

Most of the arguments for protectionism may be met with counter arguments, but underlying the economic arguments as opposed to the social, moral, political, strategic, etc, is the free trade argument.

Free trade argument

This, in brief, maintains that free trade allows all countries to specialize in producing commodities in which they have a comparative advantage. They can then produce and consume more of all commodities than would be available if specialization had not taken place.

By implication, any quotas, tariffs, other forms of import control and/or export subsidies all interfere with the overall advantages from free trade and so make less efficient use of world resources than would otherwise be the case.

Reduced output argument

It has been said that import controls will protect jobs initially, but not in the longer run. If we in the home country limit imports, then other countries will have less of our currency with which to buy our exports.

This will lead to a decline in sales and a loss of jobs in export industries. The overall effect is likely to be a redistribution of jobs from those industries in which the country has a comparative advantage to those in which it has a comparative disadvantage. The net result will be that total employment is unchanged but total output is reduced.

The infant industries seldom grow up

The infant industry argument is sometimes met with the claim that infant industries seldomadmit to growing up and cling to their protection when they are fully grown up. Most economists, however, appear to accept the infant industry argument as a valid case for protection provided it is temporary.

Gains from comparative advantage

The argument for protection against low wage foreign labour is partly a moral argument which is outside the scope of positive economics, but even the economic part of the argument that it will drag down the living standards of high wage economies can be shown to be invalid. It is true as noted above that the payment of low wages will allow a country to export their goods cheaply and so possibly undercut those of high wage countries. However, it must be noted that countries importing these cheap goods gain by virtue of their low cost in terms of the goods required to be exported in return. This again is another use of the comparative advantage argument.

No Validity in economics

The other arguments such as the need to avoid over dependence on particular industries and the defence argument are really strategic arguments which are valid in their own terms and for which economic science is largely irrelevant.

Retaliation

Advocates of free trade also believe that if one country imposes import restrictions, then those countries adversely affected will impose retaliatory restrictions on its exports, so it will not end up any better off. This could lead to a “beggar-my-neighbour” tariff war, which no one can benefit from, and which contracts the volume of world trade on which every country’s international prosperity depends.

Inflation

If key foreign goods are not free to enter the country (or cost more), this will raise their prices and worsen the rate of inflation in the country.

Inefficiency

It is argued that if home industries are sheltered from foreign competition there is no guarantee that they will become more efficient and be able to compete in world markets.

REGIONAL INTEGRATION ORGANISATIONS, COMMODITY AGREEMENTS AND THE RELEVANCE TO LESS DEVELOPED COUNTRIES(LDCs)

International Commodity Agreements (ICAS)

International Commodity Agreements (ICAS) represents attempts to modify the operation of the commodity markets so as to achieve various objectives such as price stabilization of price enhancement. Support for such intervention stems from apparent weaknesses in the operation of market forces in achieving an efficient allocation of resources, appropriate levels of privately held stocks in some commodities and an equitable distribution of income from their export as between exporters and importing countries.

ICAS are to be distinguished from producers' or exporters' cartels by the feature of consumer agreement to the scheme and representation on the governing body.

Most schemes have as their main objective to stabilize and/or increase the world price of commodity, producers' incomes, foreign exchange earnings of exporting countries and governing revenues from taxes on the commodity. More stable prices are desired because wildly fluctuating prices may cause hardship and are likely to increase the costs of both producers and consumers through increasing uncertainty and producing exaggerated responses in production and consumption. Where these responses are lagged one or more seasons behind price change they can be particularly damaging in producing 'cobweb' cycles. High current prices for coffee, for example, may stimulate planting of new coffee trees that will only bear fruit five or more years hence when the prices may become, as a result very depressed.

More stable earnings for producers becomes a particularly important objective when the producers are small farmers with low incomes and little or no reserves, though most countries have national measures such as marketing boards which try to stabilize producers' earnings.

Greater stability in export revenues should reduce uncertainty in economic planning and where taxes are geared to export revenues, as is the case for many primary exports, this objective is reinforced.

The aim of raising prices, incomes or export earnings above the levels that would prevail without intervention has to be seen as a form of disguised economic aid or as compensation for declining terms of trade. The charters of several ICAS also include the aim of expanding themarkets for their primary products by developing new uses, reducing trade barriers and increasing sales promotion.

As is often the case in economics, many of these objectives are mutually incompatible. A world price stabilized within narrow limits could cause greater instability in export earnings for some commodities, whereas a raised price may involve lower incomes and will certainly militate against expanded markets. Obviously these possibilities depend on assumptions about elasticities of demand and supply for specific commodities, but are in fact more than likely. For example, where demand shifts are the main cause of fluctuations but demand is price elastic, an export quota agreement will destabilize export earnings. Similarly, where supply variations are the basic cause, holding price stable though a buffer stock can destablise income if the price elasticity of demand is greater than 0.5. a stable price can also involve lower total export earnings. But recently research shows these results are less likely than was previously considered to be the case, particularly if the bank within which a buffer stock seeks to confine price movements is fairly wide. In practice the conflict between price stabilization and stabilization of export earnings for most countries' export earnings is unlikely.

The UN's Integrated Programme for Commodities

Most of the political pressure for ICAs comes from spokesmen for the developing countries.

This is reflected in countless resolutions in UNCTAD and in the grandiloquent mid-70s demands for ' A New International Economic Order', basically a collection of old ideas in a fashionable package. Stabilization and support for primary commodity prices remains the main objective and ICAs the main mechanism for achieving it. The only novel features in the UNCTAD proposals for an integrated programme were the suggestion for a Common Fund for financing international stocks and the simultaneous negotiation of a broad group of ICAs. The UNCTAD report stressed, 'That years of studies, discussions and consultations in various forums have succeeded in establishing international arrangements for only a few commodities, hardly any of which have proved to be effective or durable.' Instead of drawing the conclusions that such a dismal record might indicate basic flaws in these forms of market intervention UNCTAD demanded urgent negotiations for creating a package of up to eighteen ICAs with buffer socks and a Common Fund without wasting further time in research or consultation.

But of these eighteen commodities three already had existing price control agreements (tin, coffee and cocoa); two had existing and successful producer price raising schemes (bauxite and phosphates); four were unsuitable for buffer stocks scheme either because of the absence of organised markets or perishability (iron ore, bauxite, meats, bananas). Price enhancement for copper, cotton, iron ore, vegetable oils and oil seeds, sugar and meats was unlikely and inequitable because developed countries produced a large proportion of them, and for rubber, jute, hard fibres and cotton because of the ready availability of synthetic substitutes. A rather similar appraisal can be found in Rangarajan's book where he says, of the 18 commodities in the list, "the stock mechanism is suitable for four, of which tow already have operating mechanisms and one does not need to be stocked in the near future… it is difficult to avoid the conclusion that the stock mechanism was first chosen as a saleable proposition and the Integrated Programme then fitted around it."

If it is accepted the ICAs are a good thing then there is a case for a simultaneous approach and for the creation of a common fund for stocks. The attraction of dealing with a large group of commodities simultaneously is that it can have something in it for everyone. Countries which have interests in some commodities as consumers but in other as producers can offset gains from one agreement against losses on another. Against this can be set the sheer complexity of the task and tremendous demands that would be created for the simultaneous price increase (since that is the most likely effect of the start of the large number of stockpiles recommended) in a wide range of important imports in unlikely to raise much enthusiasm on their form for such proposals, it may be possible to give a little disguised aid in the form of an agreement on sugar or coffee without the electorate noticing what is afoot, but if similar transfers through raised prices are intended for ten or more commodities strong opposition form consumers is very likely.

A common fund for buffer socks offers several advantages. First, if the market behaviour for some commodities is out of phase with movements in prices of others, some buffer stocks could be selling at the same time as others buying. These offsetting movements could reduce the overall size of the required fund as compared with the aggregate of individual commodity funds required to achieve the same policy objectives. If, however, the main cause of instability was cyclical – fluctuations in demand which caused all commodity prices to rise and fall together –this economy in funds would be zero or negligible.

A large single fund might obtain finance on better terms than would several smaller ones. Lending risks would be pooled and reduced, and dealing in large sums of money would yield some economies of scale. UNCTAD envisages the buffer stocks as representing investments which could attract funds on a near commercial basis from OPEC members, but this is a very doubtful proposition. It depends on either rather wide swings between purchase and sale prices or very accurate predictions on the part of the stock managers. The combination of administrative, brokerage, storage and deterioration costs in stocks tends to absorb a very large part the gross margins between purchase and sale prices making it unlikely that the fund could support high interest charges.

Negotiations for a Common Fund (CF) were eventually concluded in1979. It was set up with 'two windows'. The first is intended to help finance international buffer stocks and international buffer stocks and internationally co-ordinated national stocks. Its send window will finance such measures as research and development, marketing and diversification. The financial structure of the CF is envisaged as government contributions of $470 million of which $400 million is for the first, and $70 million for the second window. Of the $400 million, $150 million is to be contributed in cash, $150 million on call and $100 million as on call for backing the Fund's borrowing. UNCTAD's earlier estimate of $6 billion for stocking the ten 'core' commodities (thought by may to be an underestimate) may be directly comparable to this because of differences in the financial arrangements, but the obvious disparity in size is so huge as to suggest that the CF is unlikely to have any significant impact upon commodity trade instability.

In any case it has often been marked that the main obstacle to ICAs has seldom been lack of finance. Negotiations have almost always broken down over the issues of the target price and the allocation of quotas. Even if they could be set up it is not known whether ICAs could succeed in moderating fluctuations in the export prices and revenues of developing countries.

Past experience does not justify optimism. Nor does the evidence of theoretical and empirical research, including simulation studies, suggest that the task of keeping actual prices within, say, plus or minus 15 per cent of a target price which keeps in touch with long-term trends in supply and demand, is anything but extremely difficult in technical terms, let alone in the real world of clashing interests between producers and consumers and among producing nations.

b. Compensatory Financing

Two other schemes for alleviating the effects of commodity trade instability have been operating for a number of years. These are the IMF's Compensatory Financing Facility (CFF) started in 1963 and the EEC's STABEX scheme which was established by the Lome convention between the Community and forty-six African, Caribbean and compensating countries for shortfalls in export earnings which result from fluctuations in commodity markets.

No attempt is made to intervene in the markets to influence shortfall and the loans normally have to be repaid within a few years. The IMF's CFF defines a shortfall as the gap between the current year's and forecasts for the subsequent years. Initially drawings were limited to 25 per cent of the member's quota in the IMF, were not additional to ordinary drawings and required the member to co-operate with Fund in finding a solution to its balance - of- payments difficulties. Partly because of these limitations and partly because the 1960s were a period of relative stability the CFF was little used. Over the years the scheme was liberalized. Major changes were made in1975 in the wake of the oil crisis. The limit on drawings was raised to 75 per cent of quota and could be additional to

ordinary drawings. The permitted that the amount of outstanding drawings in any twelve-month period was raised from 25 to 50 per cent of quota.

Because the calculation of the shortfall is necessarily delayed until after the end of the current year countries were permitted to draw on their ordinary quota in anticipation of a shortfall and then convert this to a CFF drawing at anytime up to eighteen months later. Up to eighteen months later. Shortfalls have to be for reasons outside the country's control and the member still has to co-operate with the IMF in finding a solution. A rule which prevented a country from borrowing if its current exports were 5 per cent or more than the average of the two previous years was eliminated. This proved crucial in the inflationary years of the 1970s.

After the 1975 reforms drawings shot up. In the subsequent sixteen months drawings by fortynine member countries reached SDR2.4 billion or twice the amount in the previous thirteen years. By April 1980 the drawings by the non-oil Less Developed Countries (LDCs) had amounted to 4.6 billion SDRs and their net outstanding credits were 2.5 billion SDRs. Nevertheless, it has been criticized for providing far too little assistance to the LDCs. UNCTAD secretariat calculations show that drawings against the CFF by the LDCs have on average not exceeded 12.5 per cent of shortfalls. Even in 1976 – the year of maximum drawings –it was only 12.7per cent.

It may well be time for the CFF to meet a much larger proportion of export shortfalls, and most suggested reforms point that way, but several factors should be borne in mind. First, the IMF assumes that most countries will use their own reserves, borrowing from other official sources and commercial sources as well as drawing upon the CFF. Secondly, the 1976 drawings were in relation to the shortfalls of 1975 which was a quite exceptional year. Primary commodities hit their peak in 1974 and their trough in 1975, recovering substantially in 1976 and 1977. Many LDCs should have accumulated reserves from the preceding commodity boom in 1973/4 and the IMF had created several emergency funds to assist in this world crisis. For example the Oil Facility and the Trust Fund. The NOLDCs did draw on these.

The CFF scheme is, in principle, a much easier system to operate than ICAs. It is much more comprehensive in that it covers all merchandise exports (and could easily include invisibles as well) and it is much less demanding of political necessity to obtain agreements. Or technical skill in forecasting future prices of individual commodities and designing optimal stocking policies than is the case for ICAs. CFF-type schemes emerge in favourable light from simulation exercises and, in practice, the IMF scheme seems to have worked in the right directions even if the amounts of compensation have seemed small in relation to the recent problems of the LDCs.

Increases the LDCs' Fund quotas, the inclusion of invisibles, and calculation of shortfalls in real terms (allowing for changes in the prices of imports) are all possible reforms which could increase the value the CFF to LDCs.

c. STABEX

The STABEX scheme was designed to stabilize earnings from exports of the African, Caribbean and Pacific (ACP) countries to the Community. It covered seventeen agricultural commodities and iron ore. The original forty-six ACP countries later rose to fifty-two so that it involves substantial number of developing countries, many of them rather small, poor and vulnerable. But the commodities whose earnings are intended to be stabilized amount to only 20 percent of the export earnings of the ACP countries. In 1976, its first year of operation, seventeen ACP countries drew SDDR 72 million. In the same year ACP counties drew

SDR124 million from the IMF scheme and LDCs total drawings for 1976 were SDDR 1,575 million.

The total sum allocated to STABEX for the whole period 1976 – 80 was only about $420 million and conditions for eligibility were quite stringent. The exports had to be crude or in very elementary processed form. Individually they had to account for at least 7.5 percent of the country's total merchandise exports to all destinations. The shortfalls, calculated in nominal terms, had to be at least 7.5 percent below the average earnings from the product the ECC over the previous four years. For the least developed, land-locked or island economies these two conditions are dropped to 2.5 per cent.

The terms for repayment are liberal. Compensation payments to the least developed countries are in the form of grants and for the others the loans are interest free and repayable as and when export earnings recover. The STABEX can be criticized for discriminating between ACP and other LDCs and for being too limited in coverage and funds. This has the effect of making it liable to political influence when decisions have to be made on rationing funds between intending borrowers. The idea of making compensation payments grants to the least developed countries is widely commended as an appropriate change for adoption by the IMF\CFF. But is it sensible to confuse transfers intended to promote development with assistance intended to deal with temporary financial imbalances? The criteria for allocating funds for each of these purposes should be quite different. Of course situations may arise where what was intended as a short-term loan has to be re-phased. Instead of exports rising in the next three years they may drop still further or there may be drop and still unforeseen events need special *ad hoc* arrangements and that basically is the attitude of the IMF.

d. A complementary facility for commodity-related shortfalls in export earnings

This is the most recent proposal of the Group of 77 at UNCTAD in June 1979. There they requested that the UNCTAD secretariat in consultation with the IMF staff carry out a detailed study for a complementary facility' to compensate for shortfalls in each commodity, taking account of its financial requirements, possible sources of financing, its financial feasibility, institutional arrangements and the modalities and considerations that would provide adequate compensation in real terms to developing countries …' it is intended that this should be additional to improvements in the CFF to the IMF and other IFC arrangements. Most of the OECD nations voted against this resolution or abstained.

If the major worry of the LDCs is fluctuations in their export earnings (and this is what has usually been maintained) the CFF approach offers much greater prospects of success. There is scope for reforming and expanding it, but not in the direction of turning it into a mechanism for long-term transfers of resources to LDCs. The criteria for long-term assistance out to differ significantly from the relatively automatic provision of short-term finance to meet balance-of payments problems induced by export instability.

e. Economic integration

It refers to the merging to various degrees of the economies and economic policies of two or more countries in a given region.

• **Free Trade Area:**

Exist when a number of countries agree to abolish tariffs, quotas and any other physical barriers trade between them, while retaining the right to impose unilaterally their own level of customs duty, etc, on trade with the rest of the world.

• **Customs Market**

Exists where a number of countries decide to permit free trade among themselves without tariff or other trade barriers, while establishing a common external tariff against imports from the rest of the world.

• **Common Market**

Exists when the countries, in addition to forming a custom union, decide to permit factors of production full mobility between them, so that citizens of one country are free to take up employment in the other, and capitalist are free to invest and to move their capital from one country to another.

• **Economic union**

Is where the countries set up joint economic institutions, involving a degree of supranational economic decision-making.

• **Common Monetary System**

Is where countries share a common currency, or ensure that each national currency can be exchanged freely at a fixed rate of exchange, and agree to keep any separate monetary policies roughly in line, to make this possible.

Benefits of integration

The formation of an economic integration could be beneficial in the light of the following aspects:

- **Enlarged market size:** Regional economic blocks provide larger markets than individual countries. Such increase in size of the market permits economies of scale, resulting in lower production costs and expansion of output. In fact, member countries are better placed to bargaining for better terms of trade with non-member countries.
- **Industrialization:** the size of the domestic market of one member country may not be sufficiently large to justify the setting up of an industry, whereas the market provided by many countries (regional market) is much more likely to be an incentive for establishment of new manufacturing industries, thus what economists consider as potentially derived industrial development.
- **Infrastructural facilities:** Jointly financed infrastructural facilities such as in the field of transport (e.g. railway systems, ports and habours, and airlines. The East African Cooperation (EAC), for instance, would reduce costs by setting up one Development
- Bank to serve all the three countries rather than each country maintaining its own.

- **Specialisation:** Each member country concentrates on production of those goods which it can produce more efficiently. Surpluses are exchanged and resources utilisation is increased - comparative advantage.
- **Increased employment opportunities and subsequent reduction in income inequalities:** Free mobility of labour leads to people moving from areas where incomes are low to areas of high labour incomes. This becomes even more beneficial if such incomes are invested back in respective countries. Furthermore, firms will have to pay highly in order to retain factor services (economic rent), thereby enhancing productivity.
- **Improvement of balance of payments (BOP):** increased market implies more exports than before and given fairly low priced imports (from member countries) relative to imports from non-member countries, balance of payments position is most likely to improve. Foreign exchange savings also arise from this situation i.e. hard currencies such as the US dollar will only be required to import what cannot be produced from within the region.
- **Competitive business environment:** Absence of trade barriers allows for free flow ofgoods and services which develops an upward pressure on competition and the driving force for relatively lower prices for higher quality products. This helps reduce or even eliminate monopoly practices, since firms can only acquire and maintain a market base by producing as efficiently as possible. Overall, there is increased variety of goods and services their consumption of which enhances living standards/development.
- **Indigenisation of economies:** Regional governments play their part by creating the right incentives for the growth of the private sector which is the prime-mover of economic activities in liberalized situations. The private sector participation should not be limited to business activities but should extend to the formation of regional professional and business associations in order to advise on the influence future cooperation policies (e.g. the East African Business Council.) This creates more awareness among potential investors to take advantage of investment opportunities available within the region to create wealth. This way, over-reliance on private foreign investments and other forms of capital inflows (such as conditional Aid from IMF and World Banks) tends to be reduced.

The African continent regional integrations have not gone far in realizing the intended objectives due to:

- **Minimal or lack of practical commitment** hence the low implementation of policies and agreements. Policy-induced factors such as inward looking policies of individual countries could result in the protection of less or uncompetitive domestic producers against imports irrespective of resources, and stringent trade and payments controls instituted to deal with the persistent balance of payments problems have adversely affected the volume of trade among African countries.
- **Indispensable high capital import content:** Most African countries are not in a position to sufficiently produce capital goods and other inputs for the production of goods hence continued vulnerability to foreign influence and dominance. This is traced to widespread poverty and minimal technical progress.
- **Neo-colonialism and dependence mentality:** The psychological influence arising from massive and persuasive advertising by the developed world has largely role-modelled the consumption pattern (tastes and preferences) of the African People. Preference has been given to products which do not originate from within the region. This then forces regional member countries to import such products in order to meet their domestic demand.

- **Trade-diversion:** countries previously importing cheap goods from outside the region switch to importing the same goods from other member countries. This is brought about by the removal of tariffs and other trade restrictions on the movement of goods between member states, while the tariffs on goods from outside remain. Depending on price/cost difference(s) such expenditure –switching may increase production cost accompanied by negative welfare implications.
- **Government loss of tax revenue:** Removal of tariffs (import duties) leads to reduced tax revenue to the government. With free trade, in countries where import duties constituted a high proportion of tax revenue; the government's spending programmes will be distorted.
- **Unequal distribution of trade benefits/gains:** Although all countries gain to a certain extent, one member country may benefit more than the others. This often arises due to high subsidization of production so that one country may succeed in attracting a more than proportionate share of new industrial development. In particular, incomes and employment opportunities will increase more than proportionately due to the multiplier effect.
- **Product similarity and duplication:** Because of product similarity (especially primary products), regional member countries have not lived to substantial benefits as would be required; what one country produces is equally produced by others so that the overall relative market share remains distinctively small.
- **Widespread internal conflicts and general political instabilities:** the most immediate result of such an atmosphere is suspicion and increase of protectionist strategies which no doubt hinders the free movement of goods and people. This then negates the intention of an economic integration. Owing to the current political and economic reforms sweeping across the economies of the developing would, it's quite evident that much cannot be done without any form of collective effort. Economic integration aspect(s) could receive the seriousness deserved now or in the immediate future. In fact, some which had collapsed or remained insensitive are getting revived e.g. the East African Co-operation.

The disadvantages of economic integration

a. **The "trade-diversion" effect has already been mentioned.** Countries previously importing cheap goods from outside the free trade area switch to importing the same goods from other member countries. This is brought about by the removal of tariffs on goods moving between member states, while the tariffs on goods from outside remain. The result is a less efficient use of resources. Further more, the goods produced in the other member states are often of inferior quality to those formerly imported from outside.\\

b. **Government suffer a loss of tax revenue from the setting up of a free trade area.**
Before a lot of tax revenue was received from import duties on goods brought into the country from overseas. If goods are imported from other member states when the free trade area is up, import duties are no longer payable and tax revenue, the effect on a government's spending programme will be substantial.

TERMS OF TRADE, BALANCE OF TRADE, BALANCE OF PAYMENTS , CAUSES AND METHODS OF CORRECTING DEFICITS IN BALANCE OF PAYMENTS

TERMS OF TRADE

The relation between the prices of a country's exports and the prices of its imports, represented arithmetically by taking the export index as a percentage of the import index. In the comparative cost model, terms of trade were, defined as the international exchange ratio between a country's export good and its import good. This is the barter terms of trade which measures the quantity of exports which have to be sacrificed to obtain a unit of imports and is easily calculated when there are just two goods traded. But in practice, countries trade hundreds of different goods and services and the concept of the terms of trade becomes more complex.

Estimates of the terms of trade are usually made by calculating an index of import prices; this gives an index of the term of trade:

Terms of trade index = Export Price Index/Import price index x 100

Thus, the price indices are essentially weighted averages of export and import pries. If these are set at 100 in the same base year, say, 1990, then the terms of trade index is also 100. If, for instance, import prices fall relative to export prices, the terms of trade will rise above 100, the terms of trade then being said to be more favourable to the country concerned since it means that it can obtain more goods from abroad than before in exchange for a given quantity of exports. On the other hand, if the terms of trade become unfavourable, the terms of trade index will fall below 100. A rise in terms of trade index is usually described as an "improvement" or as "favourable" on the grounds that a rise in export prices relative to import prices theoretically means that a country can now buy the same quantity of imports for the sacrifice of less export (or it can have more imports for the same volume of exports). Similarly, a fall in the terms of trade index is a "deterioration" or is an "unfavourable" movement.

Factors affecting the long run trend of the Terms of Trade for developing countries

Most Third World countries have been faced by a fall in their terms of trade over the long run.

There are a number of factors which contributed to this result, namely: -

The income-elasticity of demand for primary products

These countries export primary products like basic foodstuffs which may be considered to be "necessities" on which a decreasing proportion of income is likely to be spent as these incomes rise. Countries relying on basic foodstuffs and other primary product exports may therefore find their exports growing more slowly than those of individual countries exporting manufactured goods.

The discovery of synthetic materials

Over a whole range of items, the substitution of synthetic man-made products has reduced the market for particular primary products. The long term trend in the market shares of natural and synthetic products is likely to be influenced by a "ratchet" effect. When prices of natural products are high, due to cyclical fluctuations or temporary shortages, research into possible synthetic substitutes will be encouraged. When prices of natural products revert to more normal levels, these products may have permanently lost a further part of the market.

Raw material – saving innovations

This is likely to apply to technical change aimed at economizing the use of raw material's in industry. Periodic high prices will stimulate the search for and application of raw material saving process. Technical change aimed at a progressive reduction in costs per unit of output directly by permitting industrial output (and thus income) to expand in greater proportion than the demand for materials.

Agricultural protection and import substitution in developed countries

The protectionist policies within the industrial countries which aim to raise the incomes of farmers and other primary producers like fishermen by placing tariffs or quotas on competing imports have also affected the terms of trade for developing countries. As a result, many industrial countries have become almost self-sufficient in producing grains, sugar (from beet), livestock products, and even tobacco and wines. Sometimes the policies have been directed at saving foreign exchange as well as maintaining domestic incomes and employment, and have not only been confined to primary products. Indeed, restrictions on access to markets for manufactured goods by developing countries at large, or potentially large, export industries like

India's textiles have forced developed countries to sell more primary products instead. Thus even without tariffs or quotas, therefore, the expansion in primary product exports is likely to result in a decline in their commodity terms of trade in the many cases where the exports are likely to result in a decline in their commodity terms of trade in the many cases where the price elasticity of demand in industrial countries is very low. When in addition, these primary products face tariff or quota restrictions, the deterioration in the terms of trade will be greater.

Diminishing returns in agriculture and limited natural resources

Whatever the income-elasticity of demand for primary products, continuous expansion of industrial output means a continually increasing requirement of raw materials. If the supply of land suitable for various agricultural products is limited, the law of diminishing returns may apply, leading to increasing scarcity of such products, and a rise in their prices.

Technical progress in manufacturing

Although technical progress in the industrialized countries should, through the market mechanism, have been shared between the industrial producers and the producers of primary products, according to Raoul Presbich, this desirable development has been frustrated. On the one hand, industrial monopolistic practices and trade union action producing cost-push inflation in the developed countries have persistently raised money wages in these countries and, with these, the prices of manufactured goods. In contrast, competition among primary producers, and the ineffectiveness of trade unions in the agricultural sectors of these economies, has kept down the prices of raw materials. In fact the benefits of any costreducing innovation in these countries is likely to be passed on, as a result of competition, to industrial consumers in the form of reduced prices.

BALANCE OF TRADE

The balance of trade compares the value of a country's exports of goods and services against its imports. When exports are greater than imports, that's a trade surplus, which is generally considered a favorable trade balance. The opposite, when the value of imports outweighs the value of exports, is a trade deficit, and this is generally considered an unfavorable trade balance. However, it all depends on where the country is in its business cycle, how long the deficit or surplus has been ongoing, and the reasons behind it.

The trade balance subtracts imports from exports. Imports are any goods and services that are made in a foreign country and bought by a country's residents. You generally think of imports as being shipped in from a foreign country, but even if it's bought by residents while traveling abroad, it's still an import. They are just shipping the imports home themselves, in their suitcases. Services provided while traveling, such as transportation, hotels and meals, are also technically imports. It also doesn't matter if the company that makes the good or service is a domestic company.

Exports are any goods or services that are sold to a foreign resident by a domestic resident or business. Whether it's a pair of jeans you mail to a friend, or merchandise shipped from an American corporate headquarters to its foreign office, if they pay for it, then it's technically an export.

The balance of trade is important because it's a large component of the current account. Besides the trade balance, this also measures a country's net income earned on international assets, as well as payments across borders.The trade balance is the easiest to measure, since all goods and many services must pass through Customs.

BALANCE OF PAYMENTS

The Balance of Payments of a country is a record of all financial transactions between residents of that country and residents of foreign countries. (Residents in this sense does not just refer to individuals, but would also include companies, corporations and the government). Thus all transactions are recorded whether they derive from trade in goods and services or transfer of capital.

Like all balance sheets, the balance of payments is bound to balance. For if the country has "overspent", then it must have acquired the finance for this **"overspending**" from somewhere (either by running up debts or using its reserves), and when this item is included in the accounts they will balance. It follows therefore that when reference is made to a balance of payments "deficit" or "surplus", this only looks at a part of the total transactions, e.g. that part involving trade in goods and services, which is termed the "Balance of Payments on the **current account"**

If the value of exports exceed the value of imports the balance of payments is said to be in **Trade Surplus**. This is regarded as a favourable position because a persistent trade surplus means lower international debts. Also, a trade surplus is regarded as a sign of success in the country's trade with other countries and is, therefore, politically desirable.

On the other hand, if the value of imports exceed the value of exports, the balance of payments is in **trade deficit**. This is an unfavourable position because a persistent balance of payment trade deficit means the country's foreign exchange reserves are being run down and so is its ability to pay for its imports and settle its international debt. Also persistent balance of payments trade deficit is regarded as a sign of failure in the country's trade with other countries and is therefore politically undesirable

Structure of the Balance of Payments

The balance of payments is divided into three accounts:

a. The Current Account

This records all transactions involving the exchange of currently produced goods and services and is subdivided into

i. Visibles:

A record of all receipts from abroad the export of goods and all expenditures abroad on the import of goods. When these are compared, this is known as the **"balance of trade**" (though it would be properly called the "balance of visible trade").

ii. Invisibles:

A record of all receipts from abroad in return for **services rendered** and all expenditure abroad for foreign services. It also includes receipts of **profits** and **interest** earned by investments abroad, and similarly profits and interest paid abroad to foreign owners of capital in the country are included in Expenditure. The comparison of all the debits (Expenditure abroad) and credits (receipts from abroad) arising from visibles and invisibles is known as the **"balance of payments on current account"** and is the best indicator of the country's trading position.

If the value of exports exceeds the value of imports the balance of payments is said to be in **Trade Surplus**. This is regarded as a favourable position because a persistent trade surplus means the country's foreign exchange reserves are rising and so its ability to pay for its imports and settle its international debts. Also a trade surplus is regarded as a sign of success in the country's trade with other countries and is, therefore, politically desirable.

On the other hand, if the value of imports exceeds the value of exports, the balance of payments is in **trade deficit.** This is an unfavourable position because a persistent balance of payments trade deficit means that the country's foreign exchange reserves are being run down and so is its ability to pay for its imports and settle its international debts. Also a persistent balance of payments trade deficit is regarded as a sign of failure in the country's trade with other countries and is therefore politically undesirable.

b. CAPITAL ACCOUNT:

This records all transactions arising from capital movements into and out of the country. There are a variety of such capital flows recorded, namely:

i. Long term capital:

This consists of:

a. Government to Government borrowing and lending
b. Government borrowing from international organizations

c. Investment by foreigners at home in such projects as factories, mines and plantations and by nationals abroad in similar projects.
d. The buying of shares by foreigners in home companies and by nationals in foreign companies.

ii. Short term capital ("hot money")

Refers to changes in bank balances held by foreigners in home banks and by nationals in foreign banks.

If money comes into the country (e.g. deliberate borrowing abroad by a domestic company or foreign investment in the country), it is recorded as a credit item, while investments abroad etc is recorded as a debit item.

When the current account and the capital account are combined, and we compare the total debits and credits, this is termed the balance for official financing (it used to be termed the "total currency flow"). This shows the final net outflow, or inflow arising from current and capital transactions.

c. THE MONETARY ACCOUNT

Also called official financing, this comprises the financial transactions of the government (handled by the central bank) needed to offset any net outflow of money on the current and capital accounts i.e. total currency flow. It comprises of:

i. Use of the foreign exchange reserves, i.e. increasing or decreasing them.
ii. Borrowing from the IMF i.e. borrowing or paying back.
iii. Central bank transactions with other countries central banks i.e. borrowing or lending.

d. THE BALANCING ITEM

Since for ever position entry in the current and capital accounts there is a corresponding negative entry in the monetary account, and for every negative entry in the first two accounts there is a corresponding positive entry in the monetary account, it follows that the balance of payments must balance i.e. the sum of the balances of all the three accounts must add up to zero.

In practice, this is usually not the case because there are so many transactions that take place, and due to human errors some may be recorded correctly in one account but incorrect in another account with the result that the sum of the tree balances may not be zero. The actual discrepancy in the records can be calculated. The balancing item represents the sum of all errors and omissions. If it is positive, it means that there have been unrecorded net exports while a negative entry means that there have been unrecorded net imports.

Equilibrium and Disequilibrium in the Balance of Payments

If on the **current account**, the value of exports is equal to the value of imports, the balance of payments is said to be equilibrium. If the two values are not equal, the balance of payments is in

disequilibrium. This could be due to a trade surplus with the value of exports exceeding that of imports or due to a trade deficit with the value of imports exceeding that of exports.

In either case, a balance of payments disequilibrium cannot last indefinitely. For if this is due to a trade deficit, the country will try and move it. This is because a persistent trade deficit i.e. **a fundamental disequilibrium** poses several problems for an economy, namely:

- In short run a deficit allows a country's peoples to enjoy higher standard of living form the additional imports that would not be possible from that country's output alone in the longer term the decline of the country's industries in the face of international competition will inevitably result in lower living standards.
- A persistent trade deficit means that the country's foreign exchange reserves are being run down and so it its ability to pay for its imports and settle international debts.
- Also, a persistent balance of payments trade deficit is regarded as a sign of failure in the country's trade with other countries, and is therefore not politically desirable.

CAUSES AND METHODS OF CORRECTING DEFICITS IN THE BALANCE OF PAYMENT

Policies to cure Balance of Payment deficits

The measures available to tackle balance of payments deficits include short term measures such as **deflation, import controls, devaluation** of a fixed exchange rate or a managed downward float of the exchange rate in the short-run and foreign exchange controls and long term measures such as Export promotion and Import Substitution.

Short-Term Policies

Deflation is a policy of reducing expenditure with the intention of curing a deficit by reducing the demand for imports. This reduction of expenditure may be achieved by the use of either **fiscal** or **monetary** policy. In addition to reducing demand for imports however, deflationary measures may also have expenditure switching effect upon the balance of payments. The depression of demand may cause the domestic inflation rate to fall relative to that of competitor countries and thus increase the price competitiveness of exports. Consumers in other countries may then switch their demand towards the country's exports, whilst its own residents switch away from imports, preferring instead to buy home produced substitutes. The difficulty posed by deflation is that it not only reduces demand for imports but also reduces demand for domestically produced goods. This in turn can have a knock on effect in the form of lower output and higher unemployment.

Import controls have immediate effect on the balance of payments. **Quotas** and **embargos** directly prevent or reduce expenditure on imports, while import duties or tariffs discourageexpenditure by raising the price of imports, while import duties or tariffs discourage expenditure by raising the price of imports. Import controls also have their limitations and problems. They do not tackle the underlying cause of this disequilibrium i.e. the lack of competitiveness of a country's industry and what is more they are likely to invite retaliation to the long-term detriment of themselves as well as

their trading partners. It is also the case that trade agreements such as GATT limit the opportunities for member countries to make use of import controls and the use of subsidies to encourage exports.

A third option is that of **devaluation**. Devaluation of a fixed exchange rate or the downward float of a managed exchange rate is mainly expenditure switching in its effect. The cure works in a similar manner to the freely floating adjustment mechanism under a floating exchange rate system. In the case of a fixed exchange rate system devaluation consists of an administered reduction in the value of the currency against other currencies. In a managed system the authorities can engineer a downward float by temporarily reducing their support. In both cases the effect is to increase the price of imports relative to the price of exports and so switch domestic demand away from imports and towards home produced goods.

Certain conditions have to be met for devaluation to have this effect on boosting exports/curbing imports. They are:

a) Competing countries must not devalue at the same time, otherwise there would be no competitive advantage gained (exports would not become any cheaper in comparison with products of those countries).
b) The demand for exports (or for imports) must be price elastic, i.e. the sales must be affected by the change in price. Thus, when the domestic currency falls in value, the demand for exports should rise by a larger proportion in order to earn more foreign exchange.
c) There must be appropriate domestic policy.
    i. The extra exports must be available. If there were full employment in the economy, the home demand would have to be curbed to make room for the extra export production.
    ii. Inflation must not be allowed to erode the competitive advantage secured by devaluation.

A devaluation of the currency is not a soft option. There are a number of problems that will be involved, and these must be outlined:

a. To the extent that home demand has to be cured to make room for the extra exports, the domestic standard of living is reduced. This is only because, before it took place, the country was “living beyond its means”, but it does come as a shock to find the domestic “squeeze” accompanying devaluation. Yet if it does not take place, then the strategy will not have worked; the exports may not rise to meet the higher demand from abroad.
b. The larger cost of importing goods raises the domestic cost of living. This is not just inflationary in itself, but can trigger off pay claims which if settled will further worsen inflation.
c. It does not boost exports immediately. There is a period during which the balance of payments gets worse as the country faces a higher import bill. It is only when exports start rising (and there is a considerable time lag involved) that the situation improves.
d. It does not tackle the long-run problem of why exports were not doing well. The problem may be more in inefficiency and other non-price factors than in the price of the exports themselves. In which case devaluation would make little difference to the basic problem.

A fourth option is to use **Exchange control.** When this is used to deny foreign exchange to would be importers, its effects are identical to those of the various import restricdtions already discussed. There are various forms of exchange control that can be imposed by a government and enforced by

legislation. They all involve restrictions on the actions of holders of itscurrencies and residents of the country who may hold foreign currency.

ii. **Long-Term Policies**

One long term option of tackling balance of payments deficit is **export promotion**. In the longrun this is the best method of improving a balance of payments. If the general level of efficiency in an economy can be raised, then exports will benefit. Efficiency can be promoted by mergers in exporting firms (thereby reaping economies of scale), research and economic growth – for it is felt that once an economy is growing it is generating the necessary dynamism and technological improvement that will feed through into a better export performance.

A second long-term option is **Import Substitution**. The replacement of imports by home products can be achieved by economic planning. If the defects of home products can be analysed, and the likely future trends in demand can be forecast, then domestic firms can takethe necessary action both to improve their product and to expand their capacity. Government support for certain industries can also be helpful here.

EXCHANGE RATES

The value of two currencies relative to each other. For example, on a given day, one may trade one U.S. dollar for a certain number of British pounds. A currency's exchange rates may be floating (that is, they may change from day to day) or they may be pegged to another currency. A floating exchange rate is dependent on the supply and demand of the involved currencies, as well as the amount of the currency held in foreign reserves. On the other hand, a government may peg its currency to a certain amount in another currency or currency basket. For example, the Qatari riyal has been worth 0.274725 dollars since 1980.

An advantage to a floating exchange rate is the fact that it tends to be more economically efficient. However, floating exchange rates tend to be more volatile, depending on the particular currency. Pegged exchange rates are generally more stable, but, since they are set by government fiat, they may take political rather than economic conditions into account. For example, some countries peg their exchange rates artificially low with respect to a major trading partner to make their exports to that partner artificially cheap. See also: Currency pair, Eurodollar.

TYPES OF EXCHANGE RATES REGIMES

The main types are as follows:

- a floating exchange,
- a pegged floating exchange,
- a fixed rate exchange.

The floating exchange rate regime

Iis the most popular one and is adopted in many different countries. The USD, EUR, JPY, GBP are all examples of currencies that are floating freely. In these instances, market participants are

determining current rates through increasing/decreasing supply or demand. Central banks sometimes intervene in free market operations in order to minimize excessive volatility in the value of a currency. Some commentators would use the term "managed float" in these cases.

A pegged float regime

system of pegging a country's currency to a band or value, which may be fixed or periodically adjusted. A peg helps importers and exporters to minimize their exchange rate risk allowing them to plan safely enough regarding their future trade operations. It is also helps to ease the inflationary pressures. Danish krone (DKK) is an example of this type of currency. It is pegged to EUR using the so called European Exchange Rate Mechanism (ERM) II which allows a currency to float within +-15% against EUR. However, Denmark's National Bank uses a narrower range of +/-2.25% and now, with EUR depreciating they still wish to keep the peg and are not allowing DKK to appreciate. The Danish government has even suspended bond issuance in order to protect the peg.

Fixed exchange regime

Doesn't allow currency to fluctuate whatsoever. A country's Central Bank would use open market mechanisms in order to maintain the fixed pegged ratio. China was the last large economy that was using this type of exchange rate regime, but in July 2005 it decided to move towards more flexible system. It is crucial to understand the rules that governments use and how these rules affect currencies in order to spot great opportunities to enter the markets.

FACTORS INFLUENCING EXCHANGE RATES

1 Differentials in Inflation

As a general rule, a country with a consistently lower inflation rate exhibits a rising currency value, as its purchasing power increases relative to other currencies. During the last half of the twentieth century, the countries with low inflation included Japan, Germany and Switzerland, while the U.S. and Canada achieved low inflation only later. Those countries with higher inflation typically see depreciation in their currency in relation to the currencies of their trading partners. This is also usually accompanied by higher interest rates.

2. Differentials in Interest Rates

Interest rates, inflation and exchange rates are all highly correlated. By manipulating interest rates, central banks exert influence over both inflation and exchange rates, and changing interest rates impact inflation and currency values. Higher interest rates offer lenders in an economy a higher return relative to other countries. Therefore, higher interest rates attract foreign capital and cause the exchange rate to rise. The impact of higher interest rates is mitigated, however, if inflation in the country is much higher than in others, or if additional factors serve to drive the currency down. The opposite relationship exists for decreasing interest rates - that is, lower interest rates tend to decrease exchange rates

3. Current-Account Deficits

The current account is the balance of trade between a country and its trading partners, reflecting all

payments between countries for goods, services, interest and dividends. A deficit in the current account shows the country is spending more on foreign trade than it is earning, and that it is borrowing capital from foreign sources to make up the deficit. In other words, the country requires more foreign currency than it receives through sales of exports, and it supplies more of its own currency than foreigners demand for its products. The excess demand for foreign currency lowers the country's exchange rate until domestic goods and services are cheap enough for foreigners, and foreign assets are too expensive to generate sales for domestic interests

(**4. Public Debt**
Countries will engage in large-scale deficit financing to pay for public sector projects and governmental funding. While such activity stimulates the domestic economy, nations with large public deficits and debts are less attractive to foreign investors. The reason? A large debt encourages inflation, and if inflation is high, the debt will be serviced and ultimately paid off with cheaper real dollars in the future.

In the worst case scenario, a government may print money to pay part of a large debt, but increasing the money supply inevitably causes inflation. Moreover, if a government is not able to service its deficit through domestic means (selling domestic bonds, increasing the money supply), then it must increase the supply of securities for sale to foreigners, thereby lowering their prices. Finally, a large debt may prove worrisome to foreigners if they believe the country risks defaulting on its obligations. Foreigners will be less willing to own securities denominated in that currency if the risk of default is great. For this reason, the country's debt rating (as determined by Moody's or Standard & Poor's, for example) is a crucial determinant of its exchange rate.

5. Terms of Trade
A ratio comparing export prices to import prices, the terms of trade is related to current accounts and the balance of payments. If the price of a country's exports rises by a greater rate than that of its imports, its terms of trade have favorably improved. Increasing terms of trade shows greater demand for the country's exports. This, in turn, results in rising revenues from exports, which provides increased demand for the country's currency (and an increase in the currency's value). If the price of exports rises by a smaller rate than that of its imports, the currency's value will decrease in relation to its trading partners.

6. Political Stability and Economic Performance
Foreign investors inevitably seek out stable countries with strong economic performance in which to invest their capital. A country with such positive attributes will draw investment funds away from other countries perceived to have more political and economic risk. Political turmoil, for example, can cause a loss of confidence in a currency and a movement of capital to the currencies of more stable countries.

FOREIGN EXCHANGE RESERVES

Foreign-exchange reserves (also called **forex reserves** or **FX reserves**) are assets held by a central bank or other monetary authority, usually in various reserve currencies, mostly the United States dollar, and to a lesser extent the euro, the pound sterling, the Japanese yen, and used to back its liabilities—e.g., the local currency issued, and the various bank reserves deposited with the central bank by the government or by financial institutions

foreign-exchange reserves should only include foreign banknotes, foreign bank deposits, foreign treasury bills, and short and long-term foreign government securities. However, the term in popular usage commonly also adds gold reserves, special drawing rights (SDRs), and International Monetary Fund (IMF) reserve positions. This broader figure is more readily available, but it is more accurately termed **official international reserves** or **international reserves**.

Foreign-exchange reserves are called reserve assets in the balance of payments and are located in the capital account. Hence, they are usually an important part of the international investment position of a country. The reserves are labeled as reserve assets under assets by functional category. In terms of financial assets classifications, the reserve assets can be classified as Gold bullion, Unallocated gold accounts, Special drawing rights, currency, Reserve position in the IMF, interbank position, other transferable deposits, other deposits, debt securities, loans, equity (listed and unlisted), investment fund shares and financial derivatives, such as forward contracts and options. There is no counterpart for reserve assets in liabilities of the International Investment Position. Usually, when the monetary authority of a country has some kind of liability, this will be included in other categories, such as Other Investments. In the Central Bank's Balance Sheet, foreign exchange reserves are assets, along with domestic credit.

Official international reserves assets allow a central bank to purchase the domestic currency, which is considered a liability for the central bank (since it prints the money or fiat currency as IOUs). Thus, the quantity of foreign exchange reserves can change as a central bank implements monetary policy, but this dynamic should be analyzed generally in the context of the level of capital mobility, the exchange rate regime and other factors. This is known as Trilemma or Impossible trinity. Hence, in a world of perfect capital mobility, a country with fixed exchange rate would not be able to execute an independent monetary policy.

A central bank that implements a fixed exchange rate policy may face a situation where supply and demand would tend to push the value of the currency lower or higher (an increase in demand for the currency would tend to push its value higher, and a decrease lower) and thus the central bank would have to use reserves to maintain its fixed exchange rate. Under perfect capital mobility, the change in reserves is a temporary measure, since the fixed exchange rate attaches the domestic monetary policy to that of the country of the base currency. Hence, in the long term, the monetary policy has to be adjusted in order to be compatible with that of the country of the base currency. Without that, the country will experience outflows or inflows of capital. Fixed pegs were usually used as a form of monetary policy, since attaching the domestic currency to a currency of a country with lower levels of inflation should usually assure convergence of prices.

In a pure flexible exchange rate regime or floating exchange rate regime, the central bank does not intervene in the exchange rate dynamics; hence the exchange rate is determined by the market. Theoretically, in this case reserves are not necessary. Other instruments of monetary policy are generally used, such as interest rates in the context of an inflation targeting regime. Milton Friedman

was a strong advocate of flexible exchange rates, since he considered that independent monetary (and in some cases fiscal) policy and openness of the capital account are more valuable than a fixed exchange rate. Also, he valued the role of exchange rate as a price. As a matter of fact, he believed that sometimes it could be less painful and thus desirable to adjust only one price (the exchange rate) than the whole set of prices of goods and wages of the economy, that are less flexible.

Mixed exchange rate regimes ('dirty floats', target bands or similar variations) may require the use of foreign exchange operations to maintain the targeted exchange rate within the prescribed limits, such as fixed exchange rate regimes. As seen above, there is an intimate relation between exchange rate policy (and hence reserves accumulation) and monetary policy. Foreign exchange operations can be sterilized (have their effect on the money supply negated via other financial transactions) or unsterilized.

Non-sterilization will cause an expansion or contraction in the amount of domestic currency in circulation, and hence directly affect inflation and monetary policy. For example, to maintain the same exchange rate if there is increased demand, the central bank can issue more of the domestic currency and purchase foreign currency, which will increase the sum of foreign reserves. Since (if there is no sterilization) the domestic money supply is increasing (money is being 'printed'), this may provoke domestic inflation. Also, some central banks may let the exchange rate appreciate to control inflation, usually by the channel of cheapening tradable goods.

Since the amount of foreign reserves available to defend a weak currency (a currency in low demand) is limited, a currency crisis or devaluation could be the end result. For a currency in very high and rising demand, foreign exchange reserves can theoretically be continuously accumulated, if the intervention is sterilized through open market operations to prevent inflation from rising. On the other hand, this is costly, since the sterilization is usually done by public debt instruments (in some countries Central Banks are not allowed to emit debt by themselves). In practice, few central banks or currency regimes operate on such a simplistic level, and numerous other factors (domestic demand, production and productivity, imports and exports, relative prices of goods and services, etc.) will affect the eventual outcome. Besides that, the hypothesis that the world economy operates under perfect capital mobility is clearly flawed.

As a consequence, even those central banks that strictly limit foreign exchange interventions often recognize that currency markets can be volatile and may intervene to counter disruptive short-term movements (that may include speculative attacks). Thus, intervention does not mean that they are defending a specific exchange rate level. Hence, the higher the reserves, the higher is the capacity of the central bank to smooth the volatility of the Balance of Payments and assure consumption smoothing in the long term.

INTERNATIONAL MONETARY FUND AND THE WORLD BANK

The **International Monetary Fund** (**IMF**) is an international organization headquartered in Washington, D.C., of "188 countries working to foster global monetary cooperation, secure financial stability, facilitate international trade, promote high employment and sustainable economic growth, and reduce poverty around the world." Formed in 1944 at the Bretton Woods Conference, it came into formal existence in 1945 with 29 member countries and the goal of reconstructing the international payment system. Countries contribute funds to a pool through a quota system from

which countries experiencing balance of payments difficulties can borrow money. As of 2010, the fund had SDR476.8 billion, about US$755.7 billion at then-current exchange rates.

Through the fund, and other activities such as statistics-keeping and analysis, surveillance of its members' economies and the demand for self-correcting policies, the IMF works to improve the economies of its member countries. The organization's objectives stated in the Articles of Agreement are: to promote international monetary cooperation, international trade, high employment, exchange-rate stability, sustainable economic growth, and making resources available to member countries in financial difficulty.

Functions

According to the IMF itself, it works to foster global growth and economic stability by providing policy, advice and financing to members, by working with developing nations to help them achieve macroeconomic stability and reduce poverty. The rationale for this is that private international capital markets function imperfectly and many countries have limited access to financial markets. Such market imperfections, together with balance-of-payments financing, provide the justification for official financing, without which many countries could only correct large external payment imbalances through measures with adverse economic consequences. The IMF provides alternate sources of financing.

Upon the founding of the IMF, its three primary functions were: to oversee the fixed exchange rate arrangements between countries, thus helping national governments manage their exchange rates and allowing these governments to prioritise economic growth, and to provide short-term capital to aid balance of payments. This assistance was meant to prevent the spread of international economic crises. The IMF was also intended to help mend the pieces of the international economy after the Great Depression and World War II. As well, to provide capital investments for economic growth and projects such as infrastructure.

The IMF's role was fundamentally altered by the floating exchange rates post-1971. It shifted to examining the economic policies of countries with IMF loan agreements to determine if a shortage of capital was due to economic fluctuations or economic policy. The IMF also researched what types of government policy would ensure economic recovery. The new challenge is to promote and implement policy that reduces the frequency of crises among the emerging market countries, especially the middle-income countries that are vulnerable to massive capital outflows. Rather than maintaining a position of oversight of only exchange rates, their function became one of surveillance of the overall macroeconomic performance of member countries. Their role became a lot more active because the IMF now manages economic policy rather than just exchange rates.

In addition, the IMF negotiates conditions on lending and loans under their policy of conditionality, which was established in the 1950s.[] Low-income countries can borrow on concessional terms, which means there is a period of time with no interest rates, through the Extended Credit Facility (ECF), the Standby Credit Facility (SCF) and the Rapid Credit Facility (RCF). Nonconcessional loans, which include interest rates, are provided mainly through Stand-By Arrangements (SBA), the Flexible Credit Line (FCL), the Precautionary and Liquidity Line (PLL), and the Extended Fund Facility. The IMF provides emergency assistance via the Rapid Financing Instrument (RFI) to members facing urgent balance-of-payments needs

WORLD BANK

The **World Bank** is an international financial institution that provides loans to developing countries for capital programs. It comprises two institutions: the International Bank for Reconstruction and Development (IBRD), and the International Development Association (IDA). The World Bank is a component of the **World Bank Group**, which is part of the United Nations system.

The World Bank's official goal is the reduction of poverty. However, according to its Articles of Agreement, all its decisions must be guided by a commitment to the promotion of foreign investment and international trade and to the facilitation of Capital investment.

Objectives:

The following objectives are assigned by the World Bank:

1. To provide long-run capital to member countries for economic reconstruction and development.
2. To induce long-run capital investment for assuring Balance of Payments (BoP) equilibrium and balanced development of international trade.
3. To provide guarantee for loans granted to small and large units and other projects of member countries.
4. To ensure the implementation of development projects so as to bring about a smooth transference from a war-time to peace economy.
5. To promote capital investment in member countries by the following ways;
   (a) To provide guarantee on private loans or capital investment.
   (b) If private capital is not available even after providing guarantee, then IBRD provides loans for productive activities on considerate conditions.

Functions:

World Bank is playing main role of providing loans for development works to member countries, especially to underdeveloped countries. The World Bank provides long-term loans for various development projects of 5 to 20 years duration.

The main functions can be explained with the help of the following points:

1. World Bank provides various technical services to the member countries. For this purpose, the Bank has established “The Economic Development Institute” and a Staff College in Washington.
2. Bank can grant loans to a member country up to 20% of its share in the paid-up capital.
3. The quantities of loans, interest rate and terms and conditions are determined by the Bank itself.
4. Generally, Bank grants loans for a particular project duly submitted to the Bank by the member country.
5. The debtor nation has to repay either in reserve currencies or in the currency in which the loan was sanctioned.
6. Bank also provides loan to private investors belonging to member countries on its own guarantee, but for this loan private investors have to seek prior permission from those counties where this amount will be collected.

NATIONAL DEBT MANAGEMENT, CAUSES AND INTERVENTIONS

External debt refers to debt owing by one country to another. External debt is a more serious problem than internal debt because the payment of interest and repayment of the capital sum form debit items in the balance of payments.

The cause of third world debt is the unwise borrowing and lending during the 1970s. The oil crisis made conditions extremely difficult for many third world countries. For the same reason the Euro currency markets were awash with money and real interest rates were low. Therefore poor countries need to borrow and banks were anxious to lend where they could get better return than on the domestic market. The bulk of this lending was by commercial banks and not by international institutions such as the IMF and World Bank. This has the consequence that, while the international institutions could write-off the debt, it is very difficult for commercial ones to do so.

The problem turned into a crisis in the early 1980s. As the world fell into recession this hit the debtor nations particularly hard. Many developing economies are highly dependent on the export of primary products and the price of these dropped dramatically. At the same time real interest rates rose sharply and most of the debt was at variable interest. The debtor nations were caught in a vice between falling income and rising costs. When this happened the banks which had been so anxious to lend in the 1970s were no longer willing to do so.

This meant that the debtor nations had to turn to international agencies for help. However, the international agencies do not have sufficient funds to substantially affect the situation. For such help as they were able to give, they demanded very stringent conditions. These deflationary conditions have often impoverished further the debtor nations.

Much of the money that was borrowed was not used for development purposes but simply balance the books for the nations' overseas payments. Little found its way into the sort of projects which development economies would suggest.

Many measures have been suggested for instance by the World Bank. Among them has been suggestion for reducing interest rates on non-concessional debt, rescheduling with longer grace periods and maturities or outright conversion of bilateral loans to grants. International

Development Association (IDA) has converted its repayments to be used to reduce International Bank for Reconstruction and Development (IBRD) debt owed by low income third world countries.

CAUSES OF NATIONAL DEBT

Increased Government Spending

- Increased government spending is the single most important cause of public debt. For years, a debate has raged over whether tax cuts or increased government spending are more at fault for rising public debt. According to the Cato Institute, federal expenditures have increased by $800 billion between 2001 and 2006 while federal revenue has decreased by only $200 billion as a direct result of the 2001 and 2003 federal tax cuts. This 400 percent increase as a result of increased government spending is proof that increased government spending is more important to rising public debt than are tax cuts.

Increased Debt Ceilings

- The United States Congress has the ability to raise the federal debt ceiling, or the amount of debt that the U.S. government can carry at any given time. (See Reference 2.) In January 2010, Congress raised the debt ceiling from $12.4 trillion to $14.3 trillion, an increase of $1.9 trillion (15.3 percent). (See Reference 3.) This ability to raise the debt ceiling, in a seemingly arbitrary fashion, is an enabler that allows the government to increase its spending, which is the single most important factor in public debt.

Increased Budget Deficits

- The federal government is not bound by the Constitution to have a balanced budget. The 1974 Budget Reform and Impoundment Control Act was originally created to control deficit spending. In 1974, when the act was enacted, the U.S. budget deficit was only 0.41 percent of the U.S. Gross Domestic Product (GDP). In the 2010 federal budget, the budget deficit is projected to be 10.64 percent of the U.S. GDP. (See Reference 4.) When budget deficits occur, the federal government borrows, or adds to the public debt, to bridge the gap between revenues and expenditures that is created by increased government spending.

INTERVENTIONS TO NATIONAL DEBT

Interest Rate Manipulation

maintaining low interest rates is another way governments seek to stimulate the economy, generate tax revenue and, ultimately, reduce the national debt. Low interest rates make it easy for individuals and businesses to borrow money. In turn, the borrowers spend that money on goods and services, which creates jobs and tax revenues. Low interest rates have been employed by the Unites States, the European Union, the United Kingdom and other nations with some degree of success. That noted, interest rates kept at or near zero for extended periods of time have not proved to be a panacea for debt-ridden governments.

Spending Cuts

Canada faced a nearly double-digit budget deficit in the 1990s. By instituting deep budget cuts (20% or more within four years), the nation reduced its budget deficit to zero within three years and cut its public debt by one-third within five years. The country did this without raising taxes.

In theory, others countries could emulate this example. In reality, the beneficiaries of tax-payer fueled spending often balk at proposed cuts. Politicians are voted out of office when their constituents are angry, so they often lack the political will to make necessary cuts. Decades of political wrangling over the Social Security program in the United States is a prime example of this, with politicians avoiding action that would anger voters. In extreme cases, such as Greece in 2011, protesters take to the streets when then the government spigot is turned off.

Raise Taxes
Tax increases are a common tactic. Despite the frequency of the practice, most nations face large and growing debts. It is likely that this is largely due to the failure to cut spending. When cash flows increase and spending continues to rise, the increased revenues make little difference to the overall debt level.

Cut Spending and Raise Taxes
Sweden was close to financial ruin by 1994. By the late 90s the country had a balanced budget through a combination of spending cuts and tax increases. U.S. debt was paid down in 1947, 1948 and 1951 by Harry Truman. President Dwight D. Eisenhower managed to reduce government debt in 1956 and 1957. Spending cuts and tax increases play a role in both efforts.

Pro Business/Pro Trade
A pro business, pro trade approach is another way nations can reduce their debt burdens. Saudi Arabia reduced its debt burden from 80% of Gross Domestic Product in 2003 to just 10.2% in 2010 by selling oil.

Bailout
Getting rich nations to forgive your debts or hand you cash is a strategy that has been employed more than a few times. Many nations in Africa have been the beneficiaries of debt forgiveness. Unfortunately, even this strategy has its faults. For example, in the late 1980s Ghana's debt burden was significantly reduced by debt forgiveness. In 2011 the country is once again deeply in debt. Greece, which had been given billions of dollars in bailout funds in 2010-2011, was not much better after the initial rounds of cash infusions. (U.S. bailouts date all the way back to 1792. Learn how the biggest ones affected the economy.

STRUCTURAL ADJUSTMENT PROGRAMMES (SAPS) AND THEIR IMPACT ON LDCS

Structural Adjustment Programmes (SAPs) are economic policies for developing countries that have been promoted by the World Bank and International Monetary Fund (IMF) since the early 1980s by the provision of loans conditional on the adoption of such policies. Structural adjustment loans are loans made by the World Bank. They are designed to encourage the structural adjustment of an economy by, for example, removing “excess” government controls and promoting market competition as part of the neo-liberal agenda followed by the Bank. The Enhanced Structural Adjustment Facility is an IMF financing mechanism to support of macroeconomic policies and SAPs in low-income countries through loans or low interest subsidies.

SAPs policies reflect the neo-liberal ideology that drives globalization. They aim to achieve long-term or accelerated economic growth in poorer countries by restructuring the economy and reducing government intervention. SAPs policies include currency devaluation, managed balance of payments, reduction of government services through public spending cuts/budget deficit cuts,

reducing tax on high earners, reducing inflation, wage suppression, privatization, lower tariffs on imports and tighter monetary policy, increased free trade, cuts in social spending, and business deregulation. Governments are also encouraged or forced to reduce their role in the economy by privatizing state-owned industries, including the health sector, and opening up their economies to foreign competition.

One important criticism of SAPs, which emerged shortly after they were first adopted and has continued since, concerns their impact on the social sector. In health, SAPs affect both the supply of health services (by insisting on cuts in health spending) and the demand for health services (by reducing household income, thus leaving people with less money for health). Studies have shown that SAPs policies have slowed down improvements in, or worsened, the health status of people in countries implementing them. The results reported include worse nutritional status of children, increased incidence of infectious diseases, and higher infant and maternal mortality rates.

An acceptance of the problems associated with SAPs is reflected in the post-Washington consensus and similar thinking that promotes economic reform while protecting and possibly increasing social expenditure.

Stabilization measures are closely related to SAPs policies and are taken to reduce the national rate of inflation. They usually include cutting the budget deficit and improving the balance of payments. Such measures are often made a condition for HIPC countries to obtain fresh loans or further assistance and are therefore a part of the conditionality clause.

Effects on LDCs

Structural Adjustment Policies are economic policies which countries must follow in order to qualify for new World Bank and International Monetary Fund (IMF) loans and help them make debt repayments on the older debts owed to commercial banks, governments and the World Bank. Although SAPs are designed for individual countries but have common guiding principles and features which include export-led growth; privatisation and liberalisation; and the efficiency of the free market.

SAPs generally require countries to devalue their currencies against the dollar; lift import and export restrictions; balance their budgets and not overspend; and remove price controls and state subsidies.

Devaluation makes their goods cheaper for foreigners to buy and theoretically makes foreign imports more expensive. In principle it should make the country wary of buying expensive foreign equipment. In practice, however, the IMF actually disrupts this by rewarding the country with a large foreign currency loan that encourages it to purchase imports.

Balancing national budgets can be done by raising taxes, which the IMF frowns upon, or by cutting government spending, which it definitely recommends. As a result, SAPs often result in deep cuts in programmes like education, health and social care, and the removal of subsidies designed to control the price of basics such as food and milk. So SAPs hurt the poor most, because they depend heavily on these services and subsidies.

SAPs encourage countries to focus on the production and export of primary commodities such as cocoa and coffee to earn foreign exchange. But these commodities have notoriously erratic prices subject to the whims of global markets which can depress prices just when countries have invested in these so-called 'cash crops'.

By devaluing the currency and simultaneously removing price controls, the immediate effect of a SAP is generally to hike prices up three or four times, increasing poverty to such an extent that riots are a frequent result.

The term "Structural Adjustment Program" has gained such a negative connotation that the World Bank and IMF launched a new initiative, the Poverty Reduction Strategy Initiative, and makes countries develop *Poverty Reduction Strategy Papers* (PRSP). While the name has changed, with PRSPs, the World Bank is still forcing countries to adopt the same types of policies as SAPs.

QUESTIONS

1. What problems do African regional integration bodies /organizations face?

2. What factors affect the long-run trend of the terms of trade for developing countries?

CHAPTER 11

CURRENT DEVELOPMENTS

FACTORS AFFECTING ECONOMIC DEVELOPMENT

Inflation

Inflation affects both consumers and producers in the market, and poses a threat to the market stability. Throughout history, inflation has played a major role in the economy of nations. For example, after World War I, the German government printed a vast amount of money, which gave rise to major inflation in the economy of Germany. When people had money, but scarcity of products, the price of the products went sky high, as the value of money had gone down. Having plenty of money made the price go so high that people had to pay a large amount of money to by a simple loaf of bread. Currently, many nations of the world are facing similar financial crisis and economic downward trend, giving rise to a credit crunch.

When inflation affects the economy of a nation, you have to pay more money to maintain the same standards of living, and to purchase the same amount of goods and services, due to a rise in prices.

Interest rates

Interest rates can have a major impact on the growth of economy, especially for industries such as real estate, automobiles, and cruise companies. A larger interest rate would discourage customers from burrowing money and purchasing these products or services. It also discourages companies form making expansions, investing in new areas, making new capital investments, and starting new businesses.

Value of the currency

The value of the U.S dollar compared to other foreign currencies is important for all businesses, even if they do not engage in import or export with other countries. Just like people and businesses, the Government of the United States also has loans and must pay interest on them. The United States often prints more money than it actually has, to give the economy a boost. This can lower the value of the dollar. A customer can choose to buy goods and services that originated in the United States, or any other country. If the U.S dollar has more value, or purchasing power, the companies that are reliant and heavily depend on other countries for its inputs, such as raw materials or processing goods, can be more competitive in the market.

Environmental impacts

Environmental impacts have a profound effect on the growth of an industry in today's economy. It is not only the environmental effects, but also the perception of the general public that impacts businesses and the economy. For example, the demand for fur coats declined drastically in the past decade, as consumers became more aware of animal welfare and ecological balance. If the public perceives that the products or services of an industry as being environmentally unsafe or harmful, most companies within the sector will experience a decline in sales.

Government Intervention

Many industries are monitored and regulated by government agencies such as the **Environmental Protection Agency (EPA), U.S. Department of Agriculture (USDA),** and the **Food and Drug Administration (FDA),** to maintain standards of all operators in each industry. All companies must follow the government safety rules for employees, consumers, the environment, and natural resources, as per specifications and requirements of these agencies.

Economic development of an industry is also influenced by the confidence of general people on the industry, the economic state of a nation, and involvement of government in the industry. The real development cannot be achieved by depleting environment, or at the cost of our health, which can in the long run, backfire on us, and lead to economic and social disasters, creating negative impacts for general population around the globe.

INFORMAL CREDIT MARKETS

Many developing countries, despite double digit growth rates, have lagged far behind in terms of quality of living in urban areas. Problems of unemployment, inadequate housing, poor environmental quality, rampant crime, poverty, transportation etc. have plagued urban residents and decision makers alike. While the attempts of governmental agencies have been commendable, shear scale of the problem has diluted such efforts, resulting in the benefits being spread thinly among a large population, or not benefiting most of them.

Low income households in developing countries are seen as particularly vulnerable. Their personal problems of low education and skill levels, low incomes, lack of marketable assets, and uncertain job markets have been compounded by external factors that have failed to provide adequate infrastructure and social services that would have enabled them to participate in mainstream economic activities. As a result, this has affected every facet of their life: employment (predominantly in the informal sector), education (non-existent or up to primary levels only), health (low quality or traditional), housing (impermanent materials and illegal settlements) etc.

Often the problems of low-income households can be traced back to a single factor which, on a cumulative basis, has affected other factors, resulting in their present condition. For low income households, one such cumulating factor is credit, or the lack of access to adequate bank credit. This lack of access has resulted in households turning to informal credit markets (ICMs) to satisfy their need, where elaborate networks of credit delivery have been set up by different kinds of informal suppliers. Examples include money lenders, pawn brokers, ROSCAs, credit societies/unions etc.

With governmental action in ensuring an enabling policy/programme environment not forthcoming or insufficient, particularly with innovative financing mechanisms, there is a need to understand the systems of credit delivery that the low-income households themselves have established to fill the gap. This financial dualism of "formal" and "informal" is a result of both the shortcomings of the formal sector and the intrinsic condition and position of the urban low income households themselves [Germidis et al., 1991]. Delivery of credit for various purposes should be ensured not only by increased financial services of banks and other financial institutions, but more importantly, by also taking into account the existing systems of credit delivery already in place in the settlements. This calls for a better understanding of the systems of credit developed by the urban low income

households, the advantages it offers, and implications of such systems for devising appropriate microcredit programmes for their needs.

The material presented in this paper is part of a larger survey of low-income households and informal credit suppliers carried out in AprilCJune 1995 in Bangalore, India [Srinivas, 1996]. Information gleaned from interviews with settlements leaders, NGOs, traders and shopkeepers, social workers etc. was also used to substantiate the data collected. The ICM is vast, covering different types of suppliers providing credit for different purposes. However, the survey was limited to credit activities that were taking place within a settlement or community that serviced low-income households.

What is informal credit?

The line dividing "informal" credit and "formal" credit is very fine, and credit suppliers on both sides actually form a continuum that range from very informal suppliers (like friends, relatives neighbours etc.) to fully formal suppliers (banks and specialized financial institutions). Ranged in between these extremes are rotating savings and credit associations (ROSCAs), money lenders, pawn brokers, credit societies, traders etc. [Srinivas and Higuchi, 1996]. They mobilize savings and provide credit, or as in the case of ROSCAs, do both. The ICM is highly heterogenous in nature. It encompasses short term flows in funds markets, lending and borrowing of small amounts among group members, friends or relatives. It includes activities of finance companies, as well as those of pawnshops and money lenders. It runs the gamut from large wholesale traders financing trade, to village level crop buyers, to petty shopkeepers and itinerant peddlers. It also includes organized credit unions, ROSCAs (called 'chit funds' in southern India), credit societies etc. With advantages of unregulated money supply, easy accessibility, easy liquidity, low 'administrative' and procedural costs, little or no collateral/mortgage requirements, flexibility in interest rates and repayment schedules, ICMs are ideally suited to cater to the lower income groups [ADB, 1990].

Three types of informal credit suppliers can be observed in the ICM. The "transactional" credit suppliers those for whom credit is a business transaction. Examples include, pawn brokers, money lenders, traders, employers etc. The "mutual" credit suppliersCthose for whom credit supply is a give and take process. Examples include, chit funds/ROSCAs, credit unions, credit societies, people's organizations/self help groups. The "personal" credit suppliers those for whom credit supply is a private arrangement that may be reciprocated in the future. Examples include friends, relatives, co workers and neighbours [Srinivas and Igel, 1995]. Ranged along with formal commercial banks and financial institutions, transactional, mutual and personal suppliers would form a continuum. This would enable the positioning of 'fully informal' personal suppliers and the 'fully formal' banks at either end of the continuum, with the 'semi-formal' and 'semi-informal' transactional and mutual suppliers in between [Srinivas and Higuchi, 1996]. (While the definition of an NGO is broad and covers a range of organizations and institutions, it was not included here since it was observed that NGOs primarily function as catalysts, intermediaries and facilitators - aiming for a greater degree of community managed microcredit programmes).

A common thread that runs through all of them is their informality, adaptability and flexibility of operations features that reduce their transaction costs and confers upon them their comparative advantage [Ghate, 1990]. The following aspects characterize the ICM:

- Scale of operations is small, because lending is primarily based on personal knowledge of the borrower. Loans are small, repaid quickly, and are largely unsecured.

- Individual lenders operate in a sp
- ecialized and circumscribed area, or in a specific niche of the market.
- Due to the small scale of operations, intra agency problems typical of large organizations like banks are not seen in the ICM.
- Suppliers function outside the purview of regulations imposed in the formal sector in respect of capital, reserve and liquidity requirements, ceilings on lending and deposit rates, mandatory credit targets, audit and reporting requirements and so on.

If the features of ICMs mentioned above are taken into account, then an informal credit operation can be operationally defined as one that is not regulated/audited by the Federal or Central Bank, has flexible repayment schedules (in terms of periods and amounts), flexible interest rates, and demand little or no collateral. Several other criteria, such as decision making processes, proprietorship or area of operation, can be included to adopt the definition to specific local conditions. This is also important due to the vast variation in ICMs seen across the countries in the developing world [Srinivas and Igel, 1996].

Credit Demand and Supply in the Informal Sector

Stringent requirements for loans have systematically kept low income households out of the credit delivery network of commercial banks. Bank managers interviewed during field research revealed that banks required borrowers to have a job in the formal sector, with "steady" employment and an incomes at "adequate" levels, whereas most of the households surveyed had informal sector jobs (83.7 percent) or transient jobs in construction sites. This kept their incomes very irregular and uncertain. Banks required that only borrowers with valid land ownership or ownership certificates were to be eligible for loans but most of the low income households lived as squatters and did not own the land (100 percent of the sample). Banks requested collateral for loans, usually in the form of marketable assets, for example, the house to be built or the land itself. Mean household income of the samples studied was about Rs. 1000 per month (Rs. 40 = US$ 1, 1991). They owned very few assets, and their savings by themselves was insufficient to act as collateral. Most bank loans were large, for lump-sum/one-time investments, whereas the surveyed households usually staggered expenditures over a period of five to eight years.

Such conditions invariably led urban low income households to depend on ICMs to satisfy their credit needs. In the informal sector, income levels, land ownership and job status were not important for obtaining loans from the ICM. Loans were made on a face to face basis with personal knowledge and close proximity of the lenders and borrowers. Practically no collateral was demanded during such transactions (except, of course, in the case of pawn brokers, who lent amounts based on the value of items pawned by the borrower). Close acquaintance between the lender and borrower also ensured that loans were available quickly without any need for elaborate paperwork and costing little in terms of administration and information. Most of the loans were small, to finance only a particular part or fraction of the overall investment process. Expenditures were staggered and spread over a long period of time (from five to eight years) so that burdens of immediate loan repayment or interest payments were low and repaid quickly (less than eight weeks). By providing loans of small size (Rs. 2,000 to Rs.5,000), informal suppliers satisfied their credit needs. Besides, the long held belief of usurious practices in ICM can be put to question since, despite "high interest rates" on an annual basis, actual interest payment burdens are low, since loans are small in size and are repaid in a very short period.

Informal suppliers interviewed in the study did not receive any support from the government or financial institutions. They used their own resources and profits generated by increased circulation of loan funds (for money lenders, pawn brokers, employers) or used the savings of group members to lend (ROSCAs, credit societies etc.). Since their funds were not insured or were not secured by collateral in any way, risk of default was high. But close links and personal relationships were maintained between borrower and lender. This meant that there were few defaulting borrowers observed in the study (less than 2.3 percent of borrowers).

Dispensing with the elaborate requirements and paperwork of conventional banking, informal suppliers have devised their own methodologies which have worked to suit the conditions and positions of its low income clientele. Small scales of operation mean that informal suppliers function without the inter and intra agency support typical of formal financial institutions. However, a fundamental disadvantage of informal credit observed was that large capital investments or other such lump-sum investments were difficult to be made, affecting as a result, expansion of production, higher income generation or quality/quantity improvement.

DEVELOPMENT INDEX

The DI was created to emphasize that people and their capabilities should be the ultimate criteria for assessing the development of a country, not economic growth alone. The DI can also be used to question national policy choices, asking how two countries with the same level of GNI per capita can end up with different human development outcomes. These contrasts can stimulate debate about government policy priorities.

The Development Index (DI) is a summary measure of average achievement in key dimensions of human development: a long and healthy life, being knowledgeable and have a decent standard of living. The DI is the geometric mean of normalized indices for each of the three dimensions.

The health dimension is assessed by life expectancy at birth, the education dimension is measured by mean of years of schooling for adults aged 25 years and more and expected years of schooling for children of school entering age. The standard of living dimension is measured by gross national income per capita. The DI uses the logarithm of income, to reflect the diminishing importance of income with increasing GNI. The scores for the three DI dimension indices are then aggregated into a composite index using geometric mean. Refer to Technical notes for more details.

The DI simplifies and captures only part of what human development entails. It does not reflect on inequalities, poverty, human security, empowerment, etc. The HDRO offers the other composite indices as broader proxy on some of the key issues of human development, inequality, gender disparity and human poverty.

A fuller picture of a country's level of human development requires analysis of other indicators and information presented in the statistical annex of the report.

GROWTH OF MARKET STRUCTURES

An increase in the demand for a particular product or service over time. growth of market can be slow if consumers do not adopt a high demand or rapid if consumers find the product or service useful for the price level. For example, a new technology might only be marketable to a small set of consumers, but as the price of the technology decreases and its usefulness in everyday life increases, more consumers could increase demand

VOTING BEHAVIOUR

The most interesting questions about an election are not concerned with who won but with why people voted the way that they did or what the implications of the results are. These questions are not always easily answered. Looking only at the campaign events and incidents will not suffice. The unique aspects of the election must be blended with a more general understanding of electoral behavior to create a full explanation. We thus need to discuss basic concepts and ideas used in the study of voting behavior as a basis for analyzing the 2012 results.

Two major concerns characterize the study of electoral behavior. One concern is with explaining the election result by identifying the sources of individual voting behavior. We attempt to understand the election outcome by understanding how and why the voters made up their minds. Another major concern in voting research emphasizes changes in voting patterns over time, usually with an attempt to determine what the election results tell us about the direction in which American politics is moving. In this case we focus on the dynamics of electoral behavior, especially in terms of present and future developments. These two concerns are complementary, but they do emphasize different sets of research questions. For our purposes, these two concerns provide a useful basis for discussing key aspects of voting behavior.

Sources of Individual Voting Behavior

On what basis do voters decide how they will cast their ballot? Several basic factors can be identified as reasons for choosing a candidate in an presidential election. A voter may choose a candidate on the basis of one or more of the following considerations:

- Orientations on public policy issues
- Evaluations of government performance
- Evaluations of candidate characteristics

When voters are asked what they like or dislike about a specific candidate — such as what might make them vote for or against that candidate — most of their responses fall into one of the above three categories.
These orientations and evaluations in turn are influenced by two more general attitudinal factors:

- Party identification
- General ideological dispositions

Party identification and ideology are more general, long-run factors that influence voting behavior primarily by affecting the attitudes that are more immediate to the vote decision in a particular year.

The various factors that influence the vote decision vary in their stability over time. Evaluations of candidate qualities and government performance are distinctly short-term forces, capable of substantial shifts from one election to the next. Party identification and ideology are much more stable in the short term. Not many voters change their party identification or ideology from one election to the next, and the changes that do occur often are small ones. Issue orientations fall in between. While the specific issues crucial in presidential elections can change dramatically, as can how the voters evaluate the presidential candidates on the issues, many basic policy questions (e.g., defense spending, social welfare programs, abortion) stretch across several elections, with partisan differences remaining relatively constant.

The various attitudes and orientations that influence voting behavior in presidential elections are interrelated. Understanding the interrelationships among attitudes and orientations is important for a full understanding of voting behavior.

The attitudes and orientations that affect voting behavior are related to a number of social and demographic characteristics. Thus, social groups differ in their voting patterns. Some of these differences have existed for decades, but others represent more recent developments. The most important social and demographic factors in recent elections have been:

- **Race and ethnicity**. Minority groups, especially Blacks, are more Democratic in their voting than are Whites.
- **Social class or socio-economic status**. Those who are better off in income or socio-economic status are more Republican than are those who are worse off.
- **Religion**. Those who are more religious are more Republican than those who are less religious. In the past, White Catholics and Protestants differed considerably in their voting, but that distinction has declined in significance.
- **Region**. Voters in the South, Great Plains, and Rocky Mountains regions are more Republican, while those in the Northeast and on the Pacific Coast are more Democratic.
- **Gender**. Women are more Democratic than are men.
- **Marital status**. Married individuals are more Republican than are single individuals.
- **Age**. Younger voters are more Democratic than are older voters.

TECHNOLOGY TRANSFER

Technology transfer, also called **transfer of technology (TOT)**, is the process of transferring skills, knowledge, technologies, methods of manufacturing, samples of manufacturing and facilities among governments or universities and other institutions to ensure that scientific and technological developments are accessible to a wider range of users who can then further develop and exploit the technology into new products, processes, applications, materials or services. It is closely related to (and may arguably be considered a subset of) knowledge transfer. Horizontal transfer is the movement of technologies from one area to another. At present transfer of technology (TOT) is primarily horizontal. Vertical transfer occurs when technologies are moved from applied research centers to research and development departments.

Technology transfer is promoted at conferences organized by such groups as the Ewing Marion Kauffman Foundation and the Association of University Technology Managers, and at "challenge" competitions by organizations such as the Center for Advancing Innovation in Maryland. Local

venture capital organizations such as the Mid-Atlantic Venture Association (MAVA) also sponsor conferences at which investors assess the potential for commercialization of technology.

Transfer process

Many companies, universities and governmental organizations now have an Office of Technology Transfer (TTO, also known as "Tech Transfer" or "TechXfer") dedicated to identifying research which has potential commercial interest and strategies for how to exploit it. For instance, a research result may be of scientific and commercial interest, but patents are normally only issued for practical processes, and so someone—not necessarily the researchers—must come up with a specific practical process. Another consideration is commercial value; for example, while there are many ways to accomplish nuclear fusion, the ones of commercial value are those that generate more energy than they require to operate.

The process to commercially exploit research varies widely. It can involve licensing agreements or setting up joint ventures and partnerships to share both the risks and rewards of bringing new technologies to market. Other corporate vehicles, e.g. spin-outs, are used where the host organization does not have the necessary will, resources or skills to develop a new technology. Often these approaches are associated with raising of venture capital (VC) as a means of funding the development process, a practice more common in the United States than in the European Union, which has a more conservative approach to VC funding. Research spin-off companies are a popular vehicle of commercialisation in Canada, where the rate of licensing of Canadian university research remains far below that of the US.

Technology transfer offices may work on behalf of research institutions, governments and even large multinationals. Where start-ups and spin-outs are the clients, commercial fees are sometimes waived in lieu of an equity stake in the business. As a result of the potential complexity of the technology transfer process, technology transfer organizations are often multidisciplinary, including economists, engineers, lawyers, marketers and scientists. The dynamics of the technology transfer process has attracted attention in its own right, and there are several dedicated societies and journals.

There has been a marked increase in technology transfer intermediaries specialized in their field since 1980, stimulated in large part by the Bayh-Dole Act and equivalent legislation in other countries, which provided additional incentives for research exploitation.

DEMOCRACY AND DEVELOPMENT

The causal link between democracy and development is a controversial issue. For most of the twentieth century, conventional wisdom has held that autocracies are better able to marshal the resources necessary to promote economic development than are democracies, and that a certain level of economic development is necessary for democracy to take hold and flourish in a country. That view deserves a new examination in the post-Cold War world. Morton H. Halperin, Joseph Siegle, and Michael Weinstein make such an examination in their upcoming book on the subject, and their results present some interesting implications.

When the countries of the world are examined as a whole, democracies do perform better in terms of economic development than do autocracies or mixed polities. The debate concerns developing countries – specifically, can democracies develop as low income countries; can poor countries

democratize; and does democracy among low income countries make any difference for their development success?

According to data collected and analyzed for the book, there has been no advantage for autocracies over democracies for the past 40 years in terms of development. Both developing country democracies and non-democracies have grown at approximately 1.5% of GDP per capita per year during that time. When East Asia is removed from that sample, democracies have actually performed better – growing at 0.5% per capita per year faster than autocracies and mixed polities. What is more, there is no data on about 25% of autocracies – countries such as North Korea, Iraq, Afghanistan (for most of its rule), and Cuba – so actual growth figures for autocracies would likely be substantially lower if the performance of these additional countries were included.

More than simply growing at a faster rate, democracies have outperformed autocracies in the consistency of their growth. An analysis of the 80 worst economic performers of the last 40 years reveals that all but three have been autocracies. In addition, democracies have performed substantially better than autocracies in the social welfare dimension of development (life expectancy, child mortality, literacy, etc.) – in some cases up to 50% better.

It is important to note that there has been a variety of development experiences among democracies. This can be attributed to the differing success with which any country can develop institutions of accountability – checks on the chief executive, separation of politics from the civil service, independence of the judiciary, freedom of the press, and independence of the private sector, for example – which are the foundations of democratic systems of governance. Both autocracies and democracies that have developed these institutions have had better rates of economic development than countries without well established institutions of accountability. For this reason, there are some autocracies that have performed better than some democracies; the East Asian "tigers" fall into this category.

Despite the overall positive performance of democracies over autocracies in development, there has been no preference given to democracies over the past 40 years in the dispersal of development assistance. The same has been true with debt relief, even though new democracies – countries that have recently transitioned from autocratic rule – have often inherited debt that is as high as 23% of GDP, a figure higher than that for most low income countries. This problem is compounded by the fact that the lack of a "democracy dividend" can undercut popular support for democracy. Indeed, 70% of countries which experienced democratic backtracking over the past 20 years faced periods of economic stagnancy before backsliding.

ENVIRONMENTAL CONCERNS

1. Air Pollution: Pollution of air, water and soil take a huge number of years to recover. Industry and engine vehicle fumes are the most obvious toxins. Substantial metals, nitrates and plastic are poisons in charge of pollution. While water contamination is brought about by oil slicks, acid rain, and urban sprawl; air contamination is created by different gasses and poisons discharged by businesses and manufacturing plants and burning of fossil fills; soil contamination is majorly created by mechanical waste that takes supplements out of the soil.

2. Water Pollution: Clean drinking water is turning into an uncommon thing. Water is turning into a monetary and political concern as the human populace battles for this need. Waste from industrial and agricultural activities pollute the water that is used by humans, animals and plants.

3: Soil and Land Pollution: Land pollution simply means degradation of earth's surface as a result of human activities like mining, littering, deforestation, industrial, construction and agricultural activities. Land pollution can have huge environmental impact in the form of air pollution and soil pollution which in turn can have adverse effect on human health.

4. Climate Change: Climate change is yet another environmental concern that has surfaced in last couple of decades. Environmental change has different destructive impacts that include, but are not limited to, the melting of polar ice, change in seasons, new sicknesses, and change in general climate situation.

5. Global Warming: Environmental asset abuse is also an important environmental concern. Fossil fuel utilization brings about discharge of greenhouse gasses, which causes environmental change. However, individuals are taking endeavors to move to renewable energy sources.

6. Deforestation: Our woodlands create new oxygen and additionally help in managing temperature and precipitation. At present, timberlands cover 30% of the area, but wooded areas are being lost on a regular basis because people are looking for homes, food, and materials. Deforestation is a huge problem and will just continue to get worse.

7. Increased Carbon Footprint: Temperature increases, like climate change, are the consequence of human practices, including the use of greenhouse gasses. When the atmosphere changes and the heat increases, it can cause a number of problems and start to destroy the world we live in.

8. Genetic Modification: Genetic modification utilizing biotechnology is called genetic engineering. Genetic engineering of food brings about expanded poisons and sicknesses as qualities from a hypersensitive plant can exchange to target plant. Some of these crops can even be a threat to the world around us, as animals start to ingest the unnatural chemicals and such.

9. Effect on Marine Life: The amount of carbon in the water and the atmosphere is continuing to be a problem in the world around us. The primary effect is on shellfish and microscopic fish, and it has similar effects to osteoporosis in humans.

10. Public Health Issues: The current environmental concerns represent a considerable measure of danger to well-being of people, and creatures. Dirty water is the greatest well-being danger of the world and poses a risk to the health and lifespan of people and animals.

11. Overpopulation: The number of inhabitants in the planet is arriving at unsustainable levels as it confronts deficiency of assets like water, fuel and food. Overpopulation is one of the most important environmental concerns.

12: Loss of Biodiversity: Biodiversity is yet another casualty due to the impact of human beings on the environment. It is the result of 3.5 billion years of evolution. Habitat destruction is a major cause for biodiversity loss. Habitat loss is caused by deforestation, overpopulation, pollution and global warming.